



**Verified Carbon
Standard**

CULTIVAR PROGRESSO REDD+ PROJECT



Project title	<i>CULTIVAR PROGRESSO REDD+ PROJECT</i>
Project ID	TBD
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Crediting period	<i>01-january-2024 to 01-january-2043</i>
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1 PROJECT DETAILS

1.1 Summary Description of the Project

NOVAAGRI, founded in 2008, is a visionary player in the Agribusiness sector, specializing in independent storage and logistical services for agricultural commodities in domestic and international markets. The company's strong commitment to the environment is evidenced by its aim for a 50% reduction in greenhouse gas emissions by 2030 and carbon neutrality by 2050, which is supported by its carbon projects, all of which allocate a portion of its financial returns to social actions in nearby communities.

A key example of NovaAgri's strategy is its collaboration with the Grupo Progresso farms for the CULTIVAR PROGRESSO REDD+ PROJECT, located in the Piauí State across the municipalities of Uruçuí, Sebastião Leal, Landri Sales and Antônio Almeida, covering 32.549,79 hectares of native forest situated legally in the Cerrado biome, a biodiversity hotspot¹ rich in carbon reservoirs.

The project centers on AUD (Avoiding Unplanned Deforestation) mechanisms, adhering to REDD+ principles and VERRA's VM0015 methodology. The baseline acknowledges the intense deforestation pressure in Piauí's ecosystems, induced by agents such as illegal land grabbers and small-scale farmers who rely heavily on deforestation as an agricultural practice (as detailed in Section 3.4 Baseline Scenario). Additional contributing factors include drivers such as the high market value of deforested lands in the state and the elevated prices of agricultural commodities (as detailed in Section 3.4), all factors compounded by widespread non-compliance with Brazilian environmental legislation.

To address these challenges, NovaAgri invests in activities such as: the cross-referencing of public national data, to ensure the project area respects socioenvironmental compliance²; advanced GIS technology with near real-time deforestation alerts and high-resolution satellite imagery; various fire prevention and control strategies and mechanisms; and the allocation of financial returns from carbon credits to social investments in local communities.

The project expects to avoid net emissions of 2,629,994 tCO₂e, which means an average of 43,833 tCO₂e per year and 25,339 hectares preserved during a horizon of 60 years. The total GHG emission reductions generated in the monitoring period is 31,376.

¹ Cerrado ecosystem profile, available on: <<https://www.cepf.net/sites/default/files/cerrado-ecosystem-profile-updated.pdf>>.

² NovaAgri employs a tool called LYRA, developed by a company within the Imagem Group - ESRI, to facilitate internal verification activities in potential client and partner areas. LYRA is a technology that utilizes remote sensing and data intelligence to establish traceability, conduct socio-environmental diagnostics of individuals, companies, and territories, agroclimatic monitoring, and territorial intelligence. The system is built on layers of public and private socio-environmental and agricultural risk, adding value and mitigating reputational and operational risks. It features a GIS screen for navigation, providing a user-friendly interface for exploring new territories, especially in terms of socio-environmental monitoring, by checking daily compliance-related aspects for those involved and the monitored agroclimatic conditions. More information on: <https://vegamonitoramento.com.br/lyra/>

1.2 Audit History

The development and monitoring of this VCS project is under validation and first verification simultaneously.

Table 1. Audit History of Validation and Verification

Audit type	Period	Program	Validation/verification body name	Number of years
Validation and verification	TBD	VCS	RINA Services S.p.A	-

1.3 Sectoral Scope and Project Type

Sectoral scope	14. Agriculture, Forestry, Land Use (AFOLU)
AFOLU project category ³	Reduced Emissions from Deforestation and Forest Degradation (REDD+)
Project activity type	Avoided Unplanned Deforestation (AUD)

1.4 Project Eligibility

1.4.1 General eligibility

The project was first submitted to the Verra pipeline on [TO BE INCLUDED], and advanced to the status of 'Under Validation' on [TO BE INCLUDED]. The project activity is included under the scope of the VCS Program.

Table 2. Project Activities' inclusion under the VCS Program Scope

VCS Program Scope	Applicable / included in the project activities	Not applicable
The seven Kyoto Protocol greenhouse gases.	The methodologies, rationales, calculations, and other considerations of the activities of this project follow the principles of the Kyoto GHG Protocol.	-
Ozone-depleting substances (ODS).	-	The ODS gases, such as CFCs and HCFCs are not applicable for this specific project once the emissions' characteristics are solely related to REDD+ AFOLU.

³ See Appendix 1 of the VCS Standard

Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	This project uses an eligible methodology under the VCS Program. In this case, version 1.2 of the Methodology VM0015 has been fully applied, including any referenced tools or modules.	-
Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	The project activities are referenced by international consensus of relevant GHG programs and agreements.	-
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	-	Although this project is not part of a Jurisdictional REDD+ program, this is aligned to the most recent criteria and trends for an integrated network of carbon initiatives, including the latest methodologies' updates from VERRA, and the recent evolution of an official Carbon Market in Brazil.

The project meets requirements related to the pipeline listing deadline, the opening meeting with the validation/verification body, and the validation deadline, in compliance with the requirements 3.8.2 and 3.8.4 of VCS Standard v4.7:

- Pipeline listing deadline: The project start date is January 01, 2024 and the public consultation on Verra's website⁴ started in [TO BE INCLUDED] and ended in [TO BE INCLUDED]. The project listing process is within the three years of the project start date.
- The open meeting of the validation/verification process is scheduled for March 10, 2025, after the start date of the Verra's public consultation.
- Validation deadline: The project will complete validation within five years of the project start date, until January 01, 2029.

As outlined in Section 3.1, the Cultivar Progresso REDD+ Project applied the Methodology for Avoided Unplanned Deforestation, v1.1 (VM0015)⁵ and it is eligible under the VCS Program. Other relevant eligibility information is described below:

- The proponent is aware of the requirements in the document 'Registration and Issuance Process v4.4'.
- The proponent has contacted and hired a qualified validation/verification body (VVB), which will perform rigorous assessments of VCS methodologies and methodology elements to confirm that they conform with VCS Program requirements and AFOLU-specific best practices.

⁴ Verra, available on: <<https://verra.org/>>.

⁵ Verra VM0015, available on: <<https://verra.org/methodologies/vm0015-methodology-for-avoided-unplanned-deforestation-v1.2/>>.

- The proponent has agreed alongside the VVB about the validation/verification deadlines.
- The project applies the latest version of the applicable methodology in all cases, ensuring compliance unless a grace period applies.
- The project updates to the latest version of the methodology when reassessing the baseline or renewing a crediting period.
- The project and its implementation activities do not violate any applicable law, regardless of whether the law is enforced.
- The project is not a fragmented part of a larger project or activity, as demonstrated by the documents provided to the VVB, and publicly on this PD.

1.4.2 AFOLU project eligibility

The project is eligible under the AFOLU-specific matters:

VCS Standard - Eligibility requirements	Project justification
<u>VCS Standard v4.7, 3.2.1</u> There are currently six AFOLU project categories eligible under the VCS Program, as defined in Appendix 1 Eligible AFOLU Project Categories below: afforestation, reforestation and revegetation (ARR), agricultural land management (ALM), improved forest management (IFM), reduced emissions from deforestation and degradation (REDD), avoided conversion of grasslands and shrublands (ACoGS), and wetland restoration and conservation (WRC). Further specification with respect to eligible activities which may be included within methodologies approved under the VCS Program can be found in the VCS Methodology Requirements.	The project falls under the Reduced Emissions from Deforestation and Degradation (REDD) category.
<u>VCS Standard v4.7, 3.2.2</u> Where projects are located within a jurisdiction covered by a jurisdictional REDD+ program, project proponents shall follow the requirements in this document and the requirements related to nested projects set out in the Jurisdictional and Nested REDD+ Requirements.	The project is not located within a jurisdiction covered by a jurisdictional REDD+ program.
<u>VCS Standard v4.7, 3.2.3</u> Where an implementation partner is acting in partnership with the project proponent, the implementation partner shall be identified in the project description. The implementation partner shall identify its roles and responsibilities with respect to the project, including but not limited to implementation, management, and monitoring of the project, over the project crediting period.	Sections 1.6 and 1.7 identify the implementation partners involved in collaboration with the project proponent. Throughout the PD, their roles and responsibilities are delineated, encompassing implementation, management, and monitoring of the project throughout the project crediting period.

VCS Standard - Eligibility requirements	Project justification
<u>VCS Standard v4.7, 3.2.4</u> The project proponent shall demonstrate that project activities that lead to the intended GHG benefit have been implemented during each verification period in accordance with the project design. Where no new project activities have been implemented during a verification period, project proponents shall demonstrate that previously implemented project activities continued to be implemented during the verification period (e.g., forest patrols or improved agricultural practices of community members).	The project demonstrates that the implementation of activities leads to the intended GHG benefit during each verification period in alignment with the project design.
<u>VCS Standard v4.7, 3.2.5</u> For all Avoiding Unplanned Deforestation and Degradation (AUDD) (where the baseline is not allocated from jurisdictional level data), APD (where the agent is unknown), Avoiding Unplanned Conversion (AUC), and Avoiding Unplanned Wetland Degradation (AUWD) project types, the project proponent shall, for the duration of the project, reassess the baseline every six years. For Avoiding Unplanned Deforestation projects where the baseline is allocated from jurisdictional level data, the initial project baseline validity period may be as little as one year but no more than seven years. The second and subsequent project baseline validity periods shall be no more than six years. The project proponent shall reassess the baseline at the end of each baseline validity period.	The project commits to reassessing the baseline every six years. The baseline reassessment will follow specific guidelines, including the application of the latest VCS Program rules and methodologies, validation during subsequent verifications, capturing changes in drivers, and updating relevant sections of the project description to reflect changes in baseline emissions quantifications.
<u>VCS Standard v4.7, 3.2.8</u> Where ARR, ALM, IFM or REDD project activities occur on wetlands, the project shall adhere to both the respective project category requirements and the WRC requirements, unless the expected emissions from the soil organic carbon pool or change in the soil organic carbon pool in the project scenario is deemed below <i>de minimis</i> or can be conservatively excluded as set out in the <i>VCS Methodology Requirements</i>, in which case the project shall not be subject to the WRC requirements.	The project activities do not occur on wetlands, as detailed in Section 1.14.9.
<u>VCS Standard v4.7, Appendix 1, A1.5</u> Eligible REDD activities are those that reduce net GHG emissions by reducing deforestation and/or degradation of forests. Deforestation is the direct, human-induced conversion of forest land to non-forest land. Degradation is the persistent reduction of canopy cover and/or carbon stocks in a forest due to human activities such as animal grazing, fuelwood extraction, timber removal or other such activities, but which does not result in the conversion of forest to non-forest land (which would be classified as deforestation), and qualifies as	This project area consists exclusively of land that has met the criteria for being classified as a "forest" for a minimum of 10 years before the project's initiation (January 01, 2024), as demonstrated in the Land Use and Cover map between the years of 2012 and 2023 (Section 1.14.3).

VCS Standard - Eligibility requirements	Project justification
<i>forests remaining as forests</i>, such as set out under the IPCC 2003 Good Practice Guidance. The project area shall meet an internationally accepted definition of forest, such as those based on UNFCCC host-country thresholds or FAO definitions, and shall qualify as forest for a minimum of 10 years before the project start date. The definition of forest may include mature forests, secondary forests, and degraded forests. Under the VCS Program, secondary forests are considered to be forests that have been cleared and have recovered naturally and that are at least 10 years old and meet the lower bound of the forest threshold parameters at the start of the project. Forested wetlands, such as floodplain forests, peatland forests and mangrove forests, are also eligible provided they meet the forest definition requirements mentioned above.	The project area does not present forested wetlands, such as floodplain forests, peatland forests and mangrove forests, as demonstrated in 1.14.9.
<u>VCS Standard v4.7, Appendix 1, A1.6</u> Activities covered under the REDD project category are those that are designed to stop planned (designated and sanctioned) deforestation or unplanned (unsanctioned) deforestation and/or degradation. Avoided planned degradation is classified as IFM.	The project falls under the REDD project category designed to stop unplanned deforestation and/or degradation.
<u>VCS Standard v4.7, Appendix 1, A1.7</u> Activities that stop unsanctioned deforestation and/or illegal degradation (such as removal of fuelwood or timber extracted by non-concessionaires) on lands that are legally sanctioned for timber production are eligible as REDD activities. However, activities that reduce or stop logging only, followed by protection, on forest lands legally designated or sanctioned for forestry activities are included within IFM. Projects that include both avoided unplanned deforestation and/or degradation as well as stopping sanctioned logging activities, shall follow the REDD guidelines for the unplanned deforestation and/or degradation and the IFM guidelines for the sanctioned logging activities, and shall follow the requirements set out in Section 3.6.2.	The project activities are only related to avoided unplanned deforestation and/or degradation.
<u>VCS Standard v4.7, Appendix 1, A1.8</u> Eligible REDD activities include: 2) Avoiding Unplanned Deforestation and/or Degradation (AUDD): This category includes activities that reduce net GHG emissions by stopping deforestation and/or degradation of degraded to mature forests that would have occurred in any forest configuration.	The project falls under the Avoiding Unplanned Deforestation and/or Degradation (AUDD).

The project is eligible under the criteria of the Non-Permanence Risk Assessment:

- The project prepares a non-permanence risk report using the VERRA HUB / AFOLU NPRT⁶, supported by the AFOLU Non-Permanence Risk Tool v4.2 at both validation and verification.
- The project has a minimum project longevity of 40 years.
- Buffer credits will be deposited in the AFOLU pooled buffer account based on the non-permanence risk report assessed by the VVB.
- The project performs non-permanence risk analysis at every verification event and may be eligible for the release of buffer credits from the AFOLU pooled buffer account.
- Validation of non-permanence risk analyses is allowed to be conducted by the same VVB that validates or verifies the project.
- When an instance leaves a project before the end of project longevity, the Proponent is aware of assuming a loss of all previously verified reductions and removals associated with the instance or continues monitoring for the remainder of project longevity.
 - Loss events are reported according to the requirements set out in the Registration and Issuance Process.
 - At a verification event after a loss event, the monitoring report restates the loss and calculates the net GHG benefit for the monitoring period.
- Reversals are managed based on whether they are unavoidable or avoidable, and the Proponent follows buffer account reconciliation requirements.
- Carbon stock permanence is monitored for a minimum of 40 years, with the option for Verra to agree to monitor projects with a shorter crediting period.
- The remaining balance of buffer credits is canceled at the end of the project longevity or at the end of monitoring, whichever is later.

1.4.3 Transfer project eligibility

This item does not apply to the project since it is not a transfer project or CPA seeking registration.

1.5 Project Design

This project has been designed as:

- ☒ Multiple locations or project activity instances (but not a grouped project)

1.6 Project Proponent

⁶ <https://projecthub.verra.org/risk-assessment>

NOVAAGRI, founded in 2008, is a visionary player in the Agribusiness sector, specializing in independent storage services for agricultural commodities. The company handles grain origination, logistical flow, and exportation for domestic and international markets. Brazil's vast agricultural potential is advantageous for NOVAAGRI, but it also necessitates strict environmental diligence due to the country's sensitive biomes and rich biodiversity. Consequently, the company enforces rigorous control over farmer prospection and selection, ensuring compliance with national forestry legislation and socio-environmental regulations through geo-monitoring systems and satellite imagery for all commercial contracts.

NovaAgri's commitment to the environment is evidenced by certifications such as 2BSVS, Soy Moratorium Audit, and RTRS. Although carbon credit projects are not the main service in NovaAgri's portfolio (neither the main source of revenue), they are strategic and align with the Group's carbon reduction goals, which aim for a 50% reduction in greenhouse gas emissions by 2030 and carbon neutrality by 2050. These projects guide the company towards carbon neutrality and promote sustainable practices among its partners and custody chain.

NovaAgri also strives to impact social prosperity positively. The company's ethical values and human resource care ensure meaningful contributions to local communities. For instance, every carbon project developed by NovaAgri aims to allocate a portion of its financial returns to social actions in nearby communities, creating benefits beyond climate change mitigation.

Organization name	NOVAAGRI INFRA-ESTRUTURA DE ARMAZENAGEM E ESCOAMENTO AGRICOLA S.A.
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Telephone	+55 11 3048-2888
Email	carbon.credit@novaagri.com.br

1.7 Other Entities Involved in the Project

Organization name	GRUPO PROGRESSO
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Role in the project	Project area owner and responsible for protection activities
Contact person	CORNÉLIO ADRIANO SANDERS
Title	Landowner
Address	Sebastião Leal, Piauí, Rodovia PI-247 KM 50 S/N, 64873000 64873000
Telephone	-
Email	recepcao@fazendaprogresso.com.br

Organization name	SCCON
Role in the project	Monitoring support (deforestation and fire alerts)
Contact person	Vinicius Ceron Rissoli
Title	Partner and Commercial Director
Address	R. Vieira de Moraes, 420 - 12º andar - Campo Belo, São Paulo - SP, 04602-000
Telephone	+55 11 3529-0228
Contact (website)	https://alertas.scccon.com.br/novaagri/#/

1.8 Ownership

Section 3.7 of VCS Standard v4.7 states that Ownership is the legal right to control and operate the project or the program activities. Requirement 3.7.1 defines terms of project ownership, and the Cultivar Progresso REDD Project meets the following term:

“3) Project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership).”

The current Project ownership arises by virtue of a contractual right in the owners' land, which contemplates its vegetation, conservational and management process that generates GHG emission reductions and/or removals. The project area refers to 32.549,79 ha of land, in a private area belonging to Grupo Progresso. The area is located across the municipalities of Uruçuí, Sebastião Leal, Landri Sales and Antônio Almeida – in the State of Piauí (Brazil).

The location details are specified in Section 1.13 Project Location, and sensitive documentation evidence for land registration was presented directly to the VVB.

1.9 Project Start Date

Project start date	January-01-2024
Justification	According to the Section 3.8 of the VCS Standard v4.7, “The project start date of an AFOLU project is the date on which activities that lead to the generation of reductions or removals are implemented.” The start date is January 01, 2024, and the evidence is as follows: • The implementation of the MRV procedures document started being carried out by the proponent. • The evaluation of carbon stock losses through monitoring data from INPE and public satellite images started. Between January 01, 2024 and March 12, 2024, monitoring of the project was done through the use of public data from DETER/INPE⁷, which detects deforestation, degradation and wood logging daily, and public satellite imagery from the satellite LANDSAT series 7, 8 and 9⁸. An official report was generated at the end of each month. The monitoring mechanisms were enhanced as of March 13, 2024, with the implementation of new resources invested by the Project Proponent, such as the completion of the NovaAgri monitoring dashboard, along with real-time deforestation and fire alerts in partnership with the company SCCON Geospatial. Additionally, the acquisition of state-of-the-art, high-resolution satellite images (PLANET, with a resolution of 03m) was finalized. All these new resources result from the company's strategic plan for investment and improvement of internal carbon projects. It is worth noting that previous resources, such as the use of public deforestation data and LANDSAT images, remain important for double-checking and continue to be considered complementary resources and part of the MRV procedure, as it's described in the official monthly report • Grupo Progresso's employees were assigned new responsibilities regarding the project's conservation goals, such as extra fire training sessions and in-situ monitoring. The project being supported by a landowner with a structure similar to those of large corporations, including corporate departments and specialized teams, presents as an advantageous structure for conducting

⁷ DETER/INPE, available on: <https://www.gov.br/inpe/pt-br/area-conhecimento/unidade-amazonia/repositorio-de-arquivos/Apresentacao_Lancamento_DETER_B_2017.pdf>.

⁸ Landsat, available on: <<https://landsat.gsfc.nasa.gov/satellites/>>.

	<i>in-situ activities for monitoring, surveillance, and action. For example, Grupo Progreso has specific departments for sustainability/health & safety/environment and human resources, as well as a security team and a specialized fire brigade with periodically trained employees prepared for any emergency. In addition to trained teams, the farms have all the necessary infrastructure to safely respond to fire outbreaks, such as water trucks, water reservoirs, and all the required safety PPE. With the implementation of the project, these actions have been maintained and enhanced, as the project will create new revenue streams for the maintenance, replacement, and purchase of equipment, which are essential for in-situ forest preservation activities.</i>
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1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☐ Ten years, fixed ☒ Other <i>The crediting period of the Cultivar Progreso REDD Project is 20 years and meets the minimum crediting period for AFOLU project, as stated in the requirement 3.9.3 of VCS Standard v4.7: “For all AFOLU projects other than such ALM projects described in 3.9.2, the initial project crediting period shall be a minimum of 20 years up to a maximum of 100 years, which may be renewed at most four times, with a total project crediting period not to exceed 100 years.”.</i>
Start and end date of first or fixed crediting period	<i>January-01-2024 to 31-December-2043</i>

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
- ☐ ≥ 300,000 tCO₂e/year (large project)

Table 3. Estimated GHG emission reduction during the first crediting period (possible to extend due to renewals)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-January-2024 to 31-December-2024	31.376
01-January-2025 to 31-December-2025	34.674
01-January-2026 to 31-December-2026	38.812
01-January-2027 to 31-December-2027	42.546
01-January-2028 to 31-December-2028	46.216
01-January-2029 to 31-December-2029	54.565
01-January-2030 to 31-December-2030	58.274
01-January-2031 to 31-December-2031	61.919
01-January-2032 to 31-December-2032	65.502
01-January-2033 to 31-December-2033	69.950
01-January-2034 to 31-December-2034	71.364
01-January-2035 to 31-December-2035	70.796
01-January-2036 to 31-December-2036	69.815
01-January-2037 to 31-December-2037	69.759
01-January-2038 to 31-December-2038	68.800
01-January-2039 to 31-December-2039	67.380
01-January-2040 to 31-December-2040	66.159
01-January-2041 to 31-December-2041	65.808
01-January-2042 to 31-December-2042	64.614
01-January-2043 to 31-December-2043	63.440
Total estimated ERRs during the first or fixed crediting period	1.181.768
Total number of years	20
Average annual ERRs	57.983

1.12 Description of the Project Activity

1.12.1 Socioenvironmental Compliance

In the context of compliance, the monitoring system, integrated with other tools contracted by NovaAgri, facilitates the enforcement of socioenvironmental regulations. This is achieved by cross-referencing the public database in Brazil, which is built on a foundation of reliable sources and encompasses crucial socioenvironmental factors such as conservation units, delineated indigenous areas, Legal Reserve areas, historically mapped deforestation, among others, with

the project area⁹. These actions are part of NovaAgri's rigorous Socioenvironmental Policy, initially designed for implementation across its credit/risk, sustainability, and commercial departments as a conventional process for selecting new customers.

1.12.2 Monitoring System and Technology

The main objective of the Cultivar Progresso REDD+ Project is to stop the areas from being converted into non-forest lands, by conducting preservation activities and reinforcing the partnership existing from the commercial relationship between NovaAgri and the farmer(s). To accomplish the main goal for a REDD+ AUD purpose, NovaAgri is committed to investing in a trustable and modern technology of GIS with deforestation near real-time alerts alongside a high-resolution satellite imagery resource, dedicating an exclusive corporate department to this end goal.

The project is going to integrate an advanced deforestation alert system provided by SCCON, a renowned expert in dashboards, data automation, and artificial intelligence, with international recognition and extensive track record of working with various public and private entities, inspiring confidence in their capabilities. This project aims to implement a more refined version of such a system, incorporating micro-localized management for heightened alert precision and improved efficacy in deforestation prevention and remediation efforts. The integration of SSCON's expertise and technology aligns seamlessly with our commitment to modern practices and environmental stewardship in our carbon offset initiative.

By using PlanetScope satellite images, the world's biggest satellite constellation (about 180 nanosatellites), which collect daily images from any spot on the planet – generating continuous imaging generated from PlanetScope orthorectified images with a 3-meter spatial resolution, accessible with RGB viewing functionality in XYZ format, including levels 15 (4.77 meters spatial resolution) and 16 (2.38 meters spatial resolution) with an 8-bit radiometric resolution, and added up to an API (Application Programming Interface) integrated to a particular alerts' platform, the system might enable rapid detection of vegetation loss activities. This allows NovaAgri, alongside the owner and other entities, to:

- 1) move internal guards and other specialized departments into the alert's focus to proceed with the emergency mitigation; and/or
- 2) trigger authorities and other stakeholders to respond quickly to illegal logging or land clearing, preventing further damage to forests.

⁹ NovaAgri employs a tool called LYRA, developed by a company within the Imagem Group - ESRI, to facilitate internal verification activities in potential client and partner areas. LYRA is a technology that utilizes remote sensing and data intelligence to establish traceability, conduct socio-environmental diagnostics of individuals, companies, and territories, agroclimatic monitoring, and territorial intelligence. The system is built on layers of public and private socio-environmental and agricultural risk, adding value and mitigating reputational and operational risks. It features a GIS screen for navigation, providing a user-friendly interface for exploring new territories, especially in terms of socio-environmental monitoring, by checking daily compliance-related aspects for those involved and the monitored agroclimatic conditions. More information on: <https://vegamonitoramento.com.br/lyra/>

The satellites used for this purpose are modern and cover vast areas of land, including remote and inaccessible regions. This extensive coverage allows for a comprehensive assessment of deforestation trends across different landscapes, including forests in difficult-to-reach areas.

The summarized characteristics and specifications of the monitoring system is explicit below:

- i) High frequency: Daily in-season revisit rates powered by 180 satellites.
- ii) Field-level details: 3-3-meter resolution satellite images.
- iii) Deforestation daily alerts: workflow integration with APIs and the NovaAgri's dashboard, created specifically for the ends of this project. SCCON has vast experience with monitoring systems, including robust government projects used by the Federal Policy and other State security institutions.

This system covers the Project Area and all surroundings within the Jurisdictional State (beyond the project boundaries). In addition, this provides an objective and transparent source of data, reducing the potential for bias in monitoring and reporting deforestation. This helps in building trust among stakeholders by supporting evidence-based reports for the MRV periodic verification. The alerts cover the following classes: clear-cutting; access; degradation; fire scar.

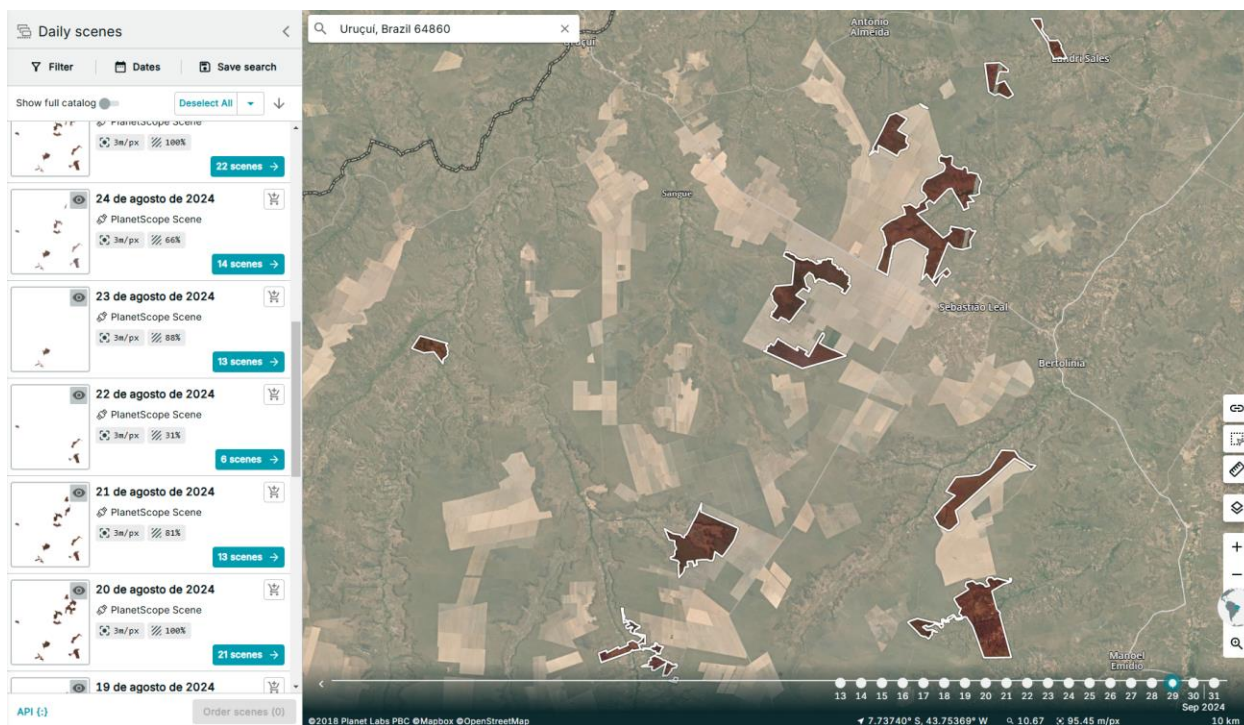


Figure 1. PLANET Satellite Image example, with Project Area overlap.

Source: PLANET¹⁰ – Satellite Images for monitoring.

¹⁰ <https://www.planet.com/>

By utilizing technology to measure VCUs (Verified Carbon Units) of the CULTIVAR PROGRESSO REDD+ PROJECT, NovaAgri will increase confidence in the Measurement, Reporting and Verification (MRV) process by recording the maintenance of intact forests in the project area and the rate of deforestation in the project area of influence (leakage belt).

Looking at the long-term scenario, this system is going to facilitate extended monitoring of deforestation trends and changes in land use over time. By analyzing historical data, the proponent can identify patterns and understand possible drivers of deforestation in the surrounding areas. This might be especially important for the premises of the leakage areas.

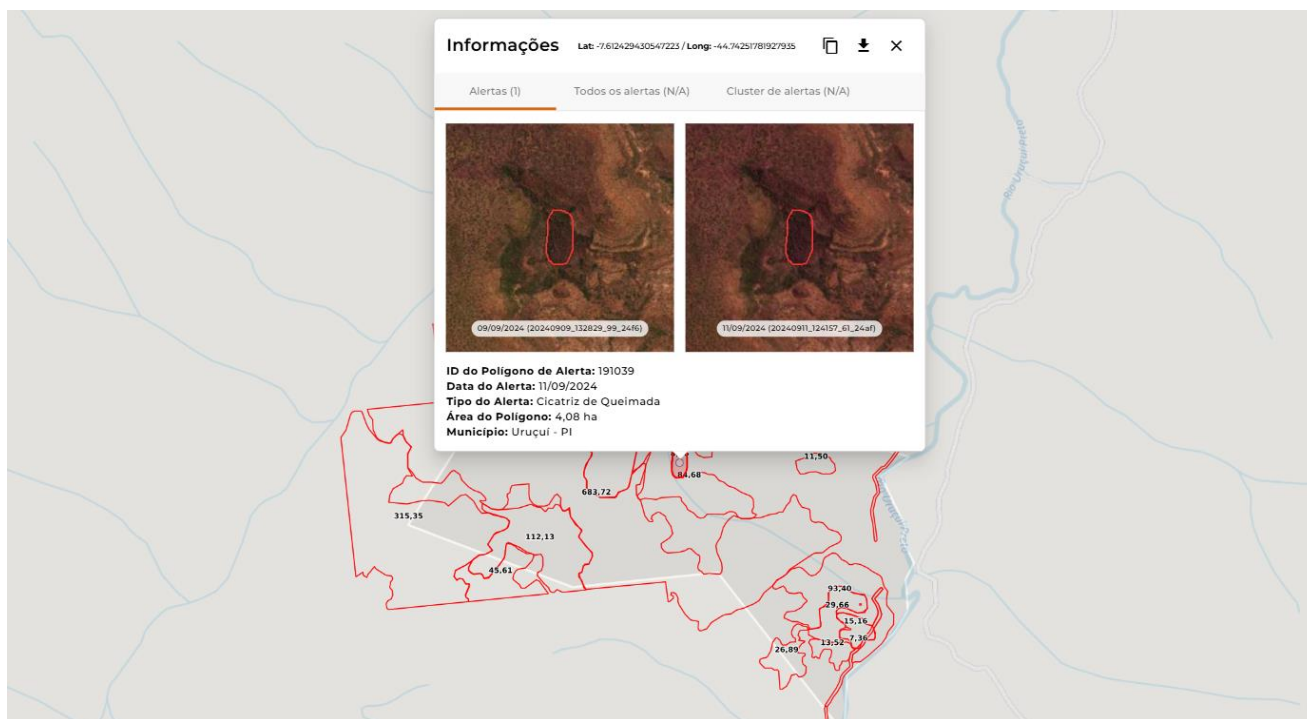


Figure 2. Deforestation alert example.

Source: SCCON – MRV dashboard deforestation alerts overview.

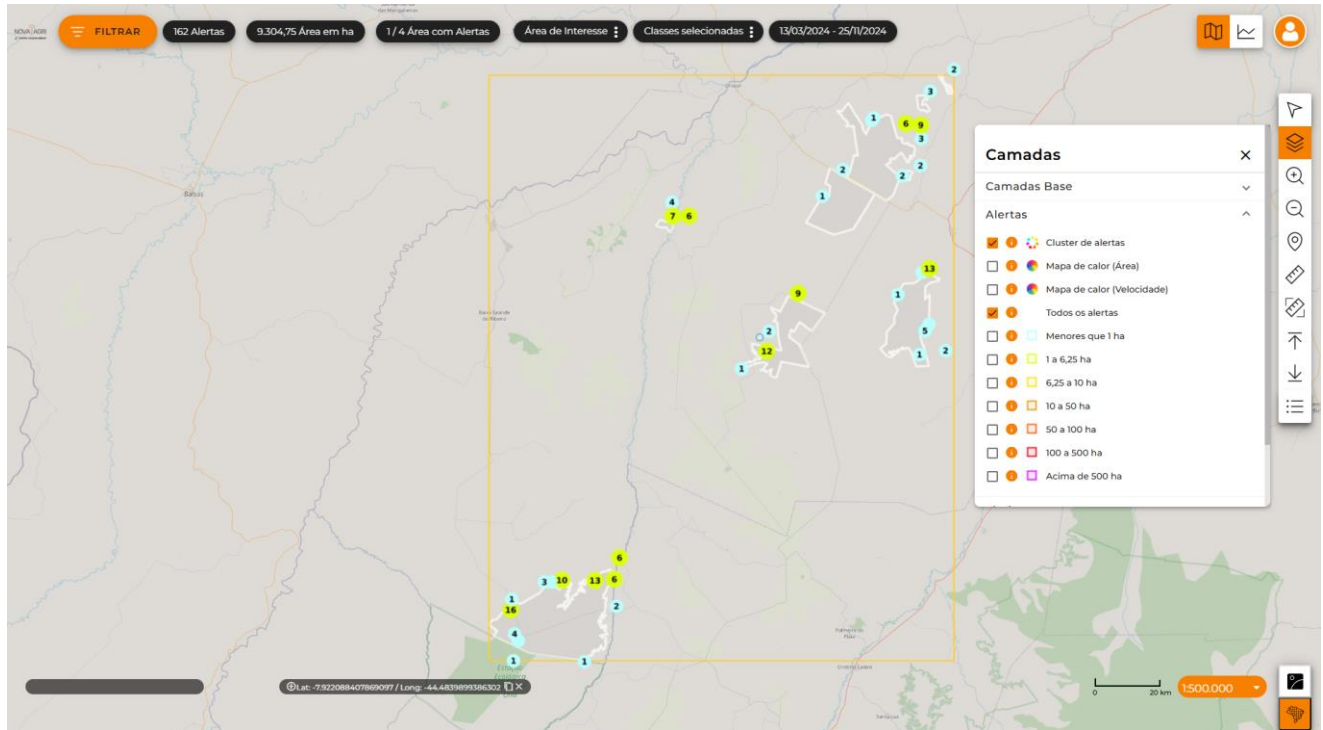


Figure 3. MRV dashboard georeferenced data layers overview.

Source: SCCON.

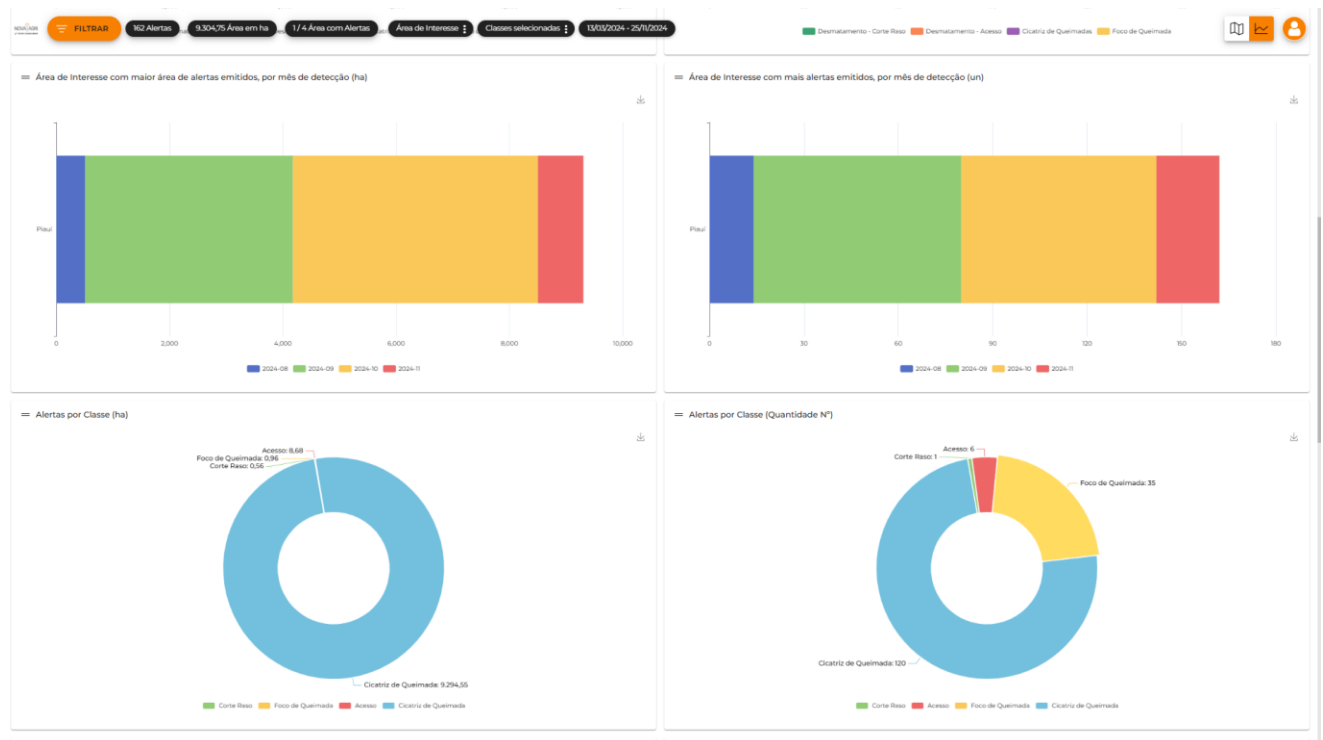




Figure 4. MRV dashboard statistics overview.

Source: SCCON.

1.12.3 Fire Control actions

Surveillance

One of the project's activities is to actively manage and reduce the risk of unexpected fires in the project area. This involves enhancing firebreak systems, strategically designed to contain and stop the spread of fires.

Additionally, the project's monitoring system can detect and alert the proponent and landowner quickly about any potential fire outbreaks, through **geospatial monitoring practices** using satellite imagery. By utilizing recent and regularly updated images, the **fire near-real-time alert system**¹¹, adapted to a web dashboard, can identify and alert the landowner and the project proponent about wildfire outbreaks near the project area, allowing for proactive measures before the fire spreads.

¹¹ Utilization of continuous generation of fire spots' alerts throughout the project period. The method involves crossing the fire alert system by INPE with the daily Planet satellite images' system in order to confirm the fire spots more precisely. Files provided by IPNPE contains information on forest fire and wildfire incidents at various intervals, ranging from annual to near real-time (10 minutes). These data are available in CSV and KML formats. A perpetual and unrestricted license for the generated alerts will be provided for use, distribution, and modification.

Informações

Lat: -7.397516241174458 / Long: -44.069582081023164



06/08/2024
(20240806_132532_74_24ee)



10/08/2024
(20240810_132459_57_24bd)

ID do Polígono de Alerta: 166816

Data do Alerta: 10/08/2024

Tipo do Alerta: Cicatriz de Queimada

Área do Polígono: 344,15 ha

Município: Landri Sales - PI

Figure 5. MRV dashboard fire alert example.

Source: SCCON.

Education and awareness

Our comprehensive fire prevention also encompasses an educational and awareness initiative, targeted at both farm employees and local communities. This program aims to instill a heightened consciousness regarding fire hazards and foster a proactive approach to fire prevention. By integrating technological solutions with educational outreach, this project aspires not only to control the immediate threat of fires but also to cultivate a culture of vigilance and responsible fire management within the project area and its surrounding communities.

Training and fire brigades

The landowner is responsible for implementing key measures to prevent and combat wildfires, including maintaining a dedicated firefighting team that undergoes **regular specialized training**. This team is equipped with **readiness vehicles, water trucks, protective gear**, and other resources essential for effective fire management.

Other employees also received training sessions and guidance to ensure they are well-prepared to respond to potential fire incidents, as the actions involve a complex range of activities and different departments. The most recent training session¹² had more than 100 employees effectively guided and trained. The property also maintains a positive relationship with local communities, keeping them informed and on alert for any fire warnings, enabling quick mitigation

¹² By the time this PD was being written (January/2025).

actions. This proactive approach demonstrates commitment to wildfire prevention and the protection of the project area.



Figure 6. Fireguard training session (practical activities).

Source: the author.



Figure 7. Fireguard training session (theoretical activities).

Source: the author.

Depending on the scale of fires on certain occasions, Grupo Progresso also has planes and pilots on-call who can be mobilized to address emergencies through aerial methods such as water spraying or using other fire-fighting elements. This ensures that the fire does not spread and/or minimizes damage to the vegetation area.

1.12.4 Engagement and Awareness actions

The project aims to foster comprehensive engagement among key stakeholders, including employees, local communities, and other participants. The strategy includes proactive initiatives that aim to raise awareness, build partnerships, and empower those involved.

A public audience was organized to provide a space for open dialogue. This was implemented on both VERRA's public platform and in a separate session, where members of the public, environmental organizations, and other interested parties were invited to learn in more detail about the project and share thoughts, concerns, and suggestions.

With the project in course, comprehensive public awareness campaigns will be launched, in order to inform a wider audience about the objectives and benefits of the REDD+ project. Through media channels, educational materials, and community events, the project aims to reach a broad spectrum of individuals and encourage their active involvement.

Other details, specifically about safeguards and engagement to avoiding negative externalities, and more information about the public consulting are presented on 'Section 2 – Safeguards' of this PD.

1.12.5 Social Responsibility

This project goes beyond forest preservation: as strong as its environmental protection aspect is, it also emphasizes social responsibility multiplication beyond mere impact mitigation, spreading positive social impacts in the same magnitude of the carbon reduction results.

Starting with the established actions of the social association managed by the Grupo Progresso itself, entitled '**Instituto Cultivar Progresso**' – which also inspired the name for this Carbon Project. The Grupo Progresso has a public commitment to share with the local people in Piauí the good results of its work and projects, contributing to the development of people and communities in the regions where they "root themselves".

The Grupo Progresso has stated that **100% of the projected financial results** from its carbon credits' revenues portion will be entirely allocated to:

- 1) Social investments in local communities, supporting a series of causes among the population of Piauí, and
- 2) Reinvesting in the project's activities themselves, such as improvements in safety and forest protection equipment, as well as technology and innovation for this purpose.

This is a major differential of this project, emerging from the motivation of a corporate entity (NovaAgri) that includes fostering positive socio-environmental impact actions in its own philosophy, and one of the largest agricultural groups in Piauí (Grupo Progresso), focusing on social transformation initiatives in the areas where it operates. All these social investments are genuinely voluntary and run beyond regulatory aspects or government requirements.

Instituto Cultivar Progresso:

Here is a summary of Instituto Cultivar Progresso's history¹³: formally established on July 8, 2022, the Instituto Cultivar Progresso was born with the mission of transforming local society's reality through the progress coming from agriculture. The Institute has four main causes for action: **Social Assistance, Education/Professionalization, Agriculture, and Environment**.

- *Social Assistance* is provided by identifying families in vulnerable situation – within the municipalities of Uruçuí, Bertolínia, Sebastião Leal, and Guadalupe –, who periodically receive food baskets and gifts for the children from the Instituto, especially during Christmas, Easter, and emergency situations like floods. The partnership with **APAEs** (Association of Parents and Friends of Disabled Individuals) in Bertolínia and Uruçuí is another way in which the Institute provides social assistance. The *flagship* initiative of the Institute is the currently undergoing construction of the APAE School Clinic in Bertolínia.
- *Education and Professionalization* are practiced through the “*Parceiros da Escola*” program (School Partners program, in English), which periodically selects young students from the Sebastião Leal region, in partnership with the Municipal Education Department, by supporting them with 1) the commuting expenses during the college/university entrance exams, and 2) when they are accepted, following up with the support and covering all other expenses. Professionalization is also achieved through the “*Academia de Operações Agrícolas*” (Agricultural Operations Academy, in English) project, offering vocational courses in agriculture-related fields. The educational aspect of the Institute is also strengthened through campaigns, lectures, and actions promoting the thematic of **responsible agriculture**, showcasing good agricultural practices, and **integrating women** in rural areas.
- *Promotion of good agricultural practices* is run through partnerships with farmers, especially supporting small farms (family agriculture) by donating agricultural input materials and seeds.
- *Environmental respect* is enhanced through actions such as the “Florir Piauí” project (Production and distribution of native seedlings) and the structuring of an own Fireguard Brigade that combats wildfires.

Focused social causes within this project:

In response to the needs noted during the public consultation with stakeholders (as detailed in Section 2. Safeguards and Stakeholder engagement), the project developer understands that the

¹³ Instituto Cultivar Progresso, available on: <<https://institutocultivar.com.br/>>.

resources will be of great value in supporting the most vulnerable populations in the region, including low-income communities. Thus, it was defined that the focus of the CULTIVAR PROGRESSO REDD+ PROJECT for social responsibility is based on two important projects encouraged by the Grupo Progreso:

1. **Rural Daycare Center:** Offering comprehensive activities for children in the region, providing a safe and nurturing environment with educational activities, while also supporting working parents.
2. **Technical School for Youth Development:** Establishing a technical school aimed at training and preparing young individuals for future employment opportunities, particularly focusing on skills relevant to the agricultural sector.

Projects related to education will receive emphasis on investment, as it is a highly requested demand for the region, which has sectors requiring specialized workforce. Recognizing that Piauí is strongly related to the agribusiness sector, the proponent understands that activities promoting good agricultural practices connected to sustainable production methods are crucial. These activities not only contribute to the local families' income but also play a vital role in dispelling the misconception of unsustainable agribusiness. Instead, the goal is to shift towards an image of a sector that generates jobs and income without the need for deforestation or degradation of forested areas.

Such initiatives or other similar ones could be supported by the project's resources, as the proponent is also committed to allocating a portion of its carbon credit-derived funds to priority social causes supported by the CULTIVAR PROGRESSO Institute. It is important to mention that any social investment will reflect the needs raised by the local population itself. Another point is related to the transparency of these actions, which will strictly be reported to the stakeholders, which includes the VVB by registering the proper accountability in further monitoring reports.

NovaAgri, as the proponent and developer of the project, adheres to a rigorous Compliance Policy expressed through its LATAM Code of Ethics from its Group. This project will be entirely governed by this instrument, and therefore, in all its activities, it will not tolerate any form of discrimination, coercion, inappropriate advertising, exposure of sensitive data, bribes, or other deviations. The company is committed to conducting a rigorous compliance assessment in all transactions, whether commercial or donation related. To achieve this, it will rely on appropriate contractual instruments and internal verifications from the company's legal team.

Transparency and Accountability

As the AUD/environmental activities, the social activities will also follow high standards of transparency as part of the company's commitment to accountability and transparency, providing regular and detailed reports on the counterparties associated with these activities.

1.12.6 Definition of project activities' location

All the activities will be implemented in locations not covered by a jurisdictional REDD+ Program. The geographic boundaries are defined and specified on Section 3 of this PD, including the separate coordinates as KML files. Further information is also provided in Section 1.13 Project Location.

Table 4. Summary of organizations, communities and other entities involved in the project's activities

Activity	Description	Timing/ Frequency	Executor/ facilitator	Stakeholders involved
Socioenvironmental Compliance	Analysis for Socio-environmental risk assessment of client portfolios and prospective Landowners for Carbon projects implementation	Pre-project Monitoring (anytime)	NovaAgri	All direct and indirect Stakeholders
Monitoring System and Technology	Providing a complete monitoring report by integrating in-loco activities with an advanced deforestation alert system within a dynamic dashboard with high-resolution satellite images	Monitoring periods	NovaAgri SCCON	All direct and indirect Stakeholders
Fire Control Actions	In-loco activities optimized by a monitoring system which detects and alerts the PP and landowner quickly to any potential fire outbreaks	Monitoring periods	NovaAgri Landowner	Landowner Local Communities
Engagement and Awareness Actions	Proactive initiatives that aim to raise awareness, build partnerships, and empower Stakeholders	Pre-project Monitoring periods	NovaAgri Landowner	Employees Local Communities (including traditional groups)
Social Responsibility	Funding and implementing impactful social initiatives within the communities surrounding the project area, aligned with local needs	Post-VCU release Monitoring periods	NovaAgri	Local Communities (including traditional groups)

1.13 Project Location

1.13.1 Macro-region description

The project area is located in the state of Piauí, which covers an area of 250,934 km², representing 16.20% of Brazil's Northeast region and 2.95% of the national territory. According to the Fundação CEPRO (1992)¹⁴, the Piauí Cerrado is the fourth most significant in Brazil and

¹⁴ FUNDAÇÃO CEPRO. Cerrados Piauienses. Teresina, 1992. Cited by Aguiar & Monteiro, 2005.

the largest in the Northeast, spanning 11.85 million hectares. This accounts for **46% of Piauí's total area**, equivalent to **5.9% of the Brazilian Cerrado** and **36.9% of the Northeast Cerrado**.

Of this area, **70% is considered core Cerrado**, while the remaining **30% consists of transitional vegetation between the Caatinga and Cerrado biomes**, extending across various points from the north to the south of the state. The densest concentration of Cerrado is found in the Southwest and Far South regions. It is estimated that around **10% of this ecosystem** is currently occupied and used for agricultural and livestock projects (Aguiar & Monteiro, 2005)¹⁵.

The project area specifically encompasses the municipalities of Uruçuí, Sebastião Leal, Landri Sales, and Antônio Almeida, in the western part of Piauí, near the border with the state of Maranhão, in the Cerrado biome. The closest airport is in Teresina, the capital of Piauí, approximately **430 km away by road**.



Figure 8. Macro location (country and state level) of the project.

Source: Google Satellite, IBGE, the author.

¹⁵ AGUIAR, Teresinha de Jesus Alves de; MONTEIRO, Maria do Socorro Lira. Modelo agrícola e desenvolvimento sustentável: a ocupação do cerrado piauiense. *Ambiente & Sociedade*, [S.L.], v. 8, n. 2, p. 161-178, dez. 2005. FapUNIFESP (SciELO). <http://dx.doi.org/10.1590/s1414-753x2005000200009>.

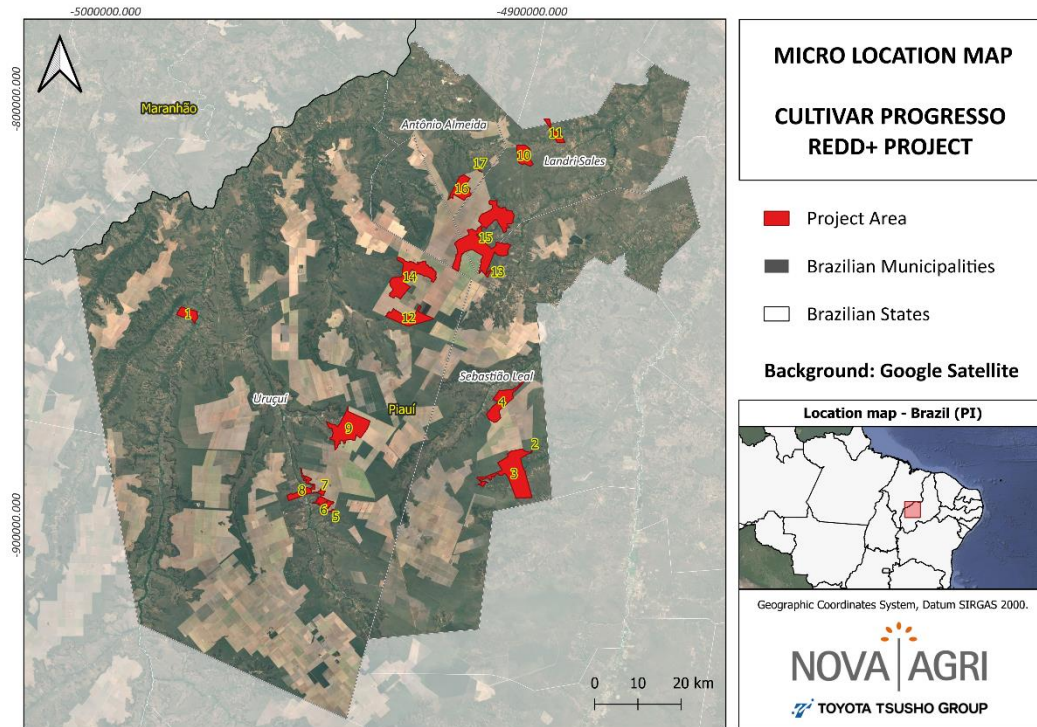


Figure 9. Micro location (municipality level) of the project area.

Source: Google Satellite, IBGE, the author.

ID	Area (ha)
1	908,08
2	9,64
3	5315,79
4	2417,06
5	38,02
6	511,38
7	146,64
8	934,81
9	4285,43
10	1074,72
11	631,75
12	2123,60
13	165,65
14	4008,04
15	8329,59
16	1641,25
17	8,35

Total	32,549.79
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1.13.2 Project Activities Instance Location

The project area is part of the Grupo Progresso farms, which is composed of more than 180,000 hectares, of which 32.549,79 hectares are native vegetation considered as the Project Area. The forested area crosses the municipalities of Uruçuí, Sebastião Leal, Landri Sales and Antônio Almeida, all of them in the State of Piauí. The area is officially registered in the CAR (Rural Socioenvironmental Registry) and the official registry Brazilian entity, which can be consulted publicly in the national websites. Thus, the property has documents that register all the exact coordinates and officialize the legal ownership in compliance with the environmental and land legislation in Brazil.

To ensure this accuracy and fidelity of landownership information, NovaAgri's Credit, Risk and Sustainability Department conducted a thorough internal audit. This meticulously scrutinized each point of the coordinates provided in the official registration documents, cross-referencing them with the visual data from the public shapefile outlining the property. As a result, the proponent can confirm that the delineations presented in the geospatial files align consistently with the coordinates documented in the official registry.

Furthermore, this audit served the additional purpose of confirming the absence of any overlap between the project area and public lands, conservation areas, traditional communities' territories, or any other areas conflicting with REDD+ methodology or VCS standards. The process strictly adheres to national and local legislation, ensuring compliance with all relevant regulations and guidelines.

Some of these documents are presented on the APPENDIX 2 of this PD, and the KML file is attached into the VERRA registry platform. The area does not overlap with the project area of any other VCS AFOLU projects.

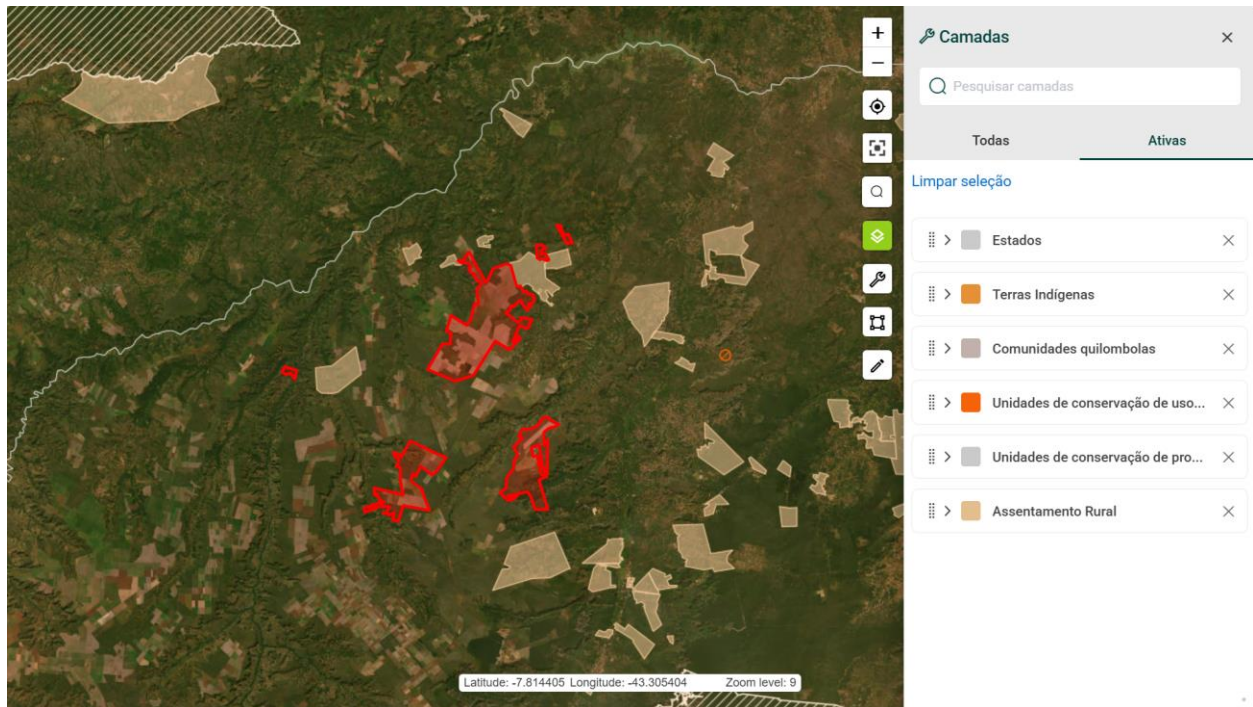


Figure 10. Map with the Project Area's Farms, showing no overlap with Indigenous Lands, Quilombola communities, Conservation Units or rural settlements.

Source: The author.



Figure 11. Idesam's Carbon Projects Dashboard, showing that there is only one other project in the Piauí State, which doesn't overlap with the Cultivar Progresso REDD+ Project.

Source: Idesam¹⁶.

Table 5. PROPERTIES' GEOGRAPHIC COORDINATES INDICATION (CENTROIDS)

Farm	Y Coordinate (South) UTM 22S	X Coordinate (West) UTM 22S
São Sebastião	9158107,024701506	528436,1092109672
Rua	9194576,44094603	606486,0892393444
Buritizinho	9199742,53872415	613941,8274593097
Progresso	9172663,219169287	588323,2155670185
Imperial	9129169,16801218	602753,4792107795
Rainha da Serra	9124591,125576876	565437,3512026299

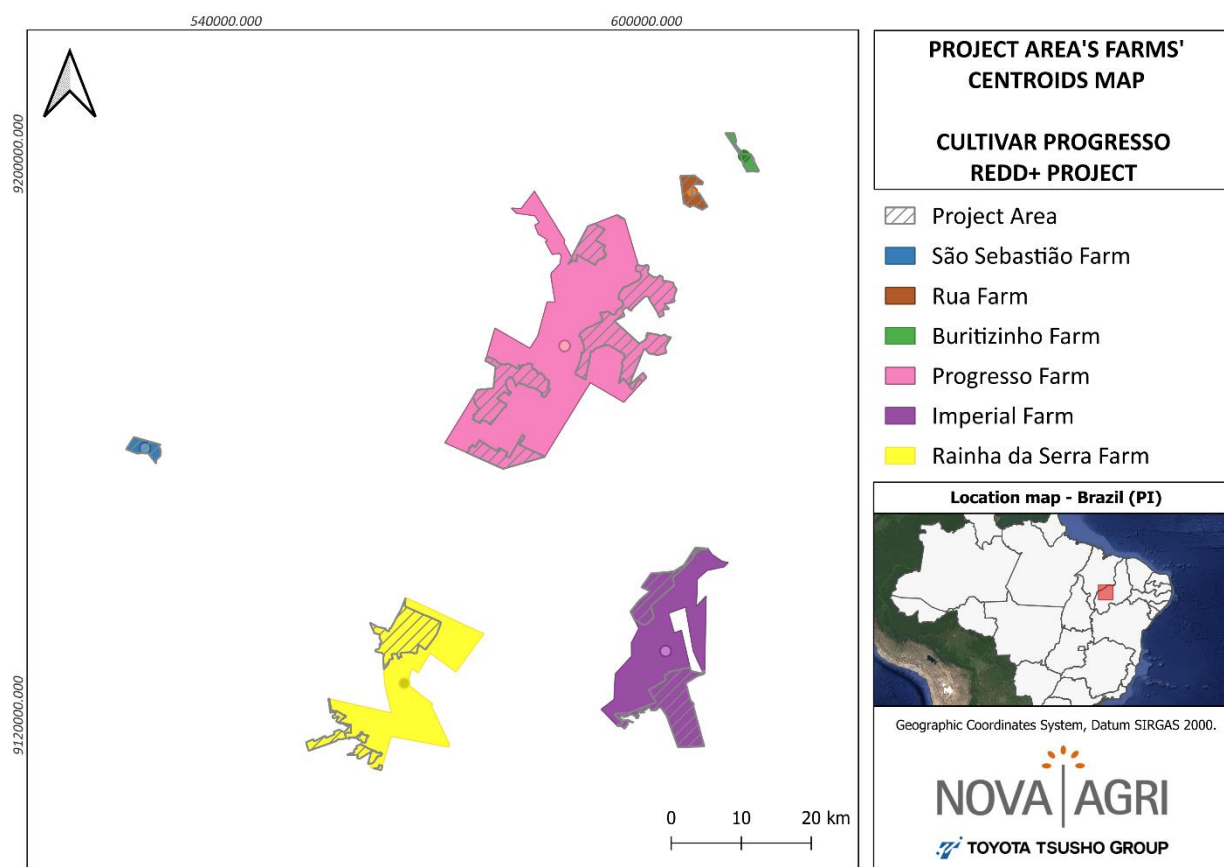


Figure 12. Cultivar Progresso's Farms and its centroids.

Source: The author.

¹⁶ Idesam's Carbon Projects dashboard, available on: <<https://idesam.org/painelprojetoscarbonoflorestal/>>.

1.14 Conditions Prior to Project Initiation

1.14.1 Deforestation historical context

Since around the 1970s, the state of Piauí has experienced accelerated occupation driven by the expansion of agricultural production, particularly soy cultivation, in the Cerrado biome — considered Brazil's last agricultural frontier. This process has been accompanied by a high and steadily increasing rate of deforestation.

According to the 2023 Annual Deforestation Report (MAPBIOMAS, 2024)¹⁷, the Cerrado biome surpassed the Amazon for the first time, recording the highest deforestation rates in Brazil. In 2023, a total of 1,110,326 hectares were cleared, representing a 67.7% increase. Agriculture and livestock production are the main drivers of this deforestation, accounting for 97% of all native vegetation loss over the past five years.

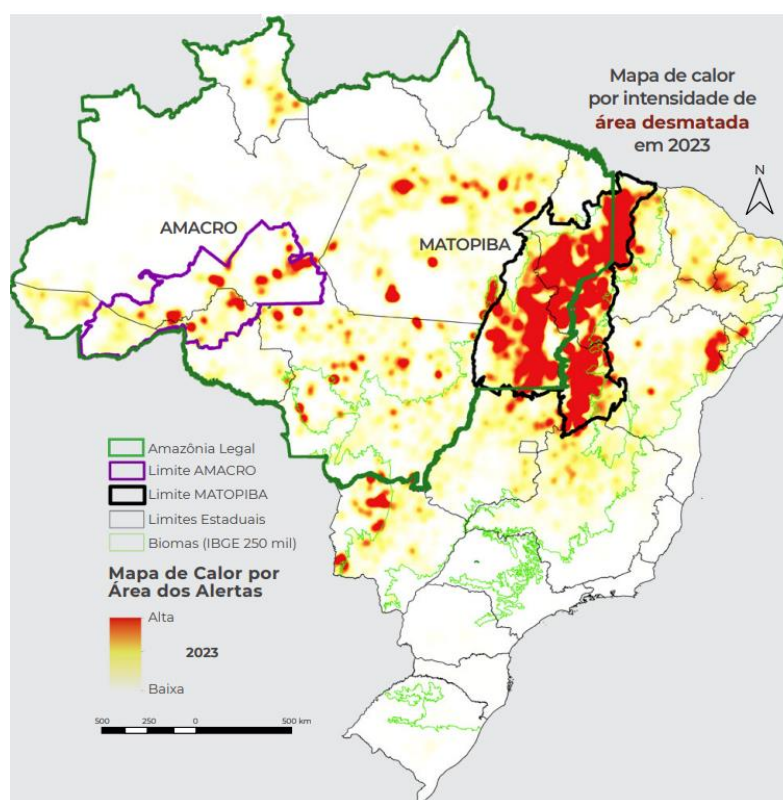


Figure 13. Heat map of deforestation intensity in Brazil in 2023, with the State of Piauí with a high amount of deforestation alerts.

Source: Mapbiomas, 2023.¹⁸

¹⁷ MapBiomas is a Brazilian collaborative initiative aimed at creating and providing a geospatial database on land cover and land use in Brazil. Utilizing satellite imagery and advanced data processing techniques, MapBiomas offers detailed and up-to-date information on land cover dynamics and environmental changes. This information supports environmental monitoring, land planning, and the sustainable management of natural resources.

¹⁸ MapBiomas 2023 Annual Deforestation Report highlights, available on: <https://alerta.mapbiomas.org/wp-content/uploads/sites/17/2024/07/RAD2023_Destaques-PT_v1-1_27-05.pdf>.

Despite the development of forestry laws in Brazil over the years, 59% of the area deforested since 2019 occurred without authorization or effective enforcement. In contrast, the share of deforested areas with authorization or subject to enforcement actions in Piauí increased from 33% in 2021 to 57.2% in 2023. On the impunity, the legal actions and embargo carried out by IBAMA and ICMBio - by May 2023 - reached only 10.2% of the deforested area identified from 2019 to 2022. That is a clear and worrying signal that the law against deforestation in Brazil is far from being entirely effective.

1.14.2 Ecosystem Type

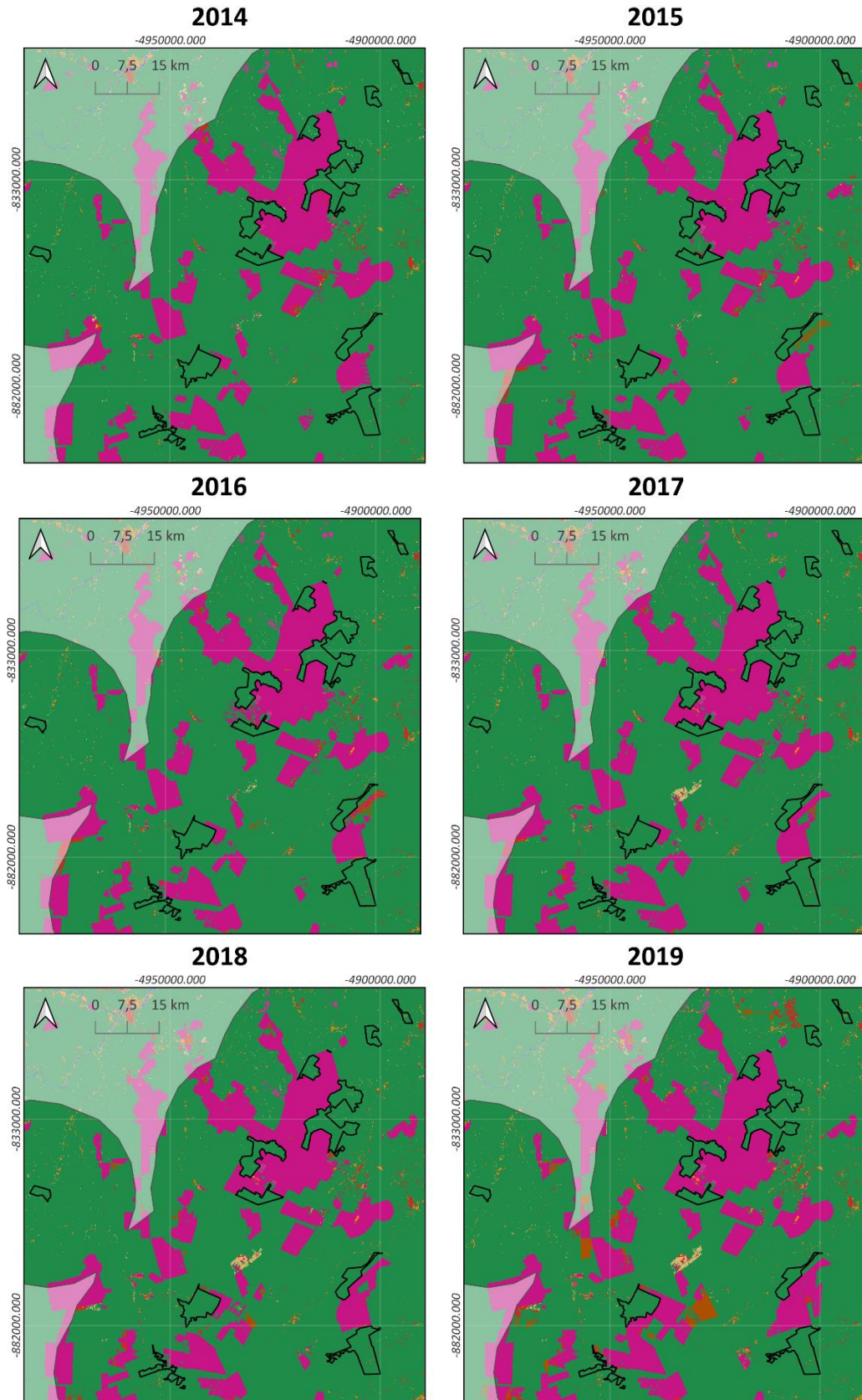
The Cerrado biome is the largest savanna region in South America and has one of the richest and most critical biodiversities in the world, holding around 6000 tree types and 800 bird species¹⁹. The biodiversity of Western Piauí has a very authentic Savanna filled by a series of habitats, fostering a wide array of terrestrial animals and bird species, bringing richness to the ecosystem.

1.14.3 Current and historical land-use

Beyond the project area, e.g. within a 20km buffer, the landscape is predominantly characterized by forested areas, agricultural lands and pasturelands, reflecting both anthropogenic and natural land uses (MapBiomas, 2023²⁰). Despite significant areas of already consolidated deforestation, recent land clearing activities can be observed, particularly since the mid 2010s, indicating ongoing land use changes even after the implementation and reinforcement of environmental laws in the country.

¹⁹ ICMBio, available on: <<https://www.icmbio.gov.br/cbc/conservacao-da-biodiversidade/biodiversidade.html>>.

²⁰ MapBiomas' Land Use and Cover collections, available on: <<https://brasil.mapbiomas.org/en/colecoes-mapbiomas/>>.



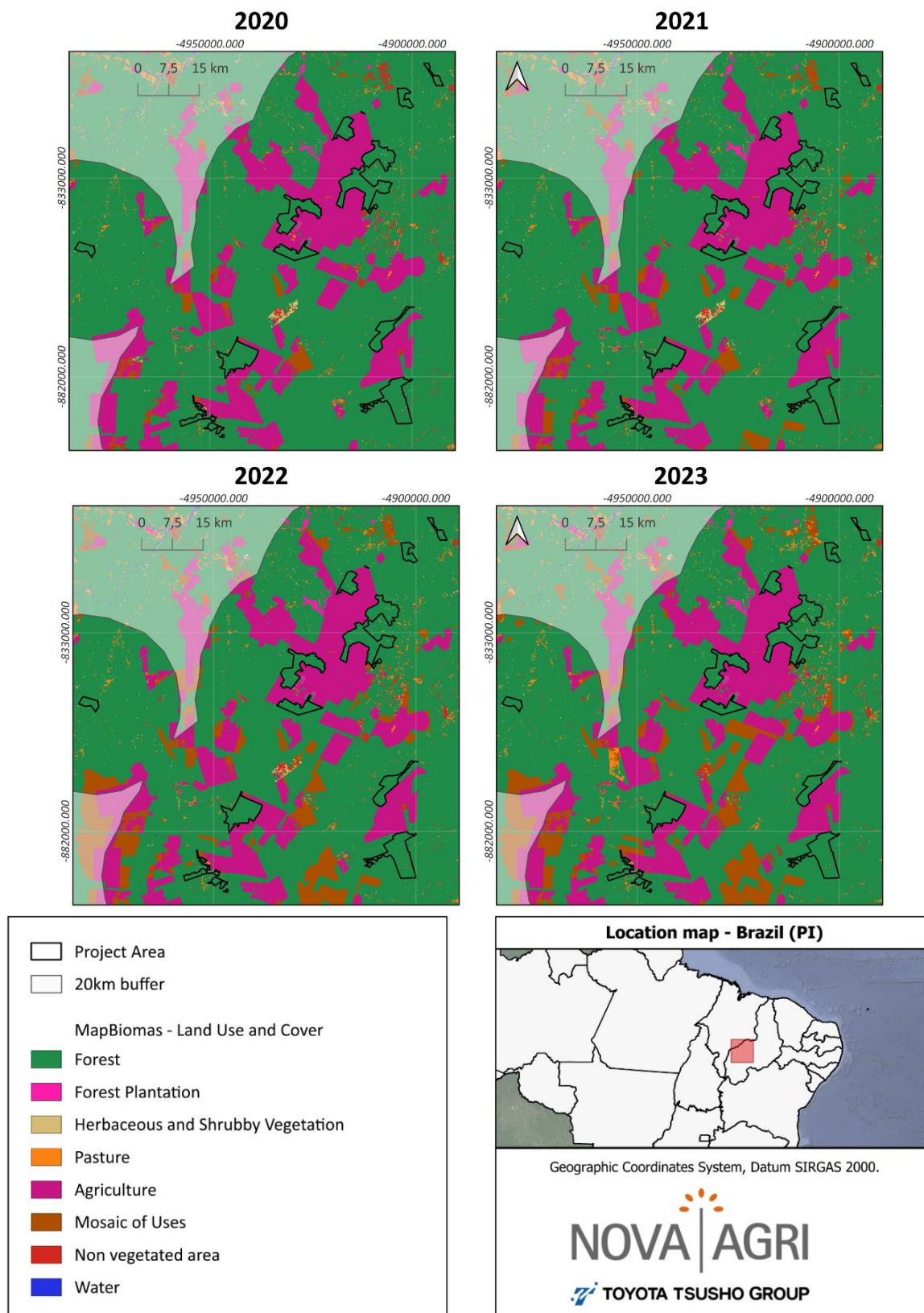


Figure 14. Land use and Cover in the Project Area, 2012-2023.

Source: MapBiomass, the author.

1.14.4 Climate

According to the IBGE^{21 22}, the area is situated in a hot equatorial climate zone, with temperatures consistently exceeding 18 degrees Celsius throughout the year. It is characterized by a Semi-arid and Semi-humid areas transition, experiencing a dry period lasting between 4 to 6 months.

Rainfall patterns in the state of Piauí are quite diverse due to its geographical position between the northeastern Northeast, the southern Northeast, the Midwest, and the Mid-North regions (Map below). While rainfall frequency decreases towards the southeastern part of the state, only 35% of Piauí's territory experiences average annual precipitation levels below 800 mm, which does not include the Project Area of the current study (average precipitation in the PA is around 940 – 1,130mm / year) (EDUFPI, 2020²³).

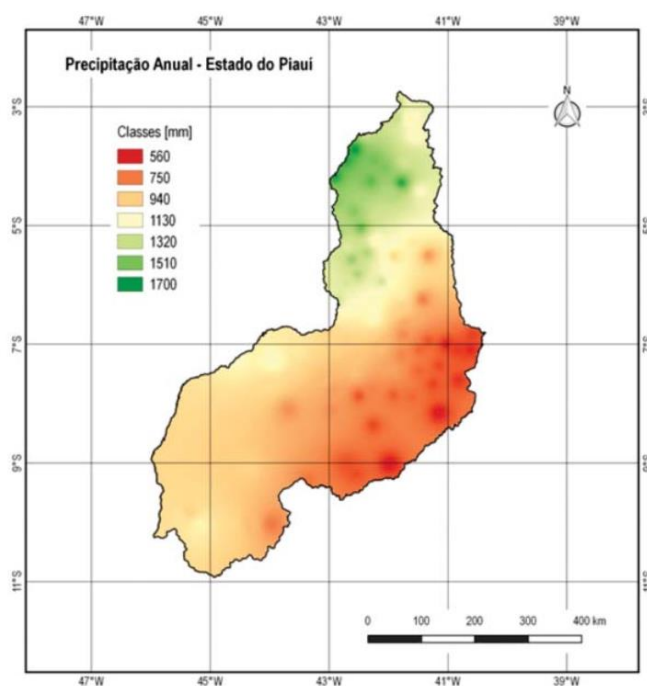


Figure 15. Characterization of rainfall in the Piauí State. Source: EDUFPI, 2020.

Based on The Köppen-Geiger Climate Classification System, one of the most common climate classification systems globally used to denote different climate regions on Earth based on local

²¹ Instituto Brasileiro de Geografia e Estatística. IBGE is the Brazilian Institute of Geography and Statistics, a public institution linked to the Ministry of Economy of Brazil. This is the main organization responsible for collecting, analyzing, and disseminating statistical, geographic, cartographic, geodetic, and environmental data about Brazil. Its information is essential for the formulation of public policies, socioeconomic studies, and to guide decisions by government, businesses, and organizations across the country.

²² IBGE, available on: <<https://www.ibge.gov.br/geociencias/informacoes-ambientais/climatologia/15817-clima.html?=&t=downloads>>.

²³ Climas do Piauí: interações com o ambiente / Organização, Milcíades Gadelha de Lima... [et al.] - Teresina: Edufpi, 2020, available on: <https://ufpi.br/arquivos_download/arquivos/LIVRO_CLIMA_E_MEIO_AMBIENTE_220201020201106.pdf>.

vegetation, The World Bank Group²⁴ shows that the Piauí State falls into the BSh/As/Aw categories, essentially a mix between the Tropical Savanna and the Hot Semi-Arid climates. It features a rainy season in the summer, and a distinct dry season in the winter, from May to October (August being the driest month). The average temperature of the coldest month is around 26,5°C. The maximum precipitation historic has indicated an average of 179mm in March (EMBRAPA)²⁵ ²⁶.

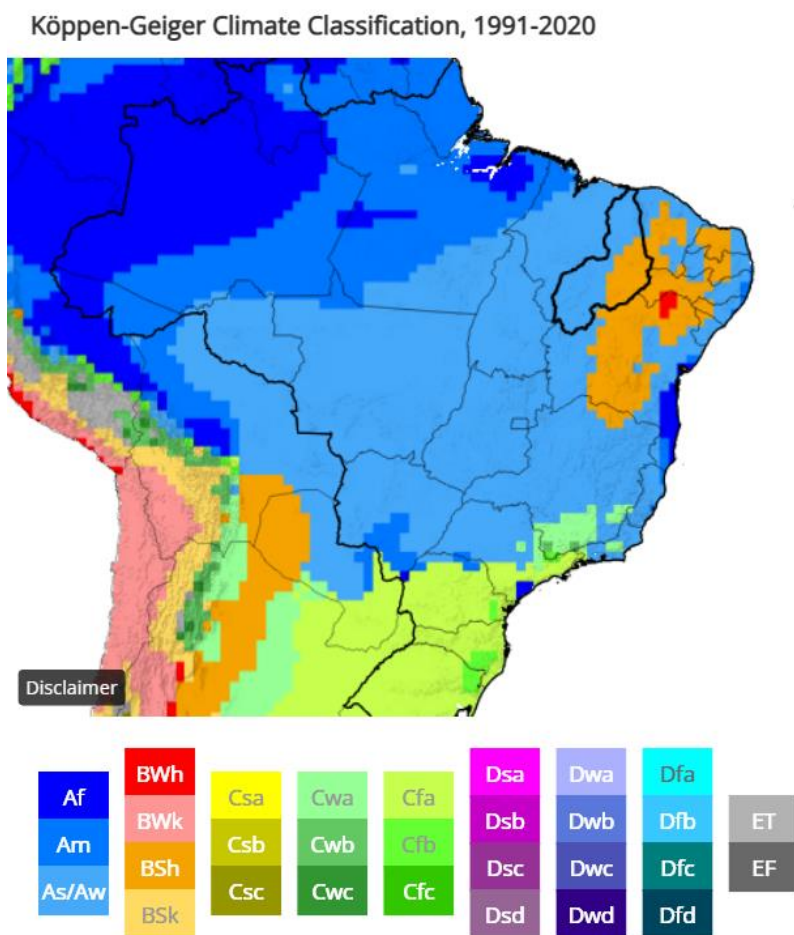


Figure 16. Köppen Climate Classification System.

Source: The World Bank.

²⁴ The World Bank Group, *Climate Change Knowledge Portal*, Köppen-Geiger Climate Classification, 1991-2020, available on: <<https://climateknowledgeportal.worldbank.org/country/brazil>>.

²⁵ Empresa Brasileira de Pesquisa Agropecuária. EMBRAPA is the Brazilian Agricultural Research Corporation, a public company affiliated with the Ministry of Agriculture, Livestock, and Food Supply (MAPA) of Brazil. Its mission is to develop technological solutions, knowledge, and information for sustainable agriculture in Brazil through scientific research in various fields, including agriculture, livestock, natural resources, and environmental sustainability.

²⁶ Pesquisa Embrapa, Clima, available on: <<https://www.cnpf.embrapa.br/pesquisa/efb/clima.htm>>.

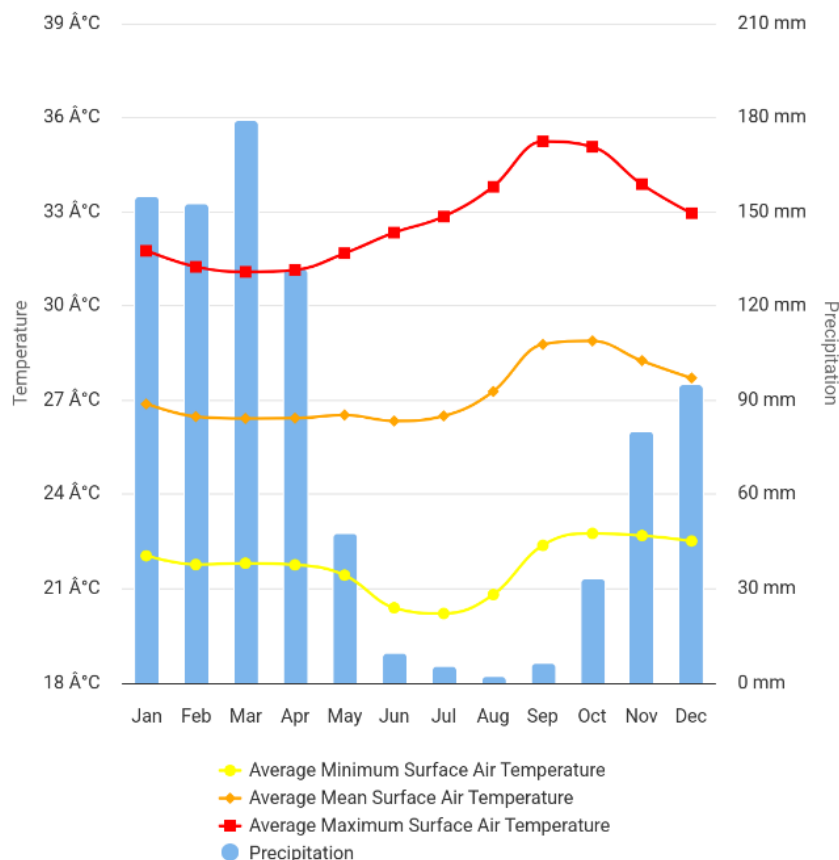


Figure 17. Temperature & Precipitation, 1991-2022, Piauí State.

Source: The World Bank.

1.14.5 Geology

The geology in the project area refers to the Paleozoic Era, a significant period in Earth's geologic time scale that spanned approximately 541 to 252 million years ago. This era was marked by the emergence of the first complex terrestrial ecosystems. Additionally, it was also marked by formation of extensive sedimentary basins, such as the Parnaíba Sedimentary Basin, present in the project area, and the deposition of important sedimentary layers, which would later give rise to resources like coal and fossil fuels. The Paleozoic Era also saw the evolution and diversification of marine life, as well as the advent of early land plants and invertebrates, laying the foundation for the rich ecosystems that would develop in subsequent eras²⁷.

²⁷ Provenance study of the Piauí formation, southern Parnaíba Basin. Available on: <<https://lume.ufrgs.br/bitstream/handle/10183/257372/001167164.pdf?sequence=1>>.

1.14.6 Topography

The geological history has directly influenced the topography of Piauí. The region includes flat to gently undulating relief, with areas with geographical depressions. The topography can influence water drainage, soil erosion, and agricultural suitability.

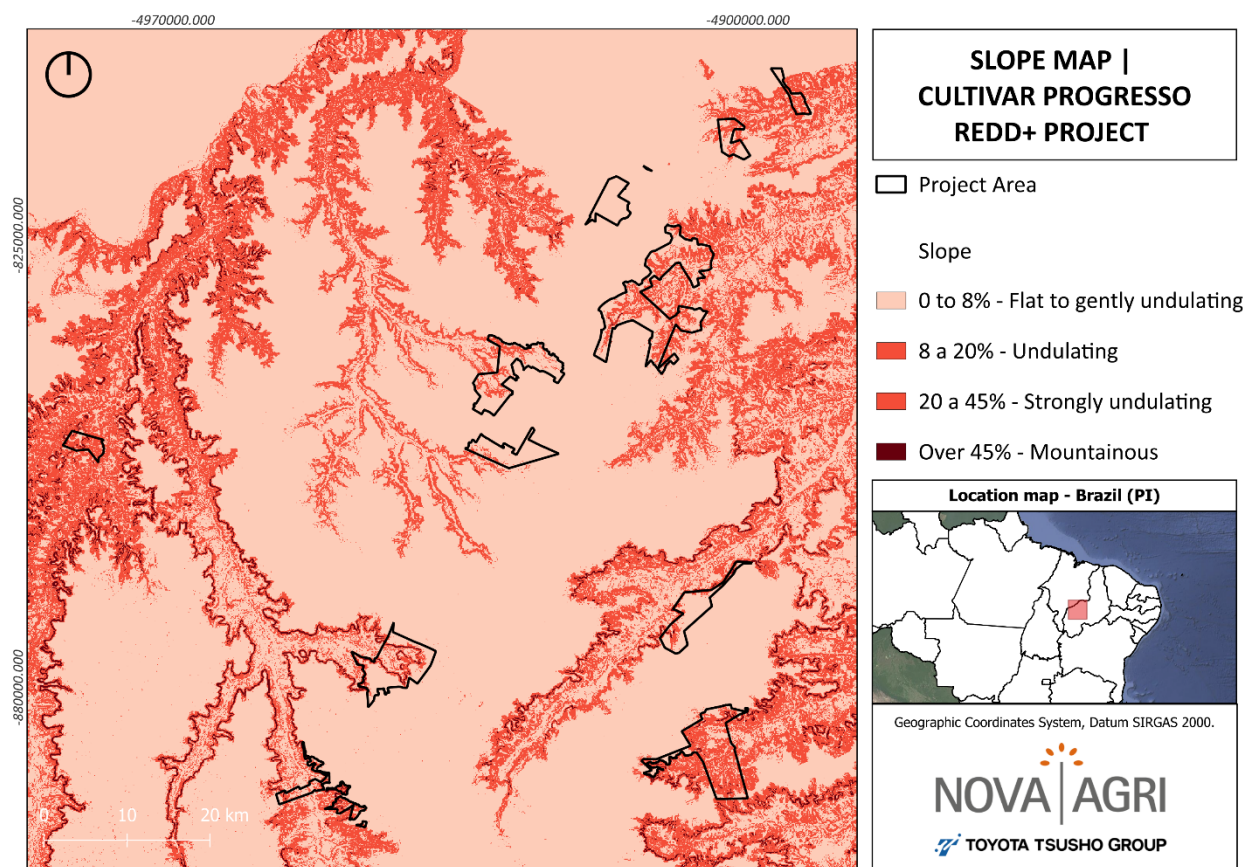


Figure 18. Map: Slope in the project area.

Source: EMBRAPA²⁸.

1.14.7 Soil

The majority of the soil in the project area consists of Dystrophic Yellow Latosols, which are deep soils with good drainage and textures ranging from medium to clayey. These soils are typical of transition zones between the Amazon and the Cerrado, with vegetation predominantly consisting of wooded savanna and forest fragments. Due to their high level of weathering, they have low natural fertility, requiring chemical amendments for agricultural use (EMBRAPA, 2021)²⁹.

²⁸ Brazilian relief, available on: <<https://www.cnpm.embrapa.br/projetos/relevobr/download/pi/pi.htm>>.

²⁹ EMBRAPA, Latosols. Available on: <<https://www.embrapa.br/agencia-de-informacao-tecnologica/tematicas/solos-tropicais/sibcs/chave-do-sibcs/latossolos>>.

Another significant portion of the area is composed of Dystrophic Lithic Neosols, which are shallow and characterized by high stoniness. These soils have moderate to good drainage and are commonly found in undulating to mountainous terrains. Due to their limited water retention capacity and low fertility, they present restrictions for intensive agricultural (EMBRAPA, 2021)³⁰.

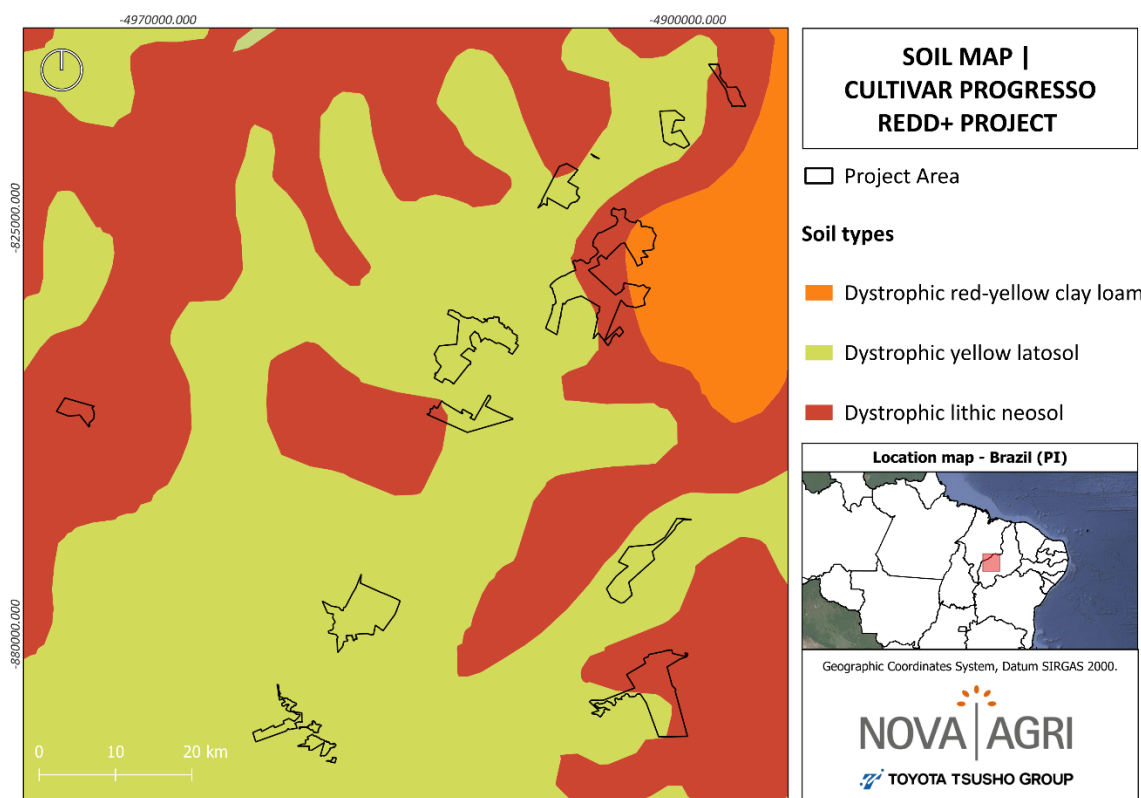


Figure 19. Soil types in the project area.

Source: EMBRAPA, IBGE³¹.

1.14.8 Hydrology

The project area is close to several water bodies, with the nearest being the Parnaíba River, one of the most significant in the region, which serves as a natural border between the states of Piauí and Maranhão. Additionally, smaller rivers and streams, such as the Uruçuí-Preto River, flow within the surrounding landscape. These watercourses maintain a permanent flow and are part of the Parnaíba River watershed, which covers an area of approximately 325.834,8 km², 279

³⁰ EMBRAPA, Neosols. Available on: <<https://www.embrapa.br/agencia-de-informacao-tecnologica/tematicas/solos-tropicais/sibcs/chave-do-sibcs/neossolos>>.

³¹ IBGE, Pedology. Available on: <https://geoftp.ibge.gov.br/informacoes_ambientais/pedologia/vetores/escala_250_mil/versao_2023/>.

municipalities and 4.800.934 people, making it a vital freshwater resource for local ecosystems and communities (CODEVASF, 2022)³².

Deforestation and land-use changes in the region have significantly impacted the hydrological and ecological balance. Loss of native vegetation over time has contributed to changes in the volume and flow of water bodies, as well as sedimentation processes (MAPBIOMAS, 2022)³³. Computer simulations and field data suggest that the combination of anthropogenic activities and climate variability has exacerbated water resource challenges. These findings underscore the importance of initiatives like REDD+ (Reducing Emissions from Deforestation and Forest Degradation) in addressing deforestation to preserve hydrological balance and ecosystem health.

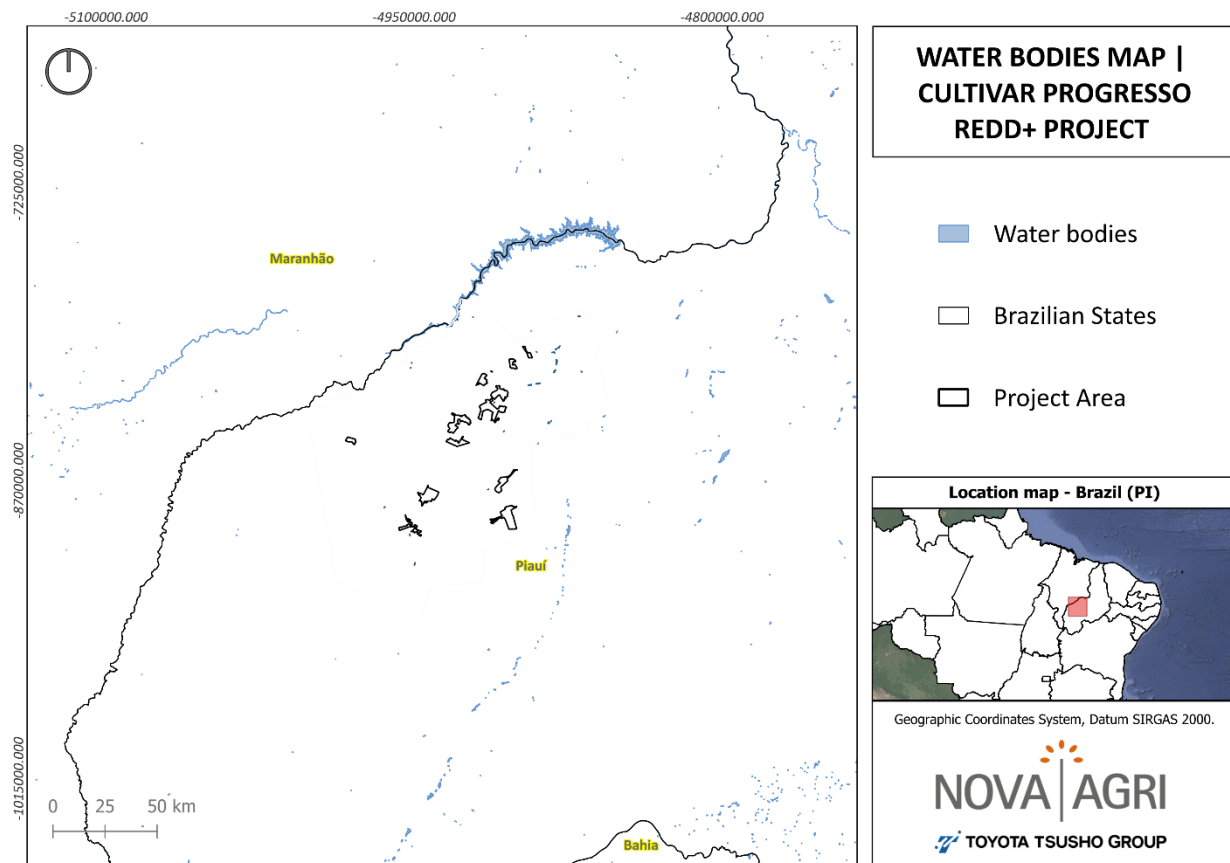


Figure 20. Water bodies in the project area.

Source: National Water Resources Information System (SNIRH)³⁴.

³² CODEVASF, available on: <<https://www.codevasf.gov.br/noticias/2014/bacia-do-parnaiba-abriga-4-8-milhoes-de-pessoas-279-municipios-e-tres-diferentes-biomas>>.

³³ MapBiomias Water, available on: <<https://plataforma.agua.mapbiomas.org/>>.

³⁴ SNIRH, Water bodies. Available on: <<https://metadados.snirh.gov.br/geonetwork/srv/api/records/7d054e5a-8cc9-403c-9f1a-085fd933610c>>.

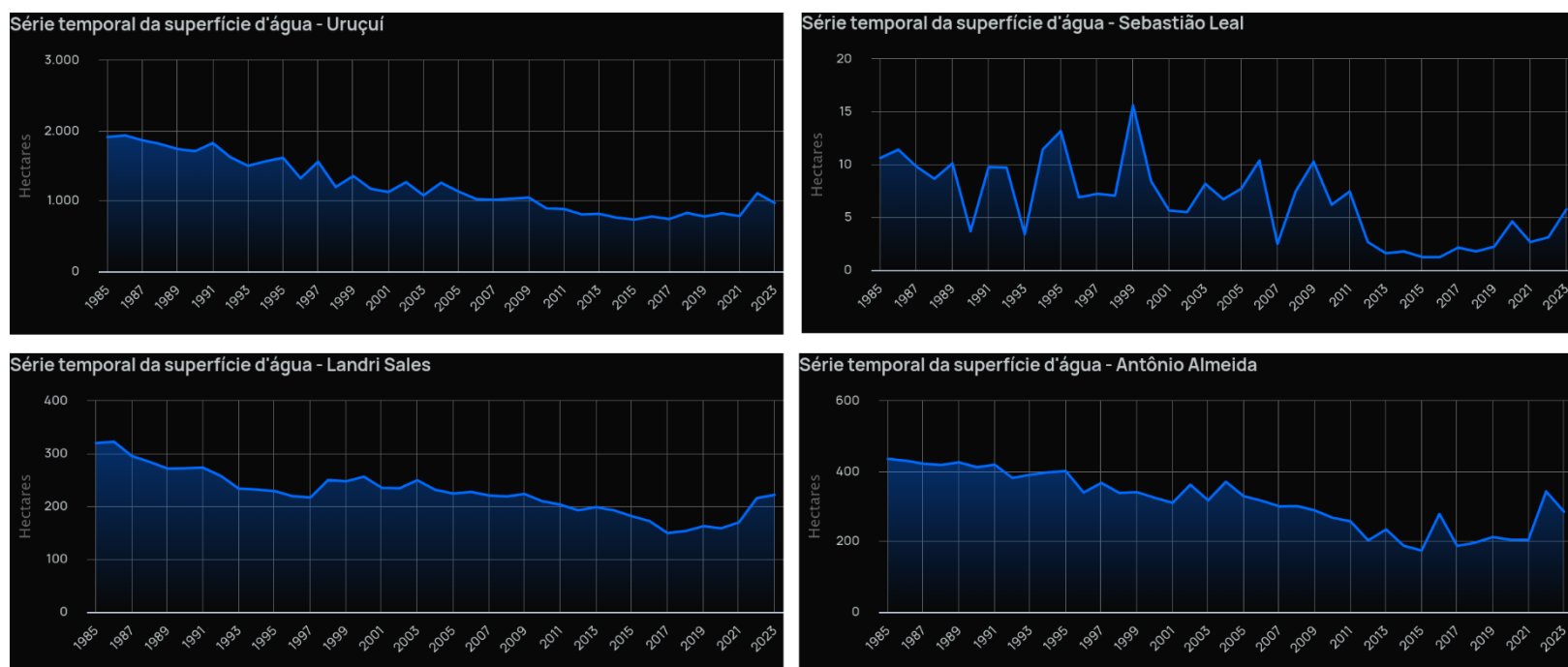


Figure 21. Water volume change, 1985 - 2023.

Source: MapBiomias Water.

1.14.9 Vegetation

The study of vegetation was based on the **Natural Resources Mapping of IBGE**, an activity developed by the Environmental Coordination and the Environmental and Geography Management of the State Superintendencies of the Institute. The classification of vegetation cover was performed at the 1:250,000 scale. At this level of precision, the mapping unit boundaries and their respective classification legends are visible. The study's methodology is detailed in the document "**Technical Manual of Brazilian Vegetation**"³⁵.

According to IBGE, the most predominant **Phyto physiognomies** that constitute the natural environment of the designated area are: Forested Savannah; Wooded Savannah; Savannah Park; Forested Seasonal Savannah; and Agriculture and/or livestock.

These vegetation types, which are characteristic of the transition zone between the Cerrado biome and seasonal forests, serve as foundational elements for our carbon project. They provide essential insights into the ecosystem structure and facilitate a comprehensive understanding of the carbon dynamics within the studied area.

³⁵ IBGE, Technical Manual of Brazilian Vegetation. Available on: <https://edisciplinas.usp.br/pluginfile.php/4228241/mod_resource/content/2/Manual%20Tecnico%20da%20Vegetacao%20Brasileira%20-%202012.pdf>.

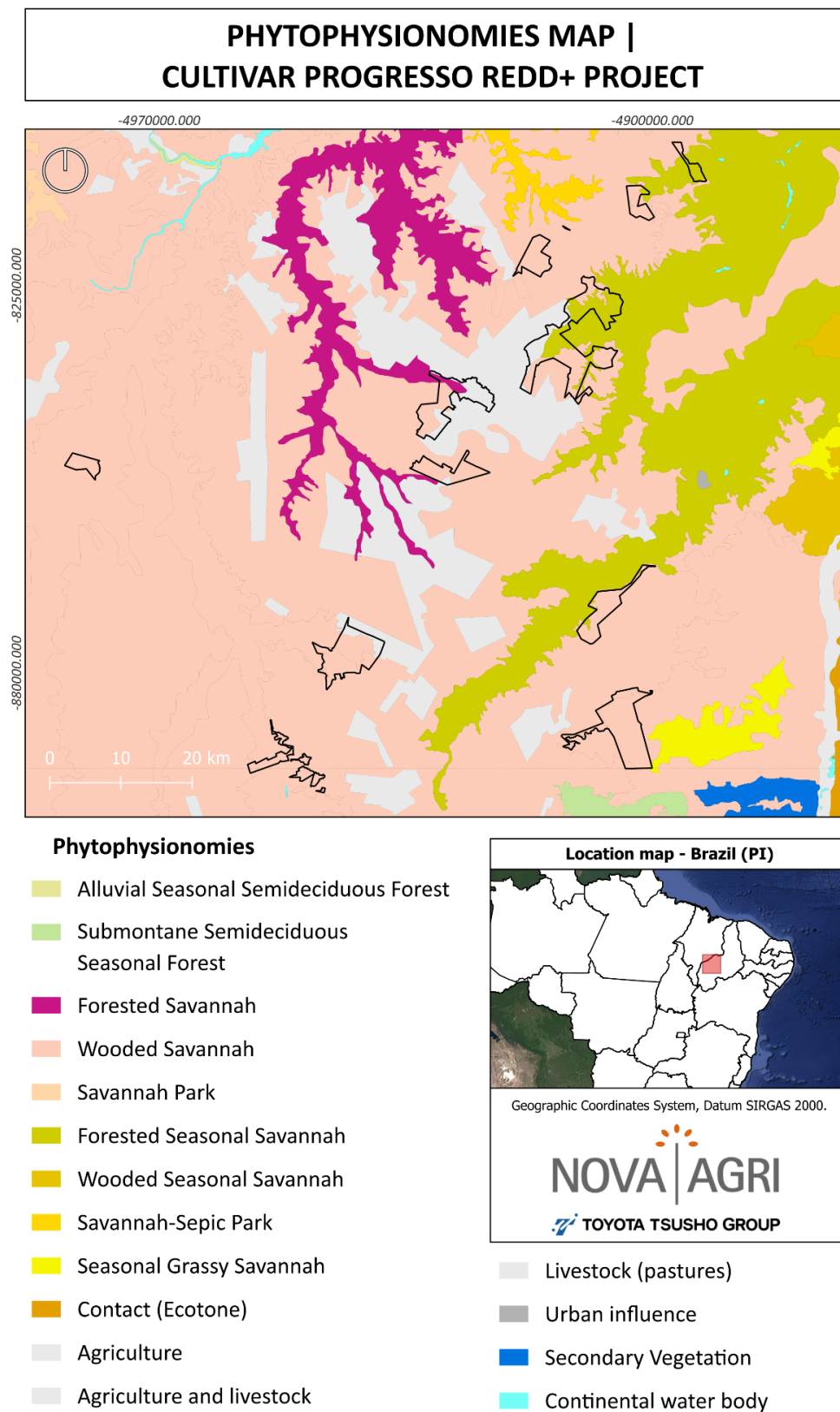


Figure 22. Phyto physiognomies in the project area and surroundings.

Source: IBGE.

Additionally, the same data was used to identify if the project area includes any type of wetland. All alluvial vegetation types, associated with riverside areas subject to periodic flooding, were grouped in the class “Alluvial formations”, and lacustrine/lake formations, which suffer constant or seasonal flooding, were grouped in the “Lake formations” class. The resultant map proves that there are no wetlands in the project area.

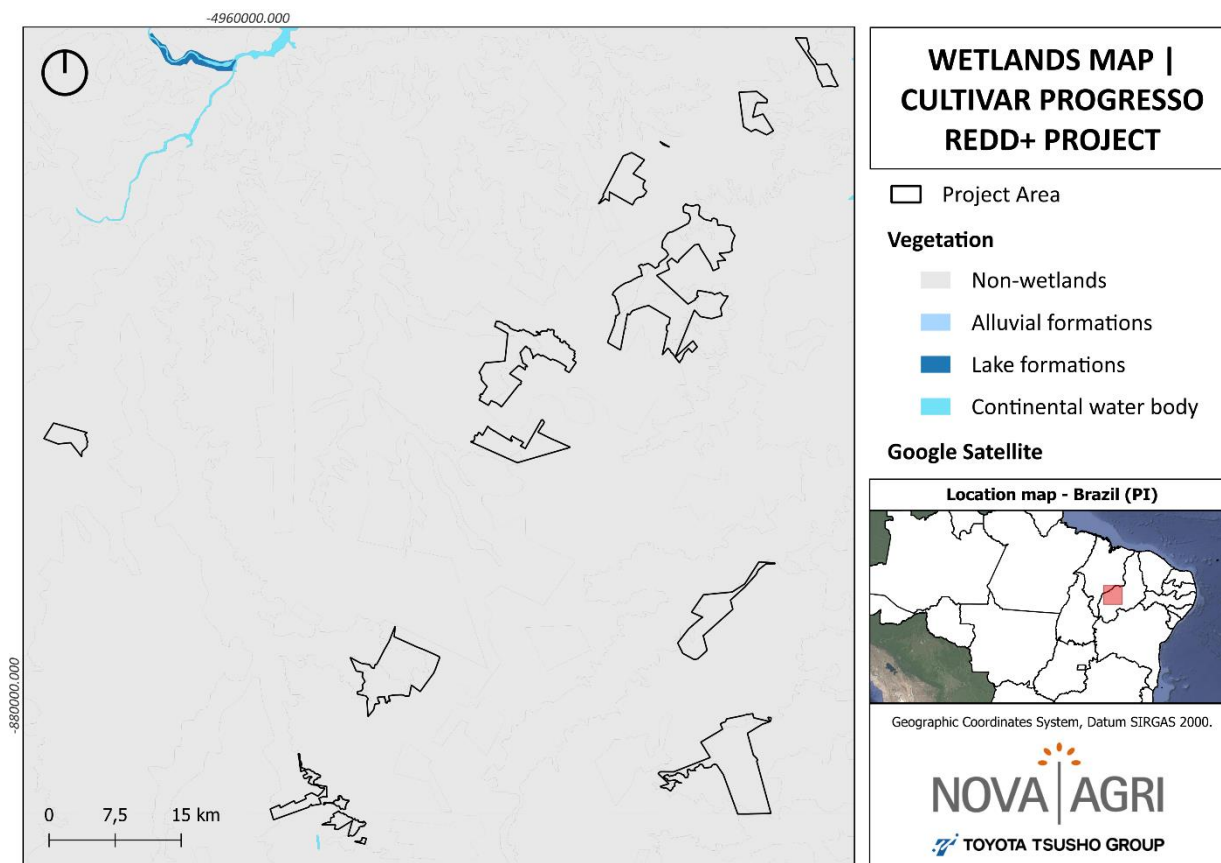


Figure 23. Wetlands map.

Source: IBGE.

1.14.10 Biodiversity

Flora (trees)

The uplands of Uruçuí, Sebastião Leal, Landri Sales, and Antônio Almeida, in the southwest of Piauí, are characterized by dense evergreen forests on fertile loamy soils, supporting species such as Brazilwood (*Caesalpinia echinata*), Jatobá (*Hymenaea* sp.), Xixá (*Sterculia* sp.), Cedro (*Cedrella* sp.), and the palm Babaçu (*Orbignya* sp.), among others. In the Cerrado areas, common

species include Pau Terra (*Qualea* spp.), Murici (*Byrsonima* spp.), Araçá (*Psidium* sp.), and palms like Pati (*Syagrus* sp.) and Babão (*Syagrus comosa*) (CASTRO et al., 2022)³⁶.

Fauna

The fauna in Uruçuí, Sebastião Leal, Landri Sales, and Antônio Almeida reflects the biodiversity of the transitional ecosystems between dense forests and Cerrado. Mammals like armadillos, anteaters, deer, tapirs, jaguars, and various monkeys are common, alongside a diverse avifauna that includes toucans, macaws, hawks, and parrots. Reptiles, such as iguanas and snakes, are also abundant. These species underscore the ecological richness of the area, supported by its mosaic of forested and savanna-like habitats (GEES, 2024)³⁷.

1.14.11 Lack of Preservation Actions Context

The conditions prevailing before the initiation of the carbon project reveal a landscape characterized by a lack of essential elements critical for effective environmental conservation and sustainable land management. These conditions are essential to understanding the baseline against which the project plans to unfold:

- The project area currently lacks a comprehensive near real-time satellite monitoring system.
- The current framework does not provide financial incentives for landowners to refrain from deforesting forested areas.
- At present, there are no financial returns or mechanisms tied to the maintenance of forests. This absence diminishes the economic value associated with forest conservation, limiting the potential for sustainable practices.
- The absence of structured financial returns from forest maintenance further translates into a deficit in resources for social responsibility initiatives. Consequently, there are limited avenues for converting forest preservation into tangible social and community benefits.
- There is a lack of targeted awareness and educational initiatives for both employees and the local community regarding the environmental significance, urgency, and future implications, particularly in the context of climate change.

³⁶ CASTRO, et al. Characterization of the flora and vegetation of the seasonal forests of Serras Gerais, in the municipalities of Eliseu Martins, Pavussu and Canto do Buriti, southwestern Piauí State. Even3, 2022. Available on: <<https://publicacoes.even3.com.br/preprint/caracterizacao-da-flora-e-da-vegetacao-de-florestas-estacionais-das-serras-gerais-municipios-de-eliseu-martins-pavussu-e-canto-do-buriti-sudoeste-do-piaui-692177>>.

³⁷ GEES Fertilizantes: Environmental Impact Study, Uruçuí (PI), 2024. Available on: <<https://siga.semapi.gov.br/media/uploads/2024/03/07/983e0cb0-5a06-4f9c-8a6f-97746de7b178.pdf>>

- Enforcement of environmental laws is currently inadequate, contributing to a permissive environment for activities such as deforestation without stringent consequences.
- Over history, including recent years, the regulatory national/local framework has demonstrated limitations in effectively avoiding deforestation. The inefficiency of existing regulations poses a significant hurdle in preventing both legal and illegal deforestation activities, contributing to spreading uncontrolled devastation.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

	Description	The project complies / is aligned with
Law n. 12.651 of 25th May 2012 Institutes the Forest Code (FEDERAL LEVEL)	The Forest Code is a federal law in Brazil that establishes the regulations for vegetation use, especially on rural properties. Its main goal is to ensure environmental balance, water quality, soil fertility, clean air, and a stable climate. The Forest Code outlines: Which areas must be preserved; Which regions can support different types of rural production; How to responsibly manage and explore forests; How to trace the origin of forest products; How to prevent and control forest fires.	The project and its activities do not violate any principles established by the Brazilian Forest Code, in any of its versions. Its primary objective is to preserve the forests protected under the Law from agents that might eventually infringe or bypass it. Therefore, the project plays a crucial role in reinforcing legal compliance by maintaining legal reserves, Permanent Preservation Areas (APPs), and forest remnants; and also by protecting the region from externalities, such as leakage of deforestation pressures.
Complementary Law 140/2011 (FEDERAL LEVEL)	This Law establishes norms for the cooperation between the Union, the States, the Federal District and the Municipalities in the administrative actions arising from the exercise of the common competence of the protection of remarkable natural landscapes, the protection of the environment, the fight against pollution in any of its forms and the preservation of forests, fauna and flora.	The project is aligned with this Complementary Law, as it actively engages in dialogue, particularly at the municipal and state levels. Deforestation is also a concern for local environmental authorities, and the project's activities contribute by enhancing surveillance and monitoring efforts when the law is not adequately enforced.

Law n. 9,985 of 18th July 2000 (FEDERAL LEVEL) Decree n. 4,340, of 22th August 2002 (FEDERAL LEVEL)	This law institutes the National System of Conservation Units (SNUC), and establishes criteria and norms for the creation, implementation, and management of protected areas, amongst other provisions. This decree regulates Articles of Law n. 9,985, of 18th July, 2000, which regards the National System of Conservation Units (SNUC) and makes other provisions. It regards the creation of Protected Areas, Management Plan, advisory management and authorization to explore goods and services.	The AFOLU activities carried out in the project are solely focused on preservation and do not take place in any public areas. Their practical impact is minimal in both the reference area studies and leakage assessments, as there are no nearby Conservation Units.
Normative Instruction 04/2009 (FEDERAL LEVEL)	This normative provides technical procedures for the use of vegetation of the Legal Reserve under a regime of sustainable forest management, and other measures.	The project is in full compliance with the regulations, as it not only adheres to the established preservation procedures but also provides new mechanisms for additional protection of legal reserves, such as continuous monitoring and the use of technology.
Decree N. 7.830, of October 17th, 2012 (FEDERAL LEVEL)	Establishes the Environmental Regularization Program - PRA, regulates the Rural Environmental Registry (CAR), the Environmental Regularization of Rural Properties, and the Environmental Licensing of polluting activities or those using natural resources, amongst other measures.	The project makes use of the CAR (Rural Environmental Registry) mechanism, a crucial governmental tool for monitoring private rural lands, to carry out its key initial and ongoing assessments. The CAR also provides transparency regarding the landowner's decisions related to land use within its territory. Its boundaries are essential for ensuring that the forest protection results are clearly and consistently demonstrated.

Law n. 15.042 of December 11h 2024: Regulatory Market for Carbon Credits (FEDERAL LEVEL)	This law establishes the Brazilian Emissions Trading System (SBCE) and regulates the carbon market in the country. The SBCE, a carbon commercialization system, aligns with global efforts to meet greenhouse gas reduction commitments under the United Nations Framework Convention on Climate Change and the Paris Agreement. The approved text indicates the adoption of cap-and-trade, setting emission limits overseen by a competent authority. Companies exceeding their allocated emission quotas may sell the surplus to those emitting above their quotas, ensuring the overall emission reduction targets are met within the established limits. Companies with emission reduction obligations can demonstrate compliance by returning initially allocated quotas or purchasing carbon credits generated through accredited methodologies.	This law does not restrict the generation of credits within the voluntary market or their participation in the regulated market. Therefore, as there are no restrictions regarding this market or AFOLU activities, the project fully adheres to the law.
CONSTITUTION OF THE STATE OF PIAUÍ (STATE LEVEL)	Article 237: 'Every person has the right to an ecologically balanced environment, which is a good for the common use of the people and essential to a healthy quality of life, imposing on the public authorities and the community the duty to defend and preserve it and to harmonize it rationally with the needs of socio-economic development for present and future generations.'	The principles of this VCS project align with the constitution of the State of Piauí, which, in addition to the federal constitution, prioritizes the protection of the environment. In addition to being compatible with and not violating any constitutional rules, the project promotes joint conservation efforts.

Normative instruction N.. 9 of March 30 th , 2023 (STATE LEVEL)	Provides for the State administrative process for investigating administrative infractions related to harmful conduct and activities to the environment, regulates the Environmental Fines Conversion Program, and establishes other provisions.	The project and its activities adhere to all environmental regulations and do not engage in any harmful practices. There are no risks associated with inappropriate conduct. On the contrary, the project reinforces positive environmental practices and is backed by a legally binding contract with the landowner, who commits to actively supporting forest protection without any changes to the current land use.
CLT – Consolidation of Labor Laws (FEDERAL LEVEL – other subjects)	The Consolidation of Labor Laws (CLT, in Portuguese), formalized by Decree-Law 5.452/1943, plays a crucial role in ensuring a range of fundamental rights for Brazilian workers. Among these rights are the maximum daily working hours, paid weekly rest, paid vacations, overtime pay, work in healthy environments, advance notice in case of dismissal, maternity and paternity leave, 13th-month salary, protection against unjustified dismissal, and the benefit of unemployment insurance. The CLT is considered one of Brazil's first instruments of social inclusion. Its promulgation was a historic milestone that established essential rights for workers, significantly contributing to the construction of a more just and egalitarian society.	The project activities do not violate the national labor laws. All direct employees working on the property are covered by the CLT (Consolidated Labor Laws) and all its regulations. Any potential risks of violations will be identified beforehand, and if there is any evidence of human rights violations such as child labor or slavery by the landowner, the project will be terminated before it even starts. After the initial screening and commencement of project activities, monitoring of these risks will continue and will be covered by a formal contract. Additionally, an appropriate compliance channel will be established to report any incidents.

1.15.1 Regulatory and controller Brazilian agencies

Brazil relies on a network of dedicated environmental control agencies. These agencies are entrusted with the responsibility of enforcing environmental regulations, managing protected forest areas, and preserving biodiversity. These agencies are crucial for the positive progress of

this project, once they are the main instruments – based on major Law – to promoting the natural resources by penalizing and restricting the deforestation drivers indicated on this PDD.

At Federal level, the following agencies can be mentioned as basis:

Table 6. Federal Agencies

Federal Agency	Attributes relative to Deforestation
IBAMA	Inspection, Penalties and Embargo.
CNAL	Coordination of actions for the Amazon preservation and protection.
SFB	Coordination of the System of the CAR (Cadastro Ambiental Rural, in English: Rural Environmental Registration).
ICMBio	Management, inspection, and protection of the 'Unidades de Conservação Federais', in English: Federal Conservation Units.
INCRA	Management of settlements from Agrarian Reform and land regularization.
FUNAI	Management and Protection of Indigenous lands.

1.15.2 Labor and Human Rights

CLT – Consolidation of Labor Laws:

The Consolidation of Labor Laws (CLT, in Portuguese), formalized by Decree-Law 5.452/1943, plays a crucial role in ensuring a range of fundamental rights for Brazilian workers. Among these rights are the maximum daily working hours, paid weekly rest, paid vacations, overtime pay, work in healthy environments, advance notice in case of dismissal, maternity, and paternity leave, 13th-month salary, protection against unjustified dismissal, and the benefit of unemployment insurance. The CLT is considered one of Brazil's first instruments of social inclusion. Its promulgation was a historic milestone that established essential rights for workers, significantly contributing to the construction of a more just and egalitarian society.

1.15.3 Carbon and Climate Change specific Frameworks

In the project's commitment to advancing sustainable development and mitigating climate change impacts, this project is resolute in adhering to recognized Climate Change studies frameworks. By adopting these frameworks, the project aims to ensure the credibility and effectiveness of its initiatives while contributing to the global effort to combat climate change. In this regard, this project acknowledges the significance of staying abreast of the most recent and influential frameworks shaping the carbon credit market, considering both technical consensus and the relation with the VERRA's methodology. Some prominent frameworks which base this project are listed below:

Intergovernmental Panel on Climate Change (IPCC)³⁸:

As a leading authority on climate science, the IPCC's assessments and reports significantly influence global climate policies and carbon market mechanisms. By leveraging the IPCC's findings, this project ensures that initiatives are rooted in the latest scientific knowledge, contributing to a well-informed approach to addressing climate change.

Greenhouse Gas Protocol (GHG Protocol)³⁹:

The GHG Protocol is widely recognized as the most widely used framework for measuring and managing greenhouse gas emissions. Adopting the GHG Protocol enables us to comprehensively assess our projects' carbon footprints and implement effective emission reduction strategies.

Recommended practices (PR) 2060 by ABNT⁴⁰:

This is a Brazilian technical standard developed by the Brazilian Association of Technical Standards (ABNT) specifically focused on carbon-related matters. This standard provides guidance for entities in Brazil to quantify, monitor, and report their greenhouse gas (GHG) emissions and removals. It aligns with international norms, including the ISO 14064 series, ensuring compatibility with global best practices in GHG accounting and reporting.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

³⁸ IPCC's Reports, available on: <<https://www.ipcc.ch/reports/>>.

³⁹ GHG Standard, available on: <<https://ghgprotocol.org/standards-guidance>>.

⁴⁰ ABNT's PR2060, available on: <<https://www.normas.com.br/visualizar/abnt-nbr-nm/13635/abnt-pr2060-especificacao-para-a-demonstracao-de-neutralidade-de-carbono>>.

☐ Yes ☒ No

The term of agreement between NovaAgri and the landowner includes not engaging in any other remunerated or non-remunerated activities in the Project Area, generating any credits/assets/values/rights, incurring to double-claiming and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

In this case, the questions regarding Scope 3 Supply Chain emissions are not applicable. Neither the landowner nor retailer(s) of impacted goods and services, nor the project proponent, have posted any public statements indicating that 'VCUs may be issued for the greenhouse gas emission reductions and removals associated'. Additionally, the term of agreement between NovaAgri and the landowner regards not engaging in any other remunerated or non-remunerated activities in the Project Area, generating any credits/assets/values/rights, incurring to double-claiming, and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines.

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The project aims to reduce deforestation and promote sustainable land use practices in Brazil's Cerrado region. Through REDD+ initiatives, it addresses climate change, biodiversity conservation, and socioeconomic challenges.

The project directly reduces greenhouse gas emissions by protecting forests and aligning with Brazil's climate targets. Conservation efforts preserve biodiversity, support local communities, and improve livelihoods. By generating employment, promoting inclusive practices, and fostering partnerships, the project contributes to economic growth and social well-being.

The company also is committed to allocating 5% of the project's financial returns – generated from the sale of Verified Carbon Units (VCUs) – to fund and implement impactful social initiatives within the communities surrounding our project area. Additionally, the landowner has committed to allocating 100% of their own financial returns to social investments within local communities and reinvestments in the project's activities themselves, ensuring its sustainability and continuity.

The project aligns with Brazil's goals of combating deforestation, conserving biodiversity, and promoting social inclusion. Monitoring and reporting mechanisms ensure compliance through using high-level technologies, a dedicated team specifically for the project, and a corporate governance structure.

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 7: Sustainable Development Contributions

Row number	SDG target	SDG indicator ⁴¹	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	1.a	1.a.1: Total official development assistance grants from all donors that focus on poverty reduction as a share of the recipient country's gross national income	Implemented activities to increase	The proponent will relate all contributions to this official SDG target and indicator after the first verification and credits sold. The purpose of monitoring this SDG indicator in this project is to highlight the financial contributions/donations made by the Project Proponent to ensure the successful implementation of the previewed social and well-being improvements. When the social services are defined, those will be presented here with the proper OECD purpose codes, which identify the sector the activity is intended to target (e.g. • Basic Education 112xx • Basic Health 122xx • Water Supply and Sanitation 140xx • Multisector aid for basic social services 16050 • Development Food Aid 52010, etc.) As this is not a government project activity, the value will result from a project's self-defined metrics, in monetary units (BRL).	No historic yet

⁴¹ SDG Indicators Metadata repository, available on: <<https://unstats.un.org/sdgs/metadata/>>.

2)	13.2	13.2.2: Total greenhouse gas emissions per year	Implemented activities to reduce	Even though Brazil is not part of the Annex 1 Parties, the country submits voluntarily its emissions for National communications (NC) and/or Biennial update reports (BUR) from non-Annex I Parties. The purpose of this indicator is to present the reduction of carbon emissions resulting from the project's activities in absolute terms, using the same metric and calendar suggested by national inventories (annual / in tCO₂e). This is aligned with the SDG and Paris Agreement objectives for tackling Climate Change, and following the framework established by the IPCC, as also outlined within the VCS methodology and AFOLU standard. The presented value refers to the Project Area scope and the monitoring period (01/Jul/2023 to 30/Jun/2024).	-31.376tCO ₂ e ⁴²
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⁴² This value is evidenced on the 'Section 7. Quantification of GHG emissions' of this PD.

3)	15.1	15.1.1: Forest area as a proportion of total land area	Implemented activities to maintain	The project's activities are directly and continuously aligned with this SDG indicator, as the main outcome of the preservation efforts is the effective area of forests that have been maintained throughout the successful implemented activities during the monitoring period. By following the metrics and rationale provided by the SDG metadata page⁴³ two key parameters are considered: 1) preserved forest area and 2) the total land area of the country⁴⁴. The result is the proportional percentage of preserved forest in relation to the total Brazilian land area, as outlined below. $\frac{\text{Forest area (reference year)}}{\text{Land area (reference year)}} \times 100$ 680 ha⁴⁵ / 851,041,777 ha = 0,000079%	0,000079%
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⁴³ SDG METADATA, available on: <<https://unstats.un.org/sdgs/metadata/>>.

⁴⁴ FAO land official data/Brazil, available on: <<https://www.fao.org/faostat/en/#data/RL>>.

⁴⁵ This value is evidenced on the 'Section 7. Quantification of GHG emissions' of this PD.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

In accordance with Methodology VM0015 v1.2, the project has defined a non-forested area within the Reference Region as the 'Leakage Management Zone' (LMZ). The attributes and characteristics of the LMZ itself are presented in Section 3.3, with its respective spatial and temporal boundaries. Regarding applicability, the area is a legitimate representative of one of the defined agents, more specifically a small-scale deforestation agent, that means a property exhibiting the same variables and drivers that influence deforestation projections in the project Project Area and Leakage Belt.

Furthermore, the LMZ, which has been historically impacted by deforestation, now contributes to the commercial scenario in the region, since it is a farm with consolidated agricultural activities; specifically, the cultivation of corn and brachiaria grass. The main objective of this plan is to enhance the farm's characteristics, in order to reinforce its potential as a revenue-generating entity, contributing to the local GDP and, consequently, fostering positive externalities for the community. This dynamic might ensure employment opportunities, enhance local infrastructure, and reduce the likelihood that activities will shift to neighboring lands, thus minimizing leakage risk. The outline for the Leakage Management Plan is presented below.

First 2 Years:

On-site Practical Activities:

Technical Consultancy: For the first two years, technical consultancy services will be provided directly to the landowner responsible for the LMZ. Since this is primarily an agricultural area, a mentorship program will be conducted by the landowner of Grupo Progreso, with support from NovaAgri, directly to the person(s) responsible for agricultural activities in the LMZ. This support will involve sharing knowledge about optimizing crop performance, covering soil balance, increased nutrient availability for plants, enhanced organic matter content, and extended productive lifespan of planting areas. Regarding practical support, the Grupo Progreso is going to provide biological additives to improve the crop performance in a more sustainable manner. This support is essential for ensuring that the chosen farmer's production is sufficiently profitable to provide adequate financial returns to the LMZ owner, thus reconciling the best use of the already deforested area for income generation. In addition to practical activities on the farm, the consultancy plan will also include issues related to finance, negotiations, supply and better decisions for the agriculture business for the LMZ. A report with reports will be presented at the end of the two-year period to demonstrate that the activities have had a practical impact on this farm. The main quantitative indicators will be (but not limited to):

- Ton of corn produced per year (@ ton/year)
- Crop productivity per area (corn ton/year/ha)

Technical Support: Throughout the Leakage Management Plan, the landowner of Grupo Progreso is willing to lend materials, biological inputs, or any other items that facilitate the execution of activities in the LMZ. It is important to note that small producers sometimes lack the necessary tools and knowledge to carry out their basic activities, something that is much more common on larger farms. This support is essential so that they do not lack the basic mechanisms to carry out their activities efficiently and safely. Due to the informal nature of this support, formal contracts or terms will not be required, but the use of these items in daily use on the LMZ farm should be recorded through photographic images.

After 2 Years of Plan Implementation:

Continuous Support and Monitoring: After the two-year period of practical consultancy, monitoring will be continuous, implemented both by the proponent and by the landowner. In addition to continuing the practices indicated so far, complementary actions related to the more sustainable Land Use practices will also be implemented. Management responsibility is divided as follows:

- **Landowner:** Continue to provide technical support (knowledge and tools) throughout the entire Leakage Management Plan. After the first two years, the LMZ owner will be able to receive new and more complex guidance, this time more related to good practices and sustainable land use, thus enabling the implementation of some good practice related to crop rotation, allowing for a more diversified and healthy use of the soil. Results reports should be prepared following the indicators established at the end of each year.
- **Proponent:** The area will be monitored through public satellite imagery (e.g. weekly SENTINEL) and GIS tools with proof of agricultural activities and their good practices throughout the Leakage Management Plan. This proof is essential to ensure that the LMZ is being effective, and that the area is being well used and does not require the opening of additional areas. In addition, an interview will be conducted with both the landowner and the LMZ owner to compile the MRV report of the effective execution of the plan.

*Note: the full information regarding the LMZ stakeholder and its boundaries is available on
Section 3.3 – Leakage Management Zone (LMZ)*

1.19.2 Commercially Sensitive Information

A few pieces of information have been excluded from this public version of the PD due to commercially sensitive data that cannot be disclosed for business security reasons. The excluded information is specifically related to the profitability figures of conventional activities, potential gains from carbon credit revenue, and the comparison of these two for calculating NPV from short to long-term. Such information is referenced in the Cost Opportunity Costs section (Section 3.4) and in the Non-Permanence Risk Analysis of Section 4.4.

It is important to highlight that information related to the determination of the baseline scenario, demonstration of additionality, and estimation and monitoring of GHG emission reductions and

removals (including operational and capital expenditures) were not considered commercially sensitive and were provided in the public versions of the project documents.

There is a confidentiality agreement for the audit process and all the information, rationales, data, documents, and supports will be duly provided to the VVB team.

1.19.3 Further Information

Contractual Safeguards

NovaAgri counts with a legal team and its own policies to ensure the most appropriate execution of its carbon credit projects. As an essential step to firm a serious engagement between NovaAgri and the landowner is the agreement made between the parties, in a contractual formalization. This comprehensive agreement delineates the roles, responsibilities, general and specific commitments of both parties towards the development and execution of this VCS project.

The collaboration underscores the necessity for adherence to NovaAgri's Compliance Policies. This includes transparency in all project-related activities and the safeguarding of confidential information. Both parties are committed to upholding the highest ethical standards. The three main terms are summarized below:

(A) Project Area integrity

With the term, the landowner formally agrees to maintain the integrity of the area subject to the Project, making it free from deforestation, degradation, invasions, fires, or any other factors that may pose a risk of damage or actual harm to the area. In addition, it includes adhering to the best and most correct soil management practices, aiming at the preservation and development of biodiversity, including fauna and flora. All this involves reporting any situation that deviates from normality promptly, from both parties, during the monitoring period.

(B) No Double-counting Credits

Another essential term of the agreement regards not engaging in any other remunerated or non-remunerated activities in the Project Area, generating any credits/assets/values/rights, incurring double-counting and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines.

(C) Ethical Governance

Regarding the ethical governance aspects, the agreement includes compulsory compliance with anti-corruption laws, regulations and policies, as well as determinations and rules with the aim of preventing corrupt practices, illegal expenses related to political activity, harmful acts, infractions or crimes against the financial system or public administration in order to avoid law laundering, hiding assets and financing illegal traffic and terrorism. Also, the parties agree not to do business with companies from countries that have suffered an international trade embargo according to the UN, OFAC, European Union or Japan list; and not make allowances for acts that result in discrimination based on race or gender, with disabled people or those with special

needs, that involve child labor, slave labor or that characterize moral or sexual harassment or disrespect for human dignity.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

No Net Harm

The REDD+ project led by NovaAgri aims primarily to tackle climate change by preventing carbon emissions resulting from deforestation/degradation. The positive impacts include contributions to global climate change mitigation, social livelihood benefits, biodiversity conservation, and support for sustainable ecosystem services. However, potential negative risks exist, including socioenvironmental and compliance risks, disruption of traditional economic practices in the region, challenges in land tenure clarity, cultural and educational aspects, and the risk of deforestation in unforeseen circumstances.

To address these risks, NovaAgri has implemented various measures. A thorough pre-project analysis was conducted, utilizing technology for a reliable assessment of socioenvironmental compliance and relevant factors, alongside an Agreement Contract with the landowner. The project also considers the economic dependencies of local populations on agribusiness-related activities and demonstrates that existing cultivated lands are sufficient to sustain economic activities without compromising local jobs. Land ownership verification is prioritized to prevent phantom credits, involving a rigorous examination of possession documents, authentication in a notary office, and checks against the CAR document. The project addresses cultural and educational gaps through awareness initiatives, meetings, and distribution of informative brochures. The implementation of the MRV model and a dedicated department for carbon credit projects ensures monitoring, reporting, and verification aligned with the company's existing processes. Additionally, buffer credits are reserved to mitigate unforeseen circumstances. These measures aim to safeguard against potential risks and enhance the integrity of the project.

2.1.1 Stakeholder Identification

Find below the description of the stakeholder identification process.

Stakeholder Identification

To identify the key stakeholders in this AFOLU project, extensive discussions were held with the Landowner and NovaAgri's regional team, which are physically present in the region. This approach ensured a more accurate understanding of the potentially impacted or related

audiences in this project. In addition to these consultations with local teams, we enlisted support from NovaAgri's internal departments directly involved in credit, risk, sustainability, and compliance issues. This collaboration allowed us to comprehend which government institutions and departments could be involved in both the Carbon Project and the company's business. For instance, environmental entities regulating and overseeing rural areas of the state. Aligning the company's environmental compliance practices with the needs of the AFOLU Project was a crucial aspect. Finally, this information was validated through public and referenced sources, including IBGE census data and georeferenced information, such as updated vector files of indigenous lands (traditional communities), conservation units, infrastructure, water net flows, among others.

Given the fact that the project area is composed of different farms, all owned by the same **landowner, he and his employees represent a common stakeholder among all the farms.** The company conducted an internal analysis to identify potential partners for the REDD+ project. Following this, the project proposal was presented to the landowner, resulting in the signing of a commitment contract that aligns with conservation objectives.

They are key stakeholders in the project, as its implementation depends entirely on their participation and consent. The landowner will benefit directly from the carbon credits generated. However, the workers may be impacted by factors such as change of routine and new monitoring and policing responsibilities; therefore, the contract includes guarantees for human rights protections, ensuring enhanced support and security for all workers involved in the project.

Another common stakeholder is **Piauí's unit for the Brazilian Institute of the Environment and Renewable Natural Resources (IBAMA)**, responsible for environmental policing and monitoring⁴⁶.

⁴⁶ IBAMA attributions, available on: <<https://www.gov.br/ibama/pt-br/aceso-a-informacao/institucional/sobre-o-ibama#atribuicoes>>. Also cited on item 1.15.

The third common stakeholder is **farmers, divided into two subgroups: small-scale, and medium to large-scale producers**. Farmers are essential to the local economy, making their engagement with the project critical to its success. However, some of these stakeholders may face land-use restrictions due to conservation efforts, emphasizing the importance of negotiation and collaboration with them to ensure mutual benefits and project viability.

Below are the stakeholders on each farm/group of farms composing the project area.

PROGRESSO/RUA/BURITIZINHO FARMS

Uruçuí and Sebastião Leal's Secretaries of the Environment, responsible for elaborating, coordinating, executing and monitoring policies related to the environment^{47,48}.

Uruçuí and Sebastião Leal's Mayors, responsible for promoting and articulating environmental policies and furthering sustainable development initiatives in the municipal level⁴⁹.

The Tamboril village, identified through consultations with the landowner's employees. The residents rely on the land for their economic activities and subsistence, and thus stand to benefit from the environmental improvements fostered by the project, as well as from the project's social investments in these communities.

RAINHA DA SERRA/SÃO SEBASTIÃO FARMS

Uruçuí's Secretary of the Environment and Mayor.

Pratinha village.

⁴⁷ Uruçuí's Secretariat for the Environment attributions, available on: <<https://urucui.pi.gov.br/wp-content/uploads/2022/07/Lei-Complementar-nA%CC%82o-827.2022-Reforma-Administrativa-Municipal.pdf>>.

⁴⁸ Sebastião Leal's Public Policy for the Environment, available on: <<https://sts-gestao.s3.amazonaws.com/uploads/leis/ad3704eacde5ca0986c5e86e86b8eaf3.pdf>>.

⁴⁹ Prefecture attributions on sustainable development and environmental management, available on: <https://www.planalto.gov.br/ccivil_03/leis/leis_2001/l10257.htm>.

	IMPERIAL FARM <i>Bertolínia's Secretary of the Environment and Mayor.</i> <i>Bertolínia's Association of Parents and Friends of Disabled Individuals (APAE), which is partially funded by and a partner of Grupo Progresso.</i>
Legal or customary tenure/access rights	<i>Following the consultation conducted and aligned with the internal due diligence process, no land ownership conflicts or land disputes involving the project area were identified. Moreover, the relationship between the landowner and neighboring rural settlements is considered friendly. The landowner also demonstrates concern for all regularization issues with local and national environmental institutions, with no pending legal actions or other irregularities with the government.</i>
Stakeholder diversity and changes over time	<i>The identified stakeholders encompass a diverse range of groups, each with distinct interests that reflect the region's diversity. In general, good relationships are observed amongst the local residents, the landowner and local government agencies.</i> <i>According to discussions held with the landowner, and the remote dates when forest clearings initially occurred, he has been established in the area for a long time and does not intend to expand his activities to neighboring areas, which helps his relationship and dialogue with neighboring settlements.</i> <i>The villages of Tamboril and Pratinha, located in the municipality of Uruçuí, may share similar interests with the small producers in the region, as both groups face economic challenges and have limited resources for their economic activities.</i> <i>Regarding rural producers in the area, both small and medium-to-large producers have shown significant growth in size, especially in the past decade, as evidenced by the increase in deforestation rates in the region. This growth reflects the expansion of agricultural and livestock activities, intensifying the need to establish sustainable</i>

	management strategies and maintain continuous dialogue among all stakeholders involved.
Expected changes in well-being	Firstly, the project directly contributes to combating climate change by reducing deforestation and forest degradation and promoting sustainable land use practices, which are relevant to the local population, once having better control of the micro-climate and future-climate is essential to guarantee predictability on any activities, including agriculture. Not only the climate-related changes, but also the natural resources' preservation is essential to guarantee well-being for locals, such as good air/water/food quality. The implementation of this REDD+ project also generates a more rigorous level of employment. NovaAgri's contractual agreement with the landowner explicitly incorporates clauses dedicated to upholding human rights. These provisions encompass a commitment to fair labor practices, non-discrimination, and the well-being of local communities. This project aims to bring substantial social benefits to local communities, addressing issues such as poverty and lack of infrastructure. By incentivizing social actions with part of the VCU's revenue, the project aims to provide economic opportunities, enhance livelihoods, and improve food security for communities that surround the project and reference areas.
Location of stakeholders	The primary stakeholders directly impacted by the project activities are within the boundaries of the property, notably the landowner himself and workers from Grupo Progreso. However, in the PA's surroundings, there are some stakeholders directly or indirectly affected, such as the Pratinhas and Tamboril villages, and Bertolín's APAE. The Prefectures are also relatively close to the PA. The exact location of each stakeholder is presented below, following this table.
Location of resources	The only territories where the project activities will be implemented are the Grupo Progreso Farms, which are



solely owned by the landowner mentioned at the beginning of this PD. All the forested area to be monitored, surveilled, and inspected is within the boundaries of this group of farms.

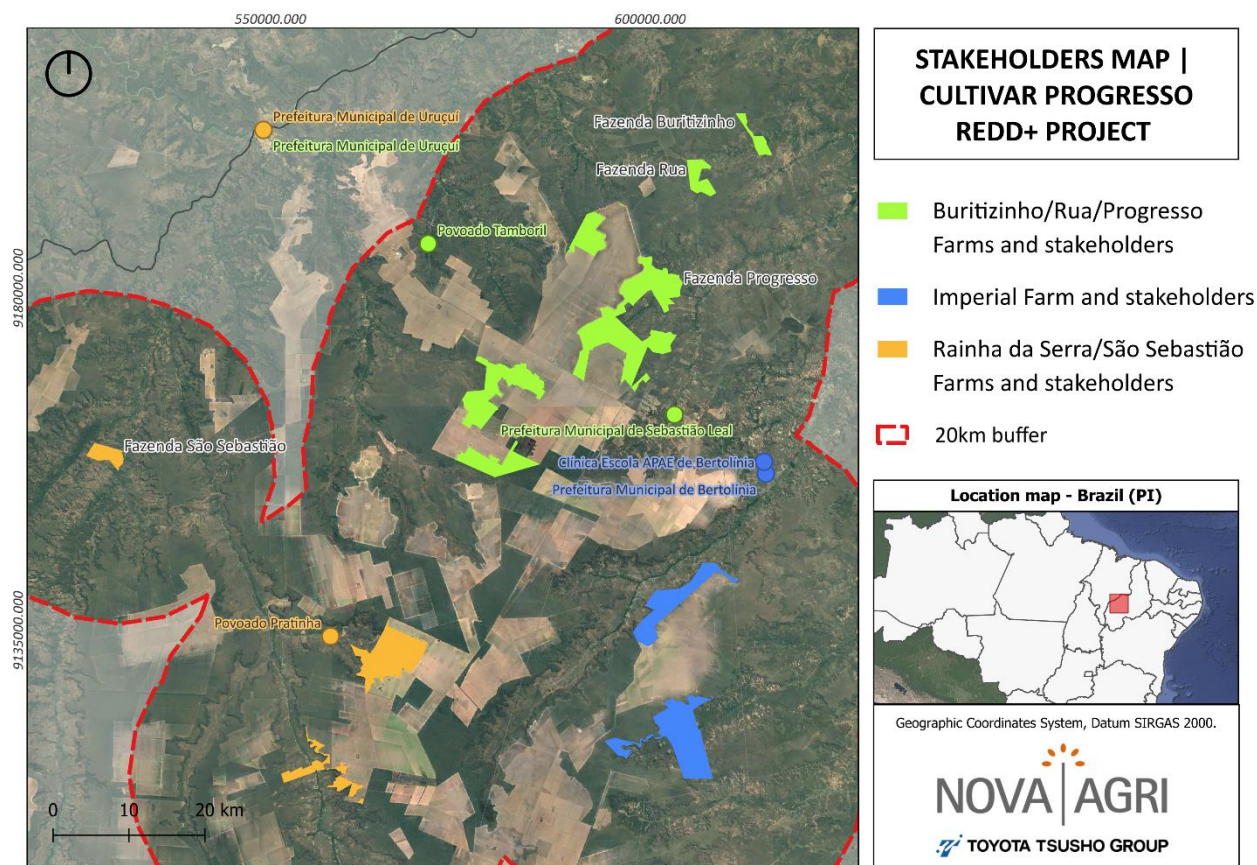


Figure 24. Location of Grupo Progresso Farms' stakeholders.

Source: the author.

2.1.2 Stakeholder Consultation and Ongoing Communication

Find below the description of the process for and the outcomes from the stakeholder consultation conducted for the project.

Date of stakeholder consultation	31-January-2024
Stakeholder engagement process	Process and methods used to document the results of the stakeholder engagement process

The project proponent took great care to ensure proper engagement with all stakeholders. Starting with communication with the landowner, this process involved several rounds of discussions — from the initial phase of project prospecting and acceptance of its conditions to the signing of the contract that binds their land to the project activities. Everything was conducted transparently, with open dialogues to clarify the conditions and responsibilities of each party.

Regarding other local stakeholders, the rules of Free, Prior, and Informed Consent (FPIC) were followed, even though in this case those might not be strictly applicable (since the land is private and the negotiations differ), because the project area is adjacent to native Indigenous tribes in the region. It is important to note that Indigenous peoples and other local residents do not have specific responsibilities in the project activities. Their importance lies primarily in being beneficiaries of the returns generated by the carbon credits, as there is a formal social responsibility component commitment within the project. However, for their representation as beneficiaries to be meaningful, it is crucial that these groups are informed from the very beginning and are also consulted about their primary needs to be addressed in their daily lives.

The FPIC process was conventionally implemented by the landowner, who maintains an open channel of communication with these groups due to a close relationship in terms of providing support when there is a technical, instrumental, resource, or other needs. In preparation for the project implementation, the leaders of these communities were informed in advance and invited by the Project Proponent to participate voluntarily in an in-person meeting for the project presentation, which was facilitated by the municipality itself. This approach was applied to local community members, members of the public sector, and nearby residents/settled people/producers.

For collecting information and reporting outcomes, such as doubts and suggestions, during the meeting, a sign-in sheet was prepared in advance, including a consent form for the use of images and voice, which was also verbally communicated before the meeting began. With this permission, it was possible to record the meeting and document the main points raised by the attendees. All this was intermediated by the municipality prefecture, which is a common place for all these publics. In addition, the methods for engagement include the use of social media (ex. WhatsApp, Facebook) which are of common use for these publics. The objective is always to reach as many representative individuals as possible within each stakeholder group.

Since the use of technology is already quite widespread among these groups, as identified previously, this was not a challenge, and invitations sent via WhatsApp, for example, worked very well. The Project Proponent will continue to use these methods for future meetings or communications, whether directly or through the local government or the landowner.

Consultation meeting with stakeholders

In order to fulfill the proponent's commitment to transparency with its stakeholders, NovaAgri conducted visits to the localities surrounding the project area. These visits included numerous discussions with the landowner as well as scheduled meetings with key representatives of each local stakeholder.

The main public disclosures and consultation meetings took place on January 31, 2024, in the morning (8:30 am BRT), as previously agreed with the Grupo Progresso and the invited prefectures. The Grupo Progresso provided space within its office for the meeting. During the afternoon, there was a visit to the local communities.

The meeting provided valuable insights as the project's general manager, the project's carbon specialist, the commercial national manager, and the regional trading representants (all of them representing NovaAgri) introduced the company, its current context, and the initiative to develop carbon projects. Emphasis was placed on climate issues, the challenges posed by carbon emissions and global climate change, and the efforts by both corporate and governmental sectors to mitigate the climate emergency. The aim was to raise awareness and disseminate information on a topic that remains largely overlooked in remote regions of Brazil. The list of present people is presented below, following this table.

Informative folders were distributed, detailing the purpose and key features of the NovaAgri's Carbon Project, presented in an accessible yet technically viewing manner. Furthermore, the environmental and social benefits of the project were highlighted for the company, landowner, and all directly and indirectly involved stakeholders, particularly the local communities.

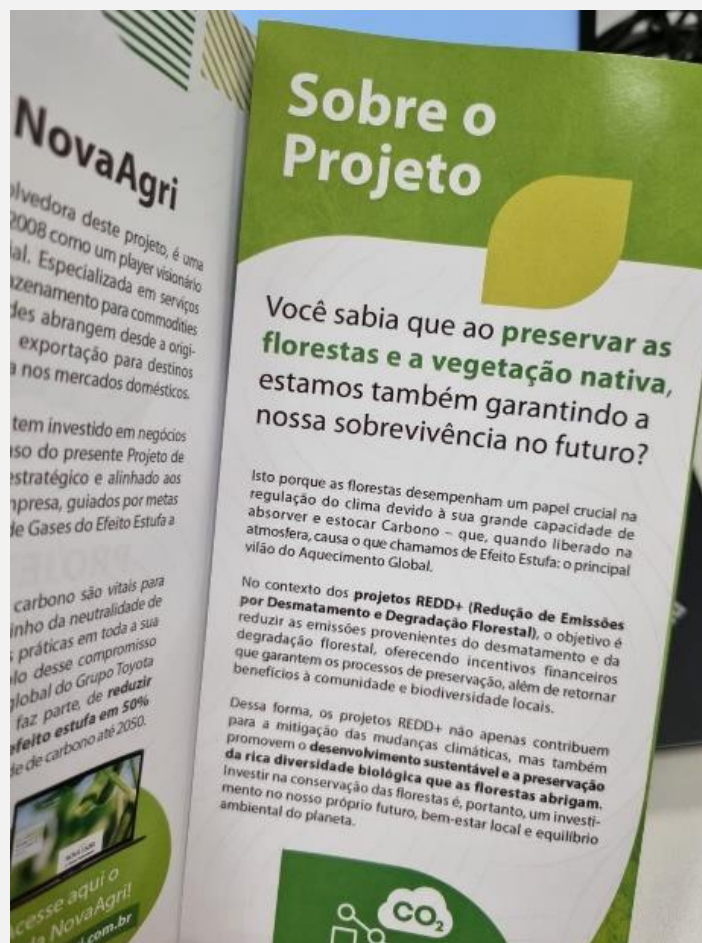


Figure 25. Brochure of the Project: 'Did you know that preserving the forests and the native vegetation also ensures our future survival? (...)'.

Source: the author.

Consultation outcome

During the consultation process, the risks, costs, and benefits of the project were presented by the Project Proponent's manager to all attendees at the consultation meeting. In addition to providing an introduction about the project's objectives, raising awareness about climate risks, and highlighting the importance of forests for the common good, the presentation also covered the technical aspects of the project activities, including investments, acquired technologies, monitoring processes, and more. The benefits were discussed in terms of the positive impacts the project could have, as it is expected to be an important source of resources that could result in significant social improvements for the region.

To achieve this, a very transparent presentation was given on the benefits-sharing agreements, which will extend beyond the stakeholders under a

legally binding contract (directly involved in conducting the project activities): benefits will also be shared with community stakeholders, such as settlers and residents of the municipality. However, it was first necessary to listen to those present, who will be the most directly impacted. Therefore, active listening was employed to understand the actual social needs of the local community for the Project Proponent to consider in this project.

Meeting Minutes

1. Individual presentation of those present at the table (Everyone)

- Table composed of 14 people + Cornélio Sanders and Ani Sanders

- Presentation and project disclosure folders were distributed to ensure transparency and public consultation for the stakeholders involved.

2. Presentation of NovaAgri, voiced by the company's commercial team (Pedro)

- The relationship of trust and respect between NovaAgri and Grupo Progresso was highlighted, historically through commercial activities, and now, with the expansion of the partnership to new business ventures — especially regarding the Carbon project.

3. Project contextualization, explanation of climate change issues, definitions, and methodologies for generating Carbon credits (Victor)

- A general context was given about the Carbon issue worldwide concerning climate change and the imminent need for the business sector to offset greenhouse gas emissions in the medium and long term.

4. Project overview presentation, proposed project activities, socio-environmental benefits, and invitation to public consultation (Mauricio)

- The general objectives of the project were presented, including the context in which NovaAgri finds itself, along with the global commitments of its economic group regarding tackling climate change; forwarding through the project's steps, ongoing progress, and current development phase; at the end of the presentation, the public consultation was introduced, along with the complementary delivery of the Invitation Letter in addition to the informative folder.

Open for discussions, feedback, suggestions, and other comments (Everyone):

At this stage, representatives from each group were able to express their opinions, feedback, suggestions, criticisms, and other relevant comments regarding the project.

Finally, the NovaAgri team was accompanied by the Sustainability and Marketing teams of Grupo Progreso to 1) take photos of the forest reserves to include in the project document; 2) visit and interview local communities. The field visit was of great importance to meet the community leaders, who represented the Tamboril village. This village has approximately 200 people and there is a strong network among the residents in the region.

At the end of the meeting, attendees were invited to the project's public consultation, where they could express their questions, opinions, suggestions, compliments, and criticisms. Additionally, a letter containing the link to the VERRA website page listing all recently listed projects in the pipeline was distributed to all present stakeholders, saying that this would be including NovaAgri's project, which would be listed furtherly.



Figure 26. Distribution of invitation letters to local stakeholders at Grupo Progreso's office.

Source: the author.



Figure 27. Distribution of informative folders to Grupo Progresso's employees.

Source: the author.

Below are some of the comments (questions and suggestions) that arose during the stakeholders' meeting, which we consider relevant to analyze for potentially shaping future project activities:

- "40% to 50% of the areas lost in the Piauí Cerrado are caused by human-induced fires. A comprehensive awareness program is needed, involving everyone."

- "If the fire spots can be identified more immediately, it's possible to prevent deforestation through the use of planes, but this is a major challenge."

- "Sparks from a fire can travel up to 1,000m away, affecting even crops."

- "In Sebastião Leal, there are two priority problems: combating forest fires—the municipality already has a firefighting brigade, but it's insufficient. There's a need for access to information, material delivery, and infrastructure improvements; the second priority problem is cattle raising, which without proper (safe and correct) management, drastically affects springs and other water bodies — often not respecting riparian areas and expanding areas too close to river networks/valleys, etc."

	- "Uruçuí has a spring fencing program, but difficulties with fire outbreaks are still a major challenge." - "There's a significant lack of infrastructure in municipalities for daycare centres and technical education programs. The latter is essential for creating qualified labour for the region." - "The previous experience with educational programs supported by Grupo Progresso is very positive, evidenced by the level of graduates — e.g., 6th place in the ranking." - "Transformative education is the only way to go forward." Multiple stakeholders present in the meeting also expressed a notable concern with illegal hunters in the area, which represent a relevant environmental and socioeconomic risk. These actors not only start dangerous fires that can spread rapidly to different properties, but also threaten local fauna. More information on them is available in Section 3.4 Baseline Scenario. *** Below are some of the comments collected during the communities' visits, which are relevant to analyze for potentially shaping future project activities: *** - "The community is quite needy, but health and education are doing okay. There's a lack of support for women who need to work and don't have anywhere to leave their children." - "People live off small farms, local commerce, restaurants, when they're not working on the larger farms nearby." - "August and September are quite critical months because when a fire starts there, it spreads here rapidly. There have been cases where it reached homes and destroyed everything. In the end, it's hard to pinpoint the cause; we just have suspicions, but we can't accuse anyone." - "Back in school, there was little talk about the environment, forests, etc."
Ongoing communication	The project establishes ways to maintain constant and meaningful communication with all stakeholders. The proponent maintains direct contact and a close relationship with both the landowner and the municipality, which is crucial for enabling and scheduling visits to the project area, to monitor production and preservation activities. Additionally, the PP also relies on its regional trading team, which is physically close to

	the project territory and has business relationships that extend beyond the carbon project itself. This allows for more effective monitoring of the project's activities and facilitates direct communication between the company and other stakeholders. Furthermore, the proponent is going to keep its internal channel open to all stakeholders. This channel aligns with the company's corporate governance of ethics and compliance, ensuring the highest level of business integrity in its relationships with stakeholders. The channel is indicated in Section 2.1.4 Grievance Redress Procedure. Official reports detailing the progress of the area monitoring are also being prepared and circulated internally within the company, ensuring project transparency. Moreover, regular updates will be shared on social media channels regarding any project changes or updates, always following the format of a PD version, which will be publicly available on the VERRA Registry website. This transparency is already guaranteed by VCS projects to their stakeholders. Finally, the Carbon department's email addresses at NovaAgri will always be available to anyone wishing to express suggestions, questions, or general comments. The addresses are listed in Section 2.1.4 Grievance Redress Procedure.
Stakeholder input	The main input received from stakeholders during the first in-person public consultation was included in this initial version of the PD. These inputs are reflected, for example, in the monitoring plan, which is based on the intrinsic needs reported during the consultation. After the public consultation on the VERRA website, this PD may undergo further updates to address or clarify the comments received.

Convite à Consulta Pública – Projeto de Carbono REDD+ NovaAgri

Prezado/a,

É com grande satisfação que a NovaAgri compartilha informações sobre nosso mais recente projeto de carbono, que representa os esforços contínuos da companhia para a promoção da sustentabilidade socioambiental nos locais em que atua.

O projeto REDD+ (Redução de Emissões por Desmatamento e Degradação Florestal) tem como objetivo reduzir as emissões de CO₂ provenientes do desmatamento e da degradação florestal, oferecendo incentivos financeiros que garantem as atividades de preservação, além de retornar benefícios à comunidade e biodiversidade locais.

Convidamos cordialmente a você participar da consulta pública para este projeto. Sua contribuição é de grande valor para nós. Junto a esta Carta estamos lhe entregando um Folder com as informações gerais do projeto, mas em breve o documento completo e detalhado estará disponível no site da Verra nos próximos meses. Para acessar, acesse o QR Code Abaixo ou acesse o site: <https://verra.org/consultations/projects-open-for-public-comment/>

Assim que divulgado, também comunicaremos em nossas redes sociais.



Caso você tenha sugestões, ou queira nos deixar qualquer comentário sobre o Projeto antes da abertura da consulta no site, entre em contato conosco pelo e-mail: carbon.credit@novaagri.com.br

Contamos com sua contribuição para este projeto. A NovaAgri agradece seu apoio!



Figure 28. Invitation letter to local stakeholders to participate in public consultation.

Source: the author.



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com ou sem fins lucrativos, no Brasil ou no Exterior.
Assino ciente abaixo, junto à lista de presença do atual encontro.

LISTA DE PRESENÇA data 31/01/24

	Nome	Grupo	Assinatura	E-mail
1	PEDRO DO NASCIMENTO	NOVA AGEI	[Redacted]	[Redacted]@novaagei.com.br
2	Ygor Garcia de Almeida	NOVA AGEI	[Redacted]	[Redacted]@novaagei.com
3	EDSON ADRIANO CARDOSO	NOVA AGEI	[Redacted]	[Redacted]@NOVAAGEI.com.br
4	ERICK PEREIRA	NOVA AGEI	[Redacted]	[Redacted]@NOVAAGEI.COM
5	LUIS SOBRINHO DE ARAUJO	Sec. Meio Amb. SAGRI	[Redacted]	[Redacted]@G.M.A.Y.COM
6	MARCIO SCHNORR SILVA	NOVA AGEI	[Redacted]	[Redacted]@NOVAAGEI.COM.BR
7	FILIZIA GABRIELADON MENDES	Progresso	[Redacted]	[Redacted]@progrs.br
8	WILSON RIBEIRO DE SOUZA	Progresso	[Redacted]	[Redacted]@progrs.br
9	LYDIA MARIA BENEDETTI	Progresso	[Redacted]	[Redacted]@progrs.br
10	RAYARA MARCELO SORDERS	Progresso	[Redacted]	[Redacted]@sorders.com.br
11	WILLIAM DE CASTRO PINTO	Progresso	[Redacted]	[Redacted]@progrs.br
12	DOMAS NUNES MOREIRA	Sec. Meio Ambiente - AGRI	[Redacted]	[Redacted]@gmail.com

Figure 29. List of present people at the meeting (p. 1).

Source: the author.



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com ou sem fins lucrativos, no Brasil ou no Exterior.
Assino ciente abaixo, junto à lista de presença do atual encontro.

[illegible]

Figure 30. List of present people at the meeting (p. 2).

Source: the author.



Figure 31. Stakeholder meeting (January 31, 2024).

Source: the author.



Figure 32. Field inspection (January 31, 2024).

Source: the author.



Figure 33. Visit to local communities (January 31, 2024).

Source: the author.



Figure 34. NovaAgri team talking to IBAMA Piauí (January 31, 2024).

Source: the author.



Figure 35. Grupo Progresso's employees' consultation (February 14, 2024)⁵⁰.

Source: the author.

2.1.3 Free Prior and Informed Consent

Obtaining consent

Consent was obtained from any key stakeholders. Especially the landowner, who is the primary stakeholder in terms of providing the land and facilitating the project implementation, is fully informed and in agreement regarding their participation, role, and responsibilities within the project. This was achieved by the proponent through multiple rounds of discussions and negotiations aimed at providing a general explanation of the project and its objectives, timeline, conditions, projections, and shared benefits. This stage ran parallel to all the documentary analyses mentioned earlier regarding property ownership

⁵⁰ During this meeting with all current employees from the landowner Grupo Progresso, the carbon project was presented to ensure everyone is on the same page about the project's specific needs regarding climate change issues, and to present the socioenvironmental benefits. Many of these employees are directly involved in the actions for maintenance and preservation of the native forests.

	<i>and cross-referencing with all public boundaries to mitigate any risk of conflicts or false information.</i>
Outcome of FPIC	<i>The entire consensus-building process between the landowner and the proponent resulted in a legally binding contract validated and signed by both parties, affirming through a formal document all the conditions accepted after the initial discussions. This project provides assurance that its implementation has not encroached on land, relocated people without consent, and forced physical or economic displacement, and all necessary evidence will be provided throughout the validation process alongside the VVB.</i>

2.1.4 Grievance Redress Procedure

Development process	<i>For issues related exclusively to the relationship between the proponent and the landowner, the methods, tools, periods, and procedures for resolution are outlined in the legal binding contract signed between the parties. These issues are limited to pre-agreed matters, primarily concerning the rules for the use of property areas and forest preservation.</i> <i>For all other types of grievances, the proponent is providing internal channels for receiving comments, feedback, and suggestions. For inquiries, suggestions, and general feedback, NovaAgri offers a direct channel to its Carbon department via email at carbon.credit@novaagri.com.br. Furthermore, for compliance issues, the company provides its corporate channel accessible via phone at +55 0800 721 5954 or through the web portal: www.canaldedenuncias.com.br/novaagri.</i> <i>The proponent has also developed a website portal exclusively to NovaAgri's projects, containing specific sections to contact the proponent about any grievance or project issues related, through the button "Comunicação com Stakeholders": https://alertas.scon.com.br/novaagri/#/</i>
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Grievance redress procedure

As soon as a complaint is received, it must be formally registered in the PP's complaints management system, assigned a protocol number for follow-up and stored.

Action 1. Confirmation receipt

Send a confirmation receipt to the complainant within 5 working days, explaining the next steps in the process and estimating a timeframe for resolution.

Action 2. Preliminary analysis

The team responsible must assess the nature of the complaint and its seriousness. If the content is technical (e.g. issues related to project mechanisms, results, general impacts), it will be forwarded to NovaAgri's Carbon Project (PP) management team. If the content is judicial, conflicts or complaints about human rights or of any other nature, it will be forwarded to the Legal and Compliance team. Complaints involving urgent or sensitive issues (such as human rights or critical environmental issues) should be prioritized.

Action 3. Appointment of person responsible

As mentioned in the previous section, a person or team should be appointed to investigate and manage the resolution of the complaint. The person responsible should have adequate knowledge of the local context and the issues under discussion.

Action 4. Investigation process

Carry out a thorough investigation, listening to all parties involved, collecting evidence, and consulting external experts if necessary.

Action 5. Culturally appropriate consultation

Ensure that the investigation process respects local cultural norms, using consultation methods that are considered fair and appropriate by the community, and involve external experts if necessary.

Action 6. Conflict resolution

If necessary, involve mediators or community leaders (e.g. town hall representatives) who can facilitate the resolution process in a culturally appropriate way.

Action 7. Solution proposal

After the investigation, the team should propose a solution that is fair and acceptable to both parties, considering the applicable cultural and legal norms.

Action 8. Formal response

The final decision must be communicated to the complainant in writing and, if necessary, verbally, in a language understood by the complainant. The communication should include the details of the proposed solution, the implementation process and the timeline.

Action 9. Corrective actions

The agreed solution should be implemented within the defined timeframe and the situation monitored to ensure that the complaint has been fully resolved.

Action 10. Satisfaction check

After implementing the solution, check with the complainant whether the solution was satisfactory and whether there are any further issues to be addressed.

Action 11. Periodic review

The PP reviews the complaints procedure annually to ensure its effectiveness and adjusts as necessary, based on feedback from stakeholders.

Action 12. Escalation / Appeal

If the complainant is not satisfied with the proposed solution, the possibility of appealing to a higher level within the organization, or to an external mediator (arbitration or the intervention of regulatory bodies) should be offered, as governed by internal company policies (PP).

During the monitoring period, no grievances were raised.

Grievances received	Resolution and outcome
NA	NA

2.1.5 Public Comments

Verra encourages public comments on the compliance of projects with program requirements, and these comments will be publicly disclosed on the Verra Registry. All the projects may be found on the webpage: <https://verra.org/consultations/projects-open-for-public-comment/>.

In addition to the public comment period on VERRA, this project maintains internal channels for receiving comments, feedback, and suggestions. For inquiries, suggestions, and general feedback, NovaAgri provides a direct channel to its Carbon department via email at carbon.credit@novaagri.com.br. Furthermore, for matters related to compliance issues, the company offers its corporate channel accessible via phone at +55 0800 721 5954 or through the web portal: www.canaldedenuncias.com.br/novaagri. This multi-faceted approach ensures transparency and accessibility for stakeholders to engage with the project.

All pertinent points and feedback raised through these channels will be thoroughly addressed and, when applicable, incorporated into this PD (following the table model below), verification reports or other suitable communication channels. NovaAgri is committed to keep its stakeholders informed of the project's progress and any adjustments made related to the Project. This collaborative approach, both internally and through the Verra public comment process, reinforces our commitment to accountability, integrity, and the overall success of the project.

Table 8. Table model for comments received on Public Consulting

Comments received	Actions taken
<i>Summary of comment received</i>	<i>To be provided a summary of actions taken and any project design updates or justify why updates were not necessary or appropriate.</i>

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Since this is a forest protection project involving technological resources, the current structure is planned so that there is one person (NovaAgri's Carbon Credit department manager, with 11 years of experience in Geosystem and Technology for Forests) responsible for overseeing the entire project and coordinating geo-monitoring functions. This is supported by its own team and also SCCON, a company that has been innovatively providing scalable analytical solutions since 2008, utilizing high-resolution daily imagery and analytics to monitor and measure changes

across all Brazilian biomes, generating up-to-date, precise, and reliable results. SCCON has also been the distributor for Planet in Brazil since 2016, adding greater technological credibility to the project.

Another key figure in the project is the carbon specialist (more than 7 years of experience in Carbon strategies and management), who not only is part of the project coordination team but also carries out the technical work of measurement and reporting according to the standards set in VCS/MRV protocols. An external consultant, with an extensive background in AFOLU, also contributes by validating studies.

The team also counts on a member with a background in sustainable development project in rural areas through an NGO specialized in social development. This team member's experience includes community mobilization, stakeholder mapping, and socioeconomic diagnostics. She is also assisting in dialogue between the NGO and NovaAgri's carbon credit team to explore a potential partnership in social matters for the project. Furthermore, the team is also in contact with other NGOs for possible partnerships, especially those which promote sustainability and citizenship through conservation efforts, environmental education and humanitarian aid.

Additionally, regional teams, who maintain close relationships with the landowner, are responsible for managing communications and relations with local stakeholders. Their expertise lies in commercial relations and agriculture, but they also possess deep knowledge of land use and local climatic conditions. The conceptual and technical aspects of the project activities are passed on to them by the coordination team, which is always informed and updated on the project's actions. The landowner also participates in regular training sessions on how to best maintain the forest, optimize protection from threats, and follow key principles such as respecting biodiversity and human rights. These training sessions are conducted by the project coordination team itself.

All teams are committed to fostering respectful and effective partnerships across all aspects of the project.⁵¹

2.2.2 Risk Assessment

As part of NovaAgri's Sustainability Policy, and as standard procedure for its general businesses, the Credit Department conducts a meticulous analysis of sensitive and critical environmental points related to the products' origin areas. This includes assessing potential issues such as regarding IBAMA/SEMA Embargoes, debts to IBAMA, and Conservation Units. These criteria are also replicated to the assessment and selection of viable landowners and its areas for a Carbon Project with NovaAgri, added to additional specific criteria (REDD+ and VCS/AFOLU criteria).

This comprehensive evaluation underscores our commitment to environmental responsibility, ensuring that the sourcing of grains aligns with legal, environmental, and conservation standards.

⁵¹ More comprehensive and sensitive information on the management team and other parties involved in the Project was presented directly to the VVB during the validation process.

Finally, all these criteria are replicated in the preliminary selection of eligible landowners entering a partnership with NovaAgri for Carbon projects. While some criteria are agribusiness specific, NovaAgri values their application for a secure and integral evaluation also in its carbon projects. Thus, the assessment is complemented with specific analyses, such as evaluations of boundaries' delimitation, phytophysiognomy types, ecosystem biodiversity, carbon stocks, higher resolution satellite analyses, and on-site assessments for data confirmation. NovaAgri remains dedicated to fostering a transparent, responsible, and environmentally conscious approach in all its services' endeavors and portfolios.

Table 9. Risks Identified for Stakeholders and the Environment

Risks identified		Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	The PP understands that in many rural areas, the livelihoods of local populations are intricately tied to the jobs generated by agricultural activities. Consequently, some carbon projects, in disrupting plans for opening new lands through deforestation, could potentially pose a threat to employment maintenance and stakeholders' income in the surrounding areas. However, the data collected in this project demonstrates that the goods produced and the financial returns from consolidated cultivated lands are sufficient to sustain activities in the long term, without compromising local jobs or the economy. Additionally, the activities conducted by the landowner indicate responsible farming practices, such as crop rotation and good soil treatment. Such practices ensure enduring activities, eliminating the need for opening new lands and migrating from an area to others periodically. Furthermore, the REDD+ Project itself aims to create a new revenue stream for the landowner, allowing him to reinvest it in their assets, workforce and social causes. This provides

		greater security for maintaining economic standards within the property. Finally, a portion of the financial returns from the project will be allocated by the PP to social initiatives, making a commitment to improving surrounding communities' wellbeing.
Risks to stakeholder participation	1) Risks associated with land loss due to allocation for Project Activities. 2) Risks related to losing project's integrity and reputation, affecting the related stakeholders – implementing and enforcing conservation measures must have sufficient resources and capabilities for a long period of monitoring and reporting. Otherwise, this can lead to potential gaps in achieving the desired environmental outcomes or even causing greenwashing issues.	1) This risk is directly related to the land tenure issues of the landowner. However, it is entirely mitigated through clear communication between the landowner and the project proponent, with agreements, responsibilities, and rules well defined in a legally binding contract signed by both parties. This contract has been reviewed by the legal teams of both parties to ensure a mutually beneficial agreement. For example, there are specific clauses that clearly outline the voluntary allocation of portions of the land for the execution of REDD+ activities, whether remotely (via satellite) or on-site. At no point are the landowner's lands exposed to any form of collateral, guarantee for penalties, or any such action as a consequence. 2) The expected results can only be guaranteed if the MRV (Monitoring, Reporting, and Verification) model is well-executed in the project. Ensuring a no-net-harm outcome for the environment and biodiversity depends on well-conducted monitoring, utilizing cutting-edge modern technology, aligned with strengthened on-site surveillance and local engagement. All of these aspects will be provided through the proposed initiatives in this Project Design (PD) and the financial incentives derived from the credits generated. With a dedicated Carbon Credit Project Development department within NovaAgri, a rigorous routine of monitoring and reporting has been established, aligning with the company's existing processes for its core business.

		Corporate goals will be assigned to the projects, along with verification and communication routines, and a technical and operational framework from various departments, providing constant support to the project, for instance the legal department. All of these measures ensure greater security and transparency, enabling the project to be conducted in the most accurate and refined manner over time.
Working conditions	Inherent risks in the work environment, especially regarding occupational health and safety	There are inherent risks in the work environment, particularly concerning occupational health and safety. These risks are present and, although minimal, must be considered, especially since the landowner is also an employer. However, the project proponent ensures that the relationships between the landowner and their employees adhere to the same policies as theirs. This is achieved through a formal agreement established in a legally binding contract, as outlined in the attached document and the example excerpt below: “The Parties acknowledge the importance of developing socially responsible practices. In this regard, the Parties declare that they operate in a socially responsible manner concerning: (a) ethical values; (b) internal public; (c) environment; (d) suppliers; (e) consumers/customers; (f) community; (g) government and society; and (h) eradication of slave, forced, child, and adolescent labor, and, accordingly: (i) do not employ minors under the age of 16 in any work - except under conditions provided by law; (ii) do not use slave, forced, child, and adolescent labor in their production chain; (iii) manage their activities in an environmentally responsible manner, aiming to identify any environmental impacts resulting from their activities and seeking to minimize negative impacts and amplify positive ones,

		while being committed to maintaining and improving environmental conditions; and (iv) invest permanently in the development of reliable products and services that minimize the risk of harm to the health and safety of users and the general community.” Regarding activities related to the project, especially for those directly involved in the monitoring activities, there will be specific instructions to follow to ensure the highest effectiveness of the project while maintaining safety. To this end, the project proponent commits to including periodic training sessions in their monitoring plan, focusing on environmental awareness and ethical conduct in combating deforestation, considering safety above all.
Safety of women and girls	No risk identified	The project does not involve any activities that expose women of any age to any type of risk. As previously mentioned, the proponent takes proper care regarding publicly disclosed lists of worker mistreatment by the government and also maintains a very serious internal policy on this topic. Moreover, the proponent provides its channels for reporting any kind of issues, concerns, and complaints.
Safety of minority and marginalized groups, including children	No risk identified	The project does not involve any activities that expose vulnerable groups of people of any age to any type of risk. As previously mentioned, the proponent takes proper care regarding publicly disclosed lists of worker mistreatment by the government and also maintains a very serious internal policy on this topic. Moreover, the proponent provides its channels for reporting any kind of issues, concerns, and complaints.
Pollutants (air, noise, discharges to water,	No risk identified	The project does not involve any activities that generate or exacerbate environmental issues related to pollutants of any kind, as all implemented tools and methods do not involve

generation of waste, release of hazardous materials)		the use of combustion of fuels, waste generation, or effluents.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Risks identified		Mitigation or preventative measure taken
Discrimination and sexual harassment	No risk identified	NovaAgri, as the proponent and developer of the project, adheres to a rigorous Compliance Policy expressed through its LATAM Code of Ethics from its Group. This project will be entirely governed by this instrument, and therefore, in all its activities, it will not tolerate any form of discrimination. Also, NovaAgri counts on a Moral Harassment and Sexual Harassment Booklet to help understanding what harassment is and thus make each of us a multiplier of awareness-raising actions on the topic. The purpose is to continuously train each of the direct and indirect employees, to not only prevent situations of moral and sexual harassment from occurring within stakeholders, but also to help friends and family members who may be suffering from it. The official channels for reporting any cases of harassment are presented in this booklet.
Management experience	No risk identified	NovaAgri has involved its corporate business departments to support the technical team with a right approach recommendation, as well as to elaborate specific clauses within the agreement between the proponent and the landowner explicitly incorporating human rights concerns. These provisions encompass a commitment to fair labor practices, non-discrimination, and the well-being of local communities. NovaAgri ensures compliance with local labor laws and international human rights standards, fostering an environment where all individuals involved

		in the project, including the landowner's employees are treated with dignity and respect. The communication department has also contributed with a proper approach incorporated on any graphical material, such as posts on LinkedIn, or paper brochures. The technical teams also contemplate experienced professionals with a sustainability background, which is an important factor when conducting in-person (e.g. meetings) or remote activities (e.g. monitoring, reporting etc.) that need a proper approach among different publics, such as local communities, government agencies, and other stakeholders. All teams are aware and considering of all aspects of a good and respectful partnership.
Gender equity in labor and work	No risk identified	The global policy from NovaAgri's Group has expressed DE&I (Diversity, Equity & Inclusion) efforts centered on the following three themes⁵². 1. Opportunities for diverse human capital; 2. Work-style improvements and work-life balance; 3. Fostering a corporate culture that leverages diversity and changes people's mindset. This reflects directly on the daily actions and decisions of the proponent⁵³, including the activities and management of this carbon project, and shared with the landowner – which is the main stakeholder of this partnership, which also employs dozens of workers.
Human trafficking, forced labor, and child labor	No risk identified	As part of NovaAgri's Sustainability Policy, and in alignment with its standard operating procedures across its diverse business operations, the Credit Department meticulously analyzes sensitive and critical social factors related to the origin areas of its products. This comprehensive assessment encompasses potential issues related to Indigenous Areas and the Registry of Employers who have subjected workers to

⁵² <https://www.toyota-tsusho.com/english/sustainability/ttc/business/case06.html>

⁵³ https://www.linkedin.com/posts/novaagri_8-de-mar%C3%A7o-dia-da-mulher-o-dia-internacional-activity-7171914789990850560-63Ky?utm_source=share&utm_medium=member_desktop

		conditions analogous to slavery. These criteria are also extended to the evaluation and selection of viable landowners and their areas for a Carbon Project with NovaAgri, supplemented by additional specific criteria in accordance with REDD+ and VCS/AFOLU standards. Indigenous Lands: As a strategy for verification, the Credit Department utilizes a geospatial information system to check the presence of official indigenous lands within the georeferenced coordinates of the analyzed areas. In the event of intersections with indigenous lands, the purchase limit for that area's owner is denied. Registry of Employers who have subjected workers to conditions analogous to slavery: During the registration process, the Credit Department checks the name and ID (In Brazil, called CPF/CNPJ) of the producer against the list of employers who have subjected workers to conditions analogous to slavery, sporadically released by the Brazilian Ministry of Labor. Approval decisions are based on the following criteria: • If the name and CPF/CNPJ are on the list, the registration is denied. • If the CPF/CNPJ is listed in the automatic system check, every time a contract is generated. This comprehensive evaluation underscores our commitment to social responsibility, ensuring that the sourcing of grains aligns with social standards and good practices. Finally, all these criteria are replicated in the preliminary selection of eligible landowners entering a partnership with NovaAgri for Carbon projects. Thus, the assessment is complemented with specific analyses, such as local public consultation, engagement and educational actions, interview with local employees, study of local investments' possibilities, among others. NovaAgri remains dedicated to fostering a transparent, responsible, and socially conscious approach in all its services' endeavors and portfolios.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Our term of agreement properly signed with the landowner regards to recognize, respect, and promote the protection of stakeholders' rights line with applicable international conventions and law. Find below specific clauses from that agreement: (...) 3.1. <i>It is the responsibility of the Parties together to:</i> (i) <i>Comply with NOVAAGRI's Compliance policies and law related to anti-corruption, the anti-trust laws, trade embargoes, laws that prohibit slave labor, child labor, any type of discrimination, money laundering, terrorism, unfair competition, and breach of human rights.</i> (...) 3.2. <i>When marketing carbon credits to third parties, the Parties shall pay attention to the status, practices, and reputation of the buyer in order to preserve the reputation of the Parties, as well as of the Project, and shall verify in advance whether the buyer:</i> (i) <i>Complies with anti-corruption laws and regulations, as well as determinations and rules to prevent corrupt practices, illegal expenses related to political activity, harmful acts, breaches or crimes against the financial system or public administration, in order to avoid money laundering, concealments of goods and financing of illegal traffic and terrorism;</i> (ii) <i>does not intend to do business with companies from countries that have suffered an international trade embargo according to the UN, OFAC, European Union or Japan list; and</i> (iii) <i>does not respond for acts that matter in discrimination of race or gender, disabled people or people with special needs, is involved in exploitation of child labor, slave labor or attitude that characterize moral or sexual harassment and disrespect for human dignity.</i> Also, the proponent's LATAM Code of Ethics ("Code") – which applies to all its employees, directors, advisors, service providers, agents, and suppliers, regardless of hierarchical level – has a main concern on the

	Principle of Inviolability of Personal Information and Human Dignity, which involves: ▪ respecting individuals, therefore, not tolerating any abuse of human rights and preserving the inviolability of information and personal data of clients, suppliers, employees, visitors, and any individuals that the Company may have access to in the course of its activities. ▪ not engaging in or being involved in any business that goes against human dignity. ▪ not discriminating against any individual based on race, color, sex, religion, or nationality. ▪ not engaging in activities that exploit child labor, human trafficking, or any other form of forced or involuntary labor. ▪ not tolerating any form of harassment. ▪ not dealing with any individuals engaged or involved in any violations of human rights. All these aspects and rules are going to be directly internalized into the ARAGUAIA REDD+ BOCALON PROJECT, extending the initial public to other stakeholders eventually involved with the project activities.
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2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	As a strategy for verification, the proponent's Credit/Risk and Sustainability Department utilizes a geospatial information system to check the presence of official indigenous lands within the georeferenced coordinates of the analyzed areas. In the event of intersections with indigenous lands, the business and/or commercial permission for that property is denied. This control is carried out through geo-monitoring systems, by using satellite images, applied to 100% of the commercial contracts carried out at NovaAgri. In addition to the use of geospatial technology, a series of documentary analyses are also conducted, because it is essential to understand whether such areas are subject to complaints,

	embargoes, or any other type of irregularity with government institutions. Once any critical cases are identified, the project is immediately stopped, and the proponent does not proceed to any subsequent phases. These actions are part of NovaAgri's rigorous Socioenvironmental Policy, aligned with other company's policies, such as the Ethic Code.
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2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	Rights to territories and resources: the right to access the property and its forested areas is fully granted to the project proponent, in accordance with the landowner through a legally binding contract, where official documents, the project area in hectares, and other pertinent information regarding the land use for project activities implementation are specified, including permission for full access by the proponent for visits and/or remote monitoring. Respect for property rights: the accurate verification of land ownership for the project is of priority importance, as the accounting and claiming of credits must be meticulously tied to genuine land ownership in order to prevent the issuance of phantom credits or credits attributed to public areas. This risk is effectively mitigated through a rigorous examination of the owner's possession documents, which are signed and authenticated in a notary (registry) office, complete with coordinates and various other official documentation. This assessment is conducted through an internal audit at NovaAgri. In addition, the CAR document (Rural Environmental Registry, in English) is also requested and verified. Although it is self-declared, CAR is a legal requirement imposed by the government, serving the specific purpose of registering and monitoring the environmental aspects of the property in question. Once done, the area can proceed to subsequent evaluations. Important note: even though the project area is within a private area, the project activities do not impact the property ownership rights or use over its resources. This is clearly defined in the legally binding contract, which states that the forested land portions are entirely owned and managed by the landowner. Although the activities are shared between the Project Proponent and the landowner, the ownership rights over the property

	remain unaffected, once the allocation of land for this purpose is entirely voluntary. Finally, the contractual binding agreement established with the landowner reaffirms its commitment to providing accurate information regarding land ownership and ensuring the periodic update of this information with NovaAgri. This comprehensive approach ensures the credibility and integrity of the land tenure information throughout the project.
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2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	<i>Not applicable since there is no property right impacted.</i>
Summary of the benefit sharing plan	<i>Not applicable since there is no property right impacted.</i>
Approval and dissemination of benefit sharing plan	<i>Not applicable since there is no property right impacted.</i>
Benefit sharing during the monitoring period	<i>Not applicable since there is no property right impacted.</i>

2.4 Ecosystem Health

In the context of AFOLU-Specific Safeguards, this project focusing on REDD+ seeks mainly to mitigate climate change by preventing carbon emissions resulting from deforestation. Thus, the main impact represents positive externalities when considering the whole Net system. However, like any large-scale project, it might comprehend a spectrum of potential impacts, encompassing environmental and socio-economic dimensions.

The positive aspects of the project are evident in its contribution to global climate change mitigation, social livelihood benefits (and consequent inequality reduction), and other positive externalities not directly observed. By conserving native forests, the initiative helps sequester carbon dioxide, fostering biodiversity conservation and supporting sustainable ecosystem services. Local communities may benefit from strengthened livelihoods through sustainable forest management practices and consequent financial returns from that. Additionally, the project has the potential to bolster Brazil's commitment to international climate goals.

Table 10. Risks Identified for the Ecosystem

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems	1) Regarding issues related to the lack of knowledge of the landowner and their employees about biodiversity management within a forest project. 2) Boundaries with physical barriers for territorial delimitation and protection of forest areas. 3) Potential excessive stress due to the use of technological resources.	1) Although the central focus of the agreement between the parties is the protection of forests within the property boundaries, there is an important factor that all participants must consider: respecting the biodiversity and ecosystems that coexist within these forests. To address this, the Project Proponent (PP) will emphasize the importance of protecting all species of terrestrial, aerial, and aquatic fauna as integral parts of the forests. These guidelines will be communicated through some strategies, including educational brochures and training sessions. 2) One of the risks, even if minimal, is the spatial restriction created for property delimitation and forest protection, which could potentially lead to the confinement of animals in the area. However, it is understood that physical barriers are primarily used to mark boundaries and to hinder vehicle passage rather than creating a fully restrictive impediment, such as walls or fences. The existing physical barriers consist of small stakes aligned, which form fences connected by smooth wires (without barbs). These stakes do not prevent animals from moving from an area to others, as it is frequently

		observed that animals of all kinds and sizes transit between forest and crop areas. Thus, no obstacles have been identified that would prevent wildlife from moving through the air, on land, or even in the water to other neighboring reserves. 3) The risks of stress to wildlife can be considered minimal, as the most intrusive technology used will be cameras that do not emit any unusual frequencies and are too small to cause any intimidation to the local animals. These cameras are placed in strategic locations, mostly hidden from view. Patrol vehicles are another resource that could potentially cause some stress; however, these have been used in the surroundings for decades, as the roads have been in place for a long time. It is understood that any stress experienced by wildlife is not a direct result of the project activities but rather a consequence of human intervention through the anthropization of the region, which the wildlife has already been used to for a long time.
Soil degradation and soil erosion	No risk identified	No risks of soil degradation or erosion resulting from the project activities were identified. Some evidence for this absence includes: 1) Activities are based on forest conservation, so there will be no invasive methods in the field, and the monitoring activities mostly utilize technology.

		2) The productive areas used for agriculture and cattle ranching follow sustainable practices, such as activity rotation (agriculture-pasture) and crop rotation (for example, soybean-corn-straw), which determine the proper land use and mitigate the risks of degradation affecting the project area.
Water consumption and stress	No risk identified	No risks involving water stress were identified, as the region benefits from an abundance of water without extensive periods of drought, as presented in Section 1.14.8 Hydrology.
Usage of fertilizers	No risk identified	No direct risks involving fertilizers were identified, as project activities are based on conservation rather than carbon capture through perennial crops or any other types of planting. It is important to mention that activities may occasionally use fertilizers, but with methods of targeted applications that do not extend to forested areas.

Sobre o Projeto

Você sabia que ao **preservar as florestas e a vegetação nativa**, estamos também garantindo a nossa sobrevivência no futuro?

Isto porque as florestas desempenham um papel crucial na regulação do clima devido à sua grande capacidade de absorver e estocar Carbono – que, quando liberado na atmosfera, causa o que chamamos de Efeito Estufa: o principal vilão do Aquecimento Global.

No contexto dos **projetos REDD+ (Redução de Emissões por Desmatamento e Degradação Florestal)**, o objetivo é reduzir as emissões provenientes do desmatamento e da degradação florestal, oferecendo incentivos financeiros que garantem os processos de preservação, além de retornar benefícios à comunidade e biodiversidade locais.

Dessa forma, os projetos REDD+ não apenas contribuem para a mitigação das mudanças climáticas, mas também promovem o **desenvolvimento sustentável e a preservação da rica diversidade biológica que as florestas abrigam**. Investir na conservação das florestas é, portanto, um investimento no nosso próprio futuro, bem-estar local e equilíbrio ambiental do planeta.

Qual o padrão de certificação utilizado?

Verified Carbon Standard
A VERRA STANDARD

A NovaAgri adota o padrão de certificação VERRA, a maior e mais consolidada certificadora do mundo, além de contar com um idôneo processo de Auditoria. Tudo isso reflete o compromisso da NovaAgri com a excelência em suas práticas de sustentabilidade, garantindo a integridade de seus projetos e serviços.

PROGRESSO GRUPO

Região: Piauí
Município: Sebastião Leal, Urucui e arredores
Preservação de +30 mil hectares de vegetação em fazendas do Grupo Progresso
Bio: Cerrado

NOVAAGRI

TOYOTA TSUSHO GROUP

PROJETO DE CARBONO REDD+
NOVAAGRI & GRUPO PROGRESSO

Sobre a NovaAgri

A NovaAgri, desenvolvedora deste projeto, é uma empresa fundada em 2008 como um player visionário no setor Agroindustrial. Especializada em serviços independentes de armazenamento para commodities agrícolas, suas atividades abrangem desde a origem de grãos até a exportação para destinos internacionais ou venda nos mercados domésticos.

Atualmente, a NovaAgri tem investido em negócios inovadores, como é o caso do presente **Projeto de Créditos de Carbono** – estratégico e alinhado aos objetivos e filosofia da empresa, guiados por metas desafiadoras de redução de Gases do Efeito Estufa a médio e longo prazo.

Os projetos de crédito de carbono são vitais para orientar a empresa no caminho da neutralidade de carbono, incentivando boas práticas em toda a sua cadeia de valor. Um exemplo desse compromisso ambiental é a própria meta global do Grupo Toyota Tsusho, da qual a NovaAgri faz parte, de **reduzir suas emissões de gases de efeito estufa em 50% até 2030**, e buscar a neutralidade de carbono até 2050.

Acesse aqui o site da NovaAgri!
www.novaagri.com.br

Sobre o Grupo Progresso

www.grupoprogresso.agr.br

Desde 2001, o Grupo Progresso atua no setor do agronegócio, com o cultivo de soja, milho e outros cereais, a produção e o beneficiamento de algodão, além de reflorestamento com eucalipto e a pecuária. Fundado pela Família Sanders, o Grupo Progresso tem investido e inovado na agricultura, colaborando para desenvolver o cerrado piauiense, onde está localizada sua sede: a Fazenda Progresso. Como missão, o grupo tem o compromisso de produzir com tecnologia inovadora, assegurar excelência na qualidade dos produtos e serviços, promovendo progresso socioeconômico sustentável. Seu propósito é plantar, colher e vender com qualidade, sustentabilidade e responsabilidade socioambiental, para alimentar e vestir o mundo, garantindo a oferta de produtos com excelência.

O que é Carbono?
Por que é tão importante realizar esforços para reduzi-lo?

O carbono é um elemento presente em diversas formas na natureza, especialmente como dióxido de carbono (CO₂). A crescente concentração de CO₂ na atmosfera contribui para o aquecimento global, resultando em mudanças climáticas prejudiciais. Reduzir suas emissões é crucial para conter esse impacto e preservar o equilíbrio ambiental, garantindo um futuro sustentável para as gerações futuras.

Acesse aqui o site do Grupo Progresso:

O que é um crédito de Carbono?

Um crédito de carbono é uma unidade que representa a redução ou remoção de uma certa quantidade de emissões de gases de efeito estufa. Projetos que conseguem preservar florestas e vegetações nativas, por exemplo, são capazes de reduzir Carbono na atmosfera e gerar créditos, incentivando assim as práticas contra o desmatamento. Esses créditos podem ser negociados no mercado, e o seu retorno financeiro pode ajudar a fomentar ações positivas à sociedade e meio ambiente.

Quais são os benefícios ambientais do Projeto?

Este projeto contribui diretamente para a redução das emissões de gases de efeito estufa, promovendo a estabilidade do clima para o futuro. Além disso, apoia a conservação da biodiversidade, preservando ecossistemas valiosos com toda a sua fauna e flora.

Quais são os benefícios sociais do Projeto?

Os resultados do Projeto serão canalizados para o fortalecimento das ações sociais que o Grupo Progresso já conduz com primor na região. O retorno financeiro gerado por meio da preservação da vegetação local contribuirá para o fomento de atividades junto ao Instituto Cultivar Progresso, que realiza ações de assistência social, alimentação, ajuda humanitária, educação e profissionalização, boas práticas agrícolas, produção de mudas nativas entre outras nobres causas.

Figure 36. Brochure's content.

Source: the author.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

Species and habitat

List of rare, threatened, or endangered species:

The RR and its internal boundaries share a rich biodiversity with the surroundings of the Piauí Region, which hosts a variety of species. Below, we highlight some of the species classified as endangered and near threatened based on the IUCN database⁵⁴, followed by their habitat types:

- *Cereus mirabella* – Endangered. Habitat Type: Savanna.
<https://www.iucnredlist.org/species/40877/2939865>
- *Ipomoea chiquitensis* – Endangered. Habitat Type: Forest, Savanna.
<https://www.iucnredlist.org/species/128224237/128245638>
- *Cafézinho (Leptolobium parvifolium)* – Endangered. Habitat Type: Savanna.
<https://www.iucnredlist.org/species/189610155/189610160>
- *Maranhão Red-handed Howler Monkey* – Endangered. Habitat Type: Forest, Savanna.
<https://www.iucnredlist.org/species/918/190435652>
- *Amburana cearensis* – Endangered. Habitat Type: Forest, Savanna.
<https://www.iucnredlist.org/species/32291/67743532>
- *Hyptidendron conspersum* – Endangered. Habitat Type: Savanna.
<https://www.iucnredlist.org/species/176131620/176131624>
- *Ficus bonijesulapensis* – Endangered. Habitat Type: Forest, Rocky areas (eg. inland cliffs, mountain

⁵⁴ IUCN 2024. The IUCN Red List of Threatened Species. Version 2024-2. Available on: <<https://www.iucnredlist.org>>

	peaks). https://www.iucnredlist.org/species/191140981/191140983 There are also species classified as Vulnerable according to the IUCN Red List, such as Brazilian Three-banded Armadillo (<i>Tolypeutes tricinctus</i>); Giant Anteater (<i>Myrmecophaga tridactyla</i>); <i>Discocactus bahiensis</i>; <i>Ouratea xerophila</i>; <i>Ficus calyptroceras</i>; and <i>Dipteryx alata</i>. As mentioned before, this rich biodiversity presents rare characteristics of ecological corridors, which allows permanence and facilitation for a series of fauna and flora species. The project has not adversely impacted habitats or areas required for habitat connectivity for rare, threatened, or endangered species during the monitoring period, evidenced by holistic factors inherent to the project. Holistic Project Factors: • No invasive activities such as road construction, clearing, or intensive use of machinery that could disturb the natural ecosystem. • No project activities involve resource extraction from natural areas. • No use of water resources, except for critical emergencies, which were not needed during the first monitoring period. • No use of chemicals for protection of the flora species nor any biological procedure.
Areas needed for habitat connectivity	The project does not negatively impact the connectivity areas of the biome, as explained in the previous section (2.4 Ecosystem Health) and summarized below. This PD brings up important aspects related to the physical spatial restrictions of the project area. These restrictions are formed by: 1) Natural physical barriers; 2) Simple, non-restrictive barriers that allow the free movement of animals of various sizes and modes of

	<i>locomotion. Additionally, no technological tools or instruments pose risks to the natural coexistence and connectivity of these species. On the contrary, the protection activities are crucial for preserving these physical environments, further facilitating the reproduction, expansion, and well-being of local wildlife.</i>
--	---

Table 11. Biodiversity Risks and Measures Taken

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	<i>Potential excessive stress due to the use of technological resources</i>	<i>The risk of stress to wildlife is minimal. The most intrusive technology used will be small, hidden cameras that do not emit unusual frequencies and are unlikely to intimidate animals. While patrol vehicles could cause some stress, they have been a common presence in the area for decades due to existing roads. Any stress on wildlife is more related to longstanding human activities in the region rather than the specific activities of the project.</i>
Areas for habitat connectivity	<i>Boundaries with physical barriers for territorial delimitation and protection of forest areas</i>	<i>The potential spatial restriction created for forest protection represents a minimal risk, being able to confine animals within the area. However, the physical barriers are primarily meant to mark boundaries and limit vehicles access rather than fully restrict movement, like walls or fences would. These barriers are made of small stakes connected by smooth wires that do not prevent animals from moving freely between forest and crop areas. Therefore, no significant obstacles hinder wildlife from moving through the air, land, or water to other neighboring reserves.</i>

2.4.2 Introduction of species

This section is N/A because this project does not have any planting or species introduction.

2.4.3 Ecosystem conversion

The project area contains only areas which were defined as “forest” for more than 10 years prior to the project start date, as described in the forest cover benchmark maps in Section 1.14.3 Current and historical land-use. That means, the project area’s ecosystem has been preserved as its natural state, with no conversion activities.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table 12. Methodology's References

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0015	Methodology for Avoided Unplanned Deforestation	version 1.2. published on 18- December- 2023
Tool	VT0001	Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities	v3.0, Sectoral Scope 14, published on 01- February- 2012
Tool	-	AFOLU Non-Permanence Risk Tool	<u>Virtual HUB</u> <u>system</u> (VERRA) 2024

This project applies the VM0015 v1.2, methodology by VCS (VERRA), specifically categorized as AFOLU REDD+. Adhering to the robust standards of the VERRA certifier, in its latest version, the project ensures heightened integrity and credibility.

Furthermore, it aligns with the recent updates of the "New Era for VERRA" regarding REDD+ methodologies, positioning itself within a thoroughly revised framework implemented by a body of qualified scientists and responsive to diverse feedback from stakeholders globally. The adoption of the VM0015 in its updated version as of December 2023 is also an important transitional step towards full compliance with the newly launched Consolidated VCS REDD+ methodology, which ensures higher integrity, usability, and consistency with market best practices (including the ICVCM's Core Carbon Principles [CCPs]) as well as latest scientific evidence. In addition, the selection and definition of the baseline are constructed to align with a jurisdictional baseline framework, (e.g. in a State level in Brazil).

According to Verra's website regarding the '[Transition Process by Project Status](#)':

*Proponents initiating the project development process after November 27, 2023, and where projects are not yet, in any form, listed on the Verra Registry, must use the new REDD methodology after jurisdictional activity data for the respective region has been available to be allocated for six months. **Until then, these projects are allowed to use VM0006, VM0007, or VM0015 for validation and verification following the grace period detailed above for projects under development.***

For this reason, the proponent is allowed to use this methodology, since the jurisdictional activity's data for Piauí State is still not released at the date of this PD's pipeline submission.

The project's interests are also linked to the goals of aligning and amplifying global and national emission reduction targets, for instance contributing to the fulfillment of nationally established objectives (NDCs).

3.2 Applicability of Methodology

The project's alignment with the applicability criteria underscores its commitment to preventing unplanned deforestation (AUD) and its associated GHG emissions. The applicability conditions of the methodology VM0015 and the tool VT0001, with their respective positive relation to the project are detailed below:

The project is eligible under the applicability conditions of the VM0015 v1.2:

- Baseline activities include unplanned agriculture and cattle ranching activities, following the category: Avoiding Unplanned Deforestation according to the most recent VCS Standard.
- The project area includes one or a combination of forest types meeting the definition of "forest."
- At the project start, the project area includes only land qualifying as "forest" for a minimum of 10 years prior to the project start date (January 01, 2024), as demonstrated in the Land Use and Cover map between the years of 2012 and 2023, inserted Section 1.14.

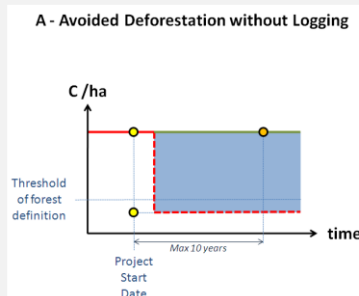
- The project area does NOT include forested wetlands growing on peat (e.g., peat swamp forests), demonstrated on Section 1.14.9.
- This PD brings evidence that native ecosystems have not been converted, cleared, drained, or degraded to generate GHG credits. The evidence is represented by historical time-series in satellite images, data from PRODES⁵⁵, georeferenced files, and other documents. Other complementary data or documents will be available for the VVB.

Table 13. Methodology's Applicability

Methodology ID	Applicability condition	Justification of compliance
VM0015	<i>Baseline activities may include planned or unplanned logging for timber, fuelwood collection, charcoal production, agricultural and grazing activities as long as the category is unplanned deforestation according to the most recent VCS AFOLU requirements.</i>	<i>The project applies to preventing unplanned deforestation caused by agricultural activities, in accordance with the most recent VCS AFOLU guidelines. The project is situated within the dynamic MATOPIBA region, encompassing the Brazilian states of Maranhão, Tocantins, Piauí and Bahia. This region, known for its burgeoning farming activities, is emblematic of the intricate interplay between agribusiness expansion and environmental preservation. The historical baseline for deforestation within this context is deeply rooted in the pursuit of economic gains through monoculture crops (e.g. soybeans and corn), grazing activities (e.g. livestock - cattle), representing the broader farming trends</i>

⁵⁵ The PRODES project conducts satellite monitoring of clear-cut deforestation in the Legal Amazon and has been producing annual deforestation rates in the region since 1988, which are used by the Brazilian government to establish public policies. <http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes>

		observed across the North of Brazil.
VM0015	Project activities may include one or a combination of the eligible categories defined in the description of the scope of the methodology (table 1 and figure 2).	This project is defined within scope A, thus considering Old growth without logging baseline and Protection without logging, fuel wood collection or charcoal production. The Project activities are essentially based on i) designing financial incentives for local communities and landowners who commit to conserving forests by generating commercial VCU's, where participants receive compensation for maintaining intact forests that sequester carbon; ii) developing robust monitoring and reporting systems that track changes in forest cover, land use and land cover, and other relevant indicators. These systems are taken with satellite technologies with a high spatial and temporal resolution, providing near-real-time data for assessing the effectiveness of REDD+ actions; iii) Engaging local communities in the design and implementation of this REDD+ project. Involving them in a public discussion and encouraging participation in forest conservation; iv) Allocating a portion of the revenue

		generated from REDD+ initiatives to fund social benefit programs aimed at enhancing the well-being of local communities. This could lead to a holistic development of the region, by directly improving the lives of those residing in the reference region. Note: This project operates with activities which do not involve logging, fuel wood collection, or charcoal production within its operational scope. This approach is in line with the project's commitment to the goals of the REDD+ initiatives. A - Avoided Deforestation without Logging 
VM0015	The project area can include different types of forest, such as, but not limited to, old growth forest, degraded forest, secondary forests, planted forests and agro-forestry systems meeting the definition of “forest”.	The project area includes old-growth forest, in line with the IPCC guidelines definition (Kyoto Forest): Minimum Canopy Cover: The IPCC criteria suggest a minimum tree canopy cover of 10-30% for a vegetation area to be classified as a forest.

		<i>Minimum Tree Crown Cover: Land of 0.05-1.0 hectares with tree crown cover (or equivalent stocking level).</i> <i>Minimum Tree Height: Trees within the area should typically reach a minimum height, which can vary but is often around 2-5 meters at maturity in situ.</i>
VM0015	<i>At project commencement, the project area shall include only land qualifying as "forest" for a minimum of 10 years prior to the project start date.</i>	<i>This project area consists exclusively of land that has met the criteria for being classified as a "forest" for a minimum of 10 years before the project's initiation (January 01, 2024), as demonstrated in the Land Use and Cover map between the years of 2014 and 2023, inserted in the item 1.14 of this document</i>
VM0015	<i>The project area can include forested wetlands (such as bottomland forests, floodplain forests, mangrove forests) as long as they do not grow on peat. Peat shall be defined as organic soils with at least 65% organic matter and a minimum thickness of 50 cm. If the project area includes forested wetlands growing on peat (e.g. peat swamp forests), this methodology is not applicable.</i>	<i>The project area does not include any forested area growing on peat. Eventually there are some lakes and other water bodies in the property. However, these areas are not permanent underwater spots. Otherwise, they would not be included within the project area.</i>
VT0001	<i>AFOLU activities the same or similar to the proposed</i>	<i>Since both scenarios presented within the</i>

	project activity⁵⁶ on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to violation of any applicable law even if the law is not enforced	additionality tool cover the same AFOLU activities as the project, and these activities are already proven to be in compliance with regulatory requirements, as demonstrated in section 1.15 (Compliance with Laws, Statutes, and Other Regulatory Frameworks), the project is therefore valid under this applicability criterion.
VT0001	The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario⁵⁷. Project proponent(s) proposing new baseline methodologies shall ensure consistency between the determination of a baseline scenario and the determination of additionality of a project activity.	This is a valid criterion for the project, since the definition of the additionality scenarios is based on the baseline scenario, which follows a coherent and plausible step-by-step approach in line with the requirements of the VM0015 methodology. The rationale behind this construction is presented in Section 3.4 (Baseline Scenario).

3.3 Project Boundary

3.3.1 The Reference Region (RR)

Reference Region Temporal Boundary

⁵⁶ In the context of this tool, activities the same or similar to the proposed project activity are used for the identification of possible land use scenarios that are allowable AFOLU activities under VCS document *AFOLU Requirements*.

⁵⁷ Please refer to the most recent version of the VCS document *AFOLU Requirements*, available on the VCS website.

The historical reference period for this project spans 10 years prior to the project's start date, from January 1, 2014, to December 31, 2023. This period is based on deforestation data from the PRODES program for 2014 to 2023.

The PRODES program monitors deforestation in the Cerrado Biome through satellite imagery. Each “PRODES year” covers the interval between annual deforestation mapping events, utilizing satellite imagery typically captured during the dry season — around August 1 of one year to July 31 of the next year. This timeframe aligns with the region’s dry season, when deforestation is most active, and avoids the rainy season, which hampers satellite visibility due to cloud cover. For example, the PRODES 2013 year began around August 1, 2012, and concluded around July 31, 2013.

Accordingly, the deforestation mapping periods for each "PRODES year" are as follows⁵⁸:

- Year 1: August 1, 2014 – July 31, 2015
- Year 2: August 1, 2015 – July 31, 2016
- Year 3: August 1, 2016 – July 31, 2017
- Year 4: August 1, 2017 – July 31, 2018
- Year 5: August 1, 2018 – July 31, 2019
- Year 6: August 1, 2019 – July 31, 2020
- Year 7: August 1, 2020 – July 31, 2021
- Year 8: August 1, 2021 – July 31, 2022
- Year 9: August 1, 2022 – July 31, 2023

Below, the baseline reassessment according to the validity period determined by the last VCS standard (4.7) – every six years.

N	Baseline Period	Reassessment (validity)
1	August 1, 2013 – July 31, 2023	July 01, 2024
2	August 1, 2019 – July 31, 2029	July 01, 2030
3	August 1, 2025 – July 31, 2035	July 01, 2036
4	August 1, 2031 – July 31, 2041	July 01, 2042
5	August 1, 2037 – July 31, 2047	July 01, 2048
6	August 1, 2043 – July 31, 2053	July 01, 2054
7	August 1, 2049 – July 31, 2059	July 01, 2060
8	August 1, 2055 – July 31, 2065	July 01, 2066
9	August 1, 2061 – July 31, 2071	July 01, 2072

⁵⁸ These dates are approximated, based on the PRODES Methodology document: <<http://mtc-m21d.sid.inpe.br/col/sid.inpe.br/mtc-m21d/2022/08.25.11.46/doc/thisInformationItemHomePage.html>>.

The most recent mapping period relies on Landsat imagery of the project area taken on July 31, 2023, in accordance with Section 5.2.1 of VM0015 v1.2. At the time of the Project Description (PD) completion, this was the latest available data from the National Institute for Space Research (INPE).

Reference Region Time-Series

For the image classification of the reference region, data from PRODES⁵⁹ (Project for Monitoring Deforestation in the Brazilian Amazon Forest by Satellite) Project were used.

PRODES Cerrado uses satellite images of the LANDSAT class (20 to 30 meters of spatial resolution and 16-day revisit rate) in a combination that seeks to minimize the cloud cover problem and ensure interoperability criteria. The images of the American satellite LANDSAT-5/TM were historically the most used by the project, but the images of the CCD sensor on board CBERS-2/2B, satellites of the Sino-Brazilian remote sensing program, were widely used. It currently makes massive use of images from LANDSAT 8/OLI, CBERS 4 and IRS-2. Regardless of the instrument used, the minimum area mapped by PRODES is 6.25 hectares.

PRODES Cerrado estimates are considered reliable by national and international scientists (KINTSCH, 2007⁶⁰). This system has been shown to be of great importance for actions and planning of public policies in the Amazon. Recent results, from analyses conducted with independent experts, indicate a level of accuracy close to 95%.

The annual deforestation rate PRODES has been used as an indicator for the proposition of public policies and for the evaluation of the effectiveness of its implementations. PRODES spatial data are used in: Certification of agribusiness production chains such as the Soy Moratorium and the Animal Husbandry Conduct Adjustment Term-TAC; (b) Intergovernmental agreements such as the United Nations Conference on Climate Change (COP 21) and the National Inventory Reports on Greenhouse Gas Emissions and (c) Monetary donations by the Amazon Fund, which use PRODES Cerrado as a reference to deforestation activity in the Legal Amazon.

The data transparency policy of forest state monitoring adopted by INPE and the federal government since 2004 allows full access to all data generated by monitoring systems, independent evaluations by the user community, including the government in its spheres and instances, the academy, the citizen and the Brazilian civil society in its various institutional arrangements.

⁵⁹ Instituto Nacional de Pesquisas Espaciais (INPE) / Coordenação Geral de Observação da Terra (OBT)
<<http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes>>

⁶⁰ Kintisch, Eli. "Improved Monitoring of Rainforests Helps Pierce Haze of Deforestation." *Science* 316 (2007): 536 - 537.

The Reference Region (RR) is an analytical domain through which information on rates, agents, drivers and underlying causes of land-use and land-cover (LU/LC) change are obtained, and subsequently used for future projection and monitoring.

Reference Region Spatial Boundary

The Reference Region lies at the Cerrado biome and features several key geographic landmarks: the Serra das Confusões National Park to the south, the Road PI-392 to the east, the Uruçuí municipal boundary to the north, and the BR-140 highway to the west. This region was defined to encompass geographic boundaries that include the main deforestation agents and drivers influencing the trends around the project area, with an emphasis on infrastructure and accessibility factors such as roads and nearby municipal centers – Uruçuí, which is a municipality with a large agriculture expansion and history of increased deforestation rate.

The RR was defined in accordance with conditions that determine the likelihood of deforestation within the project area similar or expected to become similar to those found in the reference region, considering especially on:

- a. landscape configuration and ecological conditions (elevation, slope, vegetation and precipitation);
- b. socio-economic and cultural conditions;
- c. agents and drivers of deforestation (groups of agents, infrastructure or other factors). This is the most important to adjust the RR, once it represents the dynamics of land use more accurately.

1)

a. Landscape configuration and ecological conditions

The Reference Region was defined to meet similarity percentages in line with section 5.1.1 of the VM0015 methodology. This alignment ensures comparability in terms of vegetation types, soil type, elevation, slope, and rainfall patterns, as detailed in the table below and illustrated in the following figures. These elements provide an essential basis for establishing pre-conditions consistency between the reference region and the project area.

Table 14. Similarity of the RR x Established Pre-conditions

	Reference Region	Project Area
Vegetation		
Forest	1,5%	0%
Savanna	98,0%	100%
Forest-Savanna Transition	0,5%	0%
Pedology		
Argissolo	1,4%	1,6%
Gleissolo	0,1%	0%

Latossolo	73,1%	63,2%
Neossolo	23,5%	34,6%
Plintossolo	1,8%	0,6%
Releif		
Flat	95%	96%
Montain	5%	4%
Slope		
0% to 8%	83,0%	76,7%
8% to 25%	12,2%	17,6%
25% to 45%	4,4%	5,7%
45% to 75%	0,4%	0%
Climate		
Ecuatorial	100%	100%

b. Socio-economic and cultural conditions

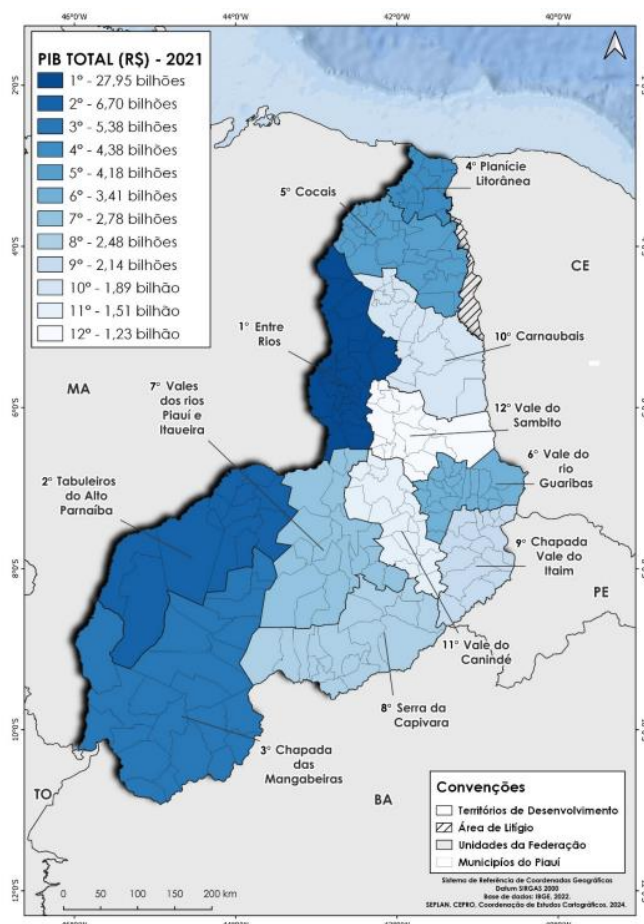
To define the boundaries of the Reference Region, a socioeconomic analysis of the region and the existence of government programs for forest preservation or even REDD projects was performed since these municipalities are within the limits of the "Amazon Arc of Deforestation" and agricultural expansion.

The Reference Region lies in the Cerrado biome and features several key geographic landmarks: state and/or federal highways connecting various municipalities in the state, including the main access routes to the capital, Teresina. This boundary was defined to encompass locations that include the main deforestation agents and drivers influencing the trends around the project area, with an emphasis on commercial farm activities, including the cultivation of soybeans and corn, cattle ranching, as well as pigs, horses, and mules, along with family farming with annual crops (e.g., beans, corn, and cassava)⁶¹. For large farms, grain production stands out, cultivated using medium to high-level technologies, typically practiced in the extensive agricultural fields implemented on both the region's plateaus and highlands. To illustrate the region's economic significance locally, a study by CEPRO (2024)⁶² provides a relevant information about the 'Tabuleiros do Alto Parnaíba' region in southwestern Piauí, *which comprises most of the Project Area of the CULTIVAR PROGRESSO REDD+ PROJECT: In 2021, the region presented the second-highest territorial GDP in the state, with BRL 6.7 billion (10.46% of the state's GDP). Several factors demonstrate that this region has progressively boosted its economy at a considerable pace in recent years. For instance, from 2020 to 2021, there was a nominal increase of 39.96%

⁶¹ MAPA/INCRA, 2023 – Relatório de Análise do Mercado de Terras do Piauí 'RAMT/PI 2022': https://www.gov.br/incra/pt-br/assuntos/governanca-fundiaria/relatorio-de-analise-de-mercados-de-terras/RAMT_SRPI_2022_corrigida_2023.pdf

⁶² CEPRO / SEPLAN, 2024 – 'Piauí - Produto Interno Bruto dos Territórios do Piauí 2021' Superintendência CEPRO/SEPLAN – Teresina: <https://www.seplan.pi.gov.br/download/157/diversos/13730/pib-do-estado-do-piaui-em-2022-relatorio.pdf>

in revenues. Its prominent activities involved soybean cultivation, dolomitic limestone production, and the manufacture of crude vegetable oils.



Fonte: Elaborado pela Superintendência CEPRO/SEPLAN (2024), a partir do IBGE em parceria com os Órgãos Estaduais de Estatística, Secretarias Estaduais de Governo e Superintendência da Zona Franca de Manaus - SUFRAMA (2023).

Figure 37. Piauí region's 2021 revenue.

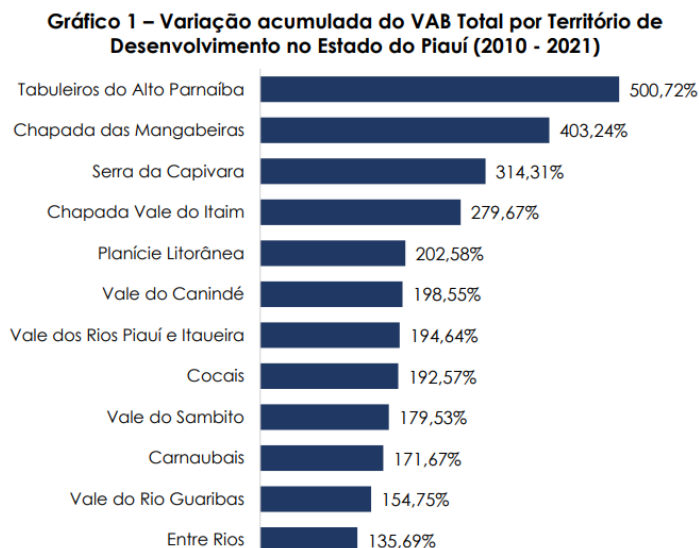
Source: CEPRO/SEPLAN (2024) – through IBGE database in partnership with other State's Statistics organizations, and other entities.

Another indicator from the same study, which better underscores the significance of the recent historical series in this specific region, is the **GVA (Gross Value Added)** – represents the contribution of each economic sector to the final value of production, deducting the value of inputs used in the production process over a given period.

Thus:

GDP = GVA + taxes – government subsidies.

Thus, analyzing the historical series between 2010 and 2021, there is an impressive growth in the GVA of this specific region (Tabuleiros do Alto Parnaíba), totaling a 500% increase, higher than any other regions of Piauí State.



Fonte: Elaborado pela Superintendência CEPRO/SEPLAN (2024), a partir do IBGE em parceria com os Órgãos Estaduais de Estatística, Secretarias Estaduais de Governo e Superintendência da Zona Franca de Manaus - SUFRAMA (2023).

Figure 38. Accumulated variation of the GVA, by development region in Piauí (2010 - 2021).

Source: CEPRO/SEPLAN (2024) – through IBGE database in partnership with other State's Statistics organizations, and other entities.

This information is relevant to the context of the Reference Region, as the boundaries of this area were defined considering the economic aspects of the macro-region of 'Tabuleiros do Alto Parnaíba' as a primary qualitative reference regarding the expansion of agricultural activities in the Piauí State, particularly those related to grain production. With the rapid and significant economic boom in the region, the still-preserved areas of native Cerrado vegetation are increasingly at risk, given the high rate of deforestation and land-use change for the production and trade of such profitable commodities.

This information is relevant to the context of the Reference Region, as the boundaries of this area were defined considering the economic aspects of the macro-region of 'Tabuleiros do Alto Parnaíba' as a primary qualitative reference regarding the expansion of agricultural activities in the Piauí State, particularly those related to grain production. With the rapid and significant economic boom in the region, the still-preserved areas of native Cerrado vegetation are increasingly at risk, given the high rate of deforestation and land-use change for the production and trade of such profitable commodities.

As a complement within this scenario, it is highly relevant to highlight social issues as well, according to official government indicators, to evaluate how wealth generated in the state is distributed and to assess its proportion in relation to the State's GDP (R\$ 22.3 BI in 2010 to R\$ 64 BI in 2021, a 187% difference). This and the next information come from CEPRO/SEPLAN

(2024)⁶³, which provided in a recent study a series of labor data that elucidates this relationship: the average real monthly income of workers in the state ranks among the worst in Brazil, with an average value of R\$1,816/month, placing it 21st out of 27 states. When considering the percentage of discouraged people – those unemployed but not actively seeking employment – the situation is even more critical. The State of Piauí ranks 2nd with an index of 10.4. Compared to 27th place (0.5), the difference is striking. Looking at the previous year (Q3, 2023), the scenario is quite similar, with Piauí showing an index of 10.5. This all illustrates a situation where, despite a massive force of production and income generation in the State's perspective as a whole, the social scenarios represented by individual and labor income do not progress proportionally, which is likely reflected to the Reference Region.

c. Agents and Drivers of Deforestation

Relationship Between Agents of the Reference Region and Project Area

To ensure the Reference Region is indeed meaningful, it is necessary to observe the direct actions of agents and drivers on the Project Area. Since there are no in-depth studies for each portion of the Project Area, which is very spread, we considered certain intrinsic characteristics of the Grupo Progresso's farms themselves and neighboring farming lands to compare and reflect them in similar areas within a larger radius – eventually supporting our Reference Region delimitation. These characteristics inevitably include the study of agents and drivers, as required by the VM0015 methodology. All the information presented in this section can be cross-checked and verified with the studies and references indicated in the Agents and Drivers section in **Section 3.4 (Baseline Scenario)**, once both sections are complementary.

- *Illegal Clear Cutters / Land-grabbers / Hunters and Fire Setters*

At the state level, there is evidence that most deforested areas in the state of Piauí lack legal mechanisms' efficiency. Illegal activities are rarely documented in their initial stages and are later investigated through initiatives and deforestation national study programs. The following statements (MAPBIOMAS⁶⁴, RAD2022⁶⁵, AGÊNCIA PÚBLICA⁶⁶, BOMBEIROS PIAUÍ⁶⁷, GOVERNO

⁶³ CEPRO/SEPLAN, 2024. Informe Socioeconômico – Dinâmica recente do Mercado de Trabalho no Piauí / Superintendência CEPRO/SEPLAN – Teresina: <https://www.seplan.pi.gov.br/download/150/informes-socioeconomicos/7119/informe-socioeconomico-n-28.pdf>

⁶⁴ Projeto MapBiomass – Mapeamento anual de cobertura e uso da terra no Brasil entre 1985 a 2022 – Coleção 8, acessado em [DATA] através do link: [<https://mapbiomas.org/download-dos-atbds>]

⁶⁵ RAD2022: Relatório Anual de Desmatamento 2022 - São Paulo, Brasil - MapBiomas, 2023 - 125 páginas <http://alerta.mapbiomas.org>

⁶⁶ Agência Pública, 2024 – “Demora da Justiça em caso de grilagem ameaça famílias no Cerrado do Piauí”: [Grilagem: Demora da Justiça em disputa ameaça famílias no PI](#)

⁶⁷ BOMBEIROS PIAUÍ, 2021 – “AÇÕES DE PREVENÇÃO DE INCÊNDIOS FLORESTAIS”: <http://www.cbm.pi.gov.br/noticia.php?id=1030/>

PIAUÍ⁶⁸) are able to demonstrate the relevance of relating the characteristics of agents from the Project Area to a broader extent, such as the Reference Region.

1. **Deforestation in Key Municipalities:** *“Among the 50 municipalities with the highest deforestation rates in 2022 – those that more than doubled their deforestation from 2021 – four are in Piauí (Santa Filomena, Sebastião Leal, Baixa Grande do Ribeiro, and Uruçuí)” (RAD, 2022)* – This statement illustrates the baseline deforestation pressures in the municipalities within the Reference Region. Notably, three of the municipalities in the Project Area are part of this alarming deforestation trend, highlighting the expansion threat in these specific areas.
2. **Increase in Embargoed Areas:** *“The number of areas embargoed for illegal deforestation also grew considerably, from 20,875 hectares in 2022 to 57,846 hectares in 2024, a jump of 177%” (GOVERNO PIAUÍ, 2024)* – This growth indicates a significant rise in illegal deforestation activities, emphasizing that the issue extends initially beyond agribusiness to include illegal clearing that is later regularized. Both Project Area and Reference Region mirror this agent’s trend, as it represents one of the locations of fastest expansion within the State.
3. **Land Grabbing Practices:** *“Land grabbers have used tractors and large steel chains to devastate native Cerrado areas. (...) Over seven years ago, the Piauí Public Prosecutor’s Office (MPPI) denounced an immense land grabbing of 124 thousand hectares, perhaps the largest in the State of Piauí. (...) The TJPI (Piauí’s Legal Court) released farm records even with evidence of connection to land grabbing.” (AGÊNCIA PÚBLICA, 2024)* – This situation in Piauí is indicative of the systemic issues also affecting the Reference Region, where illegal land registration facilitates the sale of deforested lands, a risk also presented in the Project Area, as this shows relevant conditions/drivers for this trend, such as being included within a context of great values of land (see page 36 of RAMT 2022, 2023)⁶⁹, and great geomorphological conditions for farming.
4. **Fire as a Hunting Mechanism:** *“In the Cerrado region (...) a large part of the fires is caused by hunters who set fire to the vegetation to scare away the animals or to favor their kill” (BOMBEIROS PIAUÍ, 2021, and GOVERNO PIAUÍ, 2020)* – As mentioned in Section 2.1.2, the causes of fires due to hunters are imminent and historical in the region, according to the local population and landowner themselves. This is actual evidence that part of the fire risks is a consequence of historical activities in the whole Reference Region. This risk is also mentioned and considered within the Non-Permanence-Risk of this Project.

- Small-scale farmers / rural settlements

⁶⁸ GOVERNO PIAUÍ, 2020 – “Operação Queimadas: Instituições unem esforços para reduzir efeitos das queimadas no Piauí”: <https://antigo.pi.gov.br/noticias/operacao-queimadas-instituicoes-unem-esforcos-para-reduzir-efeitos-das-queimadas-no-piauui/>

⁶⁹ RAMT-PI/2022, 2023 – Relatório de Análise do Mercado de Terras do Piauí: https://www.gov.br/incra/pt-br/assuntos/governanca-fundiaria/relatorio-de-analise-de-mercados-de-terras/RAMT_SRPI_2022_corrigida_2023.pdf

Looking at a larger scale, it is possible to find small rural farms in relevant volumes, even if more dispersed. As explained in more detail in Section 3.4, small landowners face significant challenges in generating income, which are part of broader economic, socio-cultural, and political issues. As highlighted by EMBRAPA (2018), the limited size of their farms compromises financial viability, as small-scale production presents a structural problem for these farmers. This may lead to a potential search by these smaller agents for lands still with potential for development.

These considerations apply to various preserved areas within the Reference Region, such as the Project Area itself, especially in those locations with some characteristics not fully feasible for growing grains in large scales, because of conditions such as slope – depressed areas are less likely to have an expansion of grains' monoculture because of difficulties on planting and harvesting, which require use of heavy equipments, and also contiguous areas for the crop expansion. These locations have a lower value in the market, which facilitates the smaller agents to purchasing areas. In the map below, it is possible to observe the proportion and relevance of these smaller farms, throughout portions of the Project Area. Thus, this criterion was applied to a broader view, which encompasses the Reference Region. As this is demonstrated by the map below, the same agents are a common characteristic of smaller deforestation among both areas (PA and RR):

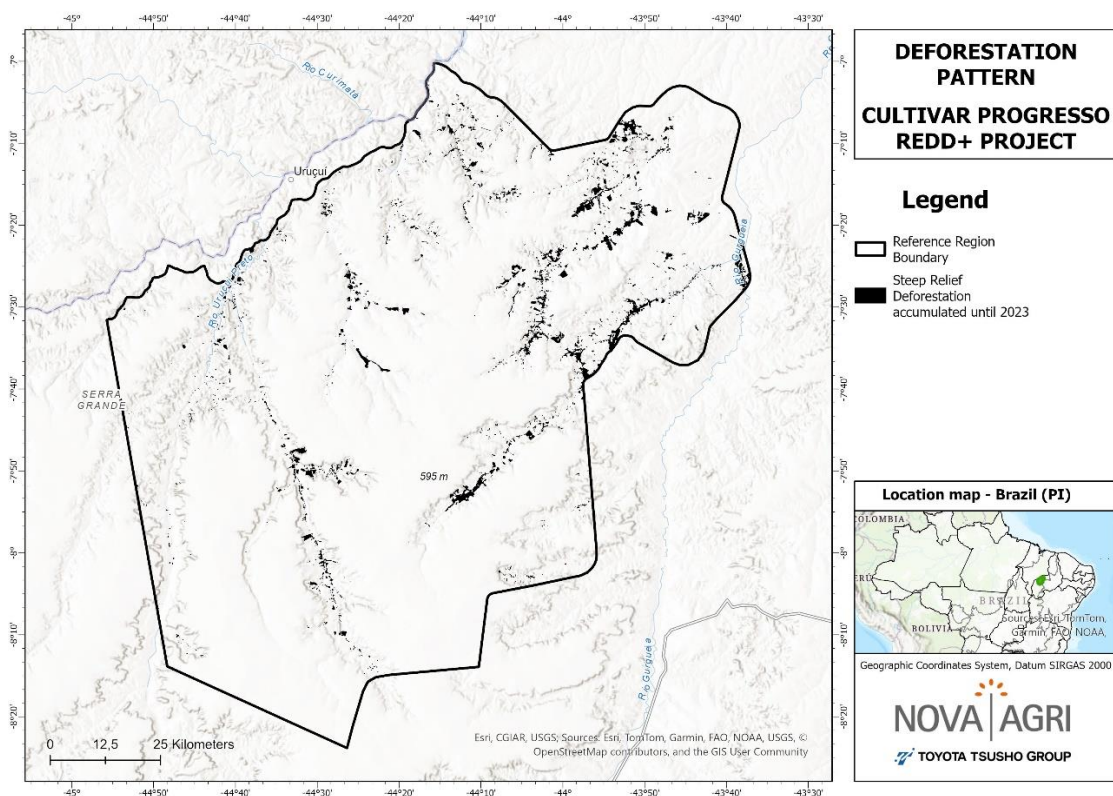


Figure 39. Dispersed and smaller deforestation, characteristic of small farmers / rural settlements, throughout the Reference Region.

Source: the author.

- Medium to large-scale farmers

Unlike small landowners, these agents have greater potential for opening new areas, *especially* large areas such as the Project Area. These are farmers who adopt an intensive production model, focusing on high-value crops such as soybeans, corn, and cotton, as well as large-scale livestock farming.

One of the crucial factors for these agents is the Market Demand for Commodities: Farmers deforest to expand land for commodity crops such as soybeans, corn, coffee, sugarcane, among others, which are in high demand in both local and global markets. Brazil's position as a leading exporter of agricultural products creates strong economic incentives for deforestation.

As the Reference Region consists mainly of rural, private areas with a higher number of fiscal modules, it is fully compatible with the Project Area, which is also a rural, private area with a very large size (+10,000 hectares). Further details on these agents and their influence on the Reference Region and Project Area can be found below in Section 3.4.

Relationship Between Drivers in the Reference Region and the Project Area

The risk of deforestation in the Project Area is linked to the presence of the same deforestation agents identified in the Reference Region and the likelihood of deforestation is intrinsically associated with the drivers of deforestation, especially the economic value and geomorphological conditions of lands.

In other words, the likelihood of deforestation in the Project Area is similar or should become similar to those found in the Reference Region based on the following drivers' criteria:

(1) the lands' geomorphological characteristics, (2) the likelihood of deforestation in the Project Area is proportional to the historical deforestation that occurred on well-valued commercial properties, and (3) rise and technological development of agriculture in the Cerrado Savanna.

In addition to the tacit evidence demonstrated through the analysis of the historical deforestation applied to each group of agents and drivers of deforestation (as presented in Section 3.4 – Baseline Scenario), the studies below substantiate the relevance of these driver's for the Reference Region delimitation and baseline construction:

(1) Lands' geomorphological characteristics

In a statistical terrain's analysis conducted in both boundaries, the Reference Region and the Project Area, it is possible to observe a clear similarity in the distribution of their areas' classifications, with varying levels of intensity. When plotting the Gaussian curve for both areas, the average is around 0.6 – on a scale ranging from 0 to 1. This indicates the similarity in geomorphological aspects, especially the relief, which is one of the characteristic drivers in the organization of the agents, as explained in the previous topic. It is also evident that neither of the two areas have extremely steep reliefs – which would correspond to areas with an index close to 1. Besides the terrain, all highlighted areas

within the Project Area also possess significant economic-commercial value due to climatic conditions, soil characteristics, and the specific activities established by each of the agents. These complementary factors will also be described as drivers in the following sections.

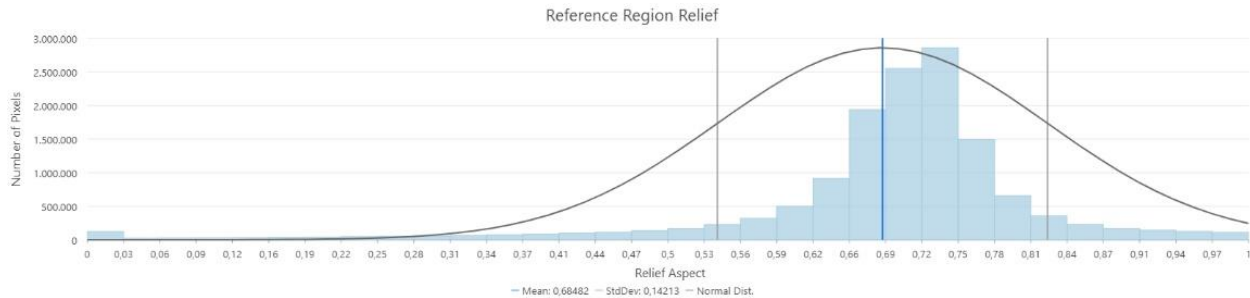


Figure 40. Gauss normal curve: analysis of the Reference Region relief.

Source: the author.

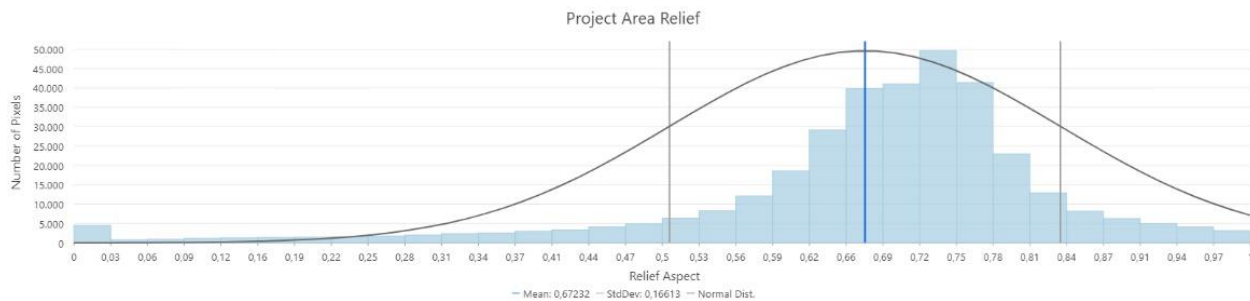


Figure 41. Gauss normal curve: analysis of the Project Area relief.

Source: the author.

(2) Economic lands' value

According to the RAMT (2020), it becomes evident that both plateau and depressed areas have commercial value due to the ease of occupation for both small (familiar) and large-scale agriculture and livestock farming. Although productivity in depressed lands is lower, it remains significant, as shown in the following table – the land's value for depressed areas with vegetation is more valuable than swamp areas, or demarcated land areas.

Tipo de área	Área – 1 Ha	Quantidade de sacas de soja	Valor – 1 Saca *R\$ 70,00
Área cultivada - Platô	1ha	200	14.000,00
Área mata virgem - Platô	1ha	100	7.000,00
Área mata virgem Baixada	1 ha	50	3.500,00
Área de Baixada - dentro do Parque Ambiental	1 ha	25	1.500,00
Área de Brejo com Produção	1ha	14 a 25	1.000,00 a 1.500,00
Área de Baixada com Demarcação	1 ha	8 a 10	600,00
Área de Baixada sem Demarcação	1 ha	4 a 6	300,00

Fonte: Corretores de Imóveis de Santa Filomena e Uruçuí.

Informante: Joildo de Oliveira Mascarenha CPF: 013.380.191.80 Bairro - Primavera – I, Nº 64, Santa Filomena – PI,

Informante: Carlos João CPF: 013.380.191.80, Rua Davis Caldas ,Nº 39, Centro - Uruçuí – PI, .

* O preço da saca varia de acordo com a cotação diária na época do estudo a cotação era de R\$ 70,00.

Figure 42. Price per Hectare of Land in the Cerrado Market of Piauí. Highlighted lines: 'Plateau Areas of Native Vegetation' and 'Depressed Areas of Native Vegetation'.

Source: RAMT (2020).

The point here is mainly to demonstrate how land's value directly influences the likelihood of clearing certain areas over others with lower value. The two characteristic classifications of the Project Area (Plateau and Depression) are suitable for planting, differing only in the nature of their products and production scale. When we look, for example, at the plateau areas of the Reference Region as a whole, we clearly observe that the majority is occupied by medium/large-scale agents, who are among the main drivers of deforestation when deciding to expand their areas (Figure 43). However, deforestation in depressed and less prominent areas cannot be disregarded, as shown on Figure 39.

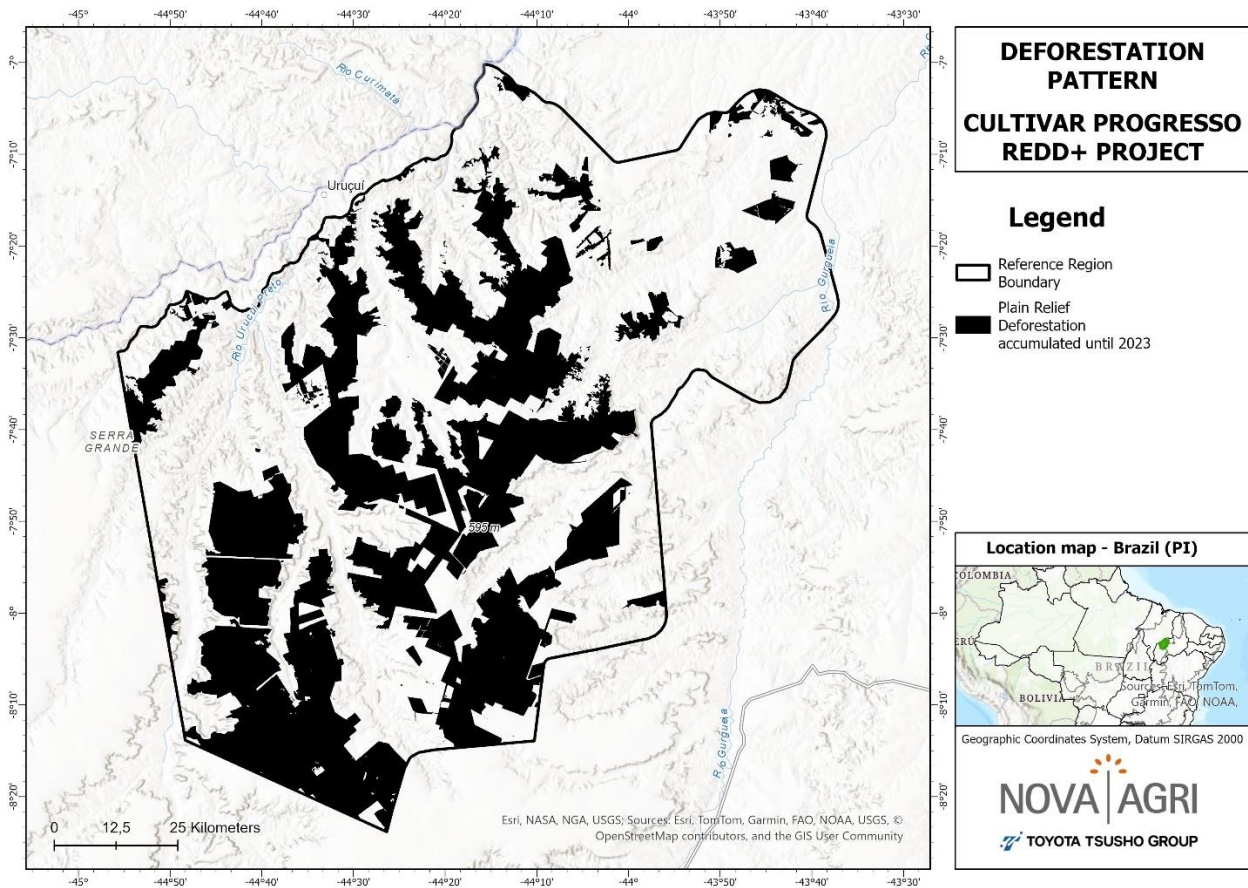


Figure 43. Bigger deforestation, characteristic of medium to large farmers, throughout the Reference Region.

Source: the author.

The dynamics of how these areas are occupied by different agents is further explored in Section 3.4.

(3) Rise and Technological Development of Agriculture in the Cerrado Savanna

As indicated in Section 3.4, an important driver to consider within the context of agricultural expansion into the Brazilian center-countryside is the recent development of technologies and practices, particularly for agriculture, which have been fostered over recent years and have contributed to the productivity boom in the Brazilian Cerrado. The Reference Region is fully integrated into this context, as shown in the studies by CEPRO/SEPLAN (2024)⁷⁰, which demonstrate how the area (south-western) encompassing the municipalities of the Project Area is categorized as the region with the most significant economic progression in recent years. This is largely due to substantial

⁷⁰ CEPRO/SEPLAN, 2024. "Piauí - Produto Interno Bruto dos Territórios do Piauí 2021 / Superintendência CEPRO/SEPLAN": <https://www.seplan.pi.gov.br/cepro/publicacoes/>

investment in research and development for agriculture, especially by EMBRAPA⁷¹, which is a national institution and a global reference in production within tropical spots. However, despite all the economic advancements and returns for the country, which are indeed meritorious, this accelerated development also brings some environmental setbacks, notably identified by uncontrolled deforestation and the lack of adaptation to appropriate local regulations. This can be seen in the recent scenario concerning the significant deforestation in the Cerrado, as deforestation in this biome is proving to be as lucrative as it once was solely in the Amazon⁷².

Final Reference Region Spatial Definition

- Thus, based on these premises, the Reference Region was delimited. It covers an area of 1.301.273 hectares, making it approximately 40 times larger than the project area, which itself covers 32.550 hectares. This extensive reference area ensures a robust comparison framework for analyzing and projecting deforestation trends that affect the project area.

3.3.2 Project Area

Project Area Temporal Boundary

The historical reference period is the period in which analysis of LULUCF (land use, land use cover and forest) within the reference region and project area is carried out. In accordance with the methodology, this period must not exceed 10 – 15 years in span and end as close as possible to the REDD project start date (≤ 2 years). The project start date is January 01, 2024 and the historical reference period comprised analysis of images from 2014 to 2023 in the project area, the same as for reference region. Sufficient satellite images referring to this period were available, with low presence of cloud cover.

Project Area Time-Series

For the image classification of the project area, satellite images from Landsat 7 (Enhanced Thematic Mapper (ETM) sensor), and images from Landsat 8 (Operational Land Imager (OLI) sensor) were found for the 2014-2024 period (Table 5 below). One image was necessary to cover the whole Project Area, being the following orbit-point respectively: 220/65. A total of 1satellite images were necessary to compose the Project Area during the historical reference period. These images were downloaded from U.S. Geological Survey (USGS)⁷³.

Project Area Spatial Boundary

⁷¹ EMBRAPA, 2004 – “Embrapa criou um novo cerrado em três décadas”: <https://www.embrapa.br/busca-de-noticias/-/noticia/17956252/embrapa-criou-um-novo-cerrado-em-tres-decadas>

⁷² REPORTER BRASIL, 2010. O “milagre do cerrado” e as escolhas da The Economist: <https://reporterbrasil.org.br/2010/08/o-quot-milagre-do-cerrado-quot-e-as-escolhas-da-the-economist/>

⁷³ U.S. Geological Survey. Available at: <<http://www.usgs.gov/>>

In accordance with VCS requirements, stipulated in Approved VCS Methodology VM0015, version 1.2, the project area may only include areas composed of “forest”⁷⁴ for a minimum of ten years prior to the project start date. Satellite images from 2014 and 2023 were analyzed and classified.

The remaining forest areas within ‘Sanders Farms’ that were defined as forest in 2023 and in 2013 were identified and utilized to compose the project area. In addition, some non-forest areas were also excluded, such as rivers and lakes.

The size of the areas that were considered “non-forest” (i.e. deforested areas and waterbodies) within the total property area at the project start date was 32,550 ha. The full contour coordinates of the project area are found previously on the Section 1.13 Project Location.

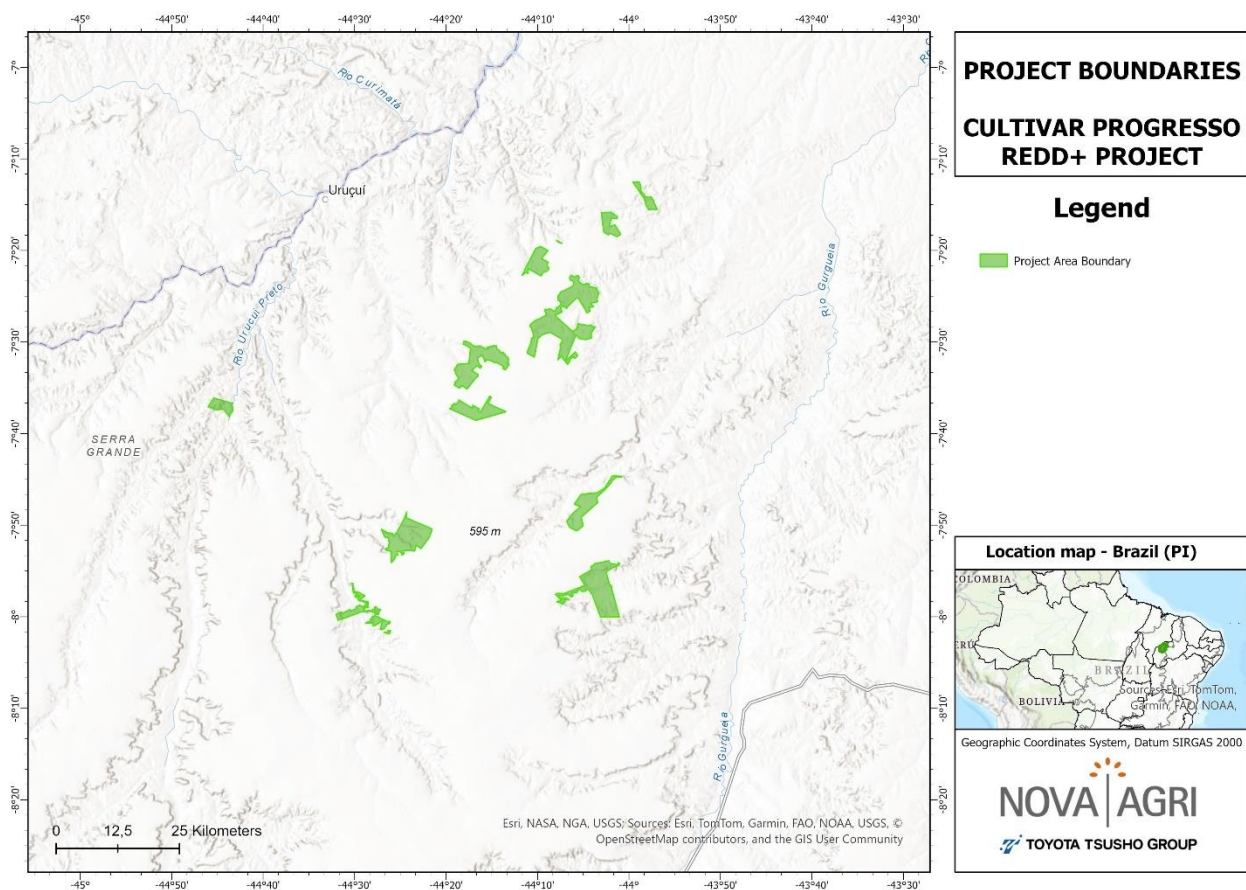


Figure 44. Project Area.

Source: the author.

⁷⁴ Brazilian Forestry Service. Brazil adopts FAO forest definition: “Forest is an area of at least 0.05-1.0 ha with canopy cover (or equivalent density) of more than 10-30%, with trees with the potential to reach the minimum height of 2-5 meters at maturity *in situ*.” Available at: < <https://snif.florestal.gov.br/pt-br/conhecendo-sobre-florestas/167-definicao-de-floresta>>.

The project area borders used in the SANDERS REDD+ PROJECT were extracted from technical appraisals (Portuguese: 'laudos') registered at an official notary and at INCRA⁷⁵. Vectorization – which is the process of converting the appraisal documents into digital shapefiles and polygons, being formats compatible with GIS. The technical appraisals were double checked against the descriptive memorandum of the property, which shows the property limits through geographic coordinates.

Table 15. Satellite images time-series used for PA classification

Satellite	Sensor	Resolution		Coverage (km ²)	Acquisition date DD/MM/YY	Scene	
		Spatial (m)	Spectral (µm)			Path/ Latitude	Row/ Longitude
Landsat 8	OLI	30	0.64 – 1.65	34,225	26/08/2014	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	13/08/2015	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	30/07/2016	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	17/07/2017	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	04/07/2018	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	10/08/2020	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	13/08/2021	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	15/07/2022	220	65
Landsat 8	OLI	30	0.64 – 1.65	34,225	18/07/2023	220	65

All the images were then processed through a band composition method (bands 1 – 5), since this composition enabled the mapping analysis in both real and false colors. The following bands and wavelengths were utilized:

- Band 1 – Visible (0.45 – 0.52 µm) = blue
- Band 2 – Visible (0.52 – 0.60 µm) = green
- Band 3 – Visible (0.63 – 0.69 µm) = red
- Band 4 – Near infrared (0.76 – 0.90 µm)
- Band 5 – Mid infrared (1.55 – 1.75 µm)

The first step of the visual image classification of land-use in the project area was done on ArcGIS Pro⁷⁶ software, using images from the Landsat 8 satellite, and in accordance with their minimal

⁷⁵ Instituto Nacional de Colonização e Reforma Agrária (INCRA): <http://www.incra.gov.br/>

⁷⁶ Environmental Systems Research Institute, Inc. (Esri) ArcGIS Pro: <https://www.esri.com/en-us/arcgis/products/arcgis-pro/overview>

30m resolution – the minimum mapping unit (MMU) was defined at 30x30m (0.09ha), therefore easily meeting the methodology requirement that the MMU cannot be larger than 1ha.

The supervised classification utilized the following three land use classes:

- 1- Forest;
- 2- Non-Forest (exposed soil, agricultural lands, pasturelands, etc);
- 3- Waterbody.

The year 2013 was then utilized to exclude forests that are less than 10 years old at the project start date from the project area. Finally, the “waterbody” class was defined as constants throughout the historical reference period, and the same class was applied for each mapped year, as those values did not significantly vary across the analyzed period.

For the present project, the process of accumulating ‘non-Forest’ areas was adopted, in such a way that areas classified as ‘non-Forest’ in one year were necessarily included in the same category in the following year.

The project area contains only areas which were defined as “forest” in 2013 (> 10 years prior to the project start date). Further analysis for this classification is expressed in the section 1.14 (Conditions Prior to Project Initiation).

3.3.3 Leakage Belt

General Introduction

The leakage belt was defined as an area of 629.311 hectares, set within a zone adjacent to the project area. This area was delineated by considering the limits of the reference region and the boundaries of homogeneous landscape units to ensure that the leakage belt accurately represents the same deforestation drivers impacting the project area.

The boundary comprising the project area – defined in accordance with the methodology’s rules governing the latter – as well as the size of the leakage belt, are displayed in *Table 16* below:

Table 16. Project Area and Leakage Belt Boundaries

Name	Area (ha)
Project Area	32.549,79 ha
Leakage Belt	629.311 ha

Current Occupation of the Deforested areas within the RR

In accordance with the VM0015 methodology, the LB has been defined using the opportunity cost analysis method (Option I), which is fully applicable as profit motivation is the primary driver

of deforestation within the reference area, affecting most agents present within it. To support this, we first confirm that at least 80% of the deforested area during the historical reference period occurred in locations that offered profitability for the agents involved. This can be demonstrated by comparing the maps of (i) historical accumulated deforestation and (ii) current occupation:

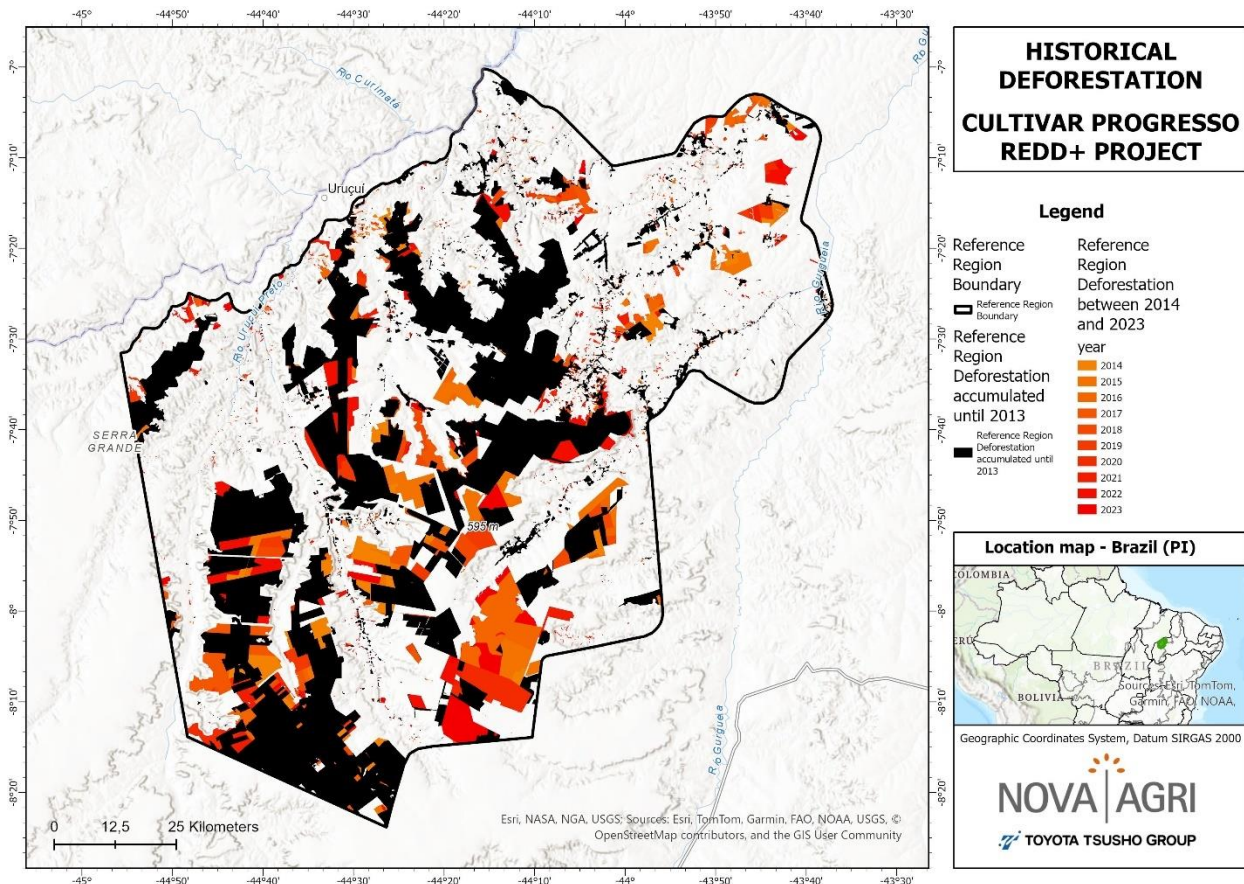


Figure 45. Historical deforestation in the Reference Region.

Source: PRODES, the author.

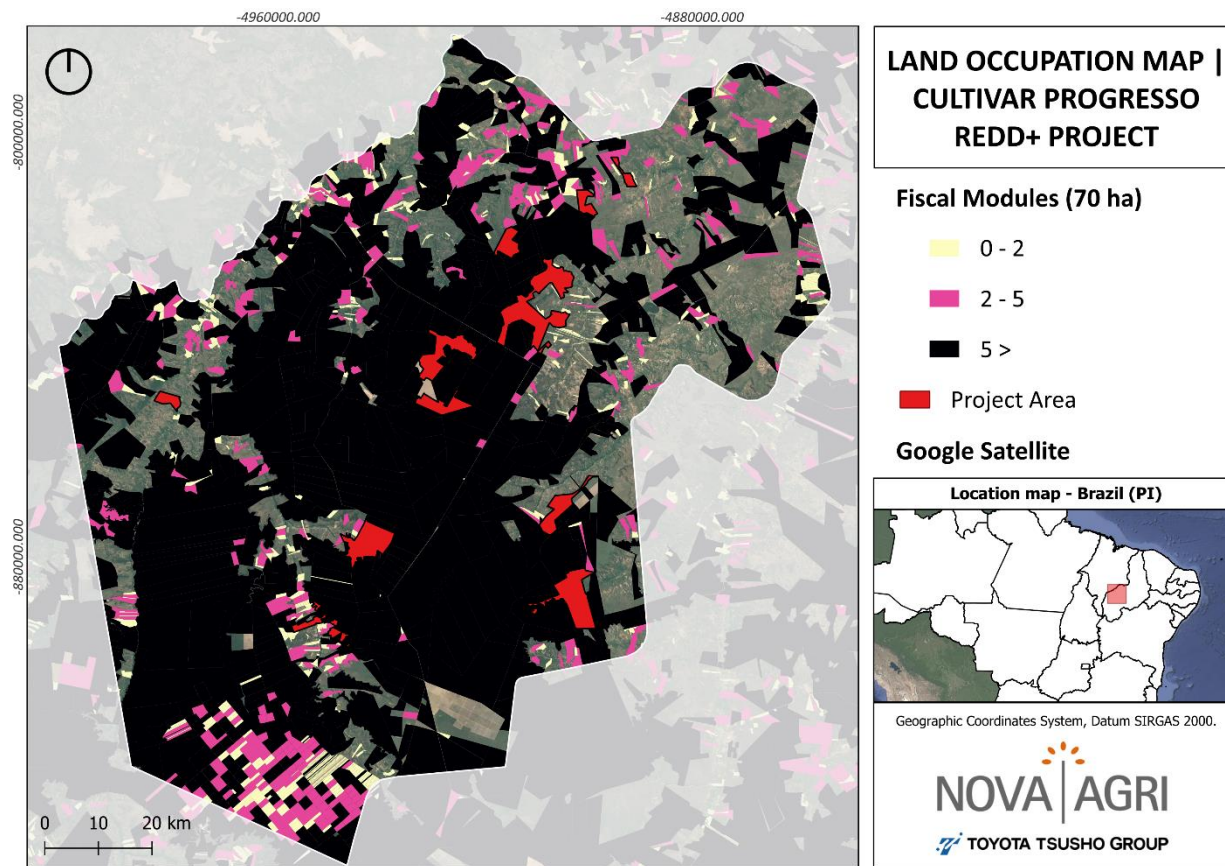


Figure 46. Land occupation in the Reference Region.

Source: EMBRAPA⁷⁷, INCRA⁷⁸, the author.

Through these maps, it is possible to clearly identify that the most part of the deforested area is currently part of a registered farm in the official government database and is included within rural farming areas. As a complement, the Land Use map from MAPBIOMAS⁷⁹ below highlights that 87% of the non-forested area in Piauí is being used for Pasture and Agriculture (“Agropecuária”, in Portuguese).

⁷⁷ The map was based on the Brazilian criteria of Módulo Fiscal (fiscal module), available on: <<https://www.embrapa.br/codigo-florestal/area-de-reserva-legal-arl/modulo-fiscal>>. One fiscal module represents 70 ha, which is the size of Sebastião Leal, Landri Sales and Antônio Almeida’s fiscal modules. Uruçuí’s fiscal module is 75 ha.

⁷⁸ INCRA – SIGEF (Sistema de Gestão Fundiária) / Brazilian Government, available on: <<https://sigef.incra.gov.br/>>.

⁷⁹ Projeto MapBiomass – Coleção 9.0 da Série Anual de Mapas de Cobertura e Uso da Terra do Brasil, acessado em 14/01/2025 através do link: <<https://plataforma.brasil.mapbiomas.org/>>.

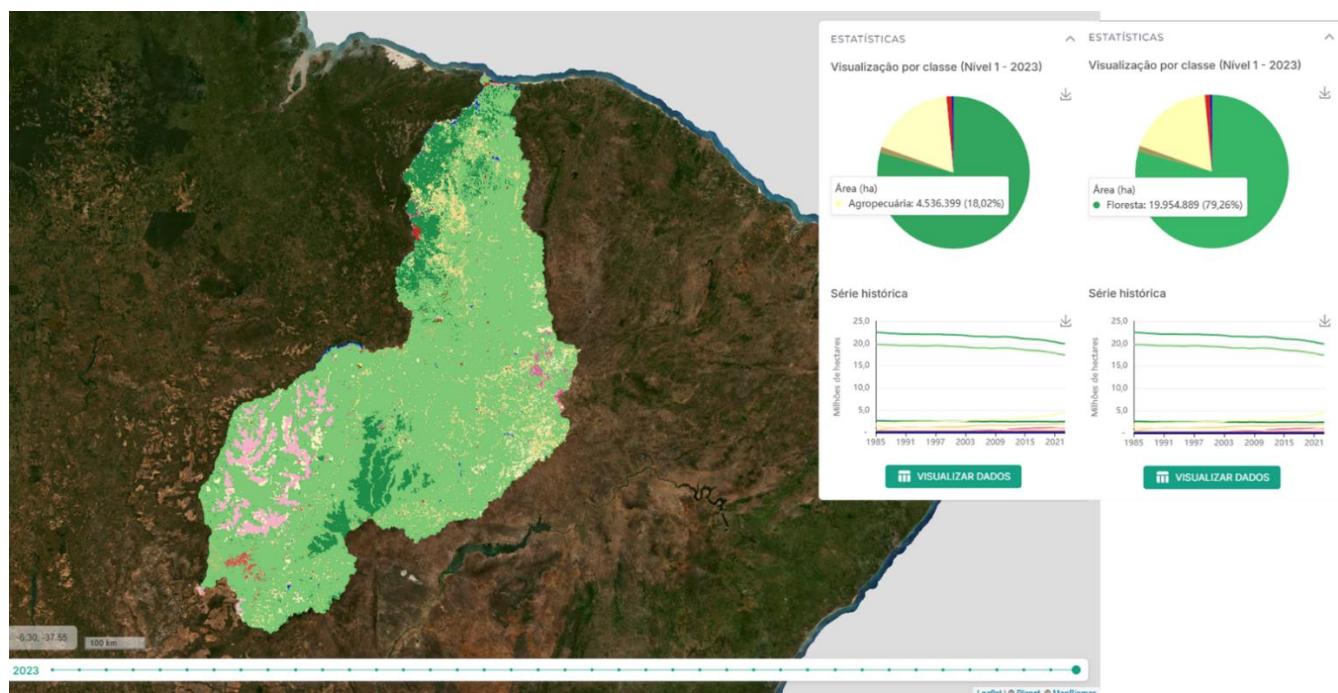


Figure 47. Land occupation in Piauí (PI) according to LU study. Forests = 79,26%; Others = 20,74%, where Agriculture and Livestock activities represents 18,02% (i.e. 87% from the non-forested area).

Source: MapBiomias (2023).

In order to compose the land tenure map above with greater detail, the proponent referred to a **targeted study lands' values evaluation within the Project Area and surroundings** (RAMT/PI⁸⁰, 2020), which shows a comparative parameter among different regions in terms of lands' values and consequently risks of deforestation leakage.

This study was particularly important to find and delineate areas with the highest likelihood of deforestation leakage, considering locally which specific areas have the highest and lowest value within the Reference Region (in accordance with the land value driver explored in Section 3.4). For this purpose, the following steps were taken:

- (1) The above-mentioned study demonstrates a particular difference in land values according to each location. The Project Area falls basically within two areas of the Southwest of the State of Piauí: MRT 03 – “Serra do Quilombo” and MRT 05 – “Vale do Rio Piauí.” These two areas, according to the same study, have distinct average land values per hectare.

⁸⁰ RAMT/PI, 2020 - Relatório de Análise de Mercado de Terras do Estado do Piauí: https://www.gov.br/incra/pt-br/assuntos/governanca-fundiaria/relatorio-de-analise-de-mercados-de-terras/ramt_sr24_2020.pdf

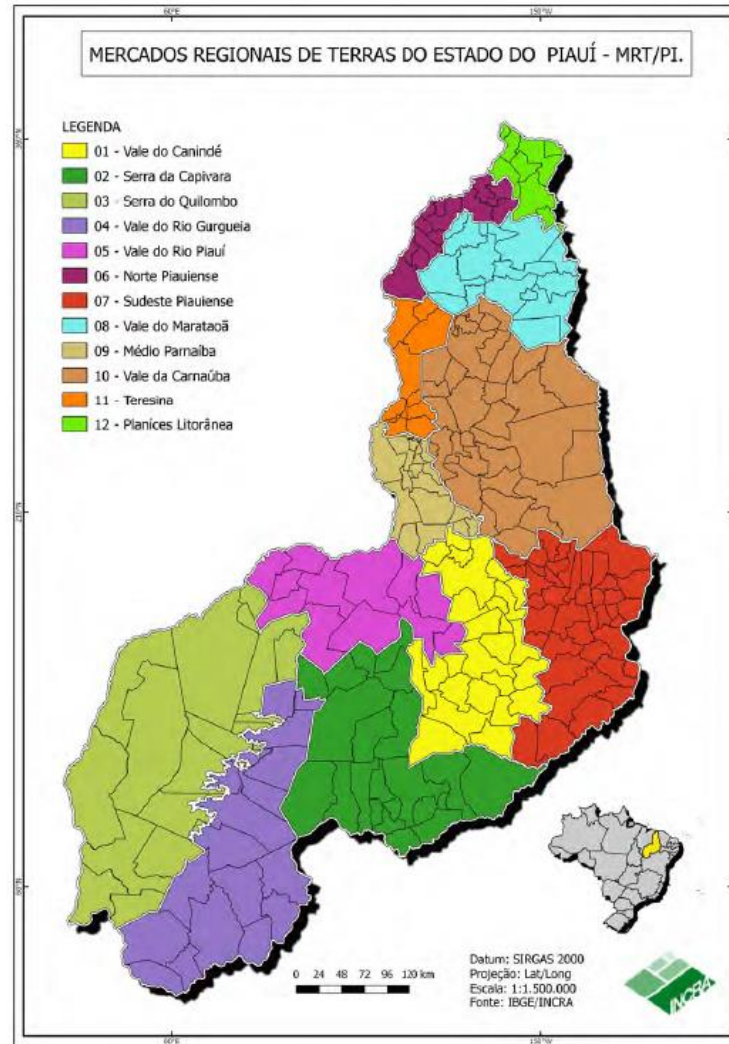


Figure 48. Division of the State regions for the economic land values study.

Source: RAMT/PI (2020).

- (2) Considering the land values of **medium/high technological level**, which are generally the most targeted for expansion of grain agriculture and monoculture, the following price configuration was established:
- MRT 03: ~ BRL 7,000/ha
 - MRT 05: ~ BRL 2,500/ha

The evidence from the mentioned study is presented below:

PPR / SR-24/PI / N° 01/ANO 2018/ MRT-03					
VALOR DE TERRA NUA POR HECTARE – VTN/ha					
TIPOLOGIAS	N° Elementos Pós Saneados	Média	Coeficiente de Variação (%)	Limite Inferior	Limite Superior
		VTN/ha (R\$)		(x -15 %)	(Média +15 %)
1º Nível categórico					
Agricultura	13	869,13	21,79	738,76	999,49
Vegetação Nativa	10	1.315,53	14,35	1.118,20	1.512,86
2º Nível categórico					
Agricultura Familiar	7	628,66	19,41	534,36	722,96
Agricultura Médio/Alto Nível Tecnológico	14	7.152,18	17,06	6.079,35	8.225,00
Vegetação Nativa Cerrado	10	1.315,53	14,35	1.118,20	1.512,86
3º Nível categórico					
-	-	-	-	-	-
MRT GERAL (R\$)					
Todas as Tipologias					
	35	1.117,91	51,23	950,22	1.285,60

Figure 49. Average prices for Farmlands with Medium/Large Technological Level within MRT 03.

Source: RAMT/PI (2020).

RESULTADO 1º NÍVEL CATEGÓRICO – AGRICULTURA			
ESTATÍSTICAS	1º NÍVEL CATEGÓRICO		
MÉDIA(VTI/ha)	2.550,75	1.621,19	1.351,49
DESVIO PADRÃO AMOSTRAL	2.343,16	618,34	157,67
COEFICIENTE DE VARIAÇÃO	91,86	38,14	11,67
MÉDIA MENOS UM DESVIO PADRÃO	207,60	1.002,86	1.193,82
MÉDIA MAIS UM DESVIO PADRÃO	4.893,91	2.239,53	1.509,16
NÚMERO DE ELEMENTOS	6	5	4
CAMPO DE ARBÍTRIO INFERIOR		1.378,01	1.148,77
CAMPO DE ARBÍTRIO SUPERIOR		1.864,37	1.554,21

Figure 50. Average prices for Farmlands with Medium/Large Technological Level within MRT 05.

Source: RAMT/PI (2020).

- (3) By consulting the land values for the whole project area, we then defined the leakage buffer according to the highest vulnerability to deforestation, considering: The MRT 03 region having a higher risk due to the higher aggregated land price of its territories; and the MRT 05 region having a lower risk due to the lower aggregated price of its territories. The initial buffer was set at 3 km, which is considered a reasonable distance in respect to the municipal boundaries. Thus, different proportions of buffer was applied according to the risk of each region: for project areas within MRT 05, a 3 km leakage buffer was adopted; for project areas within MRT 03, an 11 km leakage buffer was adopted (3.7 times greater, similar to the land value ratio compared to MRT 03). On this way, a leakage buffer become coherent with the deforestation risk, which was finally delineated within the Reference Region.

- (4) The final buffer configuration was as follows:

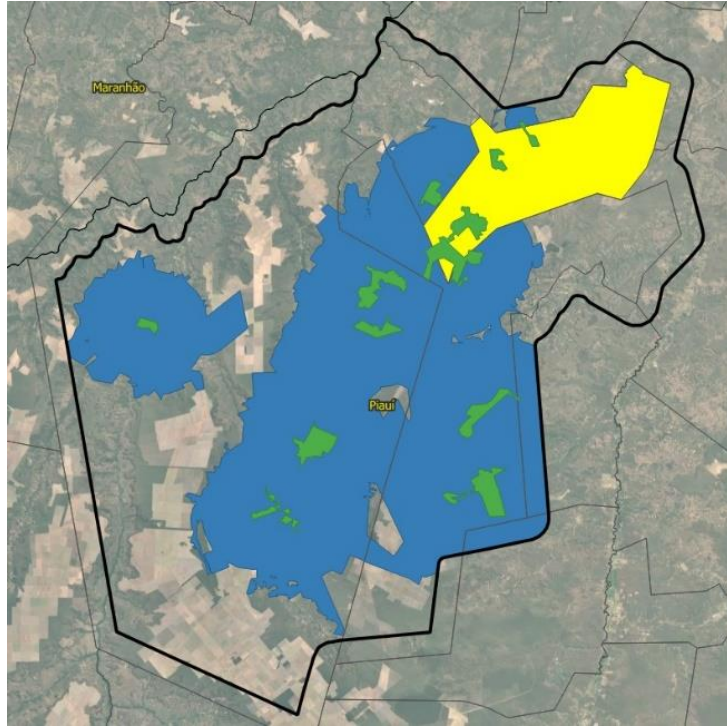


Figure 51. Buffers starting from the Project Area borders [blue (MRT 03): 11km; yellow (MRT 05): 3km].

Source: the author.

The study of Opportunity Costs will complement this study with arguments regarding profitability of the commercial activities established in these areas.

Agents and Drivers considered for the Leakage analysis

The analysis performed on the land-use/land-cover changes during the historical reference period allowed determining that illegal / landgrabbers, followed by small (including settlements) and medium/larger farmers are the main deforestation agents in the RR and the land economic values, geomorphological conditions (e.g. relief), and agribusiness expansion in the region are the main drivers of deforestation.

These are the same agents and drivers considered for the baseline construction, better detailed in Section 3.4 – Baseline Scenario. As they influence directly the RR, where the Leakage Belt is located in, this is pertinent to consider those for the LK boundary construction.

Opportunity Cost Analysis (option I – chosen)

The Opportunity Cost analysis was performed following the requirements of the VM0015 – item 5.1.3.1.

Profitability was determined using the Equation below:

$$PPx_l = S\$x - PCx_l - \sum_{v=1}^v (TDv * TCv)$$

Where:

PPxl	Potential profitability of product Px at location l (pixel or polygon);
S\$x	Selling price of product Px; \$/t
PCxi	Average in situ production costs for one ton of product Px in stratum i;
TCv	Average transport cost per kilometer for one ton of product Px on road of type v;
TDv	Transport distance on land, river or road of type v.

The formula above is used to calculate the potential profitability of a given product in a given location and the borders of the leakage belt correspond to the area where the profitability of at least one product is equal to or above 1.

Therefore, to determine the potential profitability, the above parameters were identified considering current conditions and potential markets for each activity and product. This assessment allowed concluding that farming activities are potentially profitable, especially to South-western of the Piauí State, along important routes for farming production and products' flow, as shown previously in Section 1.14.

As defined in other sections of this PD, the classes used for deforestation projection is a mix of Pasture and Agriculture lands, which includes a mixed use of land. This classification is further detailed in Section 3.4 of this PD, which provides information on land use in the post-deforestation scenario. Given that these deforestation agents in the region share common activities – primarily using their lands for large-scale agricultural and livestock production – an analysis was conducted based on external literature and profitability calculations for these main activities. These results apply to all areas where deforestation agents have similar motivations. **According to the methodology, at least one product must have a Profit Potential (PP) > 1**, and we have chosen soybean production and livestock as primary products.

1. **Analyzed Product:** Soybean production⁸¹.
2. **Key variables:**
 - **S\$x:** Average selling price per tonne (detailed below).
 - **SPxl:** Main selling point: defined port 'Cabo de Santo Agostinho' (PE)

⁸¹ Sources used for Profit analysis: FRETE ESALQ (Estimative Freight from Tocantins State): <https://sifreca.esalq.usp.br/mercado/soja>; AGROLINK (soy prices in Piauí): <https://www.agrolink.com.br/cotacoes/historico/pi/soja-em-grao-sc-60kg>; AGROLINK (soy costs – estimative for Mato Grosso State): <https://www.agrolink.com.br/noticias/custos-de-producao-de-soja-registram-reducao-em-marco-490260.html>

- **PCxi**: Average production costs per tonne (detailed below).
- **TCv**: Average transport costs.

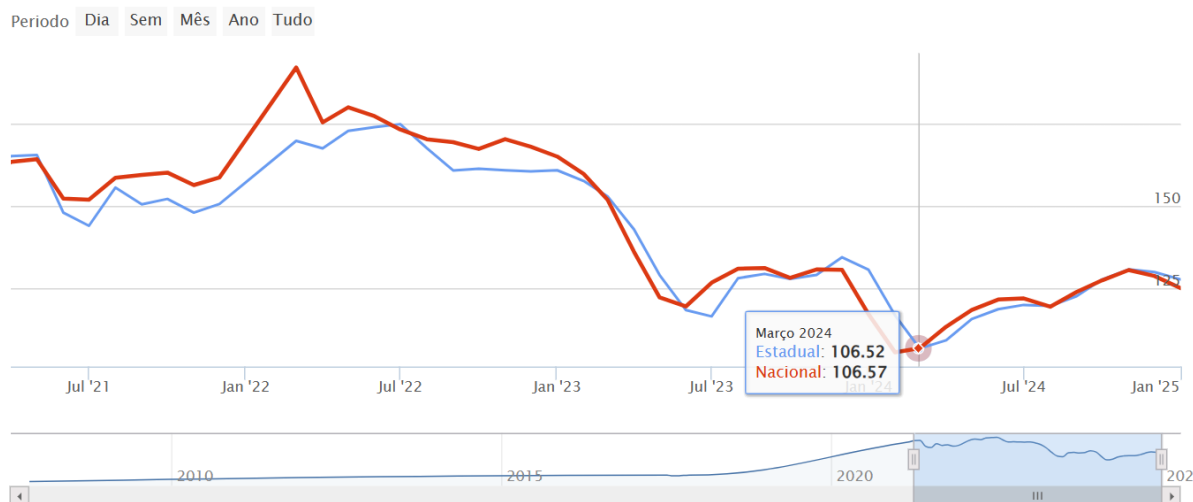
$$\begin{array}{c}
 \bullet \text{ } PP_{xl} \\
 \boxed{\text{Profit Potential}}
 \end{array}
 =
 \begin{array}{c}
 \$\$x \\
 \boxed{\text{Revenues} \\ \text{R\$ / ton}}
 \end{array}
 -
 \begin{array}{c}
 PCxi \\
 \boxed{\text{Costs} \\ \text{R\$ / ton}}
 \end{array}
 -
 \begin{array}{c}
 \Sigma (TDv) * TCv \\
 \boxed{\text{Freight}}
 \end{array}$$

$$PP_{soy} = 1.775 \text{ R\$/ton} - 1.158 \text{ R\$/ton} - 390 \text{ R\$/ton} = 227 \text{ R\$/ton (profit)} \Rightarrow PP > 1$$

The rationale and References used for these parameters and calculations are presented below:

Average Selling Price for Soybeans in Piauí

The chart below shows that the average selling price for Soybeans varies along time, but usually keep values around 100BRL/sc (sc is 'saca in Portuguese, equivalent to 'bag', in English). For this project, the reference of March, 2024 was used (price: BRL106,52/sc).



* sc unit ("saca", PT) = bag unit (EN) = 16.67 ton

* 1 sc = 60kg

* BRL 106.52/sc = BRL 106.52/60kg = BRL 1.77/kg = **BRL 1,775/ton**

Production Costs

According to AGROLINK (2024), the Average total cost (projection) for Soy production in the crop 2023/2024 was: BRL 4,098.87/ha.

$$\text{BRL } 4,098.87/\text{ha} = \text{BRL } 4,098.87/(3,540\text{kg}^*) = \text{BRL } 1.158/\text{kg} = \text{BRL } 1,158/\text{ton}$$

* Productivity (in kg/ha) in 2023/2024, according to CONAB⁸²

Freight Price

The freight price calculated by ESALQ is one of the most accurate data that can be found for the region, since it is an official data and updated constantly through actual database from all soy transportation in Brazil. The considered Route was 'Porto Nacional (TO) -> Cabo de Santo Agostinho (PE)', which represents a similar logistic flow for grains in the region (Porto Nacional, in Tocantins, is about 900km away from the RR). *This is a conservative choice, since the distance and prices from South-western Piauí (RR) to the coast are lower.*

Período: 02/12/2024 - 05/01/2025 (Outras rotas)					
Origem	UF	Destino	UF	Frete (R\$/t)	Momento (R\$/t.km)
Amambaí	MS	Paranaguá	PR	175,00	0,1900
Cascavel	PR	Imbituba	SC	175,00	0,2020
Guarapuava	PR	Paranaguá	PR	83,00	0,2325
Londrina	PR	Paranaguá	PR	102,50	0,2131
Paragominas	PA	Santos	SP	450,00	0,1686
Porto Nacional	TO	Cabo de Santo Agostinho	PE	390,00	0,1840
Rondonópolis	MT	Uberaba	MG	160,00	0,1688
São Gabriel do Oeste	MS	Três Lagoas	MS	79,50	0,1751
Sapezal	MT	Rondonópolis	MT	137,50	0,2037
Tapejara	RS	Canoas	RS	81,24	0,2328

*Variação (%) em relação à semana anterior Fonte: ESALQ-LOG Notas:

Figure 52. Defined freight, in R\$/ton.

Source: the author.

According to the VM0015 methodology, if at least one product achieves a PP > 1, it is already valid and can be considered for the conclusion of this boundary.

As it was demonstrated that at least one product has a Profit Potential (PP) substantially > 1, the LK could be delineated for all farming areas presented within the RR. Then this study was crossed with the most valuable territory and roads immediately adjacent to the project area where baseline activities may be displaced due to the project activities implemented in the project area. This considers all the variables mentioned previously:

- The same agents of deforestation of the Baseline construction, focusing on higher value's lands of medium-large proportion.

⁸² PIAUÍ NEGÓCIOS, 2025 – "Com maior área plantada, produção de soja no Piauí deve subir na safra 2024/25": <https://pinegocios.com.br/noticia/2635/com-maior-area-plantada-producao-de-soja-no-piaui-deve-subir-na-safra-2024/25>

- The same drivers of deforestation of the Baseline construction, focusing on the lands' economic value, geomorphological conditions, and expansion of local Agriculture.
- Analysis of historical and current land use in the RR and State of Piauí, evidencing that more than 80% of deforested areas are being occupied by farmers of Agriculture and Pasture activities.
- Opportunity Cost analysis, proving that at least one main product of these activities is profitable locally.

The result was the spatial configuration as demonstrated on the following images:

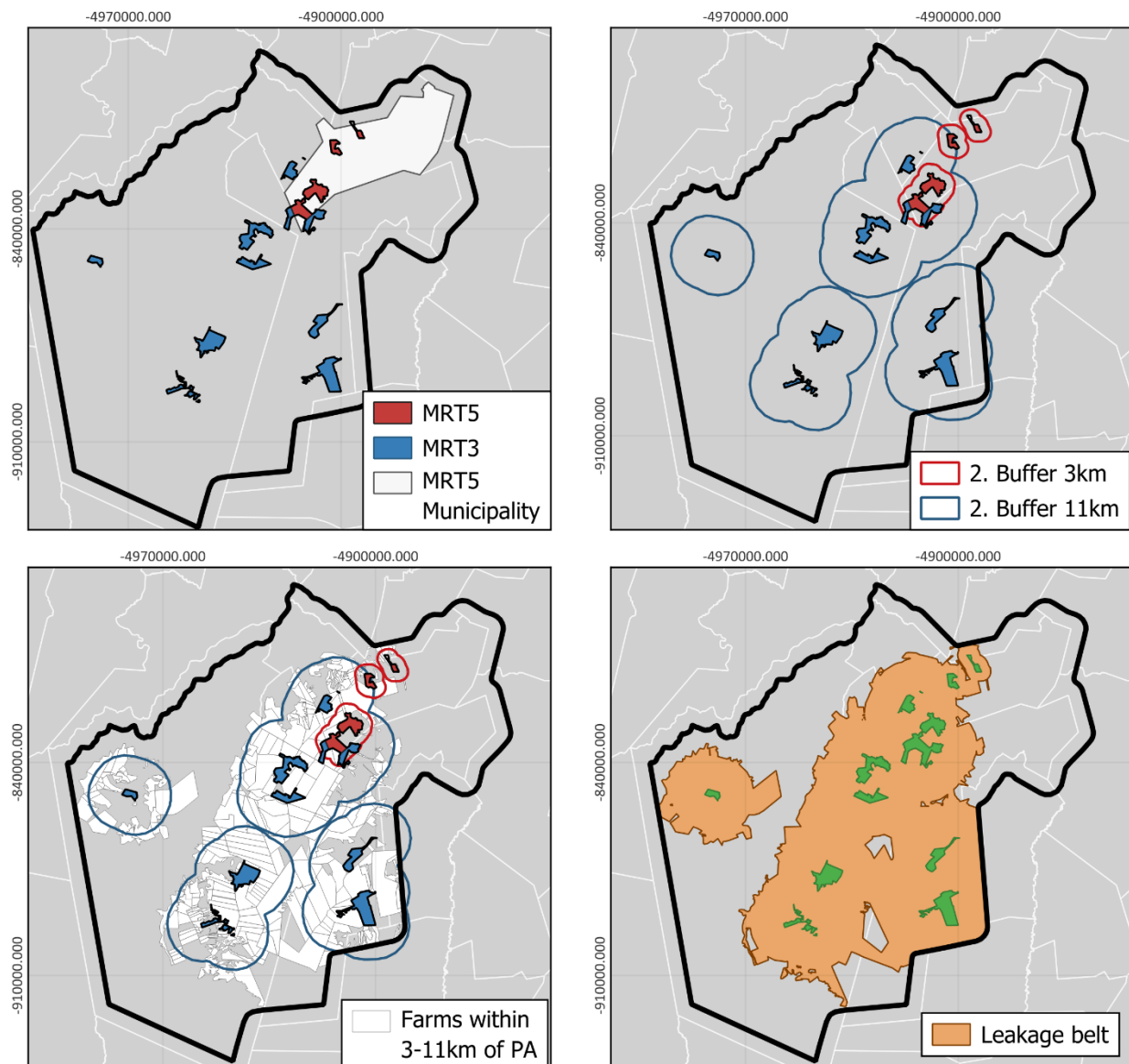


Figure 53. Leakage belt; attractiveness based on Land Economic Value and buffers of influence.

Source: the author.

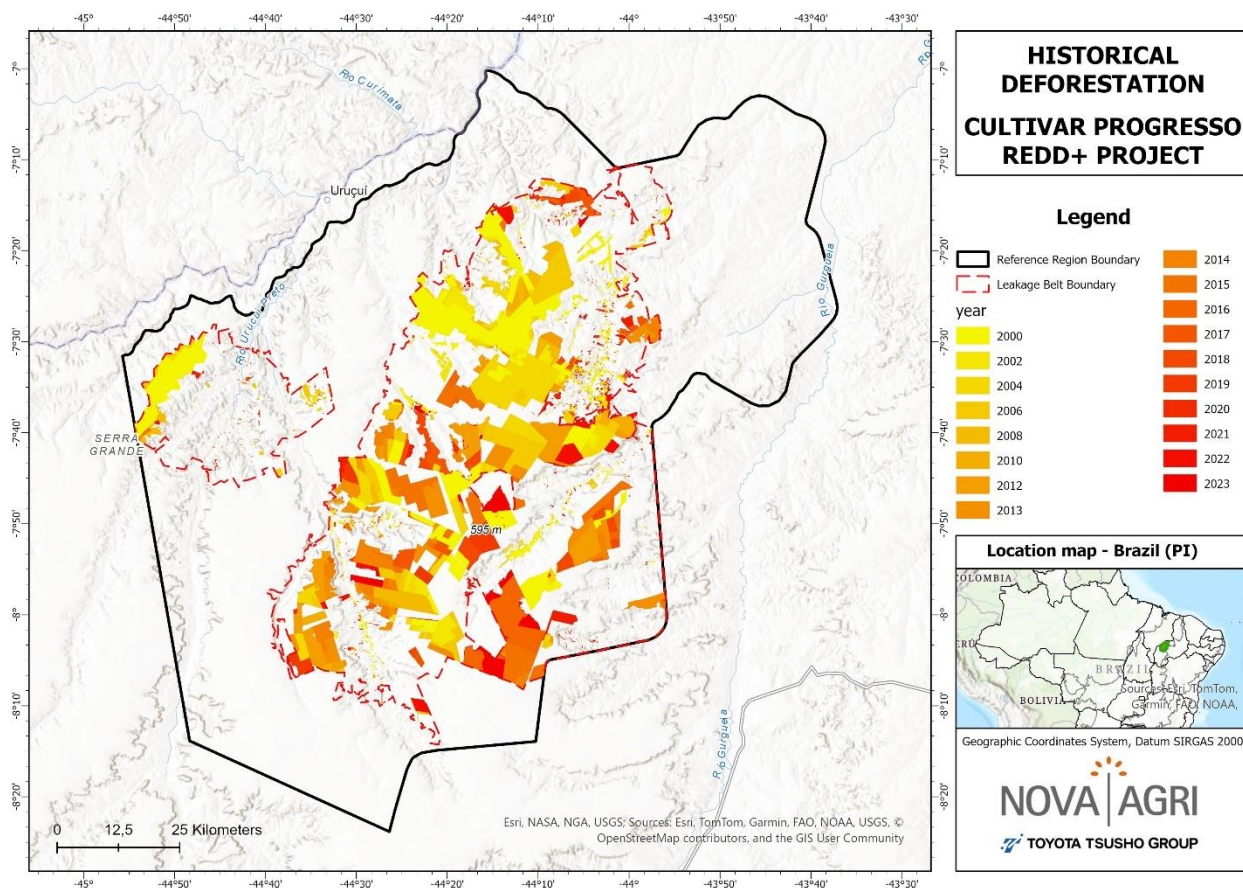


Figure 54. Leakage belt and previously deforested areas.

Source: PRODES, the author.

3.3.4 Leakage Management Zone (LMZ)

The Leakage Management Zone (LMZ) are places where the Project intends to exert the influence of its activities to reduce the risks of deforestation. These activities must include the agents of deforestation and involve seeking new sources of income which contribute to forest conservation.

Before defining the LMZ (Leakage Management Zone), it was necessary to establish a leakage management plan, which is presented in Section 1.19.1 of this PD. In this plan, we developed and outlined the rationale behind the selection of (i) the agent, (ii) the arrangement of activities, (iii) the tools required to combat risks of deforestation leakage, and (iv) the monitoring metrics to assess potential results within the LMZ, according to the requirements of the VM0015 methodology and VCS Standard 4.7.

The physical boundaries of the LMZ are based on the limits of a farm with characteristics similar to the historical patterns of deforestation in the Reference Region, considering it is a flat area, managed by a farmer engaged in grain and cattle activities, aligning with the historical context of agricultural expansion in the Brazilian Cerrado. The farm is named 'Fazenda Boa Vista,' and the

plot designated for the LMZ is called 'Plot 1'. This plot is currently 100% non-vegetated area and used for commercial cultivation of corn and brachiaria grass.

The area of this plot is approximately 225 hectares, and through the project, it will undergo monitoring and optimization of crop efficiency through the application of biological inputs. These inputs will provide several beneficial effects, such as improved soil balance, increased nutrient availability for plants, enhanced organic matter content, and extended productive lifespan of planting areas. The monitoring will be carried out by the project proponent in collaboration with the landowner from Grupo Progreso.

The LMZ property owner maintains an open dialogue and close relationship with the landowner of 'GRUPO PROGRESO', a key factor enabling activities within the LMZ to proceed with the necessary permissions and consent from both the property owner and employees.

Below is a map the LMZ for the CULTIVAR PROGRESO REDD+ PROJECT.

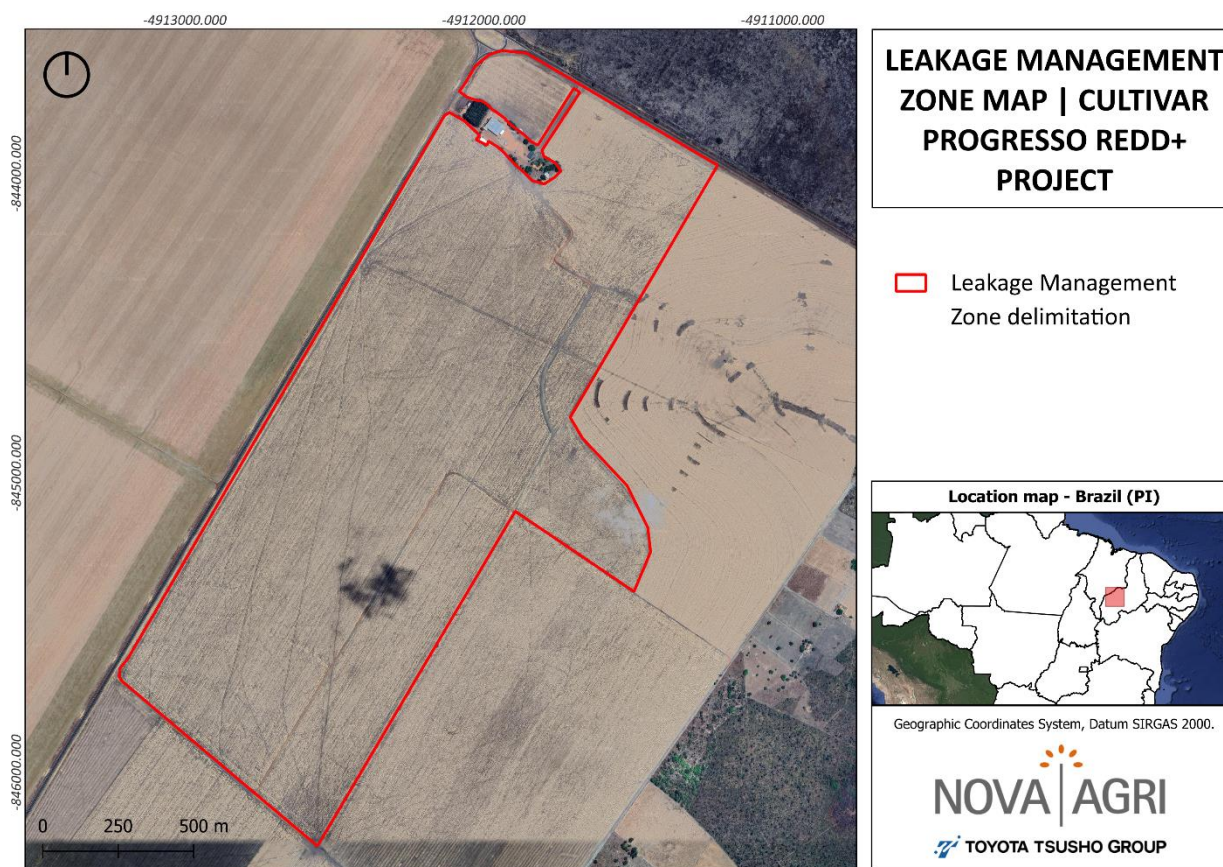


Figure 55. Leakage Management Zone within the Project Area.

Source: the author.

3.3.5 Forest

The forest boundary for the PA and LK has been delineated following a dynamic and consistent forest definition (see Section 3.2 – Applicability of Methodology) applied over the most recent satellite image (2023) near to the project start date. As evidenced on Figure 56, the Forest Boundary fully matches with the Project Area’s boundary. This approach ensures alignment with the VM0015.

The boundary was established using a forest definition suitable for measuring deforestation during the project crediting period. As the minimum mapping unit (MMU) required is of one hectare, the high-definition Land Use and Land Cover (LULC) data from MAPBIOMAS was used, since its methodology involves remote sensing imagery, in line with established guidelines: selecting of cloud-free pixels from Landsat satellite images, incorporating over 105 spectral layers per pixel – MAPBIOMAS⁸³ uses over 9 billion pixels of 30 x 30 meters to cover the entire country. These mosaics, processed using the random forest classification algorithm on the Google Earth Engine, produced detailed maps of land use and cover classes, ensuring high accuracy in delineating the forest boundary.

This map serves as the basis for tracking gross deforestation and were complemented with baseline projections for future years. The analysis will be conducted annually throughout the entire crediting period to ensure transparency in demonstrating the dynamics and any potential changes in the forest area within the Project Area (PA) and the leakage belt (LK). By systematically updating the forest boundary maps each year and presenting them to the VVB through the MRV process. On this way, the project will provide an accurate representation of deforestation trends and forest cover changes over time.

⁸³ MAPBIOMAS – ‘Método Cobertura e Uso’: https://brasil.mapbiomas.org/metodo_cobertura_e_uso/

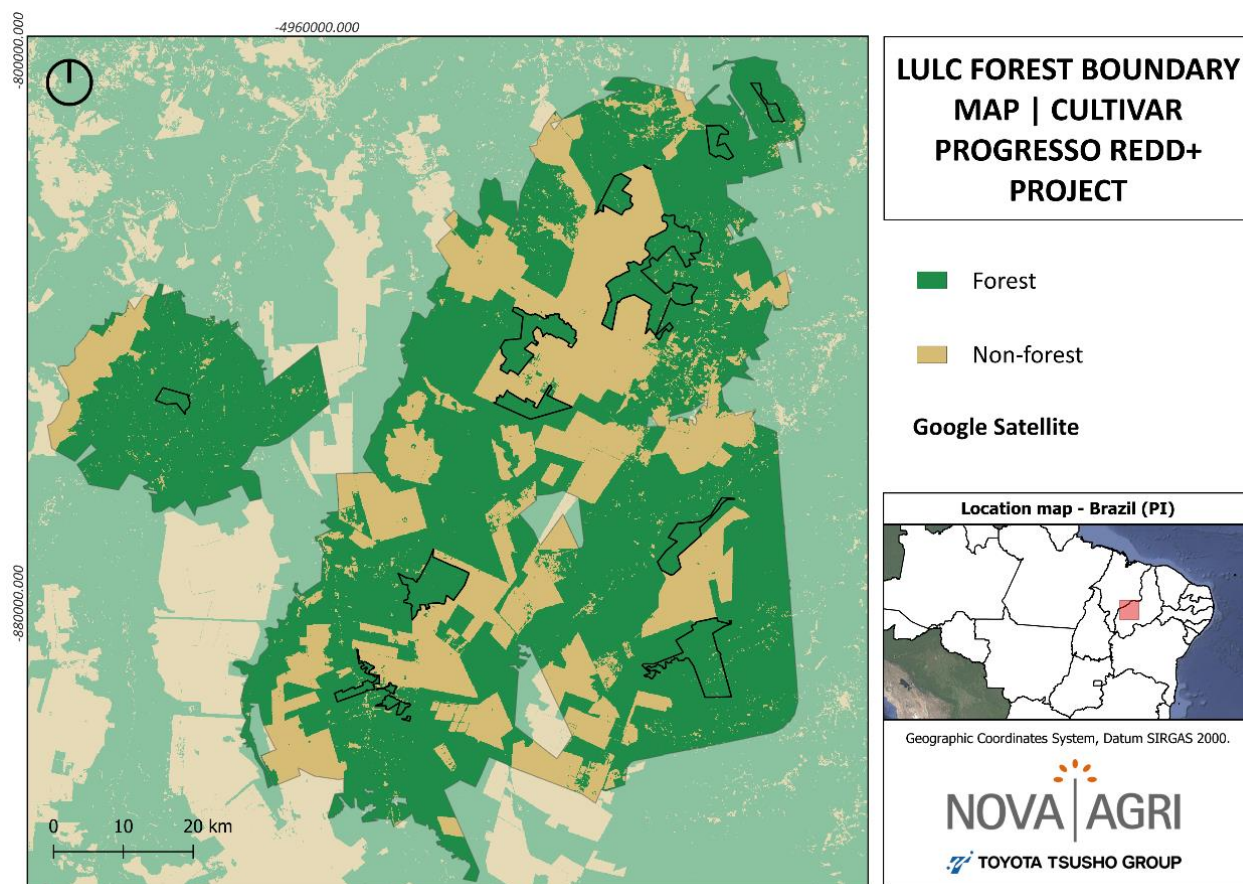


Figure 56. Forest boundary of the Project (2023 scenario).

Source: MapBiomass, the author.

Carbon Pools

The applied Methodology considers the six carbon pools listed in Table 15 below. Their inclusion or exclusion within the boundary of the proposed AUD project activity, as well as the respective justification/explanation, are described in the Table below.

Table 17. Carbon Pool Inclusion/Exclusion

Carbon Pool	Included?	Justification/Explanation of choice
Above-ground	TREE: Included	Carbon stock change in this pool is always significant
	Non-Tree: Excluded	No existence of perennial crops as final class
Below-ground	Included	Stock change in this pool is significant

Carbon Pool	Included?	Justification/Explanation of choice
Dead wood	Excluded	Excluded for simplification. This exclusion is conservative.
Harvested wood products+	Excluded	Not significant
Litter	Excluded	Not to be measured according to VCS (AFOLU) Requirements, v3.6.
Soil organic carbon+	Excluded	Excluded when the ground cover is grassland in the baseline scenario, according to VCS (AFOLU) Requirements, v3.6.

In accordance with the Methodology, approximately 1/10 of the carbon stock in the below-ground pool of the initial “forest” class will be released in a ten-year interval. This is further discussed in Section 3.4 – baseline emissions.

Table 18. Sources and GHG included or excluded within the boundary of the proposed AUD project activity (Table 4 of Methodology VM0015)

Source		Gas	Included?	Justification/Explanation
Baseline	BIOMASS BURNING	CO ₂	Excluded	Counted as carbon stock change
		CH ₄	Excluded	Excluded as recommended by the applied methodology. This exclusion is conservative and does not lead to a significant over-estimation of the net anthropogenic GHG emission reductions of the AUD project activity.
		N ₂ O	Excluded	Considered insignificant according to VCS (AFOLU) Requirements, v3.6.
		Other	Excluded	No other gases were not considered in this Scope project
	LIVESTOCK EMISSIONS	CO ₂	Excluded	Not a significant source
		CH ₄	Excluded	Not a significant source

Source		Gas	Included?	Justification/Explanation
Project		N ₂ O	Excluded	Not a significant source
		Other	Excluded	No other gases were not considered in this Scope project
	BIOMASS BURNING	CO ₂	Excluded	Counted as carbon stock change
		CH ₄	Excluded	Excluded as recommended by the applied methodology. This exclusion is conservative and does not lead to a significant over-estimation of the net anthropogenic GHG emission reductions of the AUD project activity.
		N ₂ O	Excluded	Considered insignificant according to VCS (AFOLU) Requirements, v3.6.
		Other	Excluded	No other gases were not considered in this Scope project
	LIVESTOCK EMISSIONS	CO ₂	Excluded	Not a significant source
		CH ₄	Excluded	Not a significant source
		N ₂ O	Excluded	Not a significant source
		Other	Excluded	No other gases were not considered in this Scope project

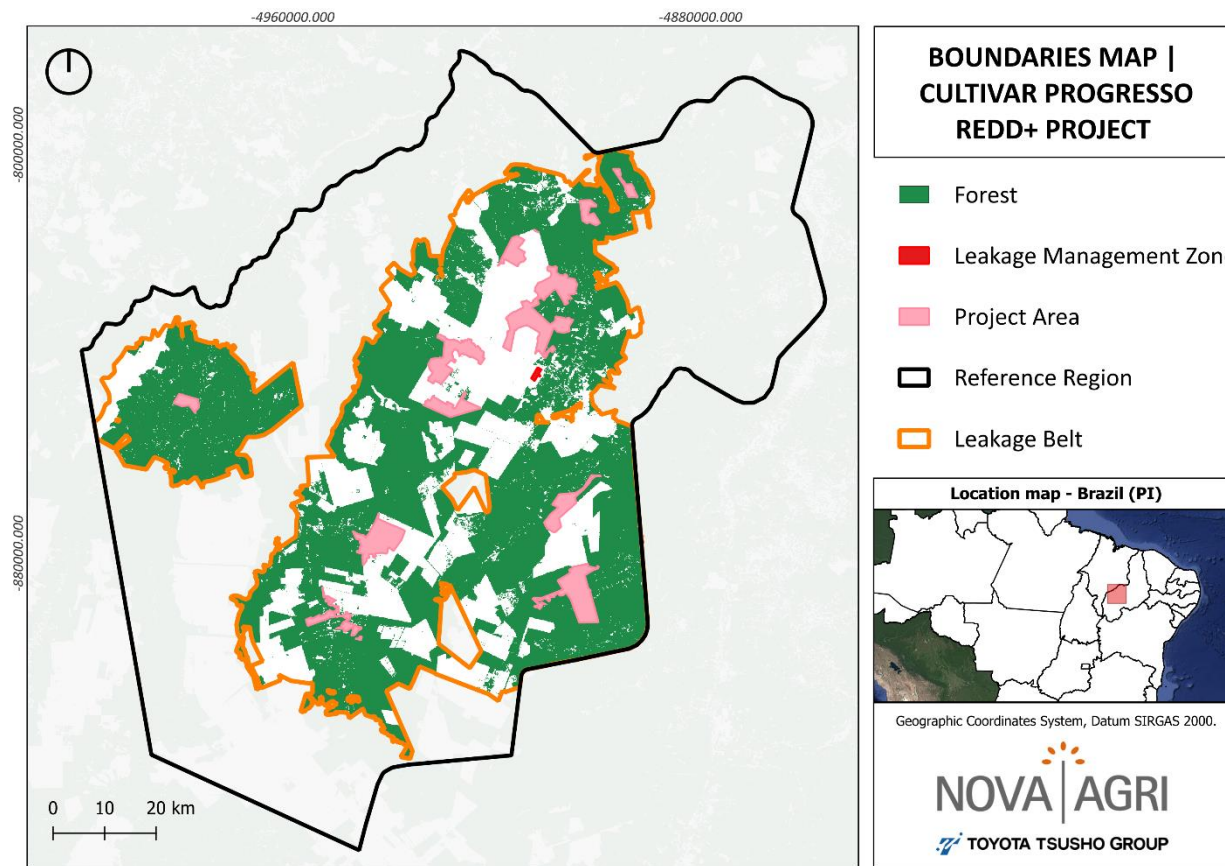


Figure 57. All project boundaries.

Source: the author.

3.4 Baseline Scenario

In the baseline scenario, forest is expected to be converted to non-forest by the agents of deforestation acting in the reference region, project area and leakage belt, as described below. Therefore, the project falls into the AFOLU-REDD category, specifically: Avoided unplanned deforestation (AUD). The revenue from the present REDD project is essential to maintain this area as a standing forest, as described under Additionality (section 3.5), as well as to carry out the present project's leakage management activities.

Degradation was not considered in the present REDD project, in accordance with methodology requirements, which define “forest” and “non-forest” as the minimum land-use and land-cover classes.

Contextualization of LULC changes in the Brazilian Amazon macro territory

An analysis conducted by Mapbiomas⁸⁴ over the historical period between 1985 and 2021 in the Brazilian Legal Amazon territory reveals notable trends. The extent of forested areas decreased from 473 million hectares to 411 million hectares, signifying a substantial decline of 12%. Similarly, savanna formations experienced a reduction of 13 million hectares, reflecting a significant decline of 26%.

Turning to the realm of human-induced land use changes, there was a remarkable increase from 36.4 million hectares to 98.4 million hectares. This highlights a marked shift in land utilization dynamics. Of particular significance is the expansion of areas dedicated to agriculture, which exhibited a momentous expansion, increasing by 15 million hectares, an astonishing surge of 1,024% during this analysis period.

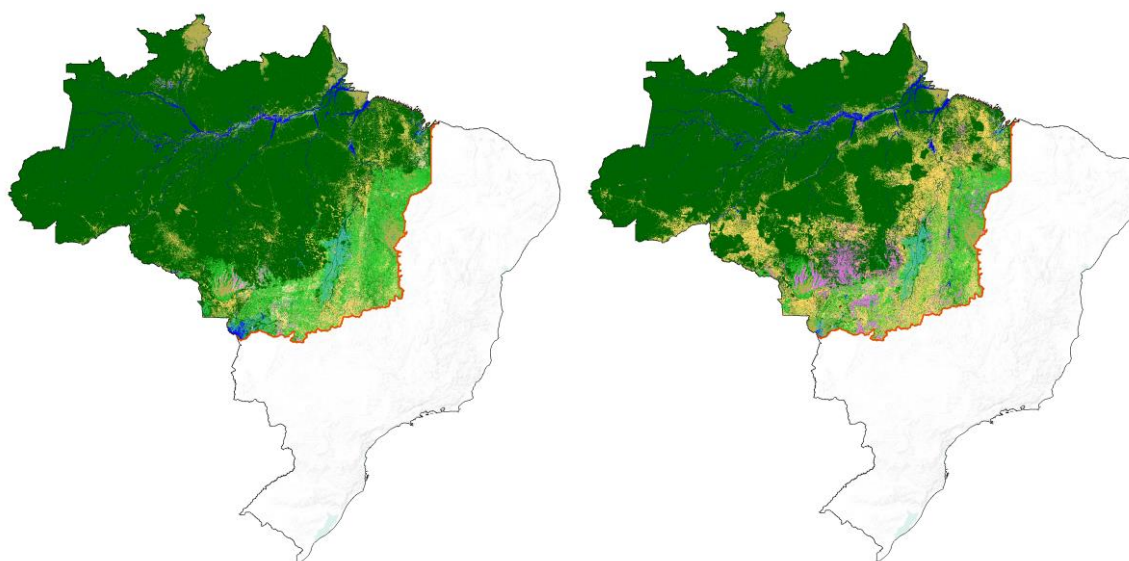


Figure 58. 1958 to 2021 LULC comparison in the Amazon national territory.

Source: MapBiomas, 2022.

Historical baseline for the reference region

For the mapping of the changes in the classes of use and soil cover, data from the PRODES Digital program (INPE, 2023) were used in vector format (shapefile) with spatial resolution of 30 meters. A total of 2 Landsat satellite images were used to map forest, hydrography, and anthropogenic vegetation (deforestation) (Table 17).

⁸⁴ Mapbiomas Amazonía (2022). Colección 4.0 de mapas anuales de cobertura y uso del suelo en la Amazonía. Accedido en 2023, através del enlace: https://s3.amazonaws.com/amazonia.mapbiomas.org/destaques/AmazoniaC4_FactSheet_Regional.pdf

Table 19. Data used to identify and map historical LU/LC change analysis in CULTIVAR PROGRESSO REDD+ Project

Vector (Satellite)	Sensor	Resolution		Coverage	Acquisition Date	Scene or Point Identifier	
		Spatial	Spectral	(km ²)	(DD/MM/AAA)	Path / Latitude	Row / Longitude
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	06/07/2013	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	26/08/2014	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	13/08/2015	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	30/07/2016	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	17/07/2017	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	04/07/2018	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	10/08/2020	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	13/08/2021	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	15/07/2022	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	18/07/2023	220/65	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	06/07/2013	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	26/08/2014	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	13/08/2015	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	30/07/2016	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	17/07/2017	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	04/07/2018	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	10/08/2020	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	13/08/2021	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	15/07/2022	220/66	
Satellite	Landsat8	30 x 30 m	0,45 – 2,35 µm	185 x 185 km	18/07/2023	220/66	

Deforestation data provided by PRODES were used to analyze the historical baseline for the reference region. The main activities carried out by the PRODES Project to monitor the forest cover loss of the Brazilian Cerrado Biomes are detailed below.

Pre-processing

The procedures of imagery preprocessing performed by the PRODES Project are constituted in the following steps (Almeida, C. A. et al., 2021⁸⁵):

⁸⁵ Almeida, C. A. et al. Metodologia para monitoramento da floresta usada nos projetos PRODES e DETER. São José dos Campos: INPE, versão: 2021-01-26. Available at <<http://mtc-m21c.sid.inpe.br/col/sid.inpe.br/mtc-m21c/2021/01.25.18.31/doc/publicacao.pdf>>

- Selection of images with lower cloud cover and acquisition date closer to dry season in the region and with adequate radiometric quality.
- Once selected, images are highlighted to highlight areas of clear-cutting by applying a contrast enhancement operation.

Interpretation and classification

The identification of deforestation is made by photointerpretation of images, performed by trained experts, that delimit the deforestation polygons directly on the computer screen. These experts identify the pattern of change of forest cover for shallow felling based on the main elements that are observable in images: hue, color, shape, texture and context.

These elements are analyzed comparatively considering images of the subsequent year, in addition to the pictures of the historical series. This procedure is performed for all images selected for PRODES year mapping.

An audit of all the scenes and 100% of the mapped area is performed by INPE's experts, in order to avoid the generation of false positives of deforestation and to reduce omissions.

Post-processing

According to VM0015, the post-processing step includes the use of non-spectral information for the stratification of the carbon density of the land cover classes. This information was generated implicitly during the next steps.

Map accuracy assessment

PRODES is validated by the Temporal Visual Inspection (TVI) tool, an open-source tool that simplifies the visualization of points in Landsat images for long time series and allows to perform analysis quickly, practically and with simultaneous supervision (Parente et al., 2021⁸⁶; Nogueira et al., 2017⁸⁷).

The accuracy assessment (global and by class) of the mapping of PRODES data is done with the random sampling procedure stratified by class that considers the classes of natural vegetation and deforestation of each year.

The sample size (n) is defined by the equation:

⁸⁶ Parente, L., Nogueira, S., Baumann, L., Almeida, C., Maurano, L., Affonso, A. G., & Ferreira, L.. Quality assessment of the PRODES Cerrado deforestation data. *Remote Sensing Applications: Society and Environment*, 21, 100444. 2021.

⁸⁷ Nogueira, S. H. M, Parente, L. L, Ferreira, L.G. Temporal visual inspection: uma ferramenta destinada à inspeção visual de pontos em séries históricas de imagens de sensoriamento remoto. In: *Anais Do XXVII Congresso Brasileiro de Cartografia e XXVI Exposita*. Rio de Janeiro, Brazil, pp. 624–628. 2017.

$$n = \left(\frac{\sum_{i=1}^H W_i * S_i}{s(\hat{\theta})} \right)^2$$

Where:

W_i is the proportion of stratum i in the target population.

$$S_i = \sqrt{U_i * (1 - U_i)};$$

U_i is the expected value of the population proportion, and the values for distant forest (Forest > 5km) is 0.9, for nearby forest (Forest 5km) 0.7 and deforestation 0.7

$s(\hat{\theta})$ is the standard error expected from sampling and was considered equal to 0.01.

Power allocation was used to determine the number of samples per stratum (n_i):

$$n_i = n * \frac{(t_{xi})^a * \sqrt{(1 - U_i)/U_i}}{\sum_{i=1}^H (t_{xi})^a * \sqrt{(1 - U_i)/U_i}}$$

Where:

$$t_{xi} = U_i * N_i$$

$a = 1/3$ (Särndal et al. (1992) recommends the use of $a = 1/2$ or $1/3$).

Based on the confusion matrix constructed from the cross tabulation of the sampling frequencies of the mapped and reference classes it was possible to obtain the values of global accuracy, of the producer and the user (Stehman, 2012⁸⁸; Olofson et al. 2014⁸⁹).

The evaluation performed by INPE showed that the overall accuracy of PRODES CERRADO 2022⁹⁰ in the Cerrado was 94.3%.

Table 20. Overall Accuracy of the Historical Baseline's Data

PRODES 2022

⁸⁸ Stehman, S. v. (2012). Impact of sample size allocation when using stratified random sampling to estimate accuracy and area of land-cover change. Remote Sensing Letters,3(2), 111–120.

⁸⁹ Olofsson, P., Foody, G. M., Herold, M., Stehman, S. v., Woodcock, C. E., Wulder, M. A. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment, 148, 42–57.

⁹⁰ Avaliação da acurácia do mapeamento do PRODES 2022. Available at <http://terrabrasilis.dpi.inpe.br/download/terrabrasilis/technicalnotes/nota_tecnica_avaliacao_PRODES2022.pdf>

Global Accuracy	Index (%)	Confidence Interval (CI±)
	94,3	1,4

Global Accuracy	PRODES 2022			
	Natural		Suppression	
	Index (%)	CI±	Index (%)	CI±
	94,4	1,4	91,9	3
	99,9	0,03	15,3	3,3

Definition of classes of Land-Use and Land-Cover

The classes of LU/LC were defined as “forest” and “non forest” in accordance with the procedures described above. These classes are the minimum classes to be considered in the present REDD project as stipulated by the Methodology. As such, degradation was not a factor.

Stratification was not carried out in either class, and therefore the categories “forest” and “non forest” have homogenous carbon stocks.

The following are the description the classes used in the Project and its area at the beginning of the historical period (2000):

The following are the description the classes used in the Project and its area at the beginning of the historical period (2000):

- **Forest (1,301,273 ha):** area of forest remnant belonging to different phytophysiognomies of the Contact Savannah-Seasonal Forest;
- **Hydrography (8 ha):** water bodies (rivers, lakes, streams, among others);
- **Anthropogenic Vegetation in equilibrium and other uses (465,111 ha):** area where there was forest, but that was removed through the clear-cutting process (totally removal of forest cover).

Analysis of the historical Land-Use and Land-Cover change

According to PRODES data, there is a large proportion of deforested areas in the 2013 land-use and land-cover data.

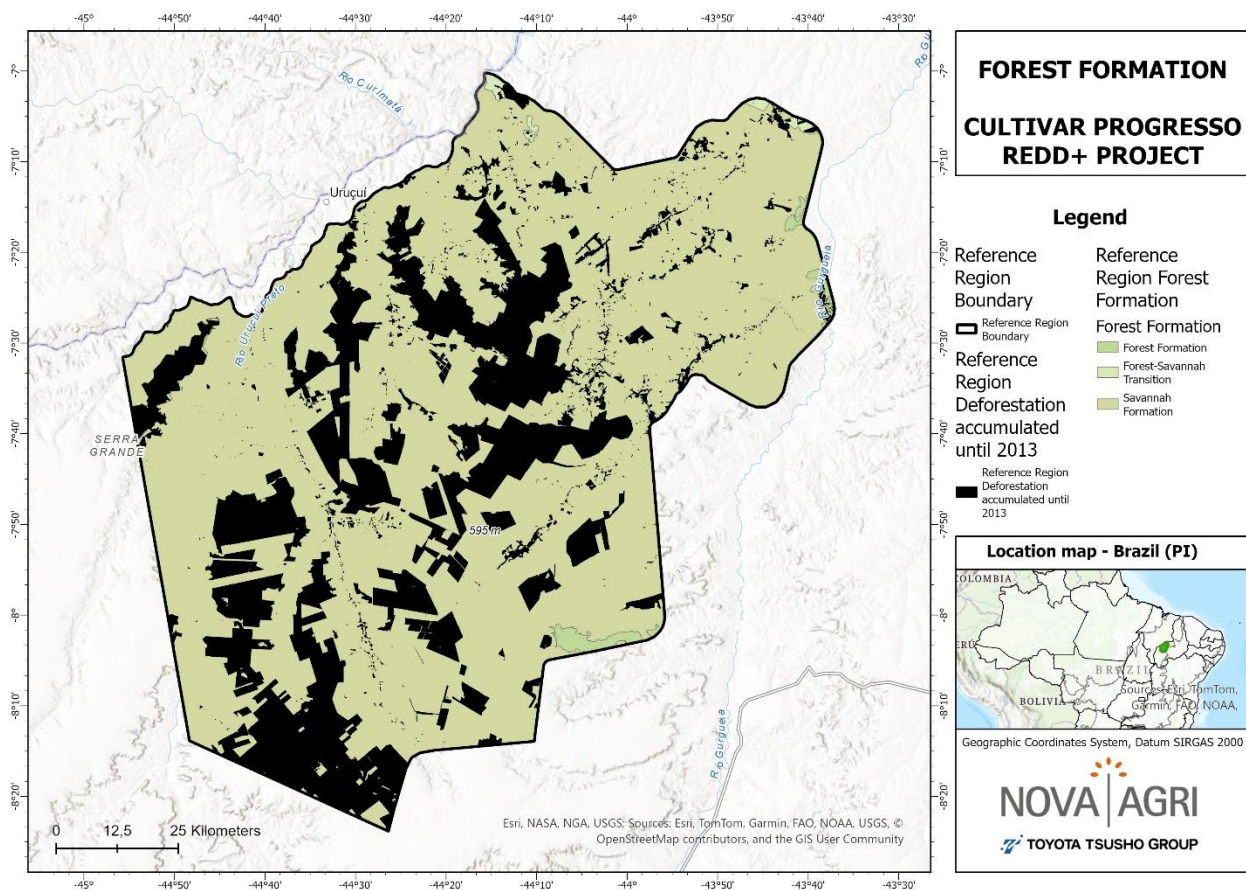


Figure 59. Map: Remaining forest areas and consolidated deforested areas in the Reference Region.

Source: PRODES, the author.

Table 21. Annual deforestation, cumulative deforestation and R in the reference region during historical reference period.

YEAR	Dominant Vegetation in Ecological Tension (ha)	Annual deforestation RR (ha)	Cumulative deforestation (ha)	R: annual rate of forest cover change
RR forest area	1.301.273			
up to 2013	966.743		334.531	25,71%
2014	947.776	18.967	353.498	2,00%
2015	924.984	22.791	376.289	2,46%
2016	905.820	19.164	395.453	2,12%
2017	889.730	16.090	411.543	1,81%
2018	877.534	12.196	423.740	1,39%

2019	867.951	9.583	433.322	1,10%
2020	862.924	5.027	438.349	0,58%
2021	856.461	6.464	444.813	0,75%
2022	842.809	13.652	458.465	1,62%
2023	836.163	6.646	465.111	0,79%

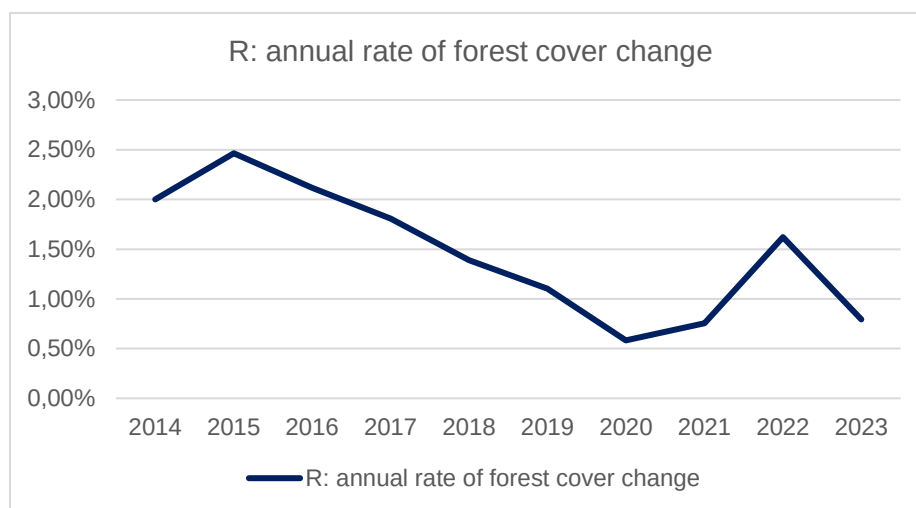


Figure 60. "r" – annual rate of forest cover change in the reference region for 2014 – 2023.

Source: the author

The annual deforestation of years analyzed within the historical reference period are also represented in the deforestation map below.

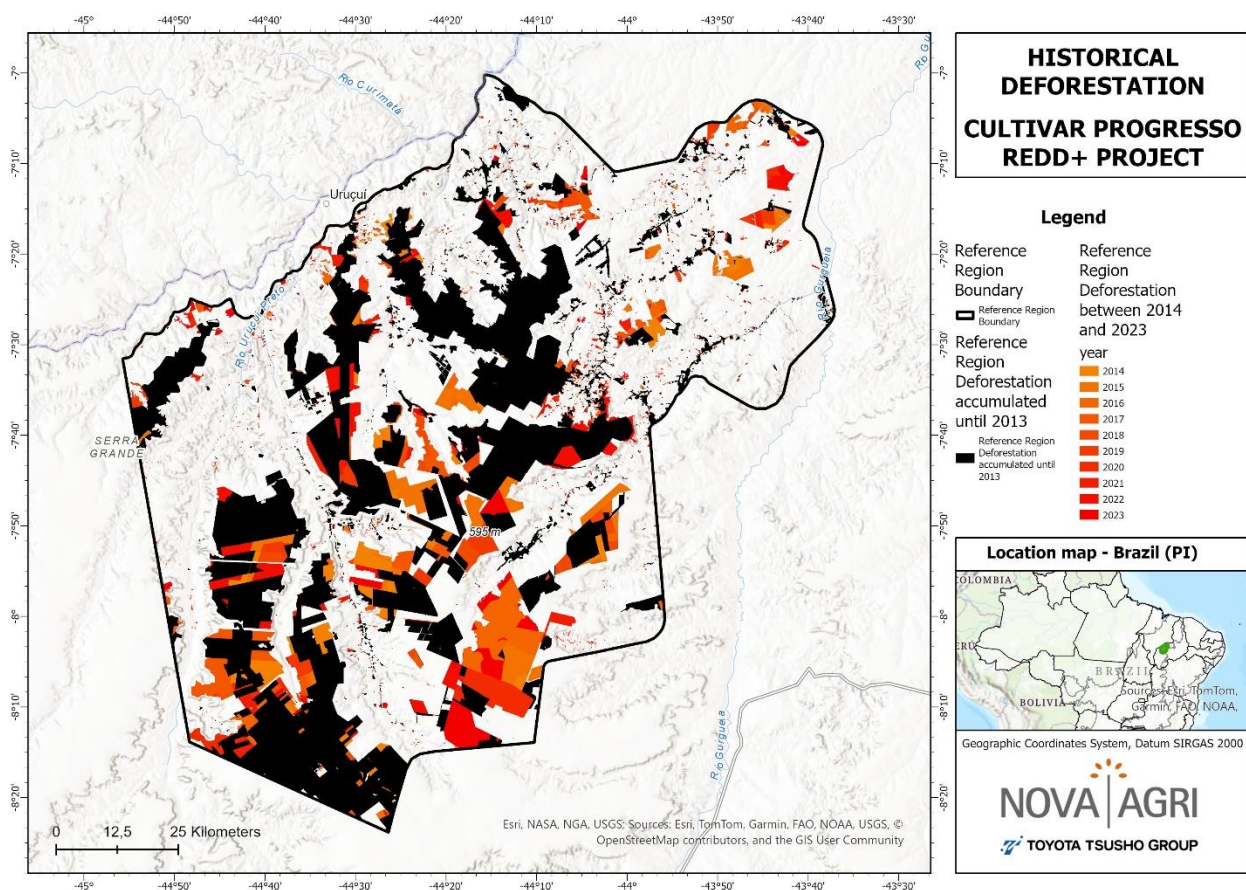


Figure 61. Deforestation within reference region over the historical reference period.

Source: the author.

Results in change history analysis in Land-Use and Land-Cover

Based on the data obtained in the previous steps, the analysis of the historical change in land cover between 2014 and 2023 was carried out in the Reference Region of the PA.

Table 19 shows the changes occurring from the Forest class to the Deforestation class, with a decrease in the carbon stock. The Map presented previously (see 'Deforestation map of within reference region over the historical reference period') shows the annual deforestation that occurred between 2013 and 2023 in the Reference Region.

Table 22. Potential LULC change matrix in the Reference Region between 2012 and 2022

Initial class (2013)					
ID _{Cl}	Name	Forest	Hydrography	Anthropized vegetation	Total (ha)

			L1	L2	L3	
Class LU/LC Final (2023)	F1	Forest	1.301.273	0	0	1.301.273
	F2		0	8	0	8
	F3	Deforestation	0	0	465.111	465.111
Total (ha)			1.301.273	8	465.111	836.154

Specifications of an alternative methodology

Methodological procedures for acquisition, pre-processing, classification, post-classification and evaluation of the accuracy of remote sensing images for analysis of changes in land use and land cover during the duration of the Project.

The official monitoring of the Brazilian Amazon conducted annually by PRODES was used for the development of the baseline and will be used to monitor the Project Area and Leakage Belt. In case PRODES data are unavailable or new sensors with better resolution are available, the following procedures will be used to maintain consistency in monitoring using remote sensing:

- Data sources:** images of satellites of optical sensors should be used. The optical images should be multispectral with a spectral resolution between 0.45 and 2.35 μm . For the mapping of forest cover and land use, images with spatial resolution equal to or greater than 30 meters should be used. The data acquisition period should be in the period of low cloudiness and rainfall in the region for the monitoring of forest cover in the Project Area and the Leakage Belt;
- Pre-processing:** the images must be geometrically corrected by means of georeferencing or orthorectification. The RMS error of the georeferencing must be less than one pixel;
- Data classification:** The identification of deforestation should be made by visual interpretation of images, carried out by trained specialists, able to delimit the deforestation polygons directly on the computer screen. These experts will identify the pattern of change from forest cover to shallow felling based on the main observable elements in optical images: hue, color, shape, texture and context;
- Post-Processing:** the result of the classification in vector format should be used for auditing the classification accuracy;
- Classification accuracy assessment:** should be performed through the analysis of the general accuracy and the kappa index obtained from a confusion matrix. The minimum accuracy of classification mapping is 80%.

Analysis of agents, drivers and underlying causes of deforestation

As specified in the methodology, the analysis of deforestation agents is important for two reasons: i) estimating the quantity and location of future deforestation; and ii) Designing effective measures to address deforestation, including leakage prevention methods.

In recent years, the project region has been deforested for the expansion of agricultural and livestock activities, mainly due to the advancement of the arc of deforestation from the south of the Amazon biome. This pressure is expected to continue, given the globalization of markets in the Amazon region and international development policies planned for the region⁹¹.

Agents of deforestation

In terms of analysis of deforestation patterns, deforestation agents are not considered separately, but as being spatially overlapping and forming a single dynamic within Homogeneous Deforestation Areas (HDAs). The dynamics of historical deforestation can be very complex in Piauí, which in recent periods has shown a very high number of illegal deforestation cases. In other words, the **current consolidation of land use** in the state does not fully reflect the **initial motivations for clearing vegetation**. That is, there is strong evidence (attached ahead) of land grabbing / illegal clearing / animal hunting, which may be motivated by logging but especially by the sale of rural areas. These areas can have high value considering their potential for agricultural and grazing uses.

The analysis performed on the historical land-use/land-cover changes within the HDAs, alongside with the evidences/references about deforestation context – during the historical reference period – allowed conclusions to be drawn related to the correlation between deforestation and the main deforestation agents in the reference region, as well as the underlying causes for deforestation.

Through this analysis, the main agents of deforestation were defined, which are:

- ***Illegal Clear Cutters / Land-grabbers***

The identification of these agents is not directly made through land-use maps or other sources of public information, mainly because illegal activities are rarely documented in their initial stages and are later investigated through initiatives and deforestation study programs, such as MAPBIOMAS⁹². One of its most recent studies highlighted alarming national and regional data regarding non-compliance with legal regulations for opening forested areas. As complement,

⁹¹ Nepstad, D. C.; C. M. Stickler e O. T. Almeida. 2006. Globalization of the Amazon Soy and Beef Industries: Opportunities for Conservation. *Conservation Biology* 20(6):1595-1603.

⁹² Projeto MapBiomas – Mapeamento anual de cobertura e uso da terra no Brasil entre 1985 a 2022 – Coleção 8, acessado em [DATA] através do link: [<https://mapbiomas.org/download-dos-atbds>]

some other key points from RAD2023⁹³, RAD2022⁹⁴, AGÊNCIA PÚBLICA⁹⁵, BOMBEIROS PIAUÍ⁹⁶, GOVERNO PIAUÍ⁹⁷ are as follows:

5. *"Among the 50 municipalities that mostly deforested in 2022 – those which more than doubled their deforested area compared to 2021 –, four are in the state of Piauí (Santa Filomena, Sebastião Leal, Baixa Grande do Ribeiro, and Uruçuí)" (RAD, 2022)* – This statement is relevant to demonstrate the baseline context of the influence of deforestation in the municipalities encompassing the spatial boundary of the Reference Region. Three of the municipalities in the Project Area are represented in this recent scenario of the highest deforestation rankings in the country as a whole. This, in itself, already raises a point of concern regarding the influence area of deforestation expansion.
6. *"The number of areas embargoed for illegal deforestation also grew considerably, from 20,875 hectares in 2022 to 57,846 hectares in 2024, a jump of 177%" (GOVERNO PIAUÍ, 2024)* – This is an alarming figure, which corroborates the increase in illegal agents and invaders within the state. Therefore, it is not just an expansion of agribusiness activities but a real expansion of illegal land clearing, which may later be regularized and commercialized for commercial purposes.
7. *"Only 9.7% of the areas deforested between 2019 and 2022 can be considered legal and planned, as they intersect with some government authorization" (RAD, 2022)* – Supporting the last statement, it is clear that there is a significant inefficiency in legal and regulatory instruments for monitoring and law enforcement, as only few deforested areas in the state have any official deforestation license. It is important to highlight that this scenario is reflected in the Reference Region, but for a more accurate view, all licenses within the project's influence areas were locally analyzed to separate planned deforestations (APD) from unplanned ones (AUD), ensuring that the baseline calculations were constructed satisfactorily according to the VM0015 methodology. The map is presented below, and the calculations are provided starting in Section 3.4.
8. *"Land grabbers have used tractors and large steel chains to devastate native Cerrado areas. (...) Over seven years ago, the Piauí Public Prosecutor's Office (MPPI) denounced an immense land grabbing of 124 thousand hectares, perhaps the largest in the State of Piauí. (...) The TJPI (Piauí's*

⁹³ RAD2023: Relatório Anual do Desmatamento no Brasil 2023 - São Paulo, Brasil - MapBiomias, 2024 - 154 páginas <http://alerta.mapbiomas.org>

⁹⁴ RAD2022: Relatório Anual de Desmatamento 2022 - São Paulo, Brasil - MapBiomias, 2023 - 125 páginas <http://alerta.mapbiomas.org>

⁹⁵ Agência Pública, 2024 – "Demora da Justiça em caso de grilagem ameaça famílias no Cerrado do Piauí": [Grilagem: Demora da Justiça em disputa ameaça famílias no PI](#)

⁹⁶ BOMBEIROS PIAUÍ, 2021 – "AÇÕES DE PREVENÇÃO DE INCÊNDIOS FLORESTAIS": <http://www.cbm.pi.gov.br/noticia.php?id=1030/>

⁹⁷ GOVERNO PIAUÍ, 2020 – "Operação Queimadas: Instituições unem esforços para reduzir efeitos das queimadas no Piauí": <https://antigo.pi.gov.br/noticias/operacao-queimadas-instituicoes-unem-esforcos-para-reduzir-efeitos-das-queimadas-no-piau/>

Legal Court) released farm records even with evidence of connection to land grabbing.” (AGÊNCIA PÚBLICA, 2024) – Information recorded by independent journalism shows how government bodies still have a lack of efficiency regarding monitoring and punishing illegal acts related to land grabbing. Another aspect that draws attention is the involvement of the government system itself in registering these grabbed areas in official land registries despite concrete evidence of illegality. This demonstrates that land instruments are also part of the illegal scheme, turning illegally opened areas into officially registered lands ready for sale.

9. *“In the Cerrado region (...) a large part of the fires is caused by hunters who set fire to the vegetation to scare away the animals or to favor their kill” (BOMBEIROS PIAUÍ, 2021, and GOVERNO PIAUÍ, 2020) – As mentioned in Section 2.1.2, the causes of fires due to hunters are imminent and historical in the region. According to the landowner and the local population themselves, these agents set fire to a strategic area to corner the animal in a specific location that would facilitate its kill. Knowing that hunting is currently an illegal activity in Brazil, these agents fall within the classification of illegal agents. These are risky activities for the region, as the fire can spread to forest reserve areas without precedent.*

Conclusion for the agent 'Illegal Clear Cutters / Land-grabbers' – It is evident, through bibliographic studies and reports, that Brazil, especially the state of Piauí, faces significant issues with illegal deforestation carried out by land grabbers and other clear cutters. Statistics show that most of the land cleared for agriculture and grazing in recent periods does not have deforestation licenses. Therefore, there is strong evidence pointing to illicit activities involving land clearing and subsequent sale to producers of various scales.

As there is no consolidated public database that qualitatively represents this complex historical context, we classify them as distinct agents, each one with its own role and specificity. Although we cannot provide detailed socioeconomic profiles of these illegal agents, it is possible to understand their underlying cause for deforestation. The primary motivation is income inequality and uneven GDP, as demonstrated in Section 3.3. This motivation is further complemented by a **cultural** aspect, marked by a deforestation culture and the awareness of an inefficient public system unable to enforce existing forest laws effectively. This results in the second underlying cause for deforestation, which is the **lack of enforcement of land-use policies**, as demonstrated through the deforestation licensing deforestation database, since the vast majority of deforestation in the RR occurred without licensing.

- ***Small-scale farmers / rural settlements***

These deforestation agents are located on small private properties, ranging between 1 and 4 fiscal modules⁹⁸, and agrarian reform areas⁹⁹. Regarding cultural and socioeconomic aspects,

⁹⁸Brazilian Government – imóveis rurais e tamanho das áreas: <https://www.gov.br/incra/pt-br/assuntos/governanca-funditaria/modulo-fiscal>

⁹⁹Brazilian Government – Política de Reforma Agrária: <https://www.gov.br/incra/pt-br/assuntos/reforma-agraria/a-politica>

SANTOS (2024)¹⁰⁰ brings a complete overview about the livelihood and work attributes of Small-Scale farmers in the Brazilian territory: they have a strong rural identity, where family labor is highly valued. Many are involved in social movements, and their income tends to be low and variable, depending on the harvest, market prices, and their capacity to commercialize their products. Agricultural production in these areas is diverse, including food crops such as rice, beans, corn, vegetables, and fruits, along with livestock raising, both small and large animals. The tools and equipment used are generally simpler, with limited access to advanced technology.

According to MAFUD et al (2009)¹⁰¹, the challenges faced by small rural farmers can be observed throughout different stages of the production process. Initially, difficulties arise in purchasing inputs, as small producers often have limited credit access and low bargaining power. This leads to a low level of technology on their farms. In the next stage, commercialization, their access to consumer markets is hindered by both a lack of physical infrastructure and insufficient information about prices and market opportunities. This inefficiency in commercialization also reflects a lack of careful planning in selecting which products to grow, taking into account the characteristics of the target markets. This scenario is still very present in the 2020's, as reinforced by SANTOS (2024). Additionally, they struggle to compete with medium and large farmers.

Small landowners face significant challenges in generating income, which are part of broader economic, socio-cultural, and political issues. As highlighted by EMBRAPA (2018)¹⁰², the limited size of their farms' compromises financial viability, as small-scale production presents a structural problem for these farmers. Studies show that the average monthly gross production value per family farm is only 0.46 of the minimum wage, leaving many producers in extreme poverty. This precarious situation often pushes these populations into such vulnerability that they feel compelled to expand into new forest areas, threatening untouched natural ecosystems. This demonstrates the primary underlying cause for deforestation by this agent, which is **Poverty and wealth**.

Innovation can create opportunities to maintain the economic viability of family farms, ensuring their sustainability as social and family units while contributing to the sector's modernization. This underscores the importance of promoting incentives to improve production and generate income, as proposed by the current project for areas such as the Leakage Management Zone, which specifically addresses these challenges.

The reference region is nearly fully partitioned into well-defined rural properties, making it unlikely that the land structure can get into a drastic transformation in the near future. As a

¹⁰⁰ SANTOS, 2024 - Gestão da pequena propriedade rural: um estudo de caso no município de Santa Rosa do Sul/SC.

¹⁰¹ MAFUD et al., 2009 – 'Desafios da pequena propriedade.':
<https://periodicos.fgv.br/agroanalysis/article/download/26972/25838/49993>

¹⁰² EMBRAPA, 20218 – 'Artigo - Agricultura familiar, desafios e oportunidades rumo à inovação':
<https://www.embrapa.br/busca-de-noticias/-/noticia/31505030/artigo---agricultura-familiar-desafios-e-oportunidades-rumo-a-inovacao>

result, land-use changes in the area are expected to follow similar patterns to those observed in the past decade among these deforestation agents. For the Reference Region of the CULTIVAR PROGRESSO REDD+ PROJECT specifically, there is a characteristic feature regarding the local topography, which consists of mixed slopes, between plateaus and depressions. This condition is important for deforestation analysis because, when examining the Project Area, it is evident that projected deforestation would be divided between these two types of terrain in the following configuration: smaller-scale deforestation, generally by smaller producers and settlers, is allocated in more depressed areas, while larger-scale deforestation occurs in the plateau areas. For this reason, small producers with smaller farm areas remain relevant and need to be considered in the context of this project (boundary definition, additionality analysis, stakeholder engagement, deforestation projection, etc.), as analyzed ahead.

Within this context, according to the RAMT (2020)¹⁰³, it becomes evident that plateau areas are much more valuable due to the ease of occupation for large-scale agriculture and livestock farming. However, depressed areas also hold commercial value, even for grains producers, for example. Although productivity is lower, it remains significant, as shown in the following table, which corresponds to the Cerrado Savanna of Piauí:

Tipo de área	Área – 1 Ha	Quantidade de sacas de soja	Valor – 1 Saca *R\$ 70,00
Área cultivada - Platô	1ha	200	14.000,00
Área mata virgem - Platô	1ha	100	7.000,00
Área mata virgem Baixada	1 ha	50	3.500,00
Área de Baixada - dentro do Parque Ambiental	1 ha	25	1.500,00
Área de Brejo com Produção	1ha	14 a 25	1.000,00 a 1.500,00
Área de Baixada com Demarcação	1 ha	8 a 10	600,00
Área de Baixada sem Demarcação	1 ha	4 a 6	300,00

Fonte: Corretores de Imóveis de Santa Filomena e Uruçuí.
Informante: Joildo de Oliveira Mascarenha CPF: 013.380.191.80 Bairro - Primavera – I, Nº 64, Santa Filomena – PI,
Informante: Carlos João CPF: 013.380.191.80, Rua Davis Caldas, Nº 39, Centro - Uruçuí – PI, .
 * O preço da saca varia de acordo com a cotação diária na época do estudo a cotação era de R\$ 70,00.

Figure 62. Price per Hectare of Land in the Cerrado Market of Piauí.

Source: RAMT (2020).

The same report provides the average value per hectare (in BRL), bringing a specific line for Family Agriculture within the MRT03 area – 'Serra dos Quilombos,' which includes the main municipalities of the Project Area and a significant part of the Reference Region. These values are lower, aligning more closely with the household income of small farmers. In summary, relating to the previous table, it is evident that depressed areas have a lower value but are the primary target for smaller farmers, who also have lower purchasing power. However, these areas remain

¹⁰³ RAMT/PI, 2020 - Relatório de Análise de Mercado de Terras do Estado do Piauí – RAMT/PI: https://www.gov.br/incra/pt-br/assuntos/governanca-fundiaria/relatorio-de-analise-de-mercados-de-terras/ramt_sr24_2020.pdf

productive and hold value, including for family agricultural production, as highlighted in the following table (VTN = Raw Land Value):

PPR / SR-24/PI/Nº 01/ANO 2018/ MRT-03					
VALOR DE TERRA NUA POR HECTARE – VTN/ha					
TIPOLOGIAS	Nº Elementos Pós Saneados	Média	Coeficiente de Variação (%)	Limite Inferior	Limite Superior
		VTN/ha (R\$)		(x -15 %)	(Média +15 %)
1º Nível categórico					
Agricultura	13	869,13	21,79	738,76	999,49
Vegetação Nativa	10	1.315,53	14,35	1.118,20	1.512,86
2º Nível categórico					
Agricultura Familiar	7	628,66	19,41	534,36	722,96
Agricultura Médio/Alto Nível Tecnológico	14	7.152,18	17,06	6.079,35	8.225,00
Vegetação Nativa Cerrado	10	1.315,53	14,35	1.118,20	1.512,86
3º Nível categórico					
-	-	-	-	-	-
MRT GERAL (R\$)					
Todas as Tipologias					
	35	1.117,91	51,23	950,22	1.285,60

Figure 63. Price per Hectare of Land in the MRT-03 Region. Highlighted word: 'Familiar Agriculture'.

Source: RAMT (2020).

When discussing the *settlers*, who mostly engage in rural farming activities, the report also provides a perspective on the number of settlement projects (P.A. and P.E.) in the region. It is observed that many projects are already established throughout the region, with 4 located in the project municipalities, covering an area of over 32k hectares and including more than 500 families.

Município	P.A's		Nº Famílias nos P.A's
	Qtde.	Área (ha)	
Barreira do Piauí	-	-	-
São Gonçalo do Gurguéia	-	-	-
Monte Alegre do Piauí	-	-	-
Gilbués	-	-	-
Redenção do Gurguéia	1PA* e 3 PE*	10.994,15	187
Bom Jesus	3PA e 2 PE	20.662,80	175
Santa Filomena	-	-	-
Currais	2 PE	67.774,97	117
Palmeira do Piauí	-	-	-
Baixa Grande do Ribeiro	-	-	-
Ribeiro Gonçalves	-	-	-
Uruçuí	2PA	14.009,00	222
Sebastião Leal	1PA e 1 PE	18.042,36	295
Alvorada do Gurguéia	-	-	-
Manoel Emídio	4 PA	7.763,91	234
Antônio Almeida	1PA	1.108,00	45
Total	12PA e 8PE	122.312,84	1.275

Fonte: INCRA/SIPRA

*PE – Projeto de Assentamento Estadual, reconhecido pelo INCRA.

*PA – Projeto de Assentamentos criados pelo INCRA.

Figure 64. Number of Settlement Projects (P.A. and P.E.) and Settled Families in the Municipalities of the Serra do Quilombo Regional Land Market - MRT - 03.

Source: RAMT (2020).

Historically, deforestation caused by these agents in the reference region has been well-documented in terms of its size and spatio-temporal distribution. Detailed statistics on historical deforestation attributed to these agents will be provided in a following section.

- **Medium to large-scale farmers**

These deforestation agents are located on medium to large private properties, typically exceeding 5 fiscal modules¹⁰⁴. Most medium and large farmers adopt an intensive production model, focusing on high-value crops such as soybeans, corn, and cotton, as well as large-scale livestock farming. These operations often rely on modern technologies and a high level of mechanization to maximize efficiency¹⁰⁵. In response to growing pressure for more sustainable production with lower environmental impact, many of these farmers have begun implementing

¹⁰⁴ Governo Brasileiro – ‘imóveis rurais e tamanho das áreas’: <https://www.gov.br/incra/pt-br/assuntos/governanca-fundiaria/modulo-fiscal>

¹⁰⁵ SENAR – Com agricultura de precisão, produtores rurais alcançam até 29% de aumento na produtividade: <https://www.cnabrazil.org.br/noticias/com-agricultura-de-precisao-produtores-rurais-alcancam-ate-29-de-aumento-na-produtividade>

practices like precision agriculture, crop rotation¹⁰⁶, and the recovery of degraded areas¹⁰⁷. The first primary underlying cause for deforestation for this agent is Poverty and wealth, as demonstrated in the topics below: “Economic factors”, and “Livestock Scenario”. The second primary cause is Land-use policies and their enforcement, demonstrated in the topic “Social and Policy-Related Factors”.

Economic factors

Agriculture Scenario

- Market Demand for Commodities: Farmers deforest to expand land for commodity crops¹⁰⁸, such as soybeans, corn, coffee, sugarcane, among others, which are in high demand in both local and global markets. Brazil's position as a leading exporter of agricultural products creates strong economic incentives for deforestation. For instance, the local context has presented an advance of the VBP (production gross value) by 76% from 2018 to 2023, as presented below:

¹⁰⁶ EMBRAPA – Rotação de Culturas: https://www.embrapa.br/agencia-de-informacao-tecnologica/cultivos/milho/producao/rotacao-de-culturas?p_auth=9O0mtIEF&p_p.id=82&p_p.lifecycle=1&p_p.state=normal&p_p.mode=view&_82_struts.action=%2Flanguage%2Fview&_82_languageId=en_US

¹⁰⁷

SENAR - Recuperação de áreas degradadas, maior produtividade e geração de renda são objetivos do programa Paisagens Rurais: <https://www.cnabrazil.org.br/noticias/recuperacao-de-areas-degradadas-maior-productividade-e-geracao-de-renda-sao-objetivos-do-programa-paisagens-rurais>

¹⁰⁸ Brazil has become the "breadbasket of the world" and is now leading global exports of seven food products, according to BTG. <https://www.cnnbrasil.com.br/economia/macroeconomia/brasil-virou-celeiro-do-mundo-e-ja-lidera-exportacoes-mundiais-de-sete-alimentos-diz-btg/>

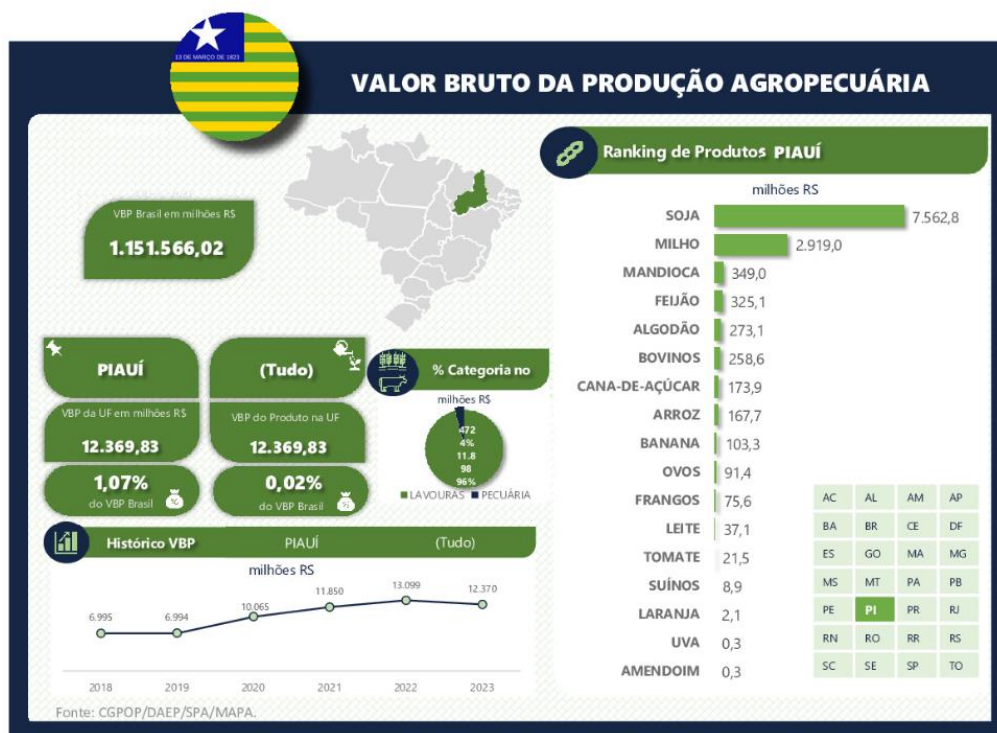


Figure 65. Gross Value for Agribusiness Production in Piauí.

Source: O PRESENTE RURAL (2023)¹⁰⁹

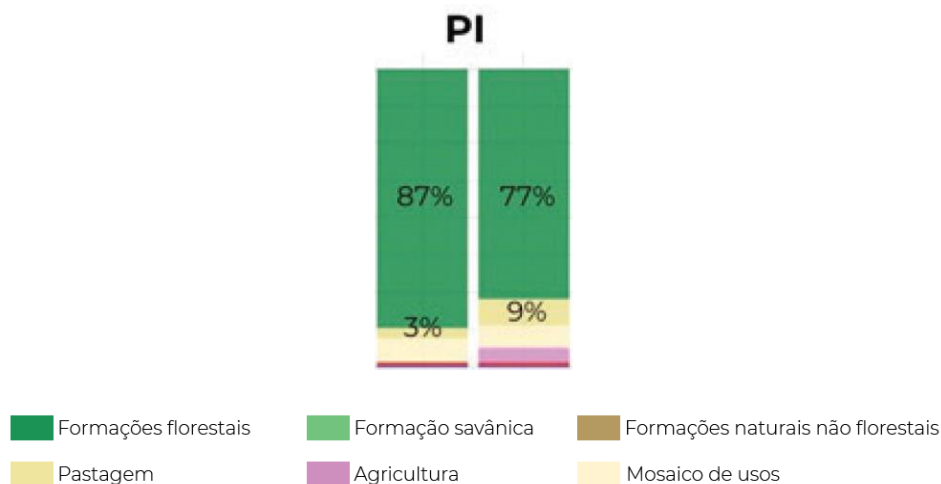


Figure 66. Annual Mapping of Land Use in Brazil between 1985 and 2022.

Source: MapBiomass¹¹⁰

¹⁰⁹ O PRESENTE RURAL, 2023 – Gross Value for Agribusiness Production in Piauí:

<https://www.flip3d.com.br/pub/opresenterrural/?numero=240&edicao=5555#page/62>

¹¹⁰ Annual Mapping of Land Use in Brazil between 1985 and 2022: https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2023/08/FACT_MapBiomass_Mapeamento-Anual-Cobertura_Colecao8_31.pdf

**Livestock, Mix Mosaic Activities, and Agriculture has led the LULC in the Piauí State between 1985 and 2022.*

- Over the last 5 years (considering current year as 2023), according to MAPBIOMAS¹¹¹ study (RAD 2023), Commercial Farming (Agriculture and Livestock) remains the primary pressure on native vegetation, accounting for over 97% of the loss.

Assessment of the most likely changes in population size of the identified main agent groups

Note: for this topic, it is important to highlight that we consider here the assessment of changes/statistics related to the agents identified in the consolidated land-use studies, since the illegal clear cutters and land-grabbers agents are not officially recorded within these databases.

Small farmers and settlers engage in continuous, small-scale deforestation. According to the map below, which shows the different deforestation patterns (from deforestation data by PRODES¹¹²), it was possible to identify these agents in 2013, covering a total of 331 ha, representing 6% of the total deforestation in the Reference Region was caused by this agent, and 28% of this area was cleared during the last 10 years.

In contrast, medium and large farmers have deforested 4.698 ha until 2023, representing 94% of the total deforestation, with 32% of this total occurring in the past decade (from deforestation data by PRODES). This reflects a significant difference in deforestation patterns by these groups. The distribution of the deforestation pattern by these agents within the project boundaries can be seen on the map below.

¹¹¹ Relatório Anual do Desmatamento no Brasil 2023 - São Paulo, Brasil - MapBiomas, 2024 - 154 páginas. <http://alerta.mapbiomas.org/>

¹¹² Instituto Nacional de Pesquisas Espaciais. Coordenação Geral de Observação da Terra. Programa de Monitoramento da Amazônia e demais Biomas. Desmatamento – Amazônia Legal – Disponível em: <http://terrabrasilis.dpi.inpe.br/downloads/>

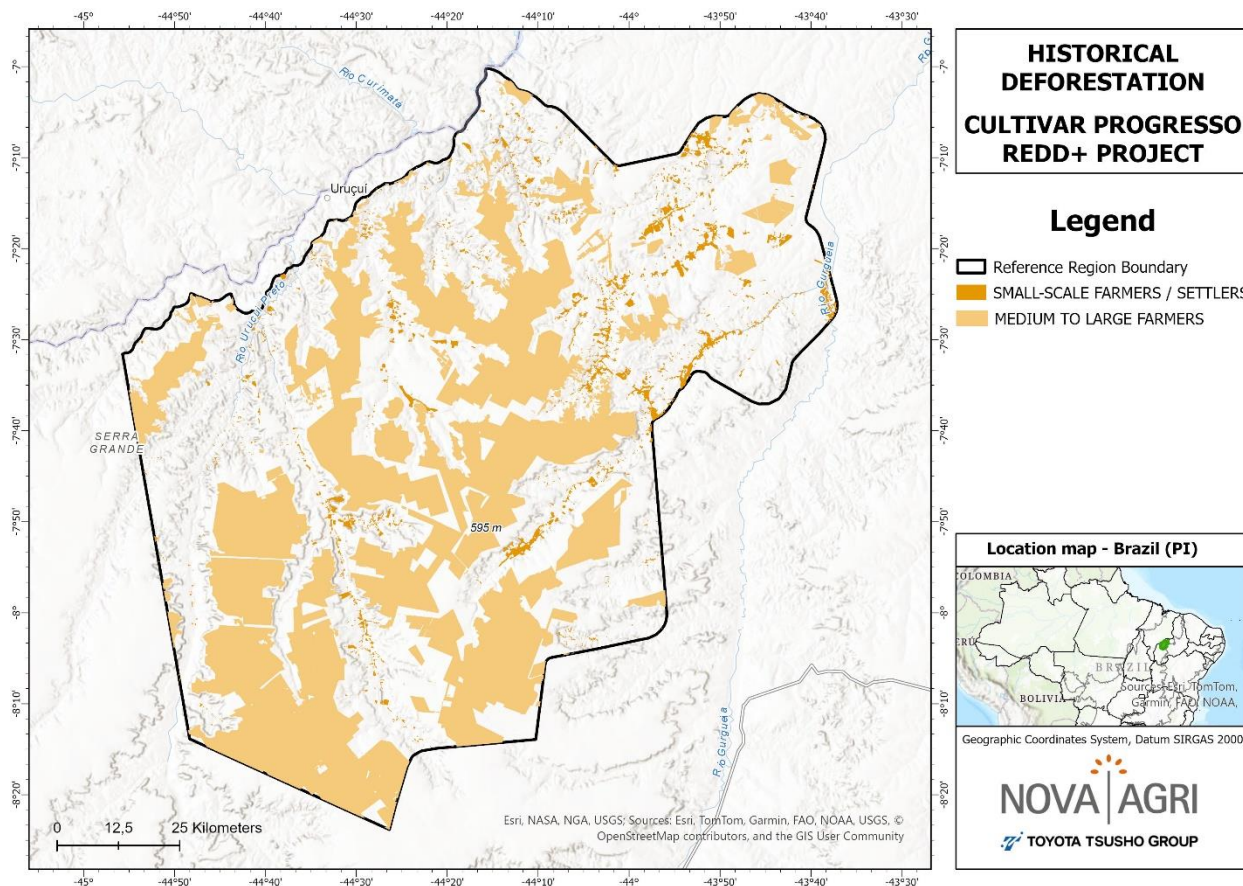


Figure 67. Distribution of agents' deforestation.

Source: PRODES, the author.

Statistics on historical deforestation attributable to each main agent group

Note: for this topic, it is important to highlight that we consider here the assessment of changes/statistics related to the agents identified in the consolidated land-use studies, since the illegal clear cutters and land-grabbers agents are not officially recorded within these databases.

The average size of deforestation during the historical reference period supports evidence that typical deforestation in the reference region, primarily carried out by small producers and settlers, is on average thirty (30) times smaller than that carried out by medium and large producers. Additionally, the maximum deforestation size by small producers and settlers during this period was 3% smaller than the largest single deforestation event carried out by medium and large farmers.

Table 24. Deforestation Proportion by Each Agent

From 2013	Average size of Deforestation	
until 2022	Small producers	Large producers

Average	5,20 ha	157,21 ha
Maximum	183.08 ha	6.671,11 ha

Deforestation drivers

The main drivers affecting the location of deforestation in the reference region are:

1- Previously deforested areas

Forested areas are influenced by their proximity to areas that have already been deforested. The distance from previously deforested areas is one of the major causes of forest degradation in the Cerrado biome and their spatial-temporal dynamics are highly influenced by annual deforestation patterns.

As described in the "future deforestation location projection" section below, a regression between the currently deforested areas and future deforestation was performed, which achieved a significant result as of projecting deforestation volume.

2- Geomorphology of the terrain

The slope divides the deforestation agents following the geomorphology of the reference region and can be considered a driver by influencing large-scale deforestation in flat areas and limiting deforestation in small portions on rugged terrain.

In addition, the spatial-temporal dynamics of deforestation in the reference region during the historical period indicates that flat terrain increased anthropogenic pressure and, consequently, the intensity of deforestation in the vicinity of these areas.

3- Lands economic values

The economic value of land is directly tied to its suitability for high-profit activities. Therefore, this driver must be included in deforestation projection analyses, especially given that agents in the region primarily use land for agriculture and pasture after clearing.

According to the RAMT (2020), it becomes evident that both plateau and depressed areas have commercial value due to the ease of occupation for both small (familiar) and large-scale agriculture and livestock farming. Although productivity in depressed lands is lower, it remains significant, as shown in the following table – the land's value for depressed areas with vegetation is more valuable than swamp areas, or demarcated land areas.

Tipo de área	Área – 1 Ha	Quantidade de sacas de soja	Valor – 1 Saca *R\$ 70,00
Área cultivada - Platô	1ha	200	14.000,00
Área mata virgem - Platô	1ha	100	7.000,00
Área mata virgem Baixada	1 ha	50	3.500,00
Área de Baixada - dentro do Parque Ambiental	1 ha	25	1.500,00
Área de Brejo com Produção	1ha	14 a 25	1.000,00 a 1.500,00
Área de Baixada com Demarcação	1 ha	8 a 10	600,00
Área de Baixada sem Demarcação	1 ha	4 a 6	300,00

Fonte: Corretores de Imóveis de Santa Filomena e Uruçuí.

Informante: Joildo de Oliveira Mascarenha CPF: 013.380.191.80 Bairro - Primavera - I, Nº 64, Santa Filomena - PI,

Informante: Carlos João CPF: 013.380.191.80, Rua Davis Caldas, Nº 39, Centro - Uruçuí - PI, .

* O preço da saca varia de acordo com a cotação diária na época do estudo a cotação era de R\$ 70,00.

Figure 68. Price per Hectare of Land in the Cerrado Market of Piauí. Highlighted lines: 'Plateau Areas of Native Vegetation' and 'Depressed Areas of Native Vegetation'.

Source: RAMT (2020).

This analysis helped highlight properties and zones with higher deforestation potential, informing future deforestation projections, as detailed in Section 7 of this PD.

Identification of distinguished driver variables for each identified agent group

- **Illegal Clear Cutters / Land-grabbers**

Quantity Relation (hectares)

The variables related to these agents can be complex due to the lack of tangible quantitative databases within official sources. However, evidence of the actions of these agents is abundant, as previously presented through articles and news reports.

Thus, among the variables indicating a relationship between the number of hectares cleared by such agents and the impunity concerning illegal deforestation in the state of Piauí stands out. Revisiting some information from the previous section, *'It is estimated that over 93% of the area deforested in Brazil in 2023 showed at least one indication of irregularity'* (RAD, 2023).

Location Relation

The variable of impunity is further evidenced by the deforestation authorization system of SINAFLOR¹¹³, which shows the number of registered authorizations, as presented on the image below:

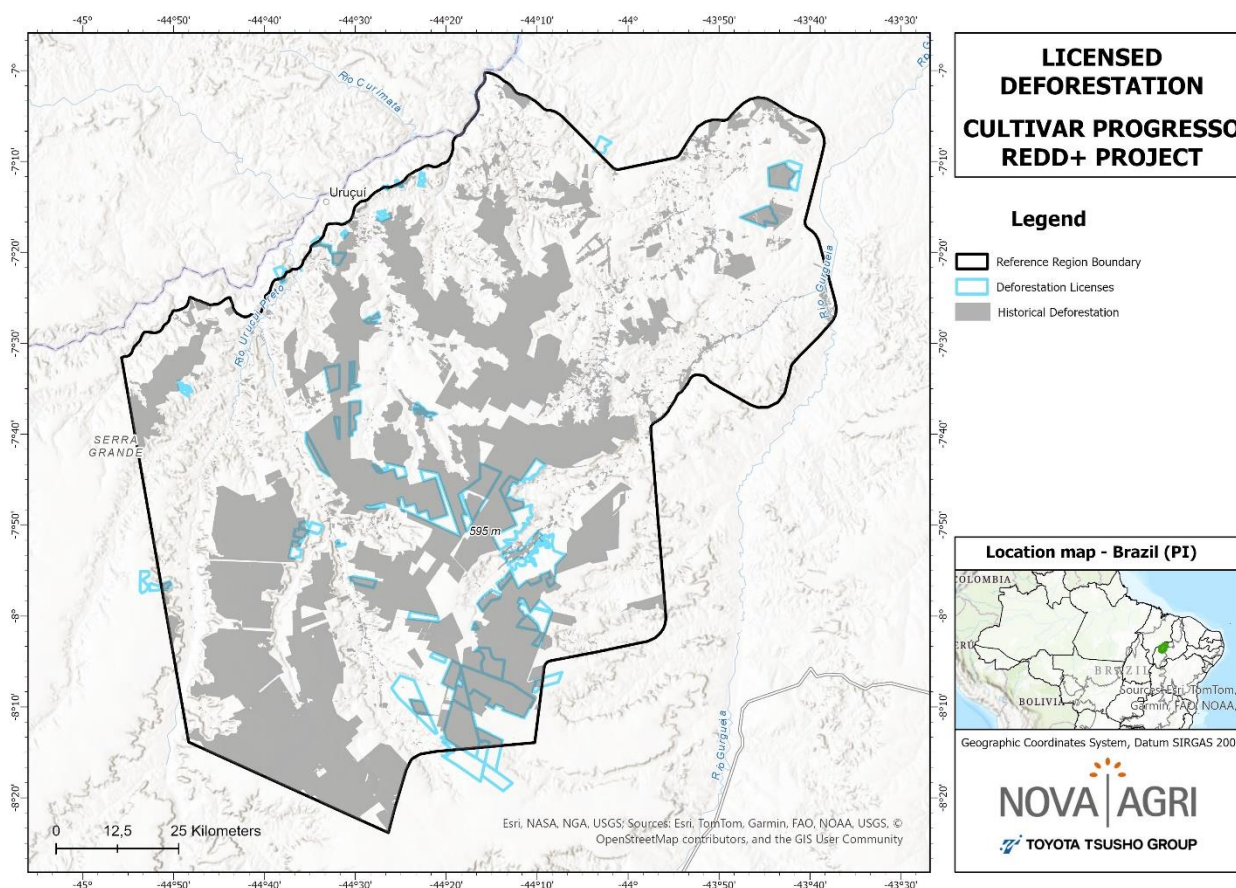


Figure 69. Area of authorized deforestation in the RR (XX% of total deforestation).

Source: SINAFLOR, the author.

Note: The information above will be important for distinguishing between planned and unplanned deforestation, which will influence the emissions calculations later.

With this, there is evidence pointing to illicit activities involving land clearing and subsequent sale to farmers of various scales, and spatially can be manifested in the whole State. Therefore, it is not possible to separate the actions of different groups of agents in both quantitative and spatial relationship of deforestation, because they are all connected.

As a conclusion regarding the variables, it is stated that there is a direct relationship between **variables related to the impunity of the law**, which do not appear in official databases and are

¹¹³ SINAFLOR – ‘Painel de Autorizações de Exploração Florestal emitidas pelo Sinaflor’ - <https://pamgia.ibama.gov.br/portal/apps/dashboards/d49b8e520f5d4938b704c39130911846>

therefore difficult to quantify. However, it is possible to affirm that the quantification of deforestation related to the other agents (small-scale farmers / rural settlements and medium-to-large farmers) is directly linked to the variables associated with illegal agents.

- ***Small-scale farmers / rural settlements***

Quantity Relation (hectares)

The driver variables present in the reference region, where deforestation agents are characterized by small-scale farmers and rural settlers, are closely linked to the land structure, which consists of small private properties ranging from 1 to 2 fiscal modules. This structure is also associated with a production model based on polyculture, typical of family farming, as mentioned in the previous topic.

In the case of these agents, the **need of land expansion** is indeed one of the predominant variables, as they are landowners with **smaller areas** and **lower productive capacity**, typically with limited financial resources. As demonstrated in the previous section, studies show that the average monthly gross production value per family farm is only 0.46 of the minimum wage, leaving many producers in extreme poverty (EMBRAPA, 2018).

Location Relation

The historical deforestation data associated with the geomorphology clearly highlights the size variable associated with these agents, which occupy their properties on rugged terrains where land is cheaper or sometimes undeveloped, promoting land-grabbing action. Over the past decade, the average deforestation observed in this part of the reference region – inhabited by small-scale farmers and rural settlers – is represented by 938 hectares per year.

The location variable connected to deforestation is explained by the dynamics of land occupation in rugged terrain, which seeks to meet the subsistence needs of these small producers. Gradually, small portions of the forest are cleared from the edges of already occupied areas (*Figure 39*).

Logically, expansion into areas already deforested by other small producers or settlers appears to follow the trend demonstrated through the historical deforestation map.

- ***Medium to large-scale producers***

Quantity Relation (hectares)

The quantitative variable associated with deforestation in this context is explained by the dynamics of land occupation aimed at meeting the increasing demand for food production. This process often starts with high-value properties, as access to credit facilitates the establishment of new agricultural areas, as evidenced in a previous topic. Consequently, large portions of the forest are primarily cleared for agriculture and livestock.

Over the past decade, the average deforestation observed in the reference region, inhabited by large and medium producers, represents 16.203 hectares per year. This figure clearly illustrates the size variable associated with these agents, who systematically clear extensive forest areas to meet their needs.

Location Relation

The driver variables present in the reference region related to the location of deforestation are associated with the flat areas where the land is more expensive and appropriate to highly mechanized farming practices. Here, the deforestation agents are characterized by large and medium-sized farmers, whose structure is linked to an intensive production model based on large-scale agriculture through ownership or leasing of properties that commonly exceed 5 fiscal modules (*Figure 43, Figure 46*).

Logically, expansion into areas already deforested by other large and medium-scale producers also appears to follow the trend demonstrated through the historical deforestation map.

Identification of Underlying Causes of Deforestation

The underlying causes of deforestation for the main agents (land-grabbers, small farmers, and medium to large-scale farmers) have all already been justified above. To recap, for land-grabbers, the primary causes for deforestation are linked to poverty and wealth, driven by economic incentives to clear and sell land, and the inefficiency of land-use policy enforcement. A lack of effective oversight and a culture of deforestation perpetuate these illegal activities, with most deforestation occurring without legal authorization. For small farmers, poverty and wealth is the main cause for deforestation, as the limited viability of their small-scale farming leaves many in extreme poverty. The average income per family farm is significantly below the minimum wage, forcing them to expand agricultural land into forested areas. For medium to large-scale farmers, the underlying causes include both poverty and wealth, influenced by economic factors and the livestock market, as well as weak land-use policies and their enforcement, which enable continued deforestation.

Analysis of events in the RR during the historical period x deforestation projection

Based on historical evidence, it was possible to analyze both qualitatively and quantitatively the spatial and temporal dynamics of land use for each deforestation agent group. This analysis

considered data on the location of deforestation observed over the past decade, paired with quantitative measures of the area deforested annually (in hectares). By examining these patterns, we could identify the drivers of deforestation as well as the underlying causes that influence each deforestation agent's characteristic land-use dynamics in terms of both space and time.

Based on these findings, criteria were established to group deforestation agents according to the geomorphology of the terrain. Projected future deforestation, derived from an interpolation of the historical deforestation series, was then spatially distributed across flat and rugged areas. This distribution takes into account both the probable location of deforestation events and the estimated area likely to be deforested in the future.

The drivers and underlying causes that help explain the sequence of events leading to deforestation in the reference region during the historical period are embedded within both flat and rugged landscapes. These landscapes were defined based on the specific characteristics of each deforestation agent group. For medium and large-scale farmers only the flat lands were considered as these geomorphological factors significantly influence land-use decisions and the spatial progression of deforestation (*Figure 43*).

Conclusion of the analysis of agents, drivers and underlying causes of deforestation and their likely future development

Based on comprehensive analyses of deforestation drivers and agents across the study areas (Reference Region, Project Area, and Leakage Belt), combined with historical statistics and specific dynamics from the region, **the most likely future deforestation trend has been decisively established**. The data and evidence presented here, drawn from both public sources and internal studies, confirm a strong link between post-deforestation land use and commercial, profit-driven activities. These trends reflect the specific proportions and characteristics of the identified deforestation agents and their related activities.

This conclusion is also supported by additional studies outlined in the Project Document, such as the Cost-Opportunity study conducted before delineating the Leakage Belt, which highlights a notable increase in land value in the Reference Region due to the high market value of the products cultivated or produced there.

Based on the information gathered above and on the baseline analysis, it is possible to find conclusive evidence that deforestation pattern explained by the relations among the agents, drivers and underlying causes present in the region during the historical period will most likely continue to cause deforestation in the future.

Furthermore, it is expected that agents and drivers of deforestation may increase because of a possible population growth in the reference region, which is also influenced by the prices of crop

commodities and livestock arroba, thus intensifying the demand to convert forested areas to productive areas in the region.

Description of baseline scenario adopted

The analysis of the chain events that led to deforestation in the reference region was based on the relations between main deforestation agents and the drivers present in each homogenous landscape that contribute to the underlying causes of deforestation and its probable development future.

Based on the historical evidence collected, the relationships among agents, key drivers and underlying causes of deforestation that typically lead to and will most likely lead to deforestation within homogenous landscape and therefore can be demonstrated by the analysis of the historical series of deforestation as a function of time and distance to the nearest road.

Selection of Baseline Approach

Establishing the baseline is an exercise of predicting the future value of a socioeconomic variable, i.e., the future value of the deforestation extension. The baseline is counterfactual, that is, something that would happen in the absence of the project implementation and above all, the baseline must be credible, identifiable, plausible and relevant to the project area.

The methodology VM0015 v1.2 suggests the use of two approaches to forecast the amount of future deforestation: (1) historical average approach and (2) the time function approach.

The approach chosen was 2 – time function.

The criteria used to define this baseline approach considered the stratification of the reference region into two types of deforestation agents, along with the subdivision of the area under the influence of medium and large producers based on variables and drivers that influence the deforestation projection in the project area, such as property value and proximity to roads. The forward materials will clarify the criteria for choosing this baseline approach, showing the evidence emerged from analyzing rates measured over the historical period in the reference region, and the agents' and drivers' dynamics, demonstrating that the **deforestation trends are decreasing and likely to continue**.

The baseline approach (1 – historical average approach) is not suitable for explaining future deforestation projections due to the heterogeneity of the reference region, clearly identified by analyzing the deforestation history over the past 10 years.

As shown in the Map previously shown (see '*Deforestation map of within reference region over the historical reference period*'), there is a clear decreasing trend in deforestation during the historical reference period within the reference region is present. As explained earlier in section

3.4 – Baseline Scenario, conclusive evidence from the analysis of the deforestation dynamic was found to suggest that this trend would continue in the future.

Overall, project proponents and specially governments (e.g. Brazilian Forest Reference Level – FREL) predict deforestation from the historical rate of deforestation and often use linear regression (i.e. average) to extrapolate the rate of deforestation into the future.

This approach, however, when analyzed from the perspective of forest cover loss (the opposite of deforestation) – without proper statistical procedure often results in a prediction of a negative value, beyond the remaining forest extension.

In addition, an inspection, albeit superficial, of the history of deforestation in the Cerrado Biome shows that, in general, as deforestation advances in each region, its speed over time decreases, because, among other reasons, the amount of forest remaining becomes increasingly scarce.

The simplest mathematical representation of this fact, which also avoids the non-plausible or credible negative values of forest cover is given by logarithmic regression of forest cover loss, which will be detailed in a next topic.

Logistic Regression

The logistic regression on the other hand, also often used by project proponents, is a statistical model that allows us to estimate the probability associated with the occurrence of a given event in the face of a set of explanatory variables.

Although logistic regression is being used in studies on the various causes of deforestation, this is based on the correlation between several external factors (predictive variables) and the rate of deforestation (explanatory variable) and, therefore, as a technique for predicting the future, suffers from the insufficiency of predictive variables that are globally comparable to feed this statistical model.

In summary, the practical application of this approach depends on the availability of the independent variables, which explain the dependent variable. The difficulty in homogenizing the predictive variables makes it impossible to apply this technique on a large scale to estimate the rate of deforestation.

The method developed by Geoconsult does not necessitate to identify agents of deforestation but is based on the simple yet powerful assumption that all factors influencing deforestation are represented within the defined homogeneous landscape (i.e. areas of homogeneous deforestation – HDAs).

An analysis of all overlapping municipality districts is undertaken to consider land tenure and its geomorphology, by which areas are individualized and defined as indigenous lands, conservation units, settlements, type of soil, climate and declivity, among other aspects, which may influence the homogeneity of the deforestation.

The homogeneous landscape is formed by the intersection of these boundaries and, therefore, formed by a combination of both their legal and geomorphological identities. Each homogeneous landscape is then used to estimate a baseline, from a historical dataset of deforestation distributed in space and time.

For these reasons, approach (2) the function approach was adopted to create the baseline. With this approach, the rate of deforestation will be estimated using a model that expresses baseline deforestation as a function of the observed time-series of deforestation multiplied by a road-distance factor, which is detailed in the next topic.

Logarithmic Regression analysis

The annual deforestation rate is considered a predictor of the likelihood of future deforestation and the distance from deforested areas to the nearest road a driver of deforestation.

Thus, the correlation between these two variables was analyzed, namely: i) observed annual rate of deforestation and ii) distance from deforested areas to the nearest road. The analysis of these variables generated data on, respectively: i) annual rate of deforestation projected for the future; and ii) multiplier factor of road distance.

It should be noted that the annual rate of deforestation is equal to the annual decrease in the remaining forest area, that is, it is equal to the time derived from the remaining forest area and that the cumulative difference in forest loss is a consequence of the annual rate of deforestation. *It is important to note that the decision to deforest is not taken every year, regardless of what happened in the previous year or in recent years.*

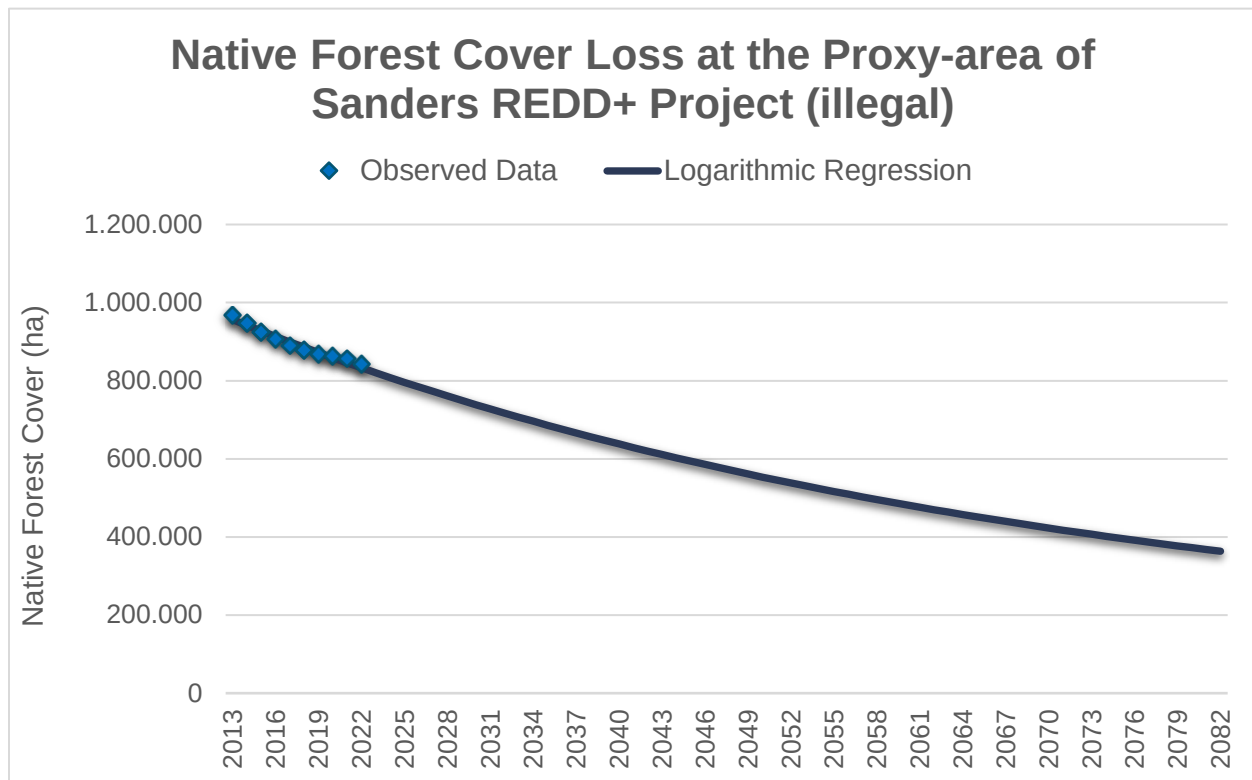


Figure 70. Logarithmic Regression of Forest Cover Loss.

Source: the author.

Another point is that in recent decades there have been important economic and political changes that have somehow affected deforestation. Thus, it can be concluded that economic changes and relevant public policies, if they affect the deforestation process, this will happen slowly, with effect in a period of four or five years.

This conclusion is because deforestation observed data follow a Markov series of 4 or 5, that is, one-year deforestation is correlated with what occurred one year earlier and two years earlier, with a lower correlation, and so on up to four or five years ago.

Markov Chains contextualization

Any real system always operates in environments where uncertainty prevails, especially when the system involves, by its nature, unpredictable human actions.

Deterministic models certainly contribute to the understanding, at a basic level, of the dynamic behavior of a system. However, because they cannot deal with uncertainty, it ends up being insufficient in decision-making processes.

Thus, stochastic processes are used as a way of quantitatively treating these phenomena, taking advantage of certain regularity characteristics that they present to be described by probabilistic

models. A stochastic process is therefore a mathematical model that evolves over time in a probabilistic way.

Markov Chains constitute a special type of stochastic process that has the property that the probabilities associated with the process at a given time in the future depend on the present state and are therefore independent of events in the past.

Although the Markov Chain is characterized by what is called “lack of memory”¹¹⁴, some Markov processes have a finite memory¹¹⁵, or finite range, and the determination of the order of the Markov chain makes explicit the magnitude of this range.

Therefore, if a dataset comes from a Markov chain, it is essential to determine the order of this dependence, that is, how much a given information, at a given instant, depends on previous information in the process.

Deforestation activity data

Earth observation data via remote sensing satellite provides information about deforestation over a given period of time. This is the source of information that is used to create a time series of deforestation, or when considered from a different perspective, it also demonstrates the loss of forest cover in space and time.

Data from PRODES¹¹⁶ produced based on satellite images with sufficient spatial resolution to identify deforestation size from 1ha were used to create an annual time series with at least 20 years of remaining forest area observations within the reference area.

The logarithm of the observed values of the remaining forest area is the object of a linear regression that is then extrapolated into the future. The extrapolated values are then the object of an exponential function, which results in the predicted values of the remaining forest area.

The difference from year to year of the predicted values of remaining forest area is the actual predicted annual forest cover loss.

Quantitative projection of future deforestation

As stated earlier, the method used to project annual deforestation areas in the baseline for the reference region is the Temporal Function approach, using a logarithmic regression equation of

¹¹⁴ Transition probability is defined as the one that characterizes the transition to the state at instant t given that the previous state was $t-1$. This is the definition of what is called a Markov chain of first order.

¹¹⁵ These models have as their basic element the Markov Chain, which is a discrete-time stochastic process in which the transition to a state at instant t depends on the last k states visited. In this case the Markov chain is called k order.

¹¹⁶ Instituto Nacional de Pesquisas Espaciais. Coordenação Geral de Observação da Terra. Programa de Monitoramento da Amazônia e demais Biomas. Desmatamento – Amazônia Legal – Disponível em: <http://terrabrasilis.dpi.inpe.br/downloads/>.

the historical data for forest cover loss (i.e., the opposite of deforestation). The use of logarithmic regression is justified by the need to project the deforestation trend in a more realistic way, considering a gradual decrease in the annual deforestation rate as the remaining forest area diminishes.

The calculation sequence begins with identifying the remaining forest area at the start of the historical reference period. This is done by subtracting the water-covered areas, non-forest vegetation, and consolidated deforestation area up until July 31, 2023 from the total area of the reference region.

Once the remaining forest area at the beginning of the reference period is determined, the observed deforestation area is subtracted year by year, creating a historical series of forest cover loss for the reference region. This historical series of forest cover loss is then interpolated using logarithmic regression to project the baseline into the future.

The equation used to project the future amount of deforestation was the logarithmic regression of historical forest cover loss data (i.e., the inverse of deforestation), which is described below:

$$y = -2247979,14 + 200797,85 * \ln(x)$$

The deforestation rate attributed to the baseline, shown on the chart below:

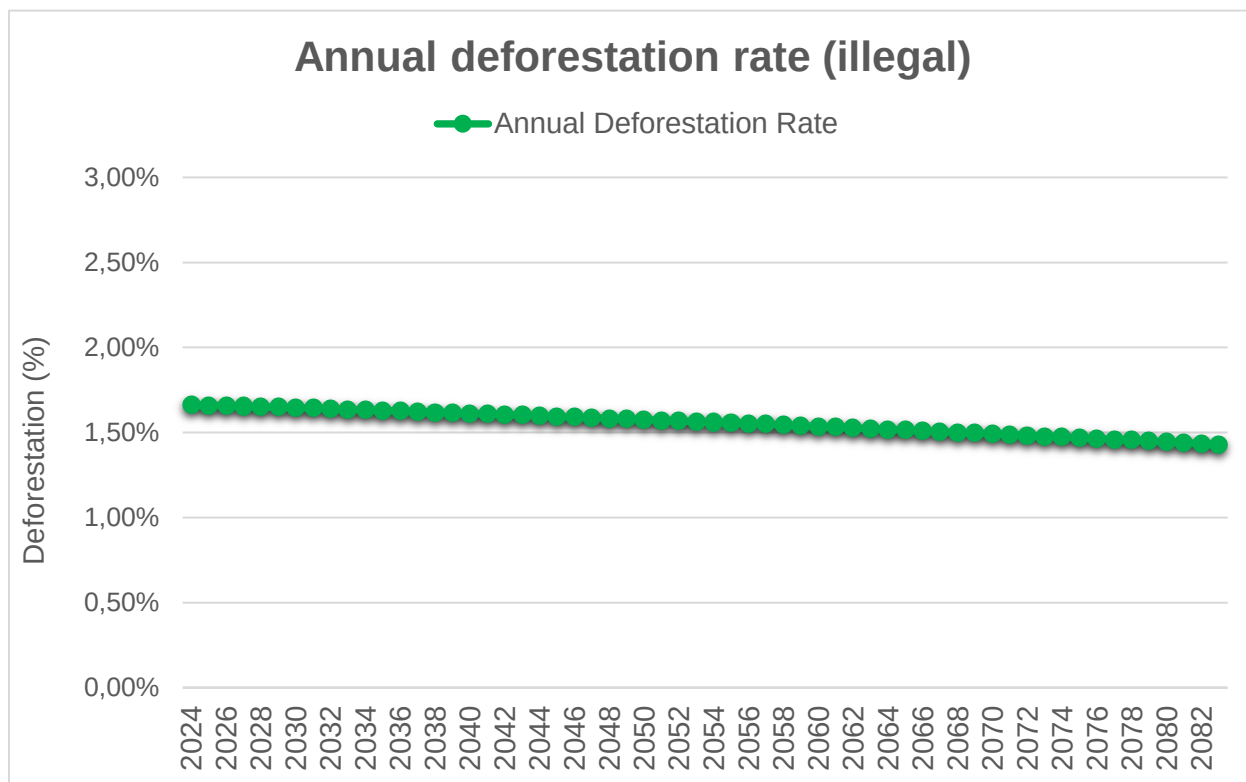


Figure 71. Final Deforestation Rate Projection based on historical baseline.

Source: the author.

The deforestation rate for the baseline is then used to estimate future deforestation by multiplying the projected annual deforestation rate by the remaining forest area in the reference region.

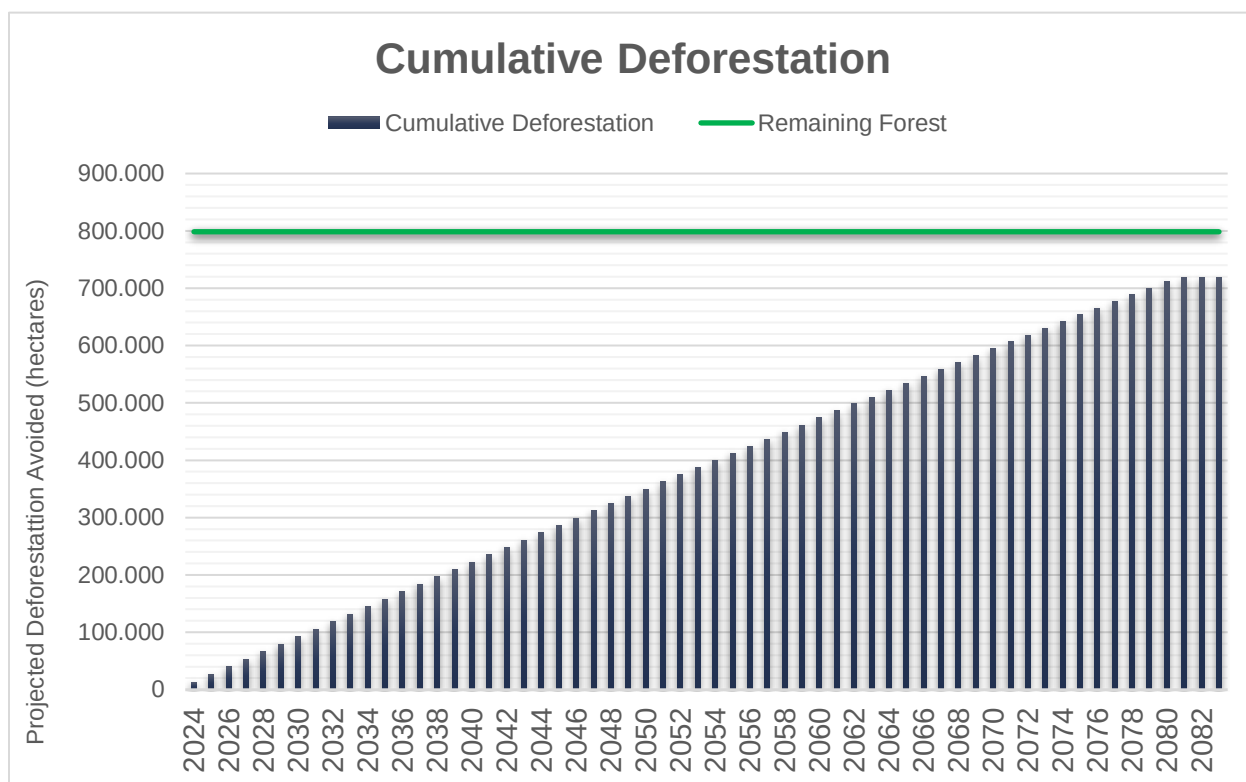


Figure 72. the author – Cumulative Deforestation in the RR.

Source: the author.

Then, the distribution among the different stratum was provided in accordance with step 5 of the Methodology - Definition of The Land-Use and Land-Cover Change Component of The Baseline.

Table 23. Annual areas of baseline deforestation in the Reference Region across the project period

Project year t	Stratum i in the reference region (ha)			Total (ha)	
	ABSLRR ₁ Forest formation	ABSLRR ₂ Savanna formation	ABSLRR ₃ Forest-Savanna Transition	ABSLRR _t Annual (ha)	ABSLRR Cumulative (ha)
2024	84	13.688	143	13.914	13.914
2025	82	13.454	140	13.676	27.590
2026	81	13.223	138	13.442	41.032
2027	79	12.996	135	13.211	54.243
2028	78	12.774	133	12.985	67.228
2029	54	12.604	104	12.761	79.989

2030	53	12.388	102	12.542	92.532
2031	52	12.175	100	12.327	104.859
2032	51	11.967	98	12.116	116.975
2033	50	11.762	97	11.909	128.884
2034	46	11.813	81	11.940	140.824
2035	46	11.610	79	11.735	152.559
2036	45	11.411	78	11.534	164.093
2037	44	11.216	77	11.336	175.430
2038	43	11.023	75	11.142	186.572
2039	54	10.582	47	10.683	197.255
2040	53	10.401	46	10.500	207.755
2041	52	10.222	45	10.320	218.075
2042	51	10.047	44	10.143	228.218
2043	51	9.875	44	9.969	238.187
2044	70	9.473	29	9.572	247.759
2045	69	9.311	29	9.408	257.167
2046	68	9.151	28	9.247	266.414
2047	66	8.994	28	9.088	275.503
2048	65	8.840	27	8.933	284.435
2049	85	8.871	28	8.984	293.419
2050	84	8.719	27	8.830	302.249
2051	82	8.570	27	8.679	310.928
2052	81	8.423	26	8.530	319.458
2053	80	8.278	26	8.384	327.842
2054	90	8.118	25	8.233	336.075
2055	89	7.979	24	8.092	344.167
2056	87	7.842	24	7.953	352.120
2057	86	7.708	24	7.817	359.937
2058	84	7.575	23	7.683	367.620
2059	87	7.351	23	7.461	375.081
2060	85	7.225	22	7.333	382.413
2061	84	7.101	22	7.207	389.621
2062	82	6.980	22	7.084	396.704
2063	81	6.860	21	6.962	403.666
2064	84	6.818	21	6.923	410.589
2065	82	6.701	20	6.804	417.393
2066	81	6.587	20	6.687	424.080
2067	79	6.474	20	6.573	430.653
2068	78	6.363	19	6.460	437.113
2069	89	6.256	16	6.362	443.475

2070	88	6.149	16	6.253	449.728
2071	86	6.044	16	6.146	455.874
2072	85	5.940	16	6.041	461.914
2073	83	5.838	15	5.937	467.852
2074	79	5.711	13	5.803	473.655
2075	77	5.613	13	5.704	479.358
2076	76	5.517	13	5.606	484.964
2077	75	5.422	13	5.510	490.474
2078	73	5.329	13	5.415	495.889
2079	73	5.122	8	5.204	501.093
2080	72	5.035	8	5.115	506.207
2081	71	4.948	8	5.027	511.234
2082	69	4.864	8	4.941	516.175
2083	68	4.780	8	4.856	521.031

Projection of the Location of Future Deforestation

Annual areas of baseline deforestation in the reference region were estimated in previous steps. In this section the analysis is about where future deforestation is most likely to happen in the baseline case in order to match the location of the projected deforestation with carbon stocks and determine the annual areas of baseline deforestation in the project area and leakage belt.

The projection of future deforestation location considers the annual deforestation estimates up to the limit of 90% of the remaining forest area, along with historical deforestation statistics applied to the stratification of the reference region into deforestation agent groups.

Based on this stratification, which organizes the territory into homogeneous landscape units for each group of agents, factor and risk maps were created. These maps consolidate the driver variables and underlying causes that define the deforestation pattern within the reference region.

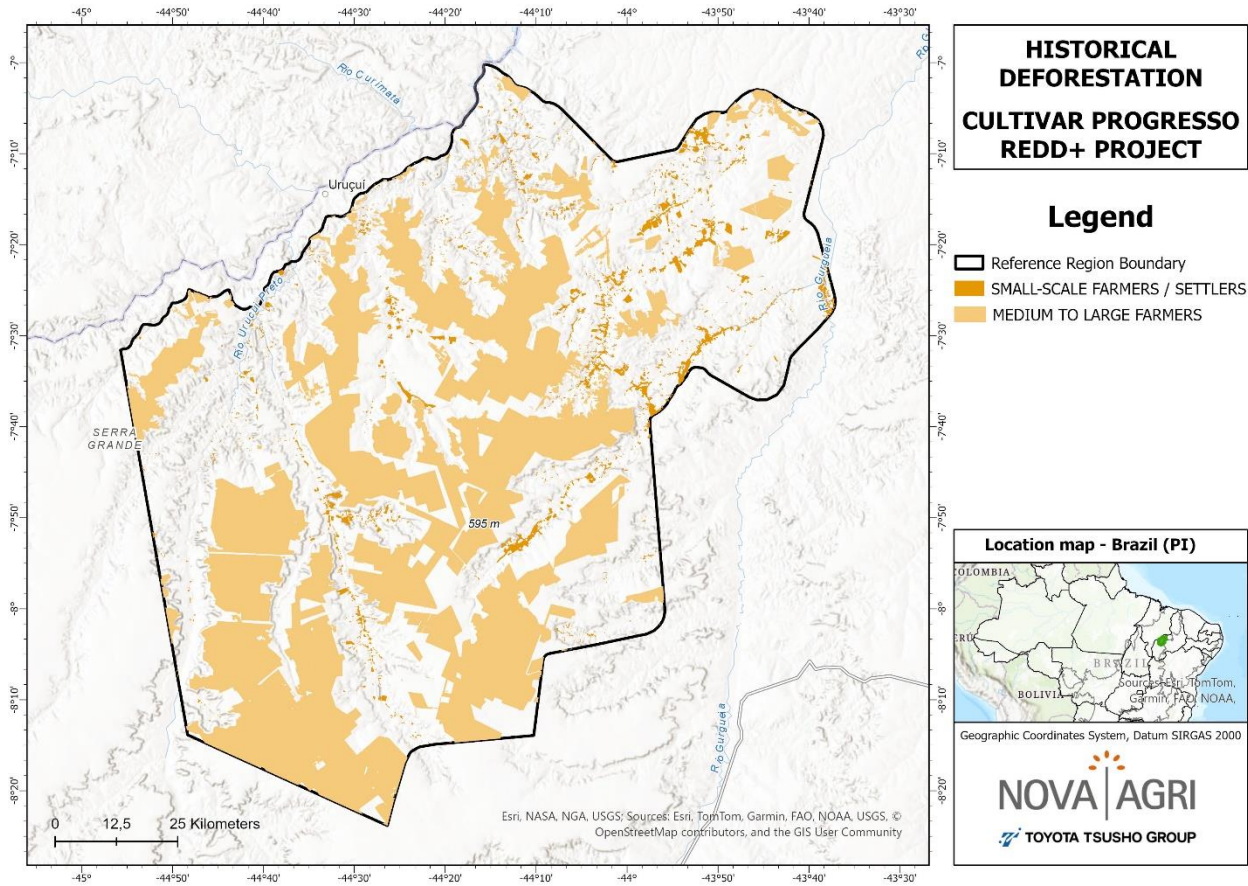


Figure 73. Agents of deforestation in the RR. Source: PRODES, the author.

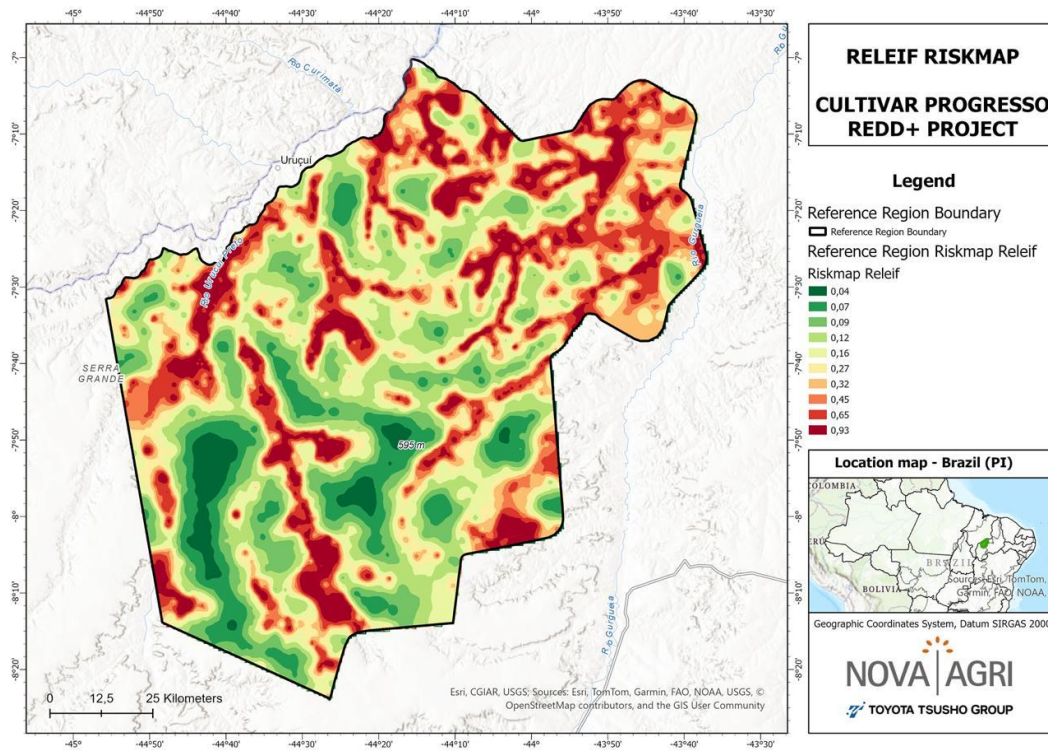


Figure 74. Deforestation risk associated with relief.

Source: the author.

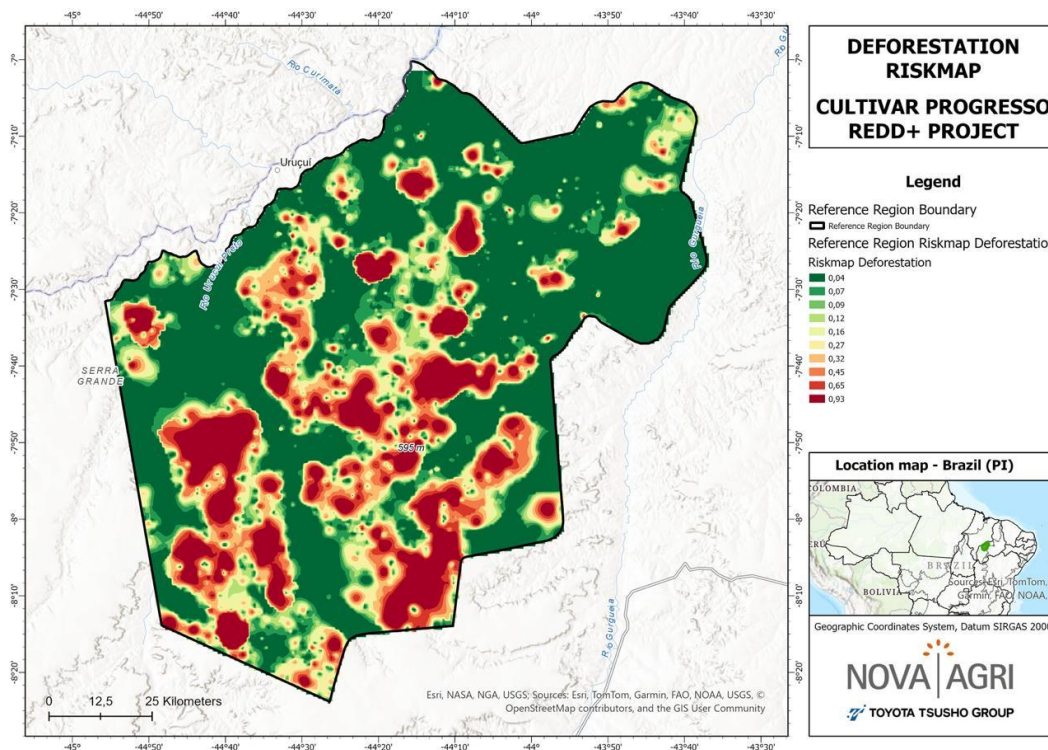


Figure 75. Deforestation Risk associated with historical deforestation.

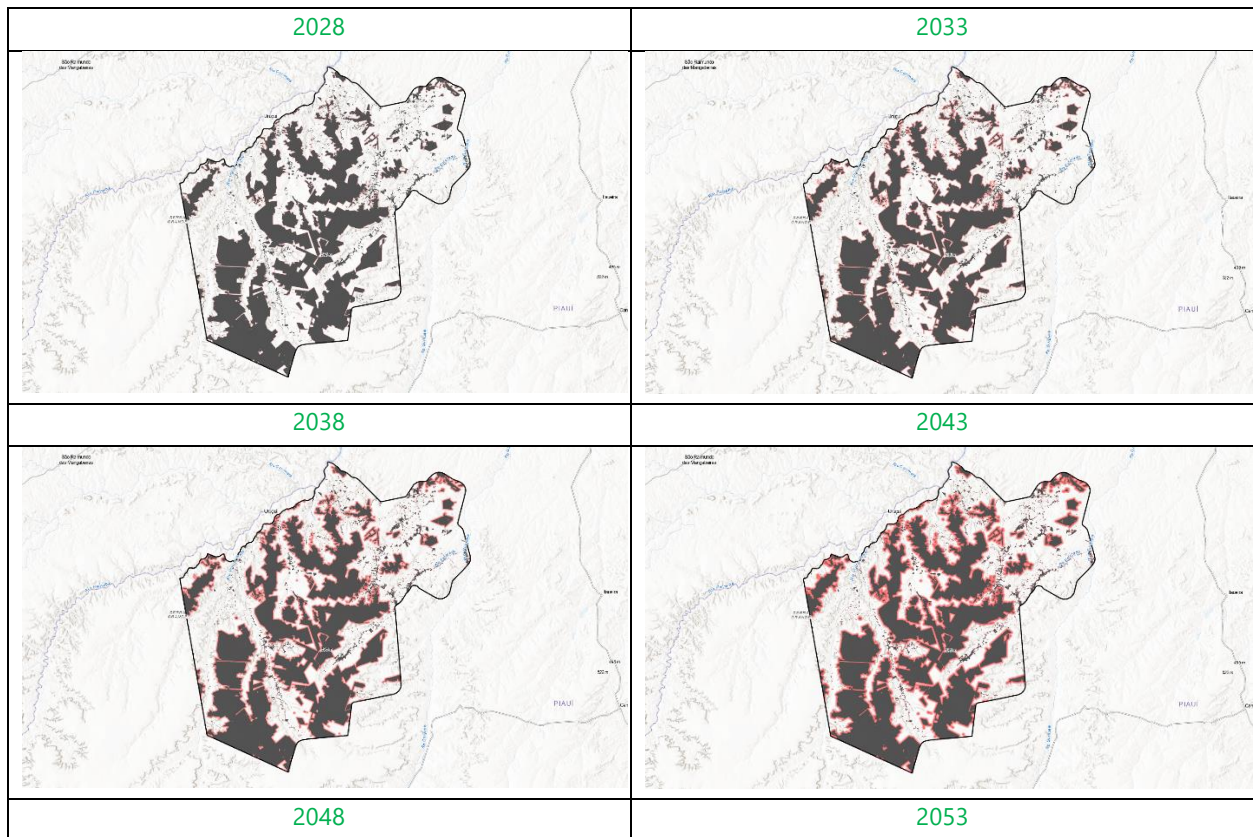
Source: the author.

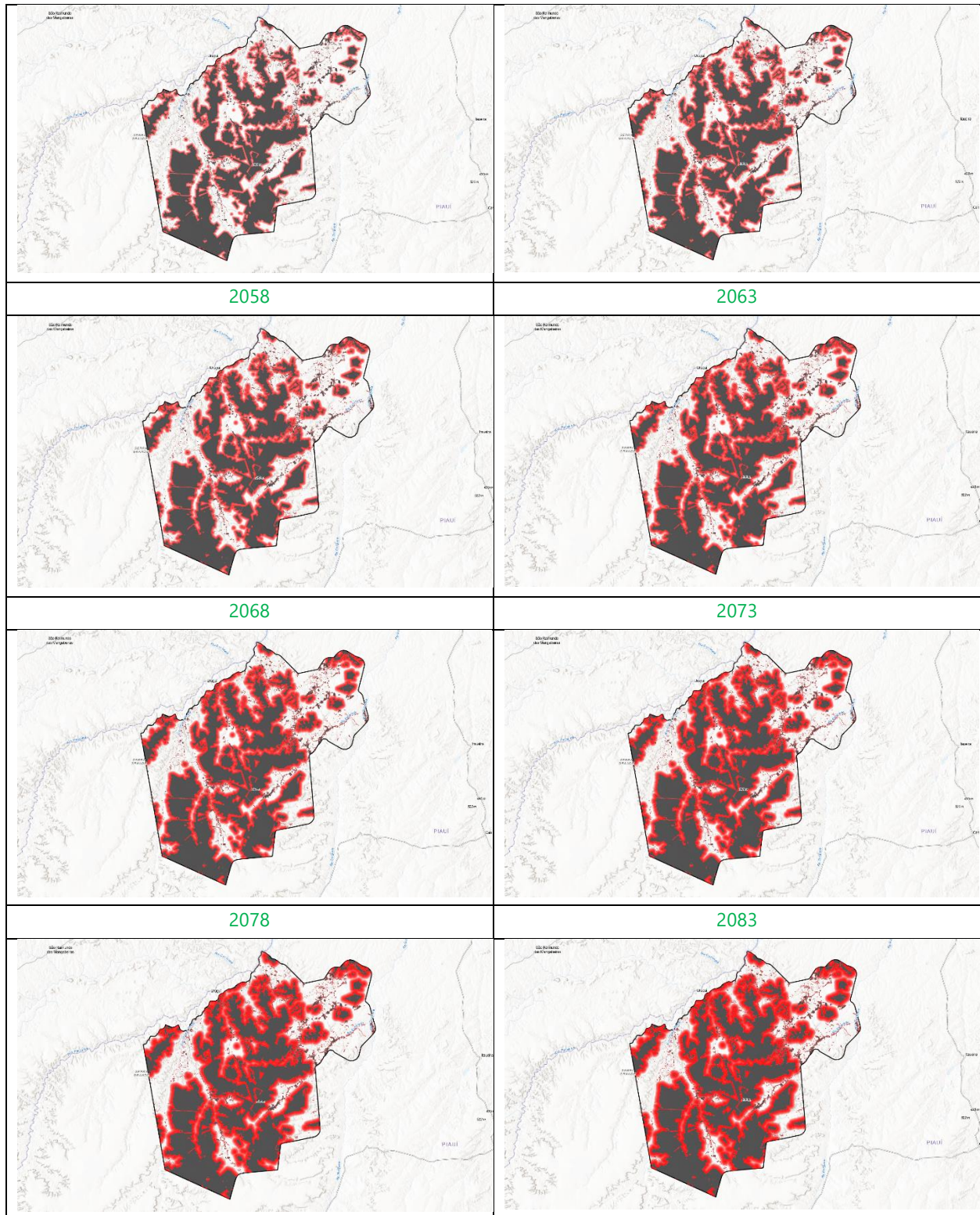
Based on the stratification of the reference region into groups of agents, the mapping of future deforestation locations was carried out using three allocation criteria that define the land use and land cover change component of the baseline: (i) future deforestation begins at the edge of already existing deforestation, (ii) the projection of future deforestation must be consistent with the baseline estimate, and (iii) the location of future deforestation must follow the proportions outlined in the deforestation risk maps.

With these criteria in place, the projection of future deforestation locations was made for a 60-year period, from 2023 to 2083 for the Reference Region as a whole.



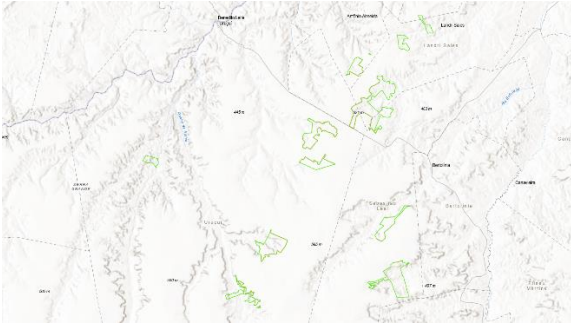
Subtitle – Mapping of Future Deforestation Location



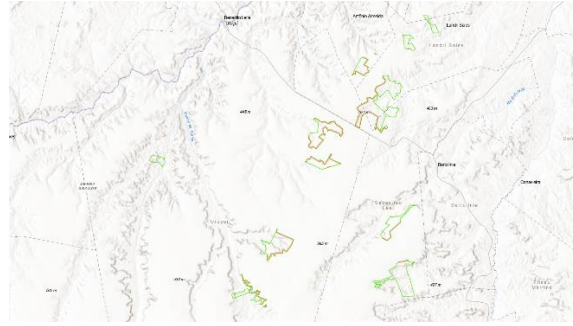


A representation of this mapping was made also focused for the Project Area and Leakage Belt as seen on the images below:

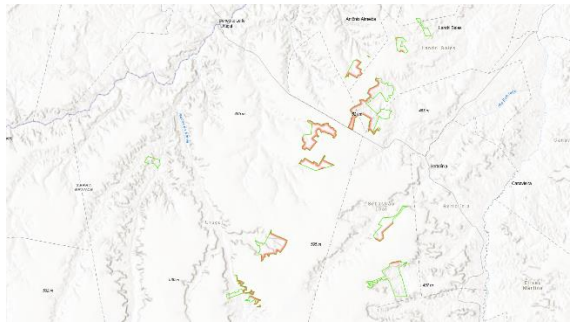
2028



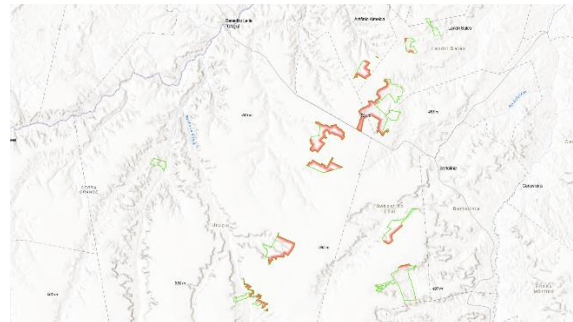
2033



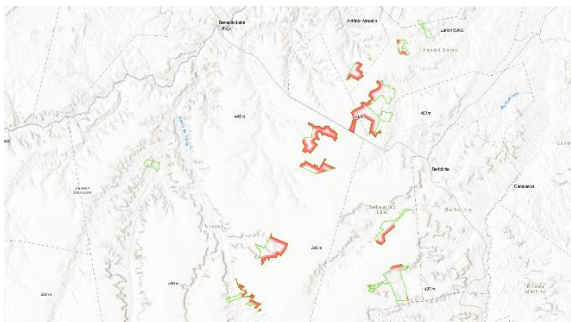
2038



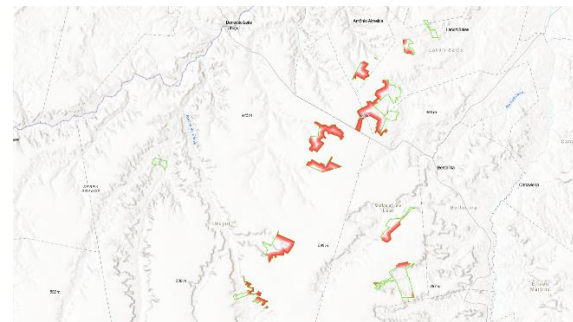
2043



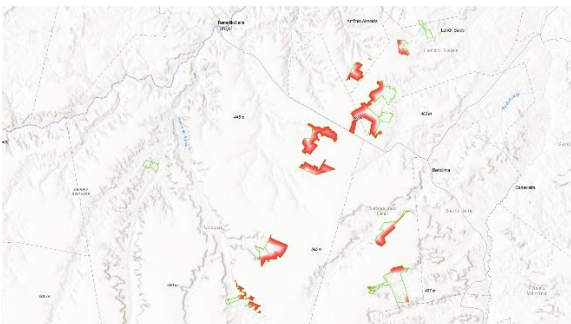
2048



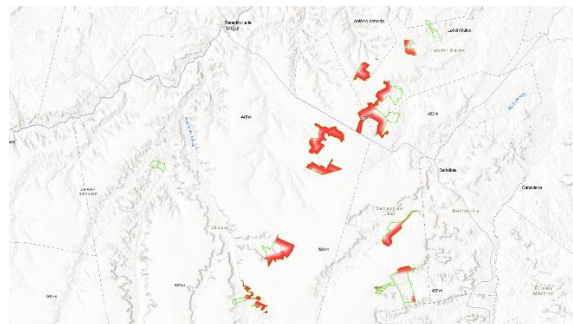
2053



2058



2063

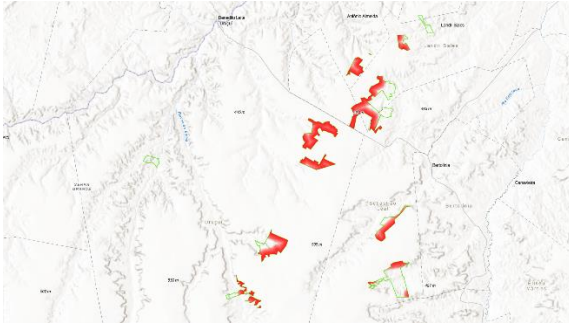


2068

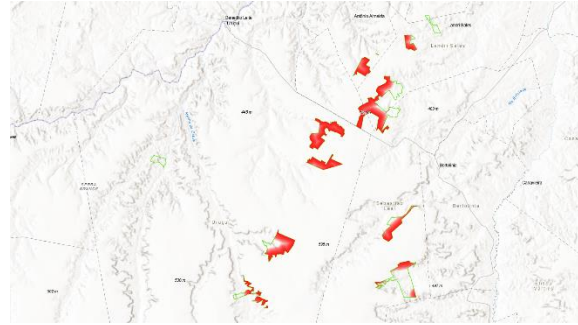


2073

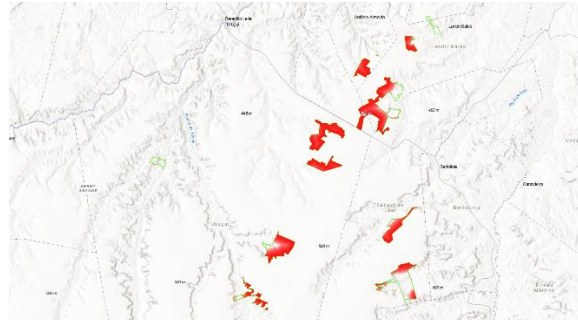
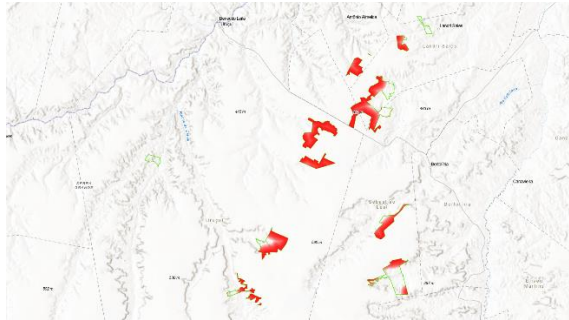




2078



2083



The application of the allocation criteria that defined the land-use and land-cover change component based on the stratification of the reference region into agent groups for mapping the location of future deforestation, when overlaid on the project area, also resulted in a volume lower than the deforestation estimate projected in the baseline of the project.

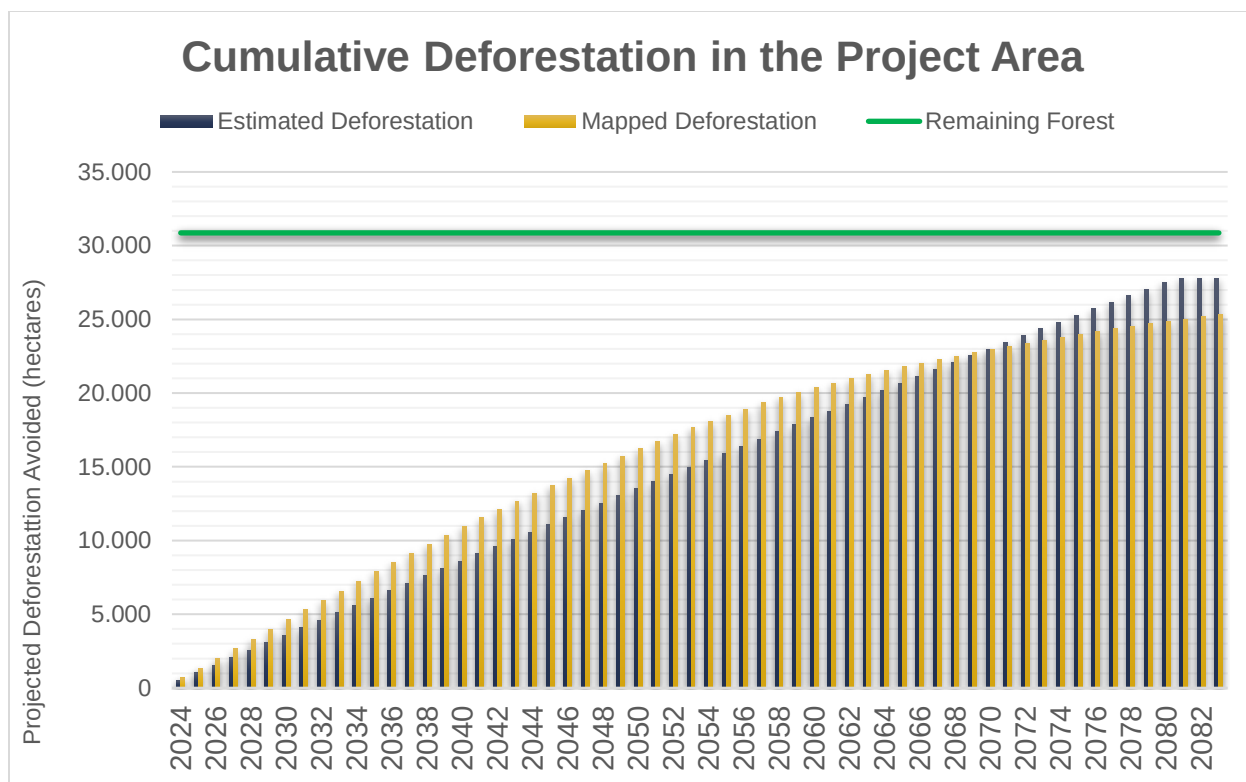


Table 24. Annual areas of baseline deforestation in the project area

Project year t	Stratum i in the project area (ha)				Total (ha)	
	ABSLPA 1 Wooded Savannah	ABSLPA 2 Savannah Forest	ABSLPA 3 Steppic-Savanna Forest	ABSLPA 4 Grassy-Woody Savannah	Annual ABSLPAt	Cumulative ABSLPA
2024	658	14	8	0	680	680
2025	647	14	8	0	668	1.348
2026	636	13	8	0	657	2.005
2027	625	13	8	0	646	2.651
2028	614	13	8	0	635	3.285
2029	622	37	21	0	681	3.966
2030	612	36	21	0	669	4.635
2031	601	36	21	0	657	5.292
2032	591	35	20	0	646	5.938
2033	581	35	20	0	635	6.573
2034	568	47	45	1	661	7.234
2035	558	46	44	1	649	7.883
2036	548	45	44	1	638	8.522

2037	539	44	43	1	627	9.149
2038	530	44	42	1	617	9.765
2039	500	41	56	2	599	10.364
2040	491	41	55	2	589	10.953
2041	483	40	54	2	579	11.532
2042	475	39	53	2	569	12.101
2043	467	39	52	2	559	12.660
2044	454	21	53	0	529	13.188
2045	446	21	52	0	520	13.708
2046	438	21	51	0	511	14.218
2047	431	20	51	0	502	14.720
2048	423	20	50	0	493	15.214
2049	425	14	65	0	504	15.717
2050	418	14	64	0	495	16.212
2051	410	13	63	0	486	16.699
2052	403	13	62	0	478	17.177
2053	396	13	61	0	470	17.647
2054	349	11	72	0	433	18.079
2055	343	11	71	0	425	18.504
2056	338	10	70	0	418	18.922
2057	332	10	69	0	411	19.333
2058	326	10	68	0	404	19.736
2059	255	0	60	0	315	20.051
2060	250	0	59	0	310	20.361
2061	246	0	58	0	304	20.665
2062	242	0	57	0	299	20.964
2063	238	0	56	0	294	21.258
2064	210	0	50	0	260	21.518
2065	206	0	49	0	255	21.773
2066	203	0	48	0	251	22.024
2067	199	0	47	0	247	22.271
2068	196	0	47	0	242	22.513
2069	174	0	45	0	219	22.732
2070	171	0	44	0	215	22.948
2071	168	0	43	0	212	23.159
2072	165	0	43	0	208	23.368
2073	163	0	42	0	205	23.572
2074	158	0	41	0	199	23.772
2075	156	0	40	0	196	23.968
2076	153	0	40	0	193	24.160

2077	150	0	39	0	189	24.350
2078	148	0	38	0	186	24.536
2079	133	0	33	0	166	24.702
2080	131	0	33	0	163	24.866
2081	128	0	32	0	161	25.026
2082	126	0	32	0	158	25.184
2083	124	0	31	0	155	25.339

Table 25 shows the projected annual deforestation for each stratum in the leakage belt across the project crediting period, represented by the variable ABSLLK.

Table 25. Annual areas of baseline deforestation in the leakage belt

Project year t	Stratum i in the leakage belt (ha)			Total (ha)	
	ABSLLK ₁ Forest formation	ABSLLK ₂ Savanna formation	ABSLLK ₃ Forest- Savanna Transition	Annual ABSLLK _t	Cumulative ABSLLK
2024	57	7.442	0	7.499	7.499
2025	56	7.314	0	7.370	14.869
2026	55	7.189	0	7.244	22.113
2027	54	7.066	0	7.120	29.233
2028	53	6.944	0	6.998	36.231
2029	31	6.999	0	7.030	43.261
2030	30	6.879	0	6.910	50.170
2031	30	6.761	0	6.791	56.962
2032	29	6.646	0	6.675	63.636
2033	29	6.532	0	6.560	70.197
2034	23	6.455	0	6.478	76.674
2035	22	6.344	0	6.367	83.041
2036	22	6.236	0	6.258	89.299
2037	22	6.129	0	6.150	95.449
2038	21	6.024	0	6.045	101.494
2039	31	5.609	0	5.640	107.134
2040	30	5.513	0	5.544	112.677
2041	30	5.419	0	5.449	118.126
2042	29	5.326	0	5.355	123.481
2043	29	5.234	0	5.263	128.744

2044	41	4.865	0	4.906	133.650
2045	40	4.781	0	4.822	138.472
2046	40	4.699	0	4.739	143.211
2047	39	4.619	0	4.658	147.868
2048	38	4.540	0	4.578	152.446
2049	54	4.489	0	4.542	156.989
2050	53	4.412	0	4.464	161.453
2051	52	4.336	0	4.388	165.841
2052	51	4.262	0	4.313	170.154
2053	50	4.189	0	4.239	174.393
2054	57	4.048	0	4.105	178.497
2055	56	3.978	0	4.034	182.532
2056	55	3.910	0	3.965	186.497
2057	54	3.843	0	3.897	190.395
2058	53	3.777	0	3.831	194.225
2059	54	3.537	0	3.590	197.815
2060	53	3.476	0	3.529	201.344
2061	52	3.416	0	3.468	204.813
2062	51	3.358	0	3.409	208.221
2063	50	3.300	0	3.350	211.572
2064	52	3.196	0	3.249	214.821
2065	52	3.141	0	3.193	218.014
2066	51	3.088	0	3.138	221.152
2067	50	3.035	0	3.084	224.236
2068	49	2.983	0	3.032	227.268
2069	52	2.886	0	2.938	230.206
2070	51	2.837	0	2.887	233.093
2071	50	2.788	0	2.838	235.931
2072	49	2.740	0	2.789	238.720
2073	48	2.693	0	2.741	241.462
2074	43	2.601	0	2.644	244.106
2075	42	2.557	0	2.599	246.705
2076	42	2.513	0	2.555	249.260
2077	41	2.470	0	2.511	251.771
2078	40	2.427	0	2.468	254.238
2079	41	2.312	0	2.352	256.591
2080	40	2.272	0	2.312	258.903
2081	39	2.233	0	2.272	261.175
2082	38	2.195	0	2.233	263.408
2083	38	2.157	0	2.195	265.604

Calculation of baseline activity data per forest class

The following is in accordance with step 5.1 of the Methodology: “Calculation of baseline activity data per forest class”, in which it is stipulated that the previously created maps of annual baseline deforestation and LU/LC map be combined, producing a map showing deforestation per class in the baseline case. The number of hectares deforested in each forest class, within the reference region, project area and leakage belt are found in *Table 26– Table 28* below.

According to the baseline projections of the present project, accumulated deforestation from 2023 – 2083 in the Project Area will sum to 23.707 ha (*Table 26* below). The LU/LC within the project period, caused by baseline deforestation, consisted of the initial class of dominant vegetation in ecological tension being converted to the final LU/LC class of “non-forest”.

Table 26. Annual areas deforested per forest class icl within the reference region in the baseline case (baseline activity data per forest class)

IDicl Name	1 Forest Formation	2 Savanna Formation	3 Forest-Savanna Transition	Total baseline deforestation in the reference region	
				ABSLRR _t Annual (ha)	ABSLRR Cumulative (ha)
Project year <i>t</i>	ha	ha	ha		
2024	84	13.688	143	13.914	13.914
2025	82	13.454	140	13.676	27.590
2026	81	13.223	138	13.442	41.032
2027	79	12.996	135	13.211	54.243
2028	78	12.774	133	12.985	67.228
2029	54	12.604	104	12.761	79.989
2030	53	12.388	102	12.542	92.532
2031	52	12.175	100	12.327	104.859
2032	51	11.967	98	12.116	116.975
2033	50	11.762	97	11.909	128.884
2034	46	11.813	81	11.940	140.824
2035	46	11.610	79	11.735	152.559
2036	45	11.411	78	11.534	164.093
2037	44	11.216	77	11.336	175.430
2038	43	11.023	75	11.142	186.572
2039	54	10.582	47	10.683	197.255
2040	53	10.401	46	10.500	207.755
2041	52	10.222	45	10.320	218.075

2042	51	10.047	44	10.143	228.218
2043	51	9.875	44	9.969	238.187
2044	70	9.473	29	9.572	247.759
2045	69	9.311	29	9.408	257.167
2046	68	9.151	28	9.247	266.414
2047	66	8.994	28	9.088	275.503
2048	65	8.840	27	8.933	284.435
2049	85	8.871	28	8.984	293.419
2050	84	8.719	27	8.830	302.249
2051	82	8.570	27	8.679	310.928
2052	81	8.423	26	8.530	319.458
2053	80	8.278	26	8.384	327.842
2054	90	8.118	25	8.233	336.075
2055	89	7.979	24	8.092	344.167
2056	87	7.842	24	7.953	352.120
2057	86	7.708	24	7.817	359.937
2058	84	7.575	23	7.683	367.620
2059	87	7.351	23	7.461	375.081
2060	85	7.225	22	7.333	382.413
2061	84	7.101	22	7.207	389.621
2062	82	6.980	22	7.084	396.704
2063	81	6.860	21	6.962	403.666
2064	84	6.818	21	6.923	410.589
2065	82	6.701	20	6.804	417.393
2066	81	6.587	20	6.687	424.080
2067	79	6.474	20	6.573	430.653
2068	78	6.363	19	6.460	437.113
2069	89	6.256	16	6.362	443.475
2070	88	6.149	16	6.253	449.728
2071	86	6.044	16	6.146	455.874
2072	85	5.940	16	6.041	461.914
2073	83	5.838	15	5.937	467.852
2074	79	5.711	13	5.803	473.655
2075	77	5.613	13	5.704	479.358
2076	76	5.517	13	5.606	484.964
2077	75	5.422	13	5.510	490.474
2078	73	5.329	13	5.415	495.889
2079	73	5.122	8	5.204	501.093
2080	72	5.035	8	5.115	506.207
2081	71	4.948	8	5.027	511.234

2082	69	4.864	8	4.941	516.175
2083	68	4.780	8	4.856	521.031

Table 27. Annual areas deforested per forest class *icl* within the project area in the baseline case (baseline activity data per forest class)

Area deforested per forest class <i>icl</i> within the project area					Total baseline deforestation in the project area	
<i>IDicl</i>	1	2	3	4	Annual ABSLPat (ha)	ABSLPat cumulative (ha)
Name	Wooded Savannah Sa	Savannah Forest Sd	Steppic-Savanna Forest - Td	Grassy-Woody Savannah - Tg		
Project year <i>t</i>	ha	ha	ha	ha		
2024	658	14	8	0	680	680
2025	647	14	8	0	668	1.348
2026	636	13	8	0	657	2.005
2027	625	13	8	0	646	2.651
2028	614	13	8	0	635	3.285
2029	622	37	21	0	681	3.966
2030	612	36	21	0	669	4.635
2031	601	36	21	0	657	5.292
2032	591	35	20	0	646	5.938
2033	581	35	20	0	635	6.573
2034	568	47	45	1	660	7.233
2035	558	46	44	1	648	7.882
2036	548	45	44	1	637	8.519
2037	539	44	43	1	626	9.145
2038	530	44	42	1	616	9.761
2039	500	41	56	2	597	10.358
2040	491	41	55	2	587	10.945
2041	483	40	54	2	577	11.522
2042	475	39	53	2	567	12.089
2043	467	39	52	2	557	12.646
2044	454	21	53	0	528	13.174
2045	446	21	52	0	519	13.694
2046	438	21	51	0	510	14.204
2047	431	20	51	0	502	14.706

2048	423	20	50	0	493	15.198
2049	425	14	65	0	504	15.702
2050	418	14	64	0	495	16.197
2051	410	13	63	0	486	16.683
2052	403	13	62	0	478	17.162
2053	396	13	61	0	470	17.632
2054	349	11	72	0	433	18.064
2055	343	11	71	0	425	18.489
2056	338	10	70	0	418	18.907
2057	332	10	69	0	411	19.318
2058	326	10	68	0	404	19.721
2059	255	0	60	0	315	20.036
2060	250	0	59	0	310	20.346
2061	246	0	58	0	304	20.650
2062	242	0	57	0	299	20.949
2063	238	0	56	0	294	21.243
2064	210	0	50	0	260	21.503
2065	206	0	49	0	255	21.758
2066	203	0	48	0	251	22.009
2067	199	0	47	0	247	22.256
2068	196	0	47	0	242	22.498
2069	174	0	45	0	219	22.717
2070	171	0	44	0	215	22.933
2071	168	0	43	0	212	23.144
2072	165	0	43	0	208	23.353
2073	163	0	42	0	205	23.557
2074	158	0	41	0	199	23.757
2075	156	0	40	0	196	23.953
2076	153	0	40	0	193	24.145
2077	150	0	39	0	189	24.335
2078	148	0	38	0	186	24.521
2079	133	0	33	0	166	24.687
2080	131	0	33	0	163	24.851
2081	128	0	32	0	161	25.011
2082	126	0	32	0	158	25.169
2083	124	0	31	0	155	25.324

Table 28. Annual areas deforested per forest class *icl* within the leakage belt in the baseline case (baseline activity data per forest class)

Area deforested per forest class <i>icl</i> within the leakage belt				Total baseline deforestation in the leakage belt	
<i>IDicl</i>	1	2	3		
Name	Forest Formation	Savanna Formation	Forest-Savanna Transition	Annual ABSLLKt (ha)	ABSLLKt cumulative (ha)
Project year <i>t</i>	ha	ha	ha		
2024	57	7.442	0	7.499	7.499
2025	56	7.314	0	7.370	14.869
2026	55	7.189	0	7.244	22.113
2027	54	7.066	0	7.120	29.233
2028	53	6.944	0	6.998	36.231
2029	31	6.999	0	7.030	43.261
2030	30	6.879	0	6.910	50.170
2031	30	6.761	0	6.791	56.962
2032	29	6.646	0	6.675	63.636
2033	29	6.532	0	6.560	70.197
2034	23	6.455	0	6.478	76.674
2035	22	6.344	0	6.367	83.041
2036	22	6.236	0	6.258	89.299
2037	22	6.129	0	6.150	95.449
2038	21	6.024	0	6.045	101.494
2039	31	5.609	0	5.640	107.134
2040	30	5.513	0	5.544	112.677
2041	30	5.419	0	5.449	118.126
2042	29	5.326	0	5.355	123.481
2043	29	5.234	0	5.263	128.744
2044	41	4.865	0	4.906	133.650
2045	40	4.781	0	4.822	138.472
2046	40	4.699	0	4.739	143.211
2047	39	4.619	0	4.658	147.868
2048	38	4.540	0	4.578	152.446
2049	54	4.489	0	4.542	156.989
2050	53	4.412	0	4.464	161.453
2051	52	4.336	0	4.388	165.841
2052	51	4.262	0	4.313	170.154
2053	50	4.189	0	4.239	174.393

2054	57	4.048	0	4.105	178.497
2055	56	3.978	0	4.034	182.532
2056	55	3.910	0	3.965	186.497
2057	54	3.843	0	3.897	190.395
2058	53	3.777	0	3.831	194.225
2059	54	3.537	0	3.590	197.815
2060	53	3.476	0	3.529	201.344
2061	52	3.416	0	3.468	204.813
2062	51	3.358	0	3.409	208.221
2063	50	3.300	0	3.350	211.572
2064	52	3.196	0	3.249	214.821
2065	52	3.141	0	3.193	218.014
2066	51	3.088	0	3.138	221.152
2067	50	3.035	0	3.084	224.236
2068	49	2.983	0	3.032	227.268
2069	52	2.886	0	2.938	230.206
2070	51	2.837	0	2.887	233.093
2071	50	2.788	0	2.838	235.931
2072	49	2.740	0	2.789	238.720
2073	48	2.693	0	2.741	241.462
2074	43	2.601	0	2.644	244.106
2075	42	2.557	0	2.599	246.705
2076	42	2.513	0	2.555	249.260
2077	41	2.470	0	2.511	251.771
2078	40	2.427	0	2.468	254.238
2079	41	2.312	0	2.352	256.591
2080	40	2.272	0	2.312	258.903
2081	39	2.233	0	2.272	261.175
2082	38	2.195	0	2.233	263.408
2083	38	2.157	0	2.195	265.604

Calculation of baseline activity data per post-deforestation forest class

The following is in accordance with step 5.2 of the Methodology: “Calculation of baseline activity data per post-deforestation forest class.” As all of the initial classes represented in the tables above were transformed into non-forest (final post-deforestation class) in the considered baseline, the annual values corresponding to the final classes are the same as those as the initial class in *Table 29– Table 31* below, which depict baseline activity data per post-deforestation forest class in the reference region, project area, and leakage belt, respectively.

Table 29. Annual areas deforested in each zone within the reference region in the baseline case (baseline activity data zone)

Area established after deforestation per zone within the reference region			
IDct Name Project year t	2 Non forest ha	Total baseline deforestation in the reference region	
		ABSLRRt Annual (ha)	ABSLRRt cumulative (ha)
2024	13.914	13.914	13.914
2025	13.676	13.676	27.590
2026	13.442	13.442	41.032
2027	13.211	13.211	54.243
2028	12.985	12.985	67.228
2029	12.761	12.761	79.989
2030	12.542	12.542	92.532
2031	12.327	12.327	104.859
2032	12.116	12.116	116.975
2033	11.909	11.909	128.884
2034	11.940	11.940	140.824
2035	11.735	11.735	152.559
2036	11.534	11.534	164.093
2037	11.336	11.336	175.430
2038	11.142	11.142	186.572
2039	10.683	10.683	197.255
2040	10.500	10.500	207.755
2041	10.320	10.320	218.075
2042	10.143	10.143	228.218
2043	9.969	9.969	238.187
2044	9.572	9.572	247.759
2045	9.408	9.408	257.167
2046	9.247	9.247	266.414
2047	9.088	9.088	275.503
2048	8.933	8.933	284.435
2049	8.984	8.984	293.419
2050	8.830	8.830	302.249
2051	8.679	8.679	310.928
2052	8.530	8.530	319.458
2053	8.384	8.384	327.842

2054	8.233	8.233	336.075
2055	8.092	8.092	344.167
2056	7.953	7.953	352.120
2057	7.817	7.817	359.937
2058	7.683	7.683	367.620
2059	7.461	7.461	375.081
2060	7.333	7.333	382.413
2061	7.207	7.207	389.621
2062	7.084	7.084	396.704
2063	6.962	6.962	403.666
2064	6.923	6.923	410.589
2065	6.804	6.804	417.393
2066	6.687	6.687	424.080
2067	6.573	6.573	430.653
2068	6.460	6.460	437.113
2069	6.362	6.362	443.475
2070	6.253	6.253	449.728
2071	6.146	6.146	455.874
2072	6.041	6.041	461.914
2073	5.937	5.937	467.852
2074	5.803	5.803	473.655
2075	5.704	5.704	479.358
2076	5.606	5.606	484.964
2077	5.510	5.510	490.474
2078	5.415	5.415	495.889
2079	5.204	5.204	501.093
2080	5.115	5.115	506.207
2081	5.027	5.027	511.234
2082	4.941	4.941	516.175
2083	4.856	4.856	521.031

Table 30. Annual areas deforested in each zone within the project area in the baseline case (baseline activity data zone)

Area established after deforestation per zone within the project area		Total baseline deforestation in the project area	
<i>IDct</i>	2		
Name	Non forest	ABSLPat Annual (ha)	ABSLPat cumulative (ha)

Project year <i>t</i>	ha		
2024	680	680	680
2025	668	668	1.348
2026	657	657	2.005
2027	646	646	2.651
2028	635	635	3.285
2029	681	681	3.966
2030	669	669	4.635
2031	657	657	5.292
2032	646	646	5.938
2033	635	635	6.573
2034	661	661	7.234
2035	649	649	7.883
2036	638	638	8.522
2037	627	627	9.149
2038	617	617	9.765
2039	599	599	10.364
2040	589	589	10.953
2041	579	579	11.532
2042	569	569	12.101
2043	559	559	12.660
2044	529	529	13.188
2045	520	520	13.708
2046	511	511	14.218
2047	502	502	14.720
2048	493	493	15.214
2049	504	504	15.717
2050	495	495	16.212
2051	486	486	16.699
2052	478	478	17.177
2053	470	470	17.647
2054	433	433	18.079
2055	425	425	18.504
2056	418	418	18.922
2057	411	411	19.333
2058	404	404	19.736
2059	315	315	20.051
2060	310	310	20.361
2061	304	304	20.665

2062	299	299	20.964
2063	294	294	21.258
2064	260	260	21.518
2065	255	255	21.773
2066	251	251	22.024
2067	247	247	22.271
2068	242	242	22.513
2069	219	219	22.732
2070	215	215	22.948
2071	212	212	23.159
2072	208	208	23.368
2073	205	205	23.572
2074	199	199	23.772
2075	196	196	23.968
2076	193	193	24.160
2077	189	189	24.350
2078	186	186	24.536
2079	166	166	24.702
2080	163	163	24.866
2081	161	161	25.026
2082	158	158	25.184
2083	155	155	25.339

Table 31. Annual areas deforested in each zone within the leakage belt in the baseline case (baseline activity data zone)

Area established after deforestation per zone within the leakage belt		Total baseline deforestation in the leakage belt	
<i>IDct</i>	2		
Name	Non forest	ABSLKt Annual (ha)	ABSLKt cumulative (ha)
Project year <i>t</i>	ABSLK _{i,t}		
2024	7.499	7.499	7.499
2025	7.370	7.370	14.869
2026	7.244	7.244	22.113
2027	7.120	7.120	29.233
2028	6.998	6.998	36.231
2029	7.030	7.030	43.261

2030	6.910	6.910	50.170
2031	6.791	6.791	56.962
2032	6.675	6.675	63.636
2033	6.560	6.560	70.197
2034	6.478	6.478	76.674
2035	6.367	6.367	83.041
2036	6.258	6.258	89.299
2037	6.150	6.150	95.449
2038	6.045	6.045	101.494
2039	5.640	5.640	107.134
2040	5.544	5.544	112.677
2041	5.449	5.449	118.126
2042	5.355	5.355	123.481
2043	5.263	5.263	128.744
2044	4.906	4.906	133.650
2045	4.822	4.822	138.472
2046	4.739	4.739	143.211
2047	4.658	4.658	147.868
2048	4.578	4.578	152.446
2049	4.542	4.542	156.989
2050	4.464	4.464	161.453
2051	4.388	4.388	165.841
2052	4.313	4.313	170.154
2053	4.239	4.239	174.393
2054	4.105	4.105	178.497
2055	4.034	4.034	182.532
2056	3.965	3.965	186.497
2057	3.897	3.897	190.395
2058	3.831	3.831	194.225
2059	3.590	3.590	197.815
2060	3.529	3.529	201.344
2061	3.468	3.468	204.813
2062	3.409	3.409	208.221
2063	3.350	3.350	211.572
2064	3.249	3.249	214.821
2065	3.193	3.193	218.014
2066	3.138	3.138	221.152
2067	3.084	3.084	224.236
2068	3.032	3.032	227.268
2069	2.938	2.938	230.206

2070	2.887	2.887	233.093
2071	2.838	2.838	235.931
2072	2.789	2.789	238.720
2073	2.741	2.741	241.462
2074	2.644	2.644	244.106
2075	2.599	2.599	246.705
2076	2.555	2.555	249.260
2077	2.511	2.511	251.771
2078	2.468	2.468	254.238
2079	2.352	2.352	256.591
2080	2.312	2.312	258.903
2081	2.272	2.272	261.175
2082	2.233	2.233	263.408
2083	2.195	2.195	265.604

Estimation of baseline carbon stock changes and non-CO2 emissions

The following is in accordance with step 6 of the methodology, specifically, 6.1.1 Estimation of the average carbon stocks of each LU/LC class, the goal of which is to finalize the baseline assessment by calculating the baseline carbon stock changes.

The estimation of the carbon stored in the vegetation present in the project area was based on information from the Fourth National Communication of Brazil to the United Nations Framework Convention on Climate Change and the vegetation map available at System of National Emissions Registration - SIRENE¹¹⁷.

Table 32 below shows the density of carbon for each type of vegetation present in the CULTIVAR PROGRESSO REDD+ Reference Region as a whole.

Table 32. Carbon Stocks for Each Vegetation Class

Vegetation acronym	Phytophysognomies	AGB (tC/ha)	BGB (tC/ha)	Sources
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¹¹⁷ The Brazilian government established the National Emissions Registration System - SIRENE as an MRV system (measurement, reporting and verification) for emissions, to give the continuity and accessibility to the results of the National Inventory of Anthropogenic Emissions by Sources and Removals by Sinks of Greenhouse Gases not Controlled by the Montreal Protocol. The methodological basis for calculating emissions are the guidance documents and methodological guidelines prepared by the Intergovernmental Panel on Climate Change - IPCC. Currently, SIRENE can access the time series of emissions referring to the latest results published in the National Inventory, as part of the Fourth National Communication of Brazil to the United Nations Framework Convention on Climate Change. Access link: <http://sirene.mctic.gov.br/>

Fa	Alluvial Semi-deciduous Seasonal Forest	65,4	11,4	Fourth National Communication of Brazil to the United Nations Framework Convention on Climate Change Brazil's National Forest Reference Emission Level for Results-based Payments for REDD+ under the United Nations Framework Convention on Climate Change (FREL Brazil, March 2024)
Fs	Sub-montane Semi-deciduous Seasonal Forest	61,3	19,3	
S	Savannah	23,6	17,8	
Sa	Wooded Savannah	13,7	28,0	
Sd	Savannah Forest	33,3	7,3	
Sp	Savannah Park	7,4	17,6	
Ta	Wooded Steppic-Savannah	9,6	5,8	
Td	Steppic-Savannah Forest	26,0	9,6	
Tg	Grassy-Woody Savannah	2,7	8,2	
Tp	Steppic-Savannah Park	6,0	19,9	
SNs	Contact Savanna / Seasonal Forest Enclave	40,7	11,6	
SNts	Contact Savanna / Seasonal Forest Ecotone	23,6	17,8	

STs	Contact Savanna / Steppic-Savannah	23,6	17,8
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This presents the forest phytophysiognomies considered in the Cerrado biome, for the construction of the FREL (Forest Reference Emission Level), and the respective carbon stocks for each main carbon pool.

Table 33 below shows the carbon density identified in the vegetation maps provided by SIRENE and the result of the weighting made based on the area of each type of vegetation present in the project area.

Table 33. Carbon Density present in the RR, the respective areas, and the result of the weighting.

Vegetation acronym		AGB (tC/ha)	BGB (tC/ha)	Area (ha)	weighted AGB (tC/ha)	weighted BGB (tC/ha)
Forest Formations	Fa	65,4	11,4	793	61,5	19,0
	Fs	61,3	19,3	19.023		
Savanna Formations	S	23,6	17,8	608	15,9	24,0
	Sa	13,7	28,0	998.744		
	Sd	33,3	7,3	53.443		
	Sp	7,4	17,6	1		
	Ta	9,6	5,8	20.472		

Forest-Savanna Transition	Td	26,0	9,6	168.907	24,5	17,5
	Tg	2,7	8,2	7.754		
	Tp	6,0	19,9	25.713		
	SNs	40,7	11,6	298		
	SNts	23,6	17,8	31		
	STs	23,6	17,8	5.485		

Thus, the Carbon stocks per hectare of initial forest classes icl existing in the project area and leakage belt are found in *Table 34* to *Table 39* below. To conclude this step, the area-weighted average carbon stocks of the post deforestation LU/LC classes existing within each zone are displayed in *table 33*.

Table 34. Carbon stocks of initial forest classes (icl) existing in the project area (Class ID 1)

Project year	Initial class Wooded Savannah - Sa 1		
	Name IDcl	Cbbicl C stock	Ctoticl C stock
	Cabicl C stock	Cbbicl C stock	Ctoticl C stock
	tCO _{2e}	tCO _{2e}	tCO _{2e}
2024	30.675	6.258	36.932
2025	30.149	12.408	42.557
2026	29.632	18.453	48.085
2027	29.124	24.394	53.518
2028	28.625	30.234	58.859
2029	29.007	36.151	65.158
2030	28.510	41.967	70.477

2031	28.021	47.684	75.705
2032	27.541	53.302	80.843
2033	27.069	58.824	85.893
2034	26.468	57.966	84.434
2035	26.014	57.122	83.137
2036	25.568	56.293	81.862
2037	25.130	55.479	80.609
2038	24.699	54.678	79.377
2039	23.308	53.515	76.823
2040	22.909	52.373	75.281
2041	22.516	51.250	73.765
2042	22.130	50.146	72.276
2043	21.751	49.061	70.812
2044	21.150	47.976	69.125
2045	20.787	46.909	67.697
2046	20.431	45.861	66.292
2047	20.081	44.831	64.912
2048	19.736	43.819	63.555
2049	19.808	43.105	62.912
2050	19.468	42.403	61.871
2051	19.134	41.713	60.847
2052	18.806	41.035	59.841
2053	18.484	40.369	58.853
2054	16.291	39.377	55.669
2055	16.012	38.403	54.415
2056	15.737	37.446	53.183
2057	15.468	36.505	51.972
2058	15.203	35.580	50.782
2059	11.871	33.961	45.832
2060	11.667	32.370	44.037
2061	11.468	30.805	42.273
2062	11.271	29.268	40.539
2063	11.078	27.757	38.835
2064	9.781	26.429	36.210
2065	9.613	25.124	34.737
2066	9.449	23.841	33.290
2067	9.287	22.580	31.867
2068	9.127	21.341	30.468
2069	8.125	20.577	28.702
2070	7.986	19.826	27.811

2071	7.849	19.087	26.936
2072	7.714	18.362	26.076
2073	7.582	17.649	25.231
2074	7.381	17.159	24.540
2075	7.254	16.678	23.932
2076	7.130	16.205	23.335
2077	7.008	15.740	22.747
2078	6.887	15.283	22.170
2079	6.199	14.890	21.089
2080	6.093	14.504	20.596
2081	5.988	14.124	20.112
2082	5.885	13.751	19.636
2083	5.785	13.384	19.169

Table 35. Carbon stocks of initial forest classes (icl) existing in the project area (Class ID 2)

Project year	Name	Initial class	
	IDcl	Savannah Forest - Sd	
		2	
	<i>Cabicl</i> C stock tCO ₂ e	<i>Cbbicl</i> C stock tCO ₂ e	<i>Ctoticl</i> C stock tCO ₂ e
2024	1.689	37	1.726
2025	1.660	74	1.733
2026	1.631	110	1.741
2027	1.603	145	1.748
2028	1.576	179	1.755
2029	4.522	279	4.801
2030	4.445	377	4.821
2031	4.368	473	4.841
2032	4.294	567	4.861
2033	4.220	660	4.880
2034	5.708	748	6.456
2035	5.610	835	6.445
2036	5.514	920	6.434
2037	5.420	1.004	6.424
2038	5.327	1.087	6.413
2039	5.058	1.099	6.157
2040	4.972	1.110	6.082

2041	4.886	1.122	6.008
2042	4.803	1.133	5.935
2043	4.720	1.144	5.864
2044	2.602	1.076	3.678
2045	2.557	1.008	3.566
2046	2.514	942	3.456
2047	2.470	878	3.348
2048	2.428	814	3.242
2049	1.684	740	2.424
2050	1.655	667	2.322
2051	1.627	595	2.222
2052	1.599	525	2.124
2053	1.572	455	2.027
2054	1.307	427	1.733
2055	1.284	399	1.683
2056	1.262	371	1.634
2057	1.241	344	1.585
2058	1.219	318	1.537
2059	40	282	321
2060	39	246	285
2061	38	211	250
2062	38	177	215
2063	37	143	180
2064	0	114	114
2065	0	86	86
2066	0	58	58
2067	0	31	31
2068	0	4	4
2069	0	3	3
2070	0	2	2
2071	0	2	2
2072	0	1	1
2073	0	0	0
2074	0	0	0
2075	0	0	0
2076	0	0	0
2077	0	0	0
2078	0	0	0
2079	0	0	0
2080	0	0	0

2081	0	0	0
2082	0	0	0
2083	0	0	0

Table 36. Carbon stocks of initial forest classes (icl) existing in the project area (Class ID 3)

Project year	Name IDcl	Initial class Steppic-Savanna Forest - Td 3		
		<i>Cabicl</i>	<i>Cbbicl</i>	<i>Ctoticl</i>
		C stock	C stock	C stock
		tCO _{2e}	tCO _{2e}	tCO _{2e}
2024	533	29	562	
2025	524	57	581	
2026	515	85	600	
2027	506	113	619	
2028	498	139	637	
2029	1.383	214	1.597	
2030	1.359	288	1.647	
2031	1.336	360	1.696	
2032	1.313	431	1.744	
2033	1.290	501	1.792	
2034	2.942	632	3.573	
2035	2.891	760	3.651	
2036	2.842	886	3.728	
2037	2.793	1.010	3.803	
2038	2.745	1.131	3.877	
2039	3.626	1.253	4.878	
2040	3.563	1.372	4.936	
2041	3.502	1.489	4.992	
2042	3.442	1.605	5.047	
2043	3.383	1.718	5.101	
2044	3.463	1.746	5.210	
2045	3.404	1.774	5.178	
2046	3.346	1.801	5.147	
2047	3.288	1.828	5.117	
2048	3.232	1.855	5.087	
2049	4.219	1.887	6.106	
2050	4.147	1.918	6.065	

2051	4.076	1.949	6.025
2052	4.006	1.980	5.986
2053	3.937	2.010	5.947
2054	4.705	2.077	6.782
2055	4.624	2.143	6.767
2056	4.545	2.208	6.753
2057	4.467	2.272	6.739
2058	4.390	2.335	6.725
2059	3.904	2.317	6.222
2060	3.838	2.301	6.138
2061	3.772	2.284	6.056
2062	3.707	2.268	5.975
2063	3.644	2.252	5.896
2064	3.243	2.173	5.416
2065	3.187	2.095	5.282
2066	3.132	2.019	5.151
2067	3.079	1.944	5.022
2068	3.026	1.870	4.896
2069	2.922	1.817	4.739
2070	2.872	1.764	4.637
2071	2.823	1.713	4.536
2072	2.775	1.662	4.437
2073	2.727	1.613	4.340
2074	2.673	1.582	4.255
2075	2.627	1.552	4.179
2076	2.582	1.522	4.104
2077	2.538	1.493	4.031
2078	2.494	1.464	3.958
2079	2.170	1.423	3.593
2080	2.133	1.383	3.516
2081	2.096	1.344	3.440
2082	2.060	1.305	3.365
2083	2.025	1.267	3.292

Table 37. Carbon stocks of initial forest classes (icl) existing in the project area (Class ID 4)

Project year	Name IDcl	Initial class
		Grassy-Woody Savannah - Tg 4

	<i>Cabicl</i> C stock tCO ₂ e	<i>Cbbicl</i> C stock tCO ₂ e	<i>Ctotal</i> C stock tCO ₂ e
2024	0	0	0
2025	0	0	0
2026	0	0	0
2027	0	0	0
2028	0	0	0
2029	0	0	0
2030	0	0	0
2031	0	0	0
2032	0	0	0
2033	0	0	0
2034	13	4	17
2035	13	9	21
2036	12	13	25
2037	12	17	29
2038	12	21	33
2039	27	30	57
2040	26	39	65
2041	26	47	73
2042	25	56	81
2043	25	64	89
2044	5	62	67
2045	5	59	64
2046	5	56	61
2047	5	54	59
2048	5	52	56
2049	0	43	43
2050	0	34	34
2051	0	25	25
2052	0	17	17
2053	0	8	8
2054	0	7	7
2055	0	5	5
2056	0	3	3
2057	0	2	2
2058	0	0	0
2059	0	0	0

2060	0	0	0
2061	0	0	0
2062	0	0	0
2063	0	0	0
2064	0	0	0
2065	0	0	0
2066	0	0	0
2067	0	0	0
2068	0	0	0
2069	0	0	0
2070	0	0	0
2071	0	0	0
2072	0	0	0
2073	0	0	0
2074	0	0	0
2075	0	0	0
2076	0	0	0
2077	0	0	0
2078	0	0	0
2079	0	0	0
2080	0	0	0
2081	0	0	0
2082	0	0	0
2083	0	0	0

Table 38. Carbon stocks of initial forest classes icl existing in the leakage belt (Class ID 1)

Project year	Initial class Forest formation 1		
	<i>Cabicl</i> C stock	<i>Cbbicl</i> C stock	<i>Ctoticl</i> C stock
	tCO ₂ e	tCO ₂ e	tCO ₂ e
2024	8.380	168	8.547
2025	8.236	332	8.568
2026	8.095	494	8.589
2027	7.956	653	8.610
2028	7.820	810	8.630
2029	4.497	899	5.396

2030	4.420	988	5.407
2031	4.344	1.075	5.419
2032	4.269	1.160	5.430
2033	4.196	1.244	5.440
2034	3.320	1.143	4.463
2035	3.263	1.043	4.307
2036	3.207	946	4.153
2037	3.152	850	4.002
2038	3.098	755	3.854
2039	4.532	756	5.288
2040	4.454	757	5.211
2041	4.378	757	5.135
2042	4.303	758	5.061
2043	4.229	759	4.988
2044	6.029	813	6.841
2045	5.925	866	6.791
2046	5.824	918	6.742
2047	5.724	970	6.693
2048	5.626	1.020	6.646
2049	7.849	1.087	8.935
2050	7.714	1.152	8.866
2051	7.582	1.216	8.798
2052	7.452	1.279	8.731
2053	7.324	1.341	8.665
2054	8.355	1.387	9.743
2055	8.212	1.433	9.645
2056	8.071	1.478	9.549
2057	7.933	1.522	9.455
2058	7.797	1.565	9.363
2059	7.861	1.566	9.427
2060	7.726	1.566	9.292
2061	7.594	1.566	9.160
2062	7.464	1.566	9.030
2063	7.336	1.567	8.902
2064	7.688	1.553	9.241
2065	7.556	1.540	9.096
2066	7.427	1.527	8.954
2067	7.299	1.515	8.814
2068	7.174	1.502	8.676
2069	7.575	1.496	9.072

2070	7.445	1.491	8.936
2071	7.318	1.485	8.803
2072	7.192	1.480	8.672
2073	7.069	1.475	8.543
2074	6.337	1.447	7.784
2075	6.228	1.421	7.649
2076	6.122	1.395	7.516
2077	6.017	1.369	7.386
2078	5.914	1.344	7.258
2079	5.941	1.311	7.252
2080	5.839	1.279	7.118
2081	5.739	1.248	6.987
2082	5.641	1.217	6.857
2083	5.544	1.186	6.730

Table 39. Carbon stocks of initial forest classes icl existing in the leakage belt (Class ID 2)

Project year	Initial class Savanna formation 2		
	Name IDcl		
	<i>Cabicl</i> C stock	<i>Cbbicl</i> C stock	<i>Ctoticl</i> C stock
	tCO _{2e}	tCO _{2e}	tCO _{2e}
2024	407.245	67.148	474.394
2025	400.265	133.146	533.411
2026	393.405	198.012	591.417
2027	386.662	261.767	648.429
2028	380.034	324.429	704.463
2029	383.038	387.586	770.624
2030	376.473	449.660	826.133
2031	370.020	510.671	880.691
2032	363.678	570.636	934.313
2033	357.444	629.573	987.017
2034	353.252	620.670	973.922
2035	347.197	611.920	959.117
2036	341.246	603.320	944.566
2037	335.397	594.867	930.265
2038	329.649	586.559	916.208
2039	306.968	574.017	880.985

2040	301.706	561.689	863.395
2041	296.535	549.572	846.108
2042	291.453	537.664	829.116
2043	286.457	525.959	812.416
2044	266.212	511.607	777.819
2045	261.649	497.502	759.150
2046	257.164	483.638	740.802
2047	252.756	470.011	722.768
2048	248.424	456.619	705.043
2049	245.646	446.508	692.153
2050	241.435	436.570	678.005
2051	237.297	426.803	664.100
2052	233.230	417.202	650.432
2053	229.232	407.767	636.999
2054	221.517	400.398	621.914
2055	217.720	393.154	610.874
2056	213.988	386.035	600.024
2057	210.321	379.038	589.359
2058	206.716	372.161	578.877
2059	193.545	363.571	557.116
2060	190.228	355.127	545.355
2061	186.967	346.829	533.796
2062	183.763	338.672	522.435
2063	180.613	330.656	511.269
2064	174.913	322.972	497.884
2065	171.915	315.419	487.334
2066	168.968	307.996	476.964
2067	166.072	300.700	466.772
2068	163.226	293.529	456.755
2069	157.939	287.658	445.598
2070	155.232	281.888	437.120
2071	152.572	276.217	428.788
2072	149.957	270.643	420.599
2073	147.386	265.164	412.550
2074	142.350	259.795	402.145
2075	139.910	254.518	394.428
2076	137.512	249.331	386.843
2077	135.155	244.233	379.388
2078	132.838	239.223	372.061
2079	126.513	234.041	360.554

2080	124.344	228.948	353.292
2081	122.213	223.942	346.155
2082	120.118	219.022	339.141
2083	118.059	214.187	332.246

Table 40. Carbon stocks of initial forest classes icl existing in the leakage belt (Class ID 3)

Project year	Initial class		
	Forest-Savanna transition		
	3		
	<i>C_{bicl}</i> C stock tCO _{2e}	<i>C_{bbicl}</i> C stock tCO _{2e}	<i>C_{toticl}</i> C stock tCO _{2e}
2024	0	0	0
2025	0	0	0
2026	0	0	0
2027	0	0	0
2028	0	0	0
2029	0	0	0
2030	0	0	0
2031	0	0	0
2032	0	0	0
2033	0	0	0
2034	0	0	0
2035	0	0	0
2036	0	0	0
2037	0	0	0
2038	0	0	0
2039	0	0	0
2040	0	0	0
2041	0	0	0
2042	0	0	0
2043	0	0	0
2044	0	0	0
2045	0	0	0
2046	0	0	0
2047	0	0	0
2048	0	0	0
2049	0	0	0

2050	0	0	0
2051	0	0	0
2052	0	0	0
2053	0	0	0
2054	0	0	0
2055	0	0	0
2056	0	0	0
2057	0	0	0
2058	0	0	0
2059	0	0	0
2060	0	0	0
2061	0	0	0
2062	0	0	0
2063	0	0	0
2064	0	0	0
2065	0	0	0
2066	0	0	0
2067	0	0	0
2068	0	0	0
2069	0	0	0
2070	0	0	0
2071	0	0	0
2072	0	0	0
2073	0	0	0
2074	0	0	0
2075	0	0	0
2076	0	0	0
2077	0	0	0
2078	0	0	0
2079	0	0	0
2080	0	0	0
2081	0	0	0
2082	0	0	0
2083	0	0	0

Carbon Stock Change in the Post-deforestation

The sources of greenhouse gas (GHG) emissions in the project scenario, outlined below, were obtained from the First Biennial Transparency Report (BTR) of Brazil to The United Nations Framework Convention on Climate Change (BRAZIL, 2024).

Emission Factor	Value	Unit	Source
Pasture stock – Cerrado	7.57	tC/ha	Brazil's National Inventory Report of Anthropogenic Emissions by Sources and Removals of Greenhouse Gases First Biennial Transparency Report of Brazil to The United Nations Framework Convention on Climate Change (Brasil, 2024)
Degraded Pasture stock – Cerrado	3.67	tC/ha	
Annual Cropland carbon stock - Cerrado	5.0	tC/ha	
Perennial Cropland stock - Cerrado	21.0	tC/ha	

The emission factor used to calculate GHG emissions in the project scenario was based on carbon removal due to the substitution of forest areas for agriculture or pasture. These factors represent the annual carbon removal per unit area for pastures and agriculture within the Cerrado Biome.

These emission factors were applied to both deforested areas converted to pasture and to annual cropland. According to TerraClass Project data, 41.4% of the deforested areas in the historical reference period were converted into some form of pasture and 35.5% into annual cropland, summing 76.9% of the deforested areas in the historical period.

The chosen factor of 7.57 tC/ha/year, representing carbon removal from pasture growth estimated annually over a period of 9 years, was applied across the total pasture-covered area in the Reference Region. This is considered conservative, as some of the pastures in the Cerrado Biome are degraded.

The chosen factor of 5.0 tC/ha/year, representing carbon removal from annual cropland growth estimated annually over a period of 9 years, was applied across the total agriculture-covered area in the Reference Region. There is no perennial agriculture in the Reference Region.

The Total Carbon Stock change of post-deforestation in the PA and LK are presented on the following tables.

Table 41. Total carbon stock change of post-deforestation in the Project Area

IDiz Name	2 Non forest $\Delta\text{CBSLPA}_{iz,t}$	Total carbon stock change of post deforestation in the project area	
Project year	tCO _{2e}	Annual (tCO _{2e})	Cumulative (tCO _{2e})
2024	0	0	0
2025	1.223	1.223	1.223

2026	1.529	1.529	2.753
2027	2.162	2.162	4.915
2028	2.784	2.784	7.699
2029	3.396	3.396	11.095
2030	4.052	4.052	15.147
2031	4.697	4.697	19.844
2032	5.330	5.330	25.174
2033	5.953	5.953	31.128
2034	6.110	6.110	37.238
2035	5.673	5.673	42.911
2036	5.666	5.666	48.576
2037	5.659	5.659	54.235
2038	5.651	5.651	59.887
2039	5.590	5.590	65.476
2040	5.522	5.522	70.999
2041	5.456	5.456	76.455
2042	5.391	5.391	81.846
2043	5.327	5.327	87.174
2044	5.233	5.233	92.406
2045	5.116	5.116	97.523
2046	5.002	5.002	102.524
2047	4.889	4.889	107.414
2048	4.779	4.779	112.193
2049	4.677	4.677	116.870
2050	4.595	4.595	121.465
2051	4.514	4.514	125.979
2052	4.432	4.432	130.411
2053	4.354	4.354	134.765
2054	4.297	4.297	139.062
2055	4.213	4.213	143.275
2056	4.131	4.131	147.406
2057	4.050	4.050	151.456
2058	3.970	3.970	155.426
2059	3.874	3.874	159.300
2060	3.700	3.700	163.001
2061	3.530	3.530	166.531
2062	3.363	3.363	169.893
2063	3.198	3.198	173.091
2064	3.064	3.064	176.156
2065	2.905	2.905	179.060

2066	2.748	2.748	181.809
2067	2.594	2.594	184.403
2068	2.443	2.443	186.846
2069	2.373	2.373	189.218
2070	2.286	2.286	191.504
2071	2.200	2.200	193.704
2072	2.116	2.116	195.820
2073	2.033	2.033	197.853
2074	1.980	1.980	199.833
2075	1.926	1.926	201.759
2076	1.873	1.873	203.632
2077	1.821	1.821	205.454
2078	1.770	1.770	207.224
2079	1.738	1.738	208.962
2080	1.691	1.691	210.653
2081	1.644	1.644	212.298
2082	1.599	1.599	213.896
2083	1.554	1.554	215.450

Table 42. Total carbon stock change of post-deforestation in the leakage belt

<i>IDiz</i> Name	2 Non forest $\Delta\text{CBSLLK}_{iz,t}$	Total carbon stock change of post deforestation in the leakage belt	
Project year	tCO ₂ e	Annual (tCO ₂ e)	Cumulative (tCO ₂ e)
2024	0	0	0
2025	13.492	13.492	13.492
2026	26.753	26.753	40.245
2027	39.787	39.787	80.032
2028	52.597	52.597	132.630
2029	65.188	65.188	197.818
2030	77.837	77.837	275.654
2031	90.269	90.269	365.923
2032	102.487	102.487	468.410
2033	114.497	114.497	582.907
2034	112.808	112.808	695.715
2035	111.202	111.202	806.917
2036	109.623	109.623	916.540

2037	108.072	108.072	1.024.612
2038	106.547	106.547	1.131.159
2039	104.775	104.775	1.235.934
2040	102.491	102.491	1.338.424
2041	100.246	100.246	1.438.671
2042	98.040	98.040	1.536.710
2043	95.872	95.872	1.632.582
2044	93.687	93.687	1.726.269
2045	91.058	91.058	1.817.327
2046	88.474	88.474	1.905.801
2047	85.935	85.935	1.991.736
2048	83.439	83.439	2.075.175
2049	81.528	81.528	2.156.703
2050	79.726	79.726	2.236.429
2051	77.956	77.956	2.314.385
2052	76.215	76.215	2.390.601
2053	74.505	74.505	2.465.106
2054	73.305	73.305	2.538.411
2055	72.016	72.016	2.610.426
2056	70.748	70.748	2.681.174
2057	69.502	69.502	2.750.677
2058	68.278	68.278	2.818.954
2059	66.997	66.997	2.885.952
2060	65.425	65.425	2.951.376
2061	63.879	63.879	3.015.255
2062	62.359	62.359	3.077.614
2063	60.866	60.866	3.138.481
2064	59.509	59.509	3.197.989
2065	58.095	58.095	3.256.084
2066	56.705	56.705	3.312.790
2067	55.340	55.340	3.368.130
2068	53.997	53.997	3.422.127
2069	52.992	52.992	3.475.119
2070	51.929	51.929	3.527.047
2071	50.883	50.883	3.577.931
2072	49.856	49.856	3.627.787
2073	48.846	48.846	3.676.633
2074	47.934	47.934	3.724.567
2075	46.947	46.947	3.771.514
2076	45.977	45.977	3.817.490

2077	45.023	45.023	3.862.513
2078	44.086	44.086	3.906.599
2079	43.240	43.240	3.949.840
2080	42.278	42.278	3.992.118
2081	41.331	41.331	4.033.449
2082	40.401	40.401	4.073.850
2083	39.487	39.487	4.113.338

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country



Brazil

Type of Party:

Non-Annex I

Ratification status:

Paris Agreement

Date of signature: 22 April 2016

Date of ratification: 21 September 2016

Kyoto Protocol

Date of signature: 29 April 1998

Date of ratification: 23 August 2002

Source: UNFCC¹¹⁸

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

Non-enforcement scenario (Brazilian GHG regulatory framework)

Brazil is not part of the UNFCCC Annex 1, and even though it has agreed GHG targets in the global community, there is currently no compulsory reduction imposed on sectors. There is an initial law discussion under consideration - at the time when the first version of this PD is being written - for

¹¹⁸ <https://unfccc.int/node/180413>

the regulation of a local carbon market. However, the project proponent understands that it will not be approved anytime soon, and even if it is approved, 1) there will be a significant delay before it is effectively implemented and 2) companies/rural producers engaged in agricultural activities are not likely to be included initially, based on discussions in parliament. The activities from this project, therefore, must be prioritized, anticipating government directives, as it understands that the climate scenario is urgent and needs to be addressed as a top priority. However, any advances in legislation will be adapted to the scope of this project when required and applicable.

Project Activities Surplus

The activities of this project are considered as additional surplus to the activities regularly consolidated in the regulatory case because:

- 1) Although there are regulatory instruments for forest protection governed by the Brazilian Forest Code, data¹¹⁹ shows that there is little effectiveness in terms of monitoring, penalties, or sanctions. In this case, the activities of this project ensure that, through the strong partnership between the private landowner and NovaAgri, which has been in place since the commercial relationship, there will be a more efficient approach to voluntary protection, based on good relations, environmental awareness, mobilization for social change, all tied to a legal binding contract that formalizes this commitment.
- 2) Another essential point of additional surplus is related to the intensive investment in technology, involving a dynamic system with deforestation alerts, fire and wildfire hotspots, and the use of state-of-the-art satellite images with excellent resolution and periodicity. All this investment is only possible due to NovaAgri's commitment to its strategic planning and relationship with its stakeholders. This investment would hardly be made to the same extent by the public sphere.
- 3) It is understood that forest legislation is very comprehensive regarding possible mechanisms for displacing deforestation in forested areas. For example, ensuring that an area has a legal reserve to be guaranteed by law today does not mean that this area cannot be deforested. This is because there are mechanisms that allow the purchase of other forested areas in other regions, as long as they are registered under the same owner, to compensate for a deforested legal reserve area. Among the activities of this project is the signing of a legal binding contract, with clear clauses regarding the restriction of this type of action. Thus, it is avoided not only the protection of the project area but also the forested areas of other regions, thus favoring non-leakage. Moreover, the project area has additional areas beyond what is required by forest law, being an important additional reserve for carbon stock preservation.

¹¹⁹ MAPBIOMAS. RAD 2022 – Relatório anual do desmatamento no Brasil (Junho, 2023).
<https://alerta.mapbiomas.org/relatorio>

4) The revenues from the sale of VCUs cannot overcome the margins of conventional agricultural and livestock activities because they are highly profitable activities in the country, especially at the reference region and State level. According to CODEVASF¹²⁰ (2024), Piauí has experienced a 384% increase in agricultural production over eight years, which represents a rise in production revenue from R\$ 2.6 billion in 2015 to nearly R\$ 13 billion in 2024. The study from Mapbiomas (2023) represented by the chart below also represents this expansion.

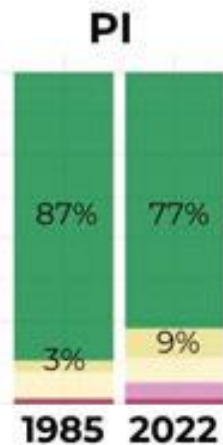


Figure 76. Expansion of anthropical areas in the PI state between 1985-2022.

Source: MapBiomas.

According to another study from MapBiomas¹²¹ (2023), some states have shown notable increases in the percentage of deforested areas that occurred either with authorization or under enforcement actions in recent years. For instance, Similarly, Piauí experienced an increase from 33% in 2021 to **57.2%** in 2023. This shows that this State has demonstrated a drastic potential for clearing expansion in very recent periods.

5) Also, there is clear evidence that public protected areas, such as delimited indigenous areas, have a proven efficiency against the deforestation, when comparing to private areas, as shown on the image below, from MAPBIOMAS.

¹²⁰ CODEVASF – ‘Piauí experiences a 384% increase in agricultural production over eight years.’
www.codevasf.gov.br/noticias/2024/piaui-cresce-384-na-producao-agricola-em-oito-anos-codevasf-contribui-para-o-crescimento-com-mecanizacao-no-campo

¹²¹ RAD2023: Relatório Anual do Desmatamento no Brasil 2023 - São Paulo, Brasil - MapBiomas, 2024 - 154 páginas.
<http://alerta.mapbiomas.org>

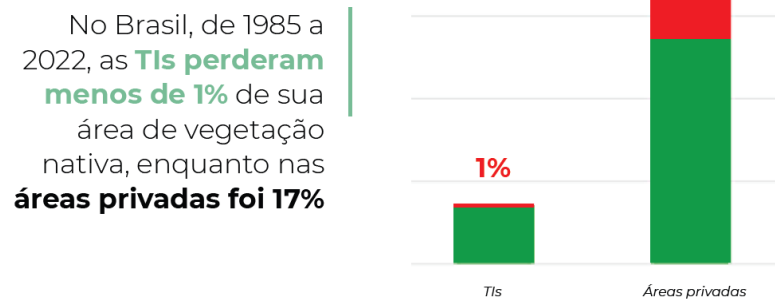


Figure 77. 'In Brazil, from 1985 to 2022, the indigenous land lost less than 1% of its native vegetation, while the private areas lost 17%'. Source: MapBiomass.

This is an important consideration because it highlights the significance of creating and maintaining preservation projects in private areas as well, something that today is more effective in public areas due to the implementation of activities that make these locations less vulnerable, such as proper demarcation and promotion of areas, research-based or ecotourism activities (e.g., parks and conservation units), or even subsistence occupation and sustainable land use, as is the case with indigenous lands, for example. In private areas, there is less surveillance over the lands or lack of attention, which increases the chances of negligence regarding the laws or even illegal invasions.

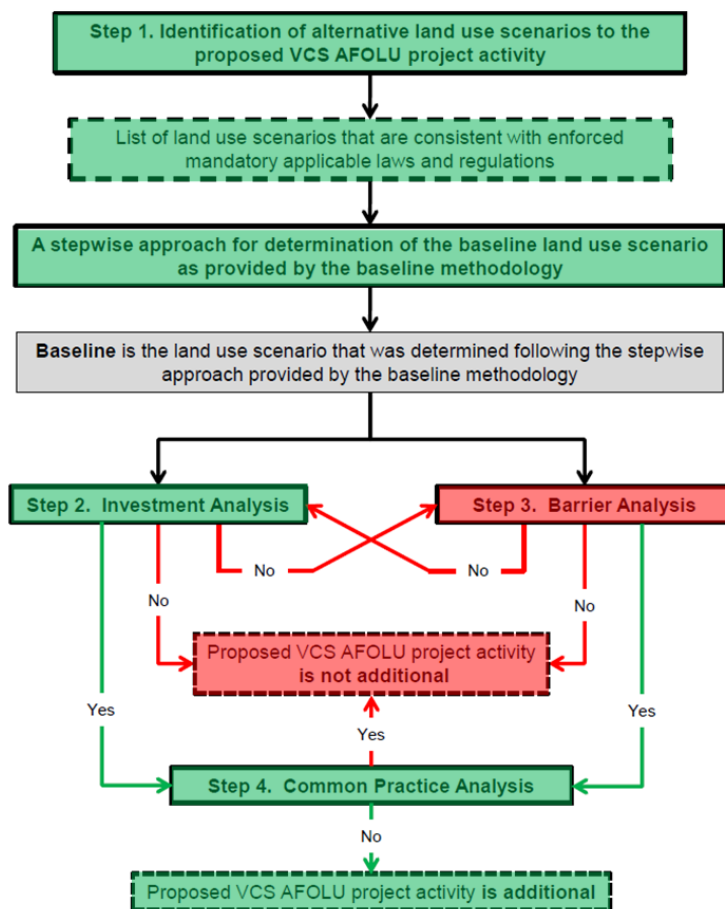
Therefore, there are many arguments which come from rural farming agents to carry out forest openings and disregarding regulatory issues. For this reason, the activities of the CULTIVAR PROGRESSO REDD+ project have a well-defined strategy, through the good commercial relationship between the parties, socio-environmental awareness, and legal agreement for no deforestation. This is a way to replicate to the maximum the protection structure of the public lands to the private lands.

5) Even if the Regulated Carbon Market was fully implemented today, there is no guarantee that the structure of this system would ensure that the financial resources (raised through carbon credits) would be reinvested in social investment. The activities of this project compulsorily establish this commitment with society, especially with the most vulnerable communities present in the surrounding area of the project, which can be considered an important additional surplus.

6) Even though the area destined for preservation was indicated in the baseline as **32,549,79 ha**, the total area subject to monitoring and protection totals **629.311 ha**. This shows that the proponent is allocating project activities against deforestation in additional portions of forested areas, which can be seen as an additional surplus.

3.5.2 Additionality Methods

As required in the VERRA Additionality tool VT0001 (v3), the additionality analysis for this project is applied following the steps below:



VT001 Indicative flowchart of the Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities.

STEP 1: Identification of alternative land use scenarios to the proposed VCS AFOLU project activity

This step serves to identify alternative land use scenarios to the proposed VCS AFOLU project activity(s) that could be the baseline scenario.

Sub-step 1a. This step aims to Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity.

The project has identified realistic and credible land-use scenarios that could/might occur on the land encompassed by the project boundary. These scenarios have been aligned with the project area's historical land uses, practices, economic trends, and relevant national and sectoral policies. The identified land use scenarios encompass:

Scenario A) The continuation of the pre-project land use.

In the absence of the REDD+ AUD project's intervention, the area would be susceptible to the continuation of conventional agricultural practices and potential deforestation, particularly considering the already high deforestation rates in the reference region. Agricultural and cattle ranching expansion driven by conventional practices could exacerbate these rates, leading to the conversion of forested land into agricultural fields for monoculture crops such as soybeans and corn, and cattle ranching activities.

The region's alarming deforestation rates accentuate the urgency of taking proactive measures. The pressures for land conversion, coupled with high economic incentives, could escalate deforestation even further, resulting in the loss of vital biodiversity, and ecosystem services. Furthermore, the prevailing financial motivation for farmers to engage in crop cultivation rather than forest preservation amplifies the risk of escalating deforestation because the prospect of immediate profit often shades the long-term environmental consequences.

The project area is included in the *Cerrado* Savanna biomes. Historically known for its significant agricultural expansion, especially in soybean and cattle ranching, the state of Piauí experienced a surge in deforestation rates during the early 2000s. If considering the historical baseline deforestation, the entire project area could be vanished from short to medium term. Even though there are legal instruments to prevent illegal deforestation in such biomes, for instance the Forest Code (Law n. 12.651, 2012). The fact is that illegal deforestation in this biome is a common practice within legal and non-legal reserve areas. That is a clear and worrying signal that the law against deforestation in Brazil is far away from being effective itself. It is important to mention that, until now, the Brazilian government has not compromised to implement a fully national monitoring system, which could support the law enforcement within a good timing precision. Therefore, feasible implementation and law enforcement is still an issue with many challenges to overcome, especially at national levels.

In summary, without implementing this REDD+ project, the area would face a worrying threat of conversion, primarily driven by the lucrative prospects offered by agribusiness activities – the profitability associated with agricultural expansion poses a substantial risk to the preservation of natural ecosystems in the absence of sustainable alternatives.

Scenario B) The project activity conducted within the project boundary, excluding registration under the VCS AFOLU project.

A scenario where REDD+ activities are introduced to the project boundary without being officially registered under a VCS AFOLU project, and absence of a structured MRV framework. This is a scenario with limited assurance for land protection, and less integrity of carbon emission reduction efforts, mainly because this leads to absence of a formal registration and consequently access to monetary incentives associated with certified carbon credits.

As shown in the Opportunity Cost analysis, even though traditional activities seem to be more profitable, a project of preservation under VCS criteria can benefit the landowner with a financial incentive, even on a smaller scale.

In this scenario, the activities that potentially benefit the people in the communities and/or the biodiversity are not put into practice or managed properly, as is the case in VCS projects. Economic incentives and effective management of activities are essential factors for positive and meaningful impacts to occur.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations

Scenario A) The continuation of the pre-project land use.

Agriculture and Pasture (Cattle Ranching) activities

This scenario is credible since there is no restriction of mandatory applicable laws and regulations for the agricultural (soybeans and corn crops) and livestock (cattle) activities.

Deforestation activities

Brazil possesses a distinctive forest law compared to other forest-developing countries. However, the country faces intrinsic difficulties in tackling deforestation due to geographical, cultural, and historical factors. Despite the existence of the 'Forest Code law' which has prevented a more significant advance of deforestation, vegetation suppression remains pervasive in Brazil, with thousands of hectares being cleared across its main biomes. In other words, even legally protected reserve areas, which should not be subject to deforestation, suffer due to a lack of oversight, weak public policies, non-compliance with laws, and a failure to enforce penalties - as mentioned earlier, the legal actions and embargoes executed by IBAMA and ICMBio (as of May 2023) only covered 10.2% of the identified deforested area from 2019 to 2022 (MAPBIOMAS, 2023)¹²². Also, there are legal instruments that permit that, even if a landowner exceeds their legal reserve through deforestation, this can offset that by reforesting, preserving, or generating/buying credits of another forested area elsewhere. In this way, adhering to the current modus operandi without a carbon project or other financial incentive, the landowner clears an area without violating the law, as long as this can be justified by this instrument.

Scenario B) The project activity conducted within the project boundary, excluding registration under the VCS AFOLU project.

Similarly, this scenario is also credible once there is no restriction of mandatory applicable laws and regulations for AFOLU activities (e.g. private monitoring in rural areas). It is important to

¹²² MAPBIOMAS. RAD 2022 – Relatório anual do desmatamento no Brasil (Junho, 2023). <https://alerta.mapbiomas.org/relatorio>

mention that this scenario is only valid if the project area is not encompassed by any jurisdictional VCS project area or demarcated area by the government, such as indigenous parks or conservation units.

Sub-step 1c. Selection of the baseline scenario

The most plausible scenario for the baseline construction is Scenario A *'The continuation of the pre-project land use'*.

The scenario described has been chosen as the most plausible baseline for the Carbon AFOLU VCS project due to:

- **Current Deforestation Trends:** The scenario acknowledges the alarming deforestation rates in the reference region. These existing trends provide a baseline that reflects the current trajectory of land-use change. By highlighting the ongoing deforestation, the baseline scenario accurately represents the business-as-usual situation without project intervention.
- **Potential for Escalation:** The potential for escalation in deforestation due to various factors such as economic incentives for agriculture / cattle ranching and the risk of immediate profit overshadowing long-term environmental consequences.
- **Historical Deforestation Data:** Referring to historical data on the reference area's forest loss rate between specific years adds empirical evidence to the baseline construction.
- **Legal Framework and Enforcement Challenges:** The acknowledgement of challenges in enforcing existing legal instruments, such as the Forest Code, contributes to the baseline's realism. The common practice of illegal deforestation despite regulatory measures suggests that, without the project, the legal framework alone may not be sufficient to stop deforestation.
- **Lack of National Monitoring System:** The absence of a fully implemented monitoring system with precision and periodicity adds another argument to the baseline construction. It underscores the uncertainty and challenges in accurately tracking and enforcing land-use changes, supporting the argument that deforestation could proceed unpredicted without the project.

In summary, it has been assessed that scenario B – even though it is credible, is unlikely to occur. This is primarily due to the absence of any types of incentives in Brazil for maintaining forests untouched. As a remote possibility, the area might be designated as part of a jurisdictional private preservation zone, by determination of the government – This is unlikely to happen given the area is part of a well-established private productive area in the region. Additionally, it is recognized that the legislation is fragile, and any gap found in it could lead to a tendency toward scenario A.

Further details will be complemented in Section 4.1, by analyzing the historical data and processing the respective deforestation images.

Step 2. Investment analysis

Sub-step 2a. Determine appropriate analysis method

The VCS AFOLU project generates no revenues other than VCU-related income. Thus, the analysis method applied was the ‘simple cost analysis’, once the sole income generated by keeping the native vegetation intact is the selling of VCU credits – when the project is fully implemented and verified. There are no plans, for instance, for activities such as agroforestry production, ecosystem service management for income generation, or the generation of credits of any other nature apart from VCU credits.

Furthermore, for the project's implementation, the proponent will allocate investments that would not materialize without the execution of this VCS project. The very approximate values of these investments¹²³ are presented on the Table below:

Table 43. Investments Estimation Analysis

Activity / System / Utility	Type of Investment (Initial or Periodic)	Investment (approx. in kR\$)
Feasibility Analysis System	Initial	200
Consulting	Initial	100
Certifier ¹²⁴	Initial	20
	Periodic	2.5
Validation	Initial	200
Verification	Periodic	100
Travels	Initial	30
Other Expenses (technology, systems etc.)	Initial	200
	Periodic	200
Total	Initial	>750
	Periodic	>300

¹²³ Note: in order to protect sensible commercial or strategic information, the values presented above do not represent the actual numbers – all the items were shown in a range basis just to demonstrate the magnitude order of those investments.

¹²⁴ Note: not considering costs for VCUs emission

Step 4. Common practice analysis

In Brazil, most conservation programs and the designation of protected areas are carried out within the public sphere boundaries, through the creation of conservation units, such as Ecological Stations, Biological Reserves, National Parks, Natural Monuments, Wildlife Refuges, and others. However, these cannot be directly compared to the activities of carbon credit projects, as they are standard government practices rather than private initiatives. Beyond the designation of protected areas, Brazil has developed mechanisms and programs aimed at promoting forest conservation and providing benefits to participants, but not necessarily comparable to carbon credits initiatives. Below are examples, along with a comparison to the additionality offered by the current REDD+ project:

- **Green CPR (CPR Verde):** Introduced in October 2021, the Green CPR is a new financial instrument for Brazilian rural landowners, enabling them to fund the conservation of native vegetation on their properties. The Green CPR builds on the well-established CPR (Rural Product Certificate) framework but aims to link it to environmental assets.

Challenges: Adoption has been limited due to low awareness and demand, and a Green CPR must not be classified as carbon credit or comparable with it. Although the Brazilian stock exchange (B3) facilitated initial issuances, there is no evidence of Green CPRs generated within the Reference Region of this project.

Comparison: While promising as a carbon-backed mechanism, the Green CPR lacks the rigorous technical and methodological framework of VCS (Verified Carbon Standard) projects, making it less impactful in delivering measurable and tradeable environmental and economic benefits.

- **Ecological ICMS (ICMS Ecológico):** This fiscal policy incentivizes municipalities to prioritize conservation by distributing tax revenues based on environmental performance, such as protected areas.

Functionality: Municipalities with larger conservation areas or superior environmental policies receive a larger share of tax revenues, reducing funds for those with less effective conservation practices.

Comparison: While the ICMS Ecológico promotes environmental protection, it primarily benefits public entities and does not link financial incentives for individual landowners. In contrast, carbon credit projects offer private stakeholders' tangible revenues, directly linking conservation to economic benefits.

- **Private Reserve of Natural Heritage (RPPN):** RPPN is a voluntary private conservation unit, legally designated in perpetuity to preserve biodiversity. Landowners retain property rights but commit to long-term conservation.

Benefits: Participants benefit from exemptions on Rural Land Tax (ITR) and can use RPPN status for marketing purposes. However, no financial assets or credits are generated, and participants are not compensated for their conservation efforts.

Comparison: Unlike REDD+ projects, RPPNs do not generate ongoing revenue or integrate conservation into global markets, limiting their ability to attract broader participation.

Conclusion

While programs like Green CPR, Eco ICMS, and RPPN play critical roles in fostering conservation, they often lack the market integration, financial sustainability, and robust monitoring frameworks of carbon credit projects. REDD+ initiatives stand out for their ability to provide quantifiable environmental and economic benefits, ensuring the preservation of critical ecosystems while addressing local development needs. This project's unique approach makes it irreplaceable in its region, considering this common practices' analysis.

For the common practice analysis, it is important to highlight the project's unique identity within this context. Even though AFOLU projects for carbon credits' generation have gained more accession and visibility recently in Latin-America¹²⁵, this might be the first REDD+ project in Piauí (or Brazil) developed by a private company whose core business is not focused on carbon credit projects. The primary objective is more aligned with reducing environmental impacts and meeting climate targets, as NovaAgri understands the necessity of addressing negative impacts within its value chain. The company is fully committed to ensuring a high-integrity project through a well-crafted Project Design (PD) and a rigorous Monitoring, Reporting, and Verification (MRV) process, supported by in-depth baseline studies and advanced technology for monitoring.

Another key differentiator lies in the geographical characteristics where the project activities are implemented. While many projects in Brazil focus on the Amazon biome¹²⁶ due to its vulnerability, this project stands out for its location in the Cerrado, an equally threatened biome with biodiversity as rich as the Amazon.

However, there are also challenges associated with a Cerrado zone, such as a higher likelihood for wildfires¹²⁷. Therefore, the project includes critical activities related to fire prevention and control. Key measures, including the use of water trucks, airplanes, and advanced fire monitoring systems with sensors and alerts, enhance the project's additionality. These activities are not

¹²⁵ IDESAM shows that the carbon projects regarding AFOLU/REDD+ are in positive ascension in Brazil and Latin America, which can be monitored in: <https://idesam.org/painelprojetoscarbonoflorestal/>

¹²⁶ IDESAM shows that the carbon projects regarding AFOLU/REDD+ are in positive ascension in Brazil, especially in the Amazonian region: <https://idesam.org/painelprojetoscarbonoflorestal/>. The CAPITAL RESET also highlights the focus from the proponents/developers on projects in the Amazon Forest: <https://capitalreset.uol.com.br/carbono/amazonas-seleciona-cinco-empresas-para-projetos-de-carbono-em-mega-area/#:~:text=O%20objetivo%20%C3%A9%20implementar%20iniciativas,na%20atmosfera%20pelo%20desmatamento%20evitado.>

¹²⁷ Article from WWF shows that the fire in Cerrado is very concerning, especially when recent cases come out: <https://www.wwf.org.br/?89520/Amazonia-ja-tem-mais-de-50-mil-focos-de-fogo-em-2024-e-fumaca-se-espalha-pelo-pais>

typically implemented by common landowners or even by the government, which faces significant challenges in predicting and combating wildfires in the State.

Considering these factors, we conclude that Step 4 is satisfied. Similar activities can be observed, and essential distinctions between the proposed VCS AFOLU project activity and comparable initiatives can be identified, confirming the project's additionality.

3.6 Methodology Deviations

There is no methodology's deviations applied in this project, only complementary rationales and processes which are not related to monitoring, and are entirely showed and discussed along this PD.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

At the time of the first validation, the initial verification monitoring is already underway, hence this Joint PD version.

Project Activity Status

The monitoring period established for the project was established as following:

- For Leakage Belt: from **January 01, 2024**, to December 31, 2024 (based on public data).
- For Project Area:
 - From January 01, 2024 to **March 12, 2024** (based on public data).
 - From March 13, 2024 to **December 31, 2024** (based on own monitoring system).

For future verifications, additional complementary tools will be applied to achieve a more precise determination of potential deforested areas. These tools are those expressed in Section 1.12 (Description of the Project Activity) of this PD. For this reason, there will indeed be a change in project activities from this first verification to the next ones, primarily based on the enhancement of monitoring techniques, using more accurate images and an exclusive dashboard for alerts (fire and deforestation), especially for the project area. However, assessing public data will also continue to be an important item as part of the assessment strategy.

All other project activities will continue to be executed as planned in this PD (according to Section 1.12). All the applicable KPIs are presented in Section 6, and the following results in both Sections 7 and 1.8.

Project validation / 1st verification

The project expects to start its validation process on Q1-2025. This is being submitted initially following the premises of VM0015, once the activity data for the State Piauí is still not available at the moment of listing this project at Verra Registry so that the applicability of the Consolidated REDD+ new methodology could be valid.

Project Activity Continuity

After the end-date for the 1st monitoring period (01-01-2024 to 12-31-2024), the project has continued implementing all its activities for the 2nd monitoring period, which begins at 01-01-2025. It is important to note that the primary activities will continue to be implemented, but new activities or an enhancement of those primary activities can be aggregated to the project while new resources may be provided for the project, such as possible revenues resulted by the VCU sales. Especially social responsibility activities will benefit from that.

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The total average biomass stock per hectare (Mg ha⁻¹) was converted to tCO₂e using the following equations:

$$Cabicl = ab \times CF \times 44/12$$

Where,

Cabicl Average carbon stock per hectare in the above-ground biomass carbon pool of initial forest class *icl*; tCO₂e ha⁻¹

ab Average biomass stock per hectare in the above-ground biomass pool of initial forest class *icl*; Mg ha⁻¹

CF Default value of carbon fraction in biomass

44/12 Ratio converting C to CO₂e

$$Cbbicl = bb \times CF \times 44/12$$

Where,

Cbbicl Average carbon stock per hectare in the below-ground biomass carbon pool of initial forest class *icl*; tCO₂e ha⁻¹

<i>bb</i>	Average biomass stock per hectare in the below-ground biomass pool of initial forest class <i>icl</i> ; Mg ha ⁻¹
<i>CF</i>	Default value of carbon fraction in biomass
<i>44/12</i>	Ratio converting C to CO ₂ e

The total baseline carbon stock change in the project area at year *t* is calculated as follows:

$$CBSLPAt = CabBSLPAicl, t + CbbBSLPAicl, t$$

Where,

<i>CBSLPAt</i>	Total baseline carbon stock changes in the project area at year <i>t</i> ; tCO ₂ e
<i>CabBSLPAicl, t</i>	Total baseline carbon stock change for the above-ground biomass pool in the project area for initial forest class at year <i>t</i> ; tCO ₂ e
<i>CbbBSLPAicl, t</i>	Total baseline carbon stock change for the below-ground biomass pool in the project area for initial forest class at year <i>t</i> ; tCO ₂ e

$$CabBSLPAicl, t = ABSLPAicl, t \times Cabicl$$

Where,

<i>CabBSLPAicl, t</i>	Total baseline carbon stock change for the above-ground biomass pool in the project area for initial forest class at year <i>t</i> ; tCO ₂ e
<i>ABSLPAicl, t</i>	Area of initial forest class <i>icl</i> deforested at time <i>t</i> within the project area in the baseline case; ha
<i>Cabicl</i>	Average carbon stock change factor per hectare in the above-ground biomass carbon pool of initial forest class <i>icl</i> ; tCO ₂ e ha ⁻¹

$$CbbBSLPAicl, t = ABSLPAicl, t \times Cbbicl$$

Where,

<i>CbbBSLPAicl, t</i>	Total baseline carbon stock change for the below-ground biomass pool in the project area for initial forest class at year <i>t</i> ; tCO ₂ e
<i>ABSLPAicl, t</i>	Area of initial forest class <i>icl</i> deforested at time <i>t</i> within the project area in the baseline case; ha
<i>Cbbicl</i>	Average carbon stock change factor per hectare in the below-ground biomass carbon pool of category <i>icl</i> ; tCO ₂ e ha ⁻¹

The Methodology step 8.1.5.1 sets out that various change factors must be applied to the baseline case initial and post-deforestation classes in above-ground and below ground biomass. The carbon stocks in various pools are specified in Section 3.3 of the present VCS-PD. As such, Table 46 to Table 46 below show carbon stock change factors for initial forest classes in above and below-ground carbon pools, which were then applied to calculate baseline carbon stock changes in various classes and pools shown in sequence.

Table 44. Carbon stock change factors for initial forest classes (icl) in below-ground carbon stocks (Method 1)

$\Delta C_{bbicl,t}$				
Year after deforestation		Forest formation	Savanna formation	Forest- savanna transition
		tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha
1	t*	-22,55	-5,82	-8,97
2	t* + 1	-22,55	-5,82	-8,97
3	t* + 2	-22,55	-5,82	-8,97
4	t* + 3	-22,55	-5,82	-8,97
5	t* + 4	-22,55	-5,82	-8,97
6	t* + 5	-22,55	-5,82	-8,97
7	t* + 6	-22,55	-5,82	-8,97
8	t* + 7	-22,55	-5,82	-8,97
9	t* + 8	-22,55	-5,82	-8,97
10	t* + 9	-22,55	-5,82	-8,97
11	t* + 10			
12	t* + 11			
13	t* + 12			
14	t* + 13			
15	t* + 14			
16	t* + 15			
17	t* + 16			
18	t* + 17			
19	t* + 18			
20	t* + 19			
21	t* + 20			
22	t* + 21			

23	$t^* + 22$
24	$t^* + 23$
25	$t^* + 24$
26	$t^* + 25$
27	$t^* + 26$
28	$t^* + 27$
29	$t^* + 28$
30	$t^* + 29$

Table 45. Carbon stock change factors for initial forest classes (icl) in above-ground carbon stocks (Method 1)

$\Delta C_{abi,icl,t}$				
Year after deforestation		Forest formation	Savanna formation	Forest-savanna transition
		tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha
1	t^*	-69,6	-88,2	-64,1
2	$t^* + 1$			
3	$t^* + 2$			
4	$t^* + 3$			
5	$t^* + 4$			
6	$t^* + 5$			
7	$t^* + 6$			
8	$t^* + 7$			
9	$t^* + 8$			
10	$t^* + 9$			
11	$t^* + 10$			
12	$t^* + 11$			
13	$t^* + 12$			
14	$t^* + 13$			
15	$t^* + 14$			
16	$t^* + 15$			
17	$t^* + 16$			
18	$t^* + 17$			
19	$t^* + 18$			
20	$t^* + 19$			

21	$t^* + 20$
22	$t^* + 21$
23	$t^* + 22$
24	$t^* + 23$
25	$t^* + 24$
26	$t^* + 25$
27	$t^* + 26$
28	$t^* + 27$
29	$t^* + 28$
30	$t^* + 29$

Table 46. Carbon stock change factors for final forest classes (fcl) or zones z (Method 1)

Year after deforestation		$\Delta C_{bbfcl,t}$ tCO ₂ e/ha
1	t^*	0
2	$t^* + 1$	1,83
3	$t^* + 2$	1,83
4	$t^* + 3$	1,83
5	$t^* + 4$	1,83
6	$t^* + 5$	1,83
7	$t^* + 6$	1,83
8	$t^* + 7$	1,83
9	$t^* + 8$	1,83
10	$t^* + 9$	1,83
11	$t^* + 10$	
12	$t^* + 11$	
13	$t^* + 12$	
14	$t^* + 13$	
15	$t^* + 14$	
16	$t^* + 15$	
17	$t^* + 16$	
18	$t^* + 17$	
19	$t^* + 18$	

20	$t^* + 19$
21	$t^* + 20$
22	$t^* + 21$
23	$t^* + 22$
24	$t^* + 23$
25	$t^* + 24$
26	$t^* + 25$
27	$t^* + 26$
28	$t^* + 27$
29	$t^* + 28$
30	$t^* + 29$

Year after deforestation		$\Delta C_{bbfcl,t}$
		tCO _{2e} /ha
1	t^*	0
2	$t^* + 1$	2,78
3	$t^* + 2$	2,78
4	$t^* + 3$	2,78
5	$t^* + 4$	2,78
6	$t^* + 5$	2,78
7	$t^* + 6$	2,78
8	$t^* + 7$	2,78
9	$t^* + 8$	2,78
10	$t^* + 9$	2,78
11	$t^* + 10$	
12	$t^* + 11$	
13	$t^* + 12$	
14	$t^* + 13$	
15	$t^* + 14$	
16	$t^* + 15$	
17	$t^* + 16$	
18	$t^* + 17$	
19	$t^* + 18$	
20	$t^* + 19$	
21	$t^* + 20$	

22	$t^* + 21$
23	$t^* + 22$
24	$t^* + 23$
25	$t^* + 24$
26	$t^* + 25$
27	$t^* + 26$
28	$t^* + 27$
29	$t^* + 28$
30	$t^* + 29$

The resulting changes in carbon stock for initial (pre-deforestation) forest classes for the Reference Region, project area and leakage belt are shown in Table 47 to Table 54 below.

Table 47. Baseline carbon stock change in the above-ground biomass in the reference region

Carbon stock change in the above-ground biomass per initial forest class					
IDcl Name	1 Forest Formation	2 Savanna Formation	3 Forest- Savanna Transition	Total carbon stock change in the above-ground biomass of initial forest class in the reference region	
Project year	tCO ₂ e	tCO ₂ e	tCO ₂ e	$\Delta\text{CabBSLRR}_{icl,t}$ Annual (tCO ₂ e)	$\Delta\text{CabBSLRR}_{icl}$ Cumulative (tCO ₂ e)
2024	18.861	796.967	12.787	828.615	828.615
2025	18.538	783.307	12.568	814.413	1.643.028
2026	18.220	769.881	12.353	800.454	2.443.482
2027	17.908	756.685	12.141	786.734	3.230.216
2028	17.601	743.716	11.933	773.249	4.003.466
2029	12.149	733.827	9.284	755.260	4.758.726
2030	11.941	721.249	9.125	742.315	5.501.041
2031	11.736	708.887	8.969	729.592	6.230.633
2032	11.535	696.736	8.815	717.087	6.947.720
2033	11.337	684.794	8.664	704.796	7.652.515
2034	10.484	687.766	7.241	705.492	8.358.007
2035	10.305	675.978	7.117	693.400	9.051.407
2036	10.128	664.392	6.995	681.515	9.732.921
2037	9.955	653.004	6.875	669.833	10.402.755

2038	9.784	641.811	6.757	658.352	11.061.107
2039	12.224	616.112	4.196	632.531	11.693.639
2040	12.014	605.552	4.124	621.690	12.315.329
2041	11.808	595.173	4.053	611.034	12.926.363
2042	11.606	584.972	3.983	600.561	13.526.924
2043	11.407	574.945	3.915	590.267	14.117.191
2044	15.792	551.539	2.631	569.962	14.687.153
2045	15.521	542.086	2.586	560.193	15.247.345
2046	15.255	532.794	2.541	550.591	15.797.936
2047	14.994	523.662	2.498	541.154	16.339.090
2048	14.737	514.687	2.455	531.878	16.870.968
2049	19.219	516.505	2.480	538.205	17.409.172
2050	18.890	507.652	2.438	528.980	17.938.152
2051	18.566	498.951	2.396	519.913	18.458.065
2052	18.248	490.399	2.355	511.002	18.969.067
2053	17.935	481.993	2.315	502.243	19.471.310
2054	20.354	472.645	2.232	495.231	19.966.540
2055	20.006	464.543	2.194	486.743	20.453.283
2056	19.663	456.581	2.156	478.400	20.931.683
2057	19.326	448.755	2.119	470.200	21.401.883
2058	18.994	441.064	2.083	462.141	21.864.023
2059	19.515	428.015	2.047	449.577	22.313.600
2060	19.180	420.679	2.012	441.871	22.755.471
2061	18.851	413.469	1.977	434.297	23.189.768
2062	18.528	406.382	1.943	426.853	23.616.621
2063	18.211	399.416	1.910	419.537	24.036.158
2064	18.877	396.977	1.850	417.704	24.453.862
2065	18.553	390.172	1.818	410.544	24.864.406
2066	18.235	383.485	1.787	403.507	25.267.913
2067	17.923	376.912	1.756	396.591	25.664.504
2068	17.616	370.452	1.726	389.794	26.054.298
2069	20.106	364.270	1.478	385.855	26.440.153
2070	19.761	358.027	1.453	379.241	26.819.394
2071	19.423	351.890	1.428	372.741	27.192.135
2072	19.090	345.859	1.404	366.352	27.558.487
2073	18.763	339.931	1.380	360.073	27.918.559
2074	17.741	332.504	1.206	351.451	28.270.011
2075	17.437	326.805	1.185	345.427	28.615.438
2076	17.138	321.204	1.165	339.507	28.954.945
2077	16.844	315.698	1.145	333.687	29.288.632

2078	16.555	310.287	1.126	327.968	29.616.600
2079	16.505	298.241	743	315.489	29.932.089
2080	16.222	293.129	730	310.081	30.242.170
2081	15.944	288.105	718	304.766	30.546.936
2082	15.671	283.166	706	299.543	30.846.479
2083	15.402	278.313	694	294.408	31.140.887

Table 48. Baseline carbon stock change in the below-ground biomass in the reference region

Carbon stock change in the below-ground biomass per initial forest class					
IDcl	1	2	3	Total carbon stock change in the below-ground biomass of initial forest class in the reference region	
Name	Forest Formation	Savanna Formation	Forest-Savanna Transition		
Project year	tCO ₂ e	tCO ₂ e	tCO ₂ e	ΔCbbBSLRRic/t Annual (tCO ₂ e)	ΔCbbBSLRRic/ Cumulative (tCO ₂ e)
2024	582	120.664	915	122.161	122.161
2025	1.154	239.259	1.814	242.227	364.388
2026	1.717	355.822	2.697	360.236	724.624
2027	2.269	470.387	3.566	476.222	1.200.846
2028	2.812	582.989	4.419	590.220	1.791.066
2029	3.187	694.093	5.083	702.363	2.493.430
2030	3.556	803.292	5.736	812.584	3.306.014
2031	3.918	910.620	6.378	920.916	4.226.930
2032	4.274	1.016.109	7.008	1.027.391	5.254.322
2033	4.624	1.119.789	7.628	1.132.041	6.386.363
2034	4.365	1.103.256	7.231	1.114.853	7.501.216
2035	4.111	1.087.006	6.841	1.097.959	8.599.174
2036	3.862	1.071.034	6.458	1.081.354	9.680.528
2037	3.616	1.055.337	6.081	1.065.034	10.745.563
2038	3.375	1.039.908	5.711	1.048.994	11.794.557
2039	3.377	1.022.085	5.347	1.030.810	12.825.367
2040	3.380	1.004.569	4.989	1.012.937	13.838.304
2041	3.382	987.352	4.638	995.371	14.833.676
2042	3.384	970.430	4.292	978.106	15.811.782
2043	3.386	953.799	3.953	961.137	16.772.919
2044	3.550	933.173	3.623	940.346	17.713.265

2045	3.711	912.902	3.299	919.911	18.633.176
2046	3.869	892.977	2.980	899.826	19.533.002
2047	4.025	873.394	2.667	880.086	20.413.088
2048	4.177	854.147	2.359	860.684	21.273.772
2049	4.393	839.066	2.236	845.696	22.119.469
2050	4.606	824.244	2.116	830.965	22.950.434
2051	4.814	809.676	1.997	816.487	23.766.921
2052	5.019	795.357	1.881	802.257	24.569.178
2053	5.221	781.284	1.766	788.271	25.357.449
2054	5.361	769.339	1.738	776.438	26.133.887
2055	5.500	757.599	1.710	764.808	26.898.695
2056	5.636	746.060	1.682	753.378	27.652.073
2057	5.769	734.718	1.655	742.143	28.394.216
2058	5.901	723.572	1.629	731.101	29.125.317
2059	5.910	710.174	1.598	717.681	29.842.999
2060	5.919	697.006	1.567	704.492	30.547.491
2061	5.928	684.064	1.537	691.528	31.239.019
2062	5.936	671.343	1.508	678.787	31.917.806
2063	5.945	658.841	1.479	666.264	32.584.070
2064	5.899	647.384	1.451	654.735	33.238.805
2065	5.855	636.124	1.425	643.403	33.882.209
2066	5.810	625.057	1.398	632.266	34.514.474
2067	5.767	614.180	1.372	621.319	35.135.794
2068	5.725	603.489	1.347	610.560	35.746.354
2069	5.743	593.838	1.306	600.887	36.347.241
2070	5.761	584.352	1.266	591.379	36.938.619
2071	5.778	575.029	1.227	582.034	37.520.653
2072	5.796	565.865	1.188	572.849	38.093.502
2073	5.813	556.859	1.150	563.822	38.657.324
2074	5.778	547.097	1.104	553.979	39.211.304
2075	5.743	537.503	1.059	544.306	39.755.609
2076	5.709	528.074	1.014	534.798	40.290.407
2077	5.676	518.806	971	525.453	40.815.860
2078	5.643	509.697	928	516.268	41.332.127
2079	5.532	499.699	875	506.107	41.838.234
2080	5.423	489.874	823	496.120	42.334.355
2081	5.316	480.216	773	486.305	42.820.659
2082	5.210	470.725	723	476.657	43.297.317
2083	5.106	461.395	674	467.176	43.764.492

Table 49. Baseline carbon stock change in the above-ground biomass in the project area

Carbon stock change in the above-ground biomass per initial forest class						
IDcl	1	2	3	4	Total carbon stock change in the above-ground biomass of initial forest class in the project area	
Name	Wooded Savannah Sa	Savannah Forest Sd	Steppic-Savanna Forest - Td	Grassy-Woody Savannah - Tg		
Project year	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	ΔCabBSLPA _i /cl,t Annual (tCO ₂ e)	ΔCabBSLPA _i /cl Cumulative (tCO ₂ e)
2024	30.675	1.689	533	0	32.896	32.896
2025	30.149	1.660	524	0	32.333	65.229
2026	29.632	1.631	515	0	31.778	97.008
2027	29.124	1.603	506	0	31.234	128.241
2028	28.625	1.576	498	0	30.698	158.940
2029	29.007	4.522	1.383	0	34.912	193.852
2030	28.510	4.445	1.359	0	34.314	228.165
2031	28.021	4.368	1.336	0	33.725	261.891
2032	27.541	4.294	1.313	0	33.147	295.038
2033	27.069	4.220	1.290	0	32.579	327.617
2034	26.468	5.708	2.942	13	35.131	362.748
2035	26.014	5.610	2.891	13	34.529	397.277
2036	25.568	5.514	2.842	12	33.937	431.214
2037	25.130	5.420	2.793	12	33.355	464.569
2038	24.699	5.327	2.745	12	32.783	497.352
2039	23.308	5.058	3.626	27	32.019	529.371
2040	22.909	4.972	3.563	26	31.470	560.841
2041	22.516	4.886	3.502	26	30.931	591.772
2042	22.130	4.803	3.442	25	30.401	622.173
2043	21.751	4.720	3.383	25	29.880	652.052
2044	21.150	2.602	3.463	5	27.220	679.272
2045	20.787	2.557	3.404	5	26.754	706.026
2046	20.431	2.514	3.346	5	26.295	732.321
2047	20.081	2.470	3.288	5	25.844	758.165
2048	19.736	2.428	3.232	5	25.401	783.566
2049	19.808	1.684	4.219	0	25.711	809.277
2050	19.468	1.655	4.147	0	25.270	834.548

2051	19.134	1.627	4.076	0	24.837	859.385
2052	18.806	1.599	4.006	0	24.411	883.796
2053	18.484	1.572	3.937	0	23.993	907.789
2054	16.291	1.307	4.705	0	22.302	930.091
2055	16.012	1.284	4.624	0	21.920	952.011
2056	15.737	1.262	4.545	0	21.544	973.556
2057	15.468	1.241	4.467	0	21.175	994.731
2058	15.203	1.219	4.390	0	20.812	1.015.543
2059	11.871	40	3.904	0	15.815	1.031.358
2060	11.667	39	3.838	0	15.544	1.046.902
2061	11.468	38	3.772	0	15.278	1.062.180
2062	11.271	38	3.707	0	15.016	1.077.196
2063	11.078	37	3.644	0	14.759	1.091.954
2064	9.781	0	3.243	0	13.024	1.104.978
2065	9.613	0	3.187	0	12.800	1.117.778
2066	9.449	0	3.132	0	12.581	1.130.359
2067	9.287	0	3.079	0	12.365	1.142.724
2068	9.127	0	3.026	0	12.153	1.154.878
2069	8.125	0	2.922	0	11.048	1.165.925
2070	7.986	0	2.872	0	10.858	1.176.784
2071	7.849	0	2.823	0	10.672	1.187.456
2072	7.714	0	2.775	0	10.489	1.197.945
2073	7.582	0	2.727	0	10.309	1.208.254
2074	7.381	0	2.673	0	10.054	1.218.308
2075	7.254	0	2.627	0	9.881	1.228.189
2076	7.130	0	2.582	0	9.712	1.237.901
2077	7.008	0	2.538	0	9.546	1.247.447
2078	6.887	0	2.494	0	9.382	1.256.829
2079	6.199	0	2.170	0	8.369	1.265.197
2080	6.093	0	2.133	0	8.225	1.273.422
2081	5.988	0	2.096	0	8.084	1.281.507
2082	5.885	0	2.060	0	7.946	1.289.452
2083	5.785	0	2.025	0	7.809	1.297.262

Table 50. Baseline carbon stock change in the below-ground biomass in the project area

Carbon stock change in the below-ground biomass per initial forest class					
IDcl	1	2	3	4	Total carbon stock change in the below-ground biomass of

Name	Wooded Savannah Sa	Savannah Forest Sd	Steppic-Savanna Forest - Td	Grassy-Woody Savannah - Tg	initial forest class in the project area	
Project year	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	ΔCbbBSLPA/cI,t Annual (tCO ₂ e)	ΔCbbBSLPA/cI Cumulative (tCO ₂ e)
2024	6.258	37	29	0	6.324	6.324
2025	12.408	74	57	0	12.539	18.862
2026	18.453	110	85	0	18.648	37.510
2027	24.394	145	113	0	24.652	62.162
2028	30.234	179	139	0	30.553	92.714
2029	36.151	279	214	0	36.644	129.359
2030	41.967	377	288	0	42.632	171.990
2031	47.684	473	360	0	48.516	220.507
2032	53.302	567	431	0	54.300	274.807
2033	58.824	660	501	0	59.985	334.792
2034	57.966	748	632	4	59.350	394.142
2035	57.122	835	760	9	58.726	452.868
2036	56.293	920	886	13	58.112	510.980
2037	55.479	1.004	1.010	17	57.509	568.490
2038	54.678	1.087	1.131	21	56.917	625.407
2039	53.515	1.099	1.253	30	55.896	681.303
2040	52.373	1.110	1.372	39	54.894	736.196
2041	51.250	1.122	1.489	47	53.908	790.104
2042	50.146	1.133	1.605	56	52.939	843.043
2043	49.061	1.144	1.718	64	51.987	895.030
2044	47.976	1.076	1.746	62	50.859	945.889
2045	46.909	1.008	1.774	59	49.751	995.640
2046	45.861	942	1.801	56	48.662	1.044.302
2047	44.831	878	1.828	54	47.591	1.091.893
2048	43.819	814	1.855	52	46.539	1.138.432
2049	43.105	740	1.887	43	45.774	1.184.205
2050	42.403	667	1.918	34	45.022	1.229.227
2051	41.713	595	1.949	25	44.282	1.273.509
2052	41.035	525	1.980	17	43.556	1.317.065
2053	40.369	455	2.010	8	42.842	1.359.907
2054	39.377	427	2.077	7	41.888	1.401.795
2055	38.403	399	2.143	5	40.950	1.442.745
2056	37.446	371	2.208	3	40.028	1.482.774
2057	36.505	344	2.272	2	39.123	1.521.896

2058	35.580	318	2.335	0	38.232	1.560.129
2059	33.961	282	2.317	0	36.560	1.596.689
2060	32.370	246	2.301	0	34.916	1.631.605
2061	30.805	211	2.284	0	33.301	1.664.906
2062	29.268	177	2.268	0	31.713	1.696.619
2063	27.757	143	2.252	0	30.153	1.726.771
2064	26.429	114	2.173	0	28.717	1.755.488
2065	25.124	86	2.095	0	27.305	1.782.793
2066	23.841	58	2.019	0	25.918	1.808.711
2067	22.580	31	1.944	0	24.555	1.833.266
2068	21.341	4	1.870	0	23.215	1.856.481
2069	20.577	3	1.817	0	22.397	1.878.877
2070	19.826	2	1.764	0	21.592	1.900.470
2071	19.087	2	1.713	0	20.802	1.921.272
2072	18.362	1	1.662	0	20.025	1.941.297
2073	17.649	0	1.613	0	19.262	1.960.558
2074	17.159	0	1.582	0	18.741	1.979.299
2075	16.678	0	1.552	0	18.229	1.997.529
2076	16.205	0	1.522	0	17.727	2.015.256
2077	15.740	0	1.493	0	17.232	2.032.488
2078	15.283	0	1.464	0	16.747	2.049.235
2079	14.890	0	1.423	0	16.313	2.065.548
2080	14.504	0	1.383	0	15.887	2.081.434
2081	14.124	0	1.344	0	15.468	2.096.902
2082	13.751	0	1.305	0	15.056	2.111.958
2083	13.384	0	1.267	0	14.651	2.126.609

Table 51. Total baseline carbon stock change of initial forest class in project area

Total carbon stock change of the initial forest class				
IDcl	1	Forest	Total carbon stock change of initial forest class in the project area	
Name	$\Delta C_{abBSLPAicl,t}$	$\Delta C_{bbBSLPAicl,t}$		
Project year	tCO _{2e}	tCO _{2e}	Annual (tCO _{2e})	Cumulative (tCO _{2e})
2024	32.896	6.324	39.220	39.220
2025	32.333	12.539	44.872	84.092

2026	31.778	18.648	50.426	134.518
2027	31.234	24.652	55.885	190.403
2028	30.698	30.553	61.251	251.654
2029	34.912	36.644	71.556	323.210
2030	34.314	42.632	76.945	400.156
2031	33.725	48.516	82.242	482.398
2032	33.147	54.300	87.448	569.845
2033	32.579	59.985	92.564	662.410
2034	35.131	59.350	94.481	756.890
2035	34.529	58.726	93.254	850.145
2036	33.937	58.112	92.049	942.194
2037	33.355	57.509	90.865	1.033.058
2038	32.783	56.917	89.700	1.122.759
2039	32.019	55.896	87.915	1.210.674
2040	31.470	54.894	86.364	1.297.038
2041	30.931	53.908	84.839	1.381.876
2042	30.401	52.939	83.340	1.465.216
2043	29.880	51.987	81.866	1.547.082
2044	27.220	50.859	78.079	1.625.161
2045	26.754	49.751	76.505	1.701.666
2046	26.295	48.662	74.957	1.776.623
2047	25.844	47.591	73.435	1.850.058
2048	25.401	46.539	71.940	1.921.998
2049	25.711	45.774	71.484	1.993.483
2050	25.270	45.022	70.292	2.063.774
2051	24.837	44.282	69.120	2.132.894
2052	24.411	43.556	67.967	2.200.861
2053	23.993	42.842	66.835	2.267.696
2054	22.302	41.888	64.190	2.331.886
2055	21.920	40.950	62.870	2.394.757
2056	21.544	40.028	61.573	2.456.329
2057	21.175	39.123	60.298	2.516.627
2058	20.812	38.232	59.044	2.575.671
2059	15.815	36.560	52.375	2.628.046
2060	15.544	34.916	50.460	2.678.507
2061	15.278	33.301	48.579	2.727.086
2062	15.016	31.713	46.729	2.773.815
2063	14.759	30.153	44.911	2.818.726
2064	13.024	28.717	41.740	2.860.466
2065	12.800	27.305	40.106	2.900.571

2066	12.581	25.918	38.499	2.939.071
2067	12.365	24.555	36.920	2.975.991
2068	12.153	23.215	35.368	3.011.359
2069	11.048	22.397	33.444	3.044.803
2070	10.858	21.592	32.451	3.077.253
2071	10.672	20.802	31.474	3.108.727
2072	10.489	20.025	30.514	3.139.242
2073	10.309	19.262	29.571	3.168.812
2074	10.054	18.741	28.795	3.197.607
2075	9.881	18.229	28.111	3.225.718
2076	9.712	17.727	27.439	3.253.157
2077	9.546	17.232	26.778	3.279.935
2078	9.382	16.747	26.129	3.306.063
2079	8.369	16.313	24.682	3.330.745
2080	8.225	15.887	24.112	3.354.857
2081	8.084	15.468	23.552	3.378.409
2082	7.946	15.056	23.002	3.401.410
2083	7.809	14.651	22.461	3.423.871

Table 52. Baseline carbon stock change in the above-ground biomass in the leakage belt area

Carbon stock change in the above-ground biomass per initial forest class					
IDcl	1	2	3	Total carbon stock change in the	
Name	Forest Formation	Savanna Formation	Forest-Savanna Transition	above-ground biomass of initial forest class in the leakage belt	
Project year	tCO _{2e}	tCO _{2e}	tCO _{2e}	ΔCabBSLLKicl,t Annual (tCO _{2e})	ΔCabBSLLKicl Cumulative (tCO _{2e})
2024	8.380	407.245	0	415.625	415.625
2025	8.236	400.265	0	408.501	824.127
2026	8.095	393.405	0	401.500	1.225.626
2027	7.956	386.662	0	394.618	1.620.244
2028	7.820	380.034	0	387.854	2.008.098
2029	4.497	383.038	0	387.535	2.395.633
2030	4.420	376.473	0	380.892	2.776.525
2031	4.344	370.020	0	374.364	3.150.889
2032	4.269	363.678	0	367.947	3.518.836

2033	4.196	357.444	0	361.641	3.880.477
2034	3.320	353.252	0	356.572	4.237.049
2035	3.263	347.197	0	350.461	4.587.510
2036	3.207	341.246	0	344.454	4.931.964
2037	3.152	335.397	0	338.550	5.270.513
2038	3.098	329.649	0	332.747	5.603.260
2039	4.532	306.968	0	311.500	5.914.760
2040	4.454	301.706	0	306.161	6.220.921
2041	4.378	296.535	0	300.913	6.521.834
2042	4.303	291.453	0	295.755	6.817.589
2043	4.229	286.457	0	290.686	7.108.275
2044	6.029	266.212	0	272.240	7.380.515
2045	5.925	261.649	0	267.574	7.648.089
2046	5.824	257.164	0	262.988	7.911.077
2047	5.724	252.756	0	258.480	8.169.557
2048	5.626	248.424	0	254.050	8.423.606
2049	7.849	245.646	0	253.494	8.677.101
2050	7.714	241.435	0	249.149	8.926.250
2051	7.582	237.297	0	244.879	9.171.129
2052	7.452	233.230	0	240.682	9.411.811
2053	7.324	229.232	0	236.556	9.648.367
2054	8.355	221.517	0	229.872	9.878.240
2055	8.212	217.720	0	225.932	10.104.172
2056	8.071	213.988	0	222.060	10.326.232
2057	7.933	210.321	0	218.254	10.544.485
2058	7.797	206.716	0	214.513	10.758.998
2059	7.861	193.545	0	201.406	10.960.404
2060	7.726	190.228	0	197.954	11.158.359
2061	7.594	186.967	0	194.561	11.352.920
2062	7.464	183.763	0	191.227	11.544.146
2063	7.336	180.613	0	187.949	11.732.095
2064	7.688	174.913	0	182.601	11.914.696
2065	7.556	171.915	0	179.471	12.094.167
2066	7.427	168.968	0	176.395	12.270.562
2067	7.299	166.072	0	173.371	12.443.933
2068	7.174	163.226	0	170.400	12.614.333
2069	7.575	157.939	0	165.515	12.779.848
2070	7.445	155.232	0	162.678	12.942.525
2071	7.318	152.572	0	159.889	13.102.415
2072	7.192	149.957	0	157.149	13.259.564

2073	7.069	147.386	0	154.455	13.414.019
2074	6.337	142.350	0	148.687	13.562.706
2075	6.228	139.910	0	146.138	13.708.844
2076	6.122	137.512	0	143.633	13.852.477
2077	6.017	135.155	0	141.171	13.993.648
2078	5.914	132.838	0	138.752	14.132.400
2079	5.941	126.513	0	132.453	14.264.854
2080	5.839	124.344	0	130.183	14.395.037
2081	5.739	122.213	0	127.952	14.522.989
2082	5.641	120.118	0	125.759	14.648.747
2083	5.544	118.059	0	123.603	14.772.350

Table 53. Baseline carbon stock change in the below-ground biomass in the leakage belt area

Carbon stock change in the below-ground biomass per initial forest class					
IDcl	1	2	3	Total carbon stock change in the	
Name	Forest Formation	Savanna Formation	Forest-Savanna Transition	below-ground biomass of initial forest class in the leakage belt	
Project year	tCO _{2e}	tCO _{2e}	tCO _{2e}	$\Delta C_{bbBSLLKicl,t}$ Annual (tCO _{2e})	$\Delta C_{bbBSLLKicl}$ Cumulative (tCO _{2e})
2024	168	67.148	0	67.316	67.316
2025	332	133.146	0	133.478	200.794
2026	494	198.012	0	198.507	399.301
2027	653	261.767	0	262.420	661.721
2028	810	324.429	0	325.238	986.959
2029	899	387.586	0	388.485	1.375.444
2030	988	449.660	0	450.648	1.826.093
2031	1.075	510.671	0	511.746	2.337.838
2032	1.160	570.636	0	571.796	2.909.634
2033	1.244	629.573	0	630.817	3.540.451
2034	1.143	620.670	0	621.813	4.162.264
2035	1.043	611.920	0	612.963	4.775.227
2036	946	603.320	0	604.266	5.379.493
2037	850	594.867	0	595.717	5.975.210
2038	755	586.559	0	587.315	6.562.524
2039	756	574.017	0	574.773	7.137.297
2040	757	561.689	0	562.445	7.699.742

2041	757	549.572	0	550.330	8.250.072
2042	758	537.664	0	538.421	8.788.493
2043	759	525.959	0	526.717	9.315.211
2044	813	511.607	0	512.420	9.827.631
2045	866	497.502	0	498.368	10.325.998
2046	918	483.638	0	484.556	10.810.554
2047	970	470.011	0	470.981	11.281.535
2048	1.020	456.619	0	457.639	11.739.174
2049	1.087	446.508	0	447.594	12.186.768
2050	1.152	436.570	0	437.722	12.624.490
2051	1.216	426.803	0	428.018	13.052.508
2052	1.279	417.202	0	418.481	13.470.989
2053	1.341	407.767	0	409.108	13.880.097
2054	1.387	400.398	0	401.785	14.281.882
2055	1.433	393.154	0	394.587	14.676.469
2056	1.478	386.035	0	387.513	15.063.982
2057	1.522	379.038	0	380.560	15.444.543
2058	1.565	372.161	0	373.727	15.818.269
2059	1.566	363.571	0	365.136	16.183.406
2060	1.566	355.127	0	356.693	16.540.099
2061	1.566	346.829	0	348.395	16.888.494
2062	1.566	338.672	0	340.239	17.228.733
2063	1.567	330.656	0	332.223	17.560.955
2064	1.553	322.972	0	324.525	17.885.480
2065	1.540	315.419	0	316.959	18.202.439
2066	1.527	307.996	0	309.523	18.511.963
2067	1.515	300.700	0	302.215	18.814.177
2068	1.502	293.529	0	295.031	19.109.208
2069	1.496	287.658	0	289.155	19.398.363
2070	1.491	281.888	0	283.379	19.681.742
2071	1.485	276.217	0	277.702	19.959.444
2072	1.480	270.643	0	272.122	20.231.566
2073	1.475	265.164	0	266.639	20.498.205
2074	1.447	259.795	0	261.242	20.759.447
2075	1.421	254.518	0	255.939	21.015.386
2076	1.395	249.331	0	250.726	21.266.112
2077	1.369	244.233	0	245.603	21.511.714
2078	1.344	239.223	0	240.567	21.752.281
2079	1.311	234.041	0	235.352	21.987.634
2080	1.279	228.948	0	230.227	22.217.861

2081	1.248	223.942	0	225.190	22.443.051
2082	1.217	219.022	0	220.239	22.663.290
2083	1.186	214.187	0	215.373	22.878.663

Table 54. Total baseline carbon stock change of initial forest class in leakage belt area

Total carbon stock change of the initial forest class				
IDcl	1	Forest	Total carbon stock change of initial forest class in the leakage belt	
Name	$\Delta C_{abBSLLKicl,t}$	$\Delta C_{bbBSLLKicl,t}$		
Project year	tCO _{2e}	tCO _{2e}	Annual (tCO _{2e})	Cumulative (tCO _{2e})
2024	415.625	67.316	482.941	482.941
2025	408.501	133.478	541.980	1.024.921
2026	401.500	198.507	600.006	1.624.927
2027	394.618	262.420	657.038	2.281.965
2028	387.854	325.238	713.092	2.995.058
2029	387.535	388.485	776.020	3.771.078
2030	380.892	450.648	831.540	4.602.618
2031	374.364	511.746	886.109	5.488.727
2032	367.947	571.796	939.743	6.428.470
2033	361.641	630.817	992.457	7.420.928
2034	356.572	621.813	978.385	8.399.313
2035	350.461	612.963	963.424	9.362.737
2036	344.454	604.266	948.719	10.311.456
2037	338.550	595.717	934.267	11.245.723
2038	332.747	587.315	920.062	12.165.785
2039	311.500	574.773	886.272	13.052.057
2040	306.161	562.445	868.606	13.920.663
2041	300.913	550.330	851.243	14.771.905
2042	295.755	538.421	834.177	15.606.082
2043	290.686	526.717	817.403	16.423.485
2044	272.240	512.420	784.660	17.208.145
2045	267.574	498.368	765.941	17.974.087
2046	262.988	484.556	747.544	18.721.631
2047	258.480	470.981	729.461	19.451.092
2048	254.050	457.639	711.689	20.162.780

2049	253.494	447.594	701.089	20.863.869
2050	249.149	437.722	686.871	21.550.740
2051	244.879	428.018	672.897	22.223.637
2052	240.682	418.481	659.163	22.882.800
2053	236.556	409.108	645.664	23.528.464
2054	229.872	401.785	631.657	24.160.121
2055	225.932	394.587	620.520	24.780.641
2056	222.060	387.513	609.573	25.390.214
2057	218.254	380.560	598.814	25.989.028
2058	214.513	373.727	588.240	26.577.267
2059	201.406	365.136	566.543	27.143.810
2060	197.954	356.693	554.648	27.698.458
2061	194.561	348.395	542.956	28.241.414
2062	191.227	340.239	531.465	28.772.879
2063	187.949	332.223	520.171	29.293.051
2064	182.601	324.525	507.126	29.800.176
2065	179.471	316.959	496.430	30.296.606
2066	176.395	309.523	485.918	30.782.524
2067	173.371	302.215	475.586	31.258.110
2068	170.400	295.031	465.431	31.723.541
2069	165.515	289.155	454.669	32.178.211
2070	162.678	283.379	446.057	32.624.267
2071	159.889	277.702	437.591	33.061.858
2072	157.149	272.122	429.271	33.491.130
2073	154.455	266.639	421.094	33.912.224
2074	148.687	261.242	409.929	34.322.153
2075	146.138	255.939	402.077	34.724.230
2076	143.633	250.726	394.359	35.118.589
2077	141.171	245.603	386.774	35.505.363
2078	138.752	240.567	379.319	35.884.681
2079	132.453	235.352	367.806	36.252.487
2080	130.183	230.227	360.410	36.612.898
2081	127.952	225.190	353.142	36.966.040
2082	125.759	220.239	345.998	37.312.037
2083	123.603	215.373	338.976	37.651.014

5.2 Project Emissions

Some unplanned deforestation may happen in the project area despite the implemented REDD project activity. The level at which deforestation will actually be reduced in the project case

depends on the effectiveness of the proposed activities, which cannot be measured *ex ante*. *Ex post* measurements of the project results will be important to determine actual emission reductions.

To allow *ex ante* projections to be made, a conservative assumption was made about the effectiveness of the proposed project activities in order to define the Effectiveness Index (*EI*). The estimated value of *EI* is used to multiply the baseline projections by the factor $(1 - EI)$ and the result was considered to be the *ex-ante* estimated emissions from unplanned deforestation in the project case. This is calculated as follows:

$$\Delta CUDdPat = \Delta CBSLPat \times (1 - EI)$$

Where:

$\Delta CUDdPat$	Total ex ante actual carbon stock change due to unavoided unplanned deforestation at year <i>t</i> in the project area; tCO ₂ e
$\Delta CBSLPat$	Total baseline carbon stock change in the project area at year <i>t</i> ; tCO ₂ e
<i>EI</i>	Ex ante estimated Effectiveness Index; %
<i>t</i>	1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless

Based on the historical of deforestation happened in the area before Project start, the Effectiveness Index (*EI*) of project activities was conservatively assumed as 85%, conservatively based on the internal risks of the project. The whole assessment of internal and other risks can be found on NPR Assessment filled in the VERRA's Hub / Risk Analysis, which is dully assessed by the VVB of this Project.

This range of values represents the efficiency of the CULTIVAR PROGRESSO REDD+ PROJECT activities in reducing deforestation within the project area. It was then applied to calculate the *ex-ante* estimate of net carbon stock change in the project area under the project scenario ($\Delta CPSPat$), as follows.

$$\Delta CPSPat = (\Delta CPAdPat + \Delta CUDdPat) - \Delta CPAiPat$$

Where:

$\Delta CPSPat$	Sum of ex ante estimated actual carbon stock changes in the project area at year <i>t</i> ; tCO ₂ e
$\Delta CPAdPat$	Total decrease in carbon stock due to all planned activities at year <i>t</i> in the project area; tCO ₂ e

$\Delta CUDdPA_t$	Total ex ante actual carbon stock change due to unavoided unplanned deforestation at year t in the project area; tCO ₂ e
$\Delta CPAiPA_t$	Total increase in carbon stock due to all planned activities at year t in the project area; tCO ₂ e

Forest fires have not been included in the baseline scenario, thus non-CO₂ emissions from biomass burning shall not be included in the project scenario. Therefore, EBBSP_{PA}_t equals to zero.

Ex ante estimated net carbon stock change in the project area under the project scenario is showed in *Table 55* below.

Table 55. Total ex ante estimated actual net carbon stock changes and emissions of non-CO₂ gases in the Project Area

Project year	Total carbon stock decrease due to planned activities		Total carbon stock increase due to planned activities		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock decrease change in the project case		Total ex ante estimated actual non-CO ₂ emissions from forest fires in the project area	
	$\Delta CPA_d PA_t$ annual	$\Delta CPA_d PA_t$ cumulative	$\Delta CPA_i PA_t$ annual	$\Delta CPA_i PA_t$ cumulative	$\Delta CUD_d PA_t$ annual	$\Delta CUD_d PA_t$ cumulative	$\Delta CPS PA_t$ annual	$\Delta CPS PA_t$ cumulative	EBBPS PA _t annual	EBBPS PA _t cumulative
	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e
2024	0,00	0,00	0,00	0,00	5.883	5.883	5.883	5.883	0,00	0,00
2025	0,00	0,00	0,00	0,00	6.731	12.614	6.731	12.614	0,00	0,00
2026	0,00	0,00	0,00	0,00	7.564	20.178	7.564	20.178	0,00	0,00
2027	0,00	0,00	0,00	0,00	8.383	28.560	8.383	28.560	0,00	0,00
2028	0,00	0,00	0,00	0,00	9.188	37.748	9.188	37.748	0,00	0,00
2029	0,00	0,00	0,00	0,00	10.733	48.482	10.733	48.482	0,00	0,00
2030	0,00	0,00	0,00	0,00	11.542	60.023	11.542	60.023	0,00	0,00
2031	0,00	0,00	0,00	0,00	12.336	72.360	12.336	72.360	0,00	0,00
2032	0,00	0,00	0,00	0,00	13.117	85.477	13.117	85.477	0,00	0,00
2033	0,00	0,00	0,00	0,00	13.885	99.361	13.885	99.361	0,00	0,00

2034	0,00	0,00	0,00	0,00	14.172	113.534	14.172	113.534	0,00	0,00
2035	0,00	0,00	0,00	0,00	13.988	127.522	13.988	127.522	0,00	0,00
2036	0,00	0,00	0,00	0,00	13.807	141.329	13.807	141.329	0,00	0,00
2037	0,00	0,00	0,00	0,00	13.630	154.959	13.630	154.959	0,00	0,00
2038	0,00	0,00	0,00	0,00	13.455	168.414	13.455	168.414	0,00	0,00
2039	0,00	0,00	0,00	0,00	13.187	181.601	13.187	181.601	0,00	0,00
2040	0,00	0,00	0,00	0,00	12.955	194.556	12.955	194.556	0,00	0,00
2041	0,00	0,00	0,00	0,00	12.726	207.281	12.726	207.281	0,00	0,00
2042	0,00	0,00	0,00	0,00	12.501	219.782	12.501	219.782	0,00	0,00
2043	0,00	0,00	0,00	0,00	12.280	232.062	12.280	232.062	0,00	0,00
2044	0,00	0,00	0,00	0,00	11.712	243.774	11.712	243.774	0,00	0,00
2045	0,00	0,00	0,00	0,00	11.476	255.250	11.476	255.250	0,00	0,00
2046	0,00	0,00	0,00	0,00	11.243	266.493	11.243	266.493	0,00	0,00
2047	0,00	0,00	0,00	0,00	11.015	277.509	11.015	277.509	0,00	0,00
2048	0,00	0,00	0,00	0,00	10.791	288.300	10.791	288.300	0,00	0,00
2049	0,00	0,00	0,00	0,00	10.723	299.022	10.723	299.022	0,00	0,00
2050	0,00	0,00	0,00	0,00	10.544	309.566	10.544	309.566	0,00	0,00
2051	0,00	0,00	0,00	0,00	10.368	319.934	10.368	319.934	0,00	0,00
2052	0,00	0,00	0,00	0,00	10.195	330.129	10.195	330.129	0,00	0,00
2053	0,00	0,00	0,00	0,00	10.025	340.154	10.025	340.154	0,00	0,00
2054	0,00	0,00	0,00	0,00	9.629	349.783	9.629	349.783	0,00	0,00
2055	0,00	0,00	0,00	0,00	9.431	359.213	9.431	359.213	0,00	0,00
2056	0,00	0,00	0,00	0,00	9.236	368.449	9.236	368.449	0,00	0,00
2057	0,00	0,00	0,00	0,00	9.045	377.494	9.045	377.494	0,00	0,00
2058	0,00	0,00	0,00	0,00	8.857	386.351	8.857	386.351	0,00	0,00
2059	0,00	0,00	0,00	0,00	7.856	394.207	7.856	394.207	0,00	0,00
2060	0,00	0,00	0,00	0,00	7.569	401.776	7.569	401.776	0,00	0,00
2061	0,00	0,00	0,00	0,00	7.287	409.063	7.287	409.063	0,00	0,00

2062	0,00	0,00	0,00	0,00	7.009	416.07 2	7.009	416.07 2	0,00	0,00
2063	0,00	0,00	0,00	0,00	6.737	422.80 9	6.737	422.80 9	0,00	0,00
2064	0,00	0,00	0,00	0,00	6.261	429.07 0	6.261	429.07 0	0,00	0,00
2065	0,00	0,00	0,00	0,00	6.016	435.08 6	6.016	435.08 6	0,00	0,00
2066	0,00	0,00	0,00	0,00	5.775	440.86 1	5.775	440.86 1	0,00	0,00
2067	0,00	0,00	0,00	0,00	5.538	446.39 9	5.538	446.39 9	0,00	0,00
2068	0,00	0,00	0,00	0,00	5.305	451.70 4	5.305	451.70 4	0,00	0,00
2069	0,00	0,00	0,00	0,00	5.017	456.72 0	5.017	456.72 0	0,00	0,00
2070	0,00	0,00	0,00	0,00	4.868	461.58 8	4.868	461.58 8	0,00	0,00
2071	0,00	0,00	0,00	0,00	4.721	466.30 9	4.721	466.30 9	0,00	0,00
2072	0,00	0,00	0,00	0,00	4.577	470.88 6	4.577	470.88 6	0,00	0,00
2073	0,00	0,00	0,00	0,00	4.436	475.32 2	4.436	475.32 2	0,00	0,00
2074	0,00	0,00	0,00	0,00	4.319	479.64 1	4.319	479.64 1	0,00	0,00
2075	0,00	0,00	0,00	0,00	4.217	483.85 8	4.217	483.85 8	0,00	0,00
2076	0,00	0,00	0,00	0,00	4.116	487.97 4	4.116	487.97 4	0,00	0,00
2077	0,00	0,00	0,00	0,00	4.017	491.99 0	4.017	491.99 0	0,00	0,00
2078	0,00	0,00	0,00	0,00	3.919	495.90 9	3.919	495.90 9	0,00	0,00
2079	0,00	0,00	0,00	0,00	3.702	499.61 2	3.702	499.61 2	0,00	0,00
2080	0,00	0,00	0,00	0,00	3.617	503.22 9	3.617	503.22 9	0,00	0,00
2081	0,00	0,00	0,00	0,00	3.533	506.76 1	3.533	506.76 1	0,00	0,00
2082	0,00	0,00	0,00	0,00	3.450	510.21 2	3.450	510.21 2	0,00	0,00
2083	0,00	0,00	0,00	0,00	3.369	513.58 1	3.369	513.58 1	0,00	0,00

5.3 Leakage Emissions

This step provides an ex-ante estimate of the possible decrease in carbon stock and increase in GHG emissions (other than carbon stock change) due to leakage. According to the applied methodology, two sources of leakage are considered: a) decrease in carbon stocks and increase in GHG emissions associated with leakage prevention measures; and b) decrease in carbon stocks and increase in GHG emissions associated with activity displacement leakage.

Ex ante estimation of decrease in carbon stocks and increase in GHG emissions due to leakage prevention measures.

Initially, it is expected that leakage prevention measures will be employed within the limits of Fazenda Tapirapé, conducting courses and training related to sustainable development and conservation and environmental awareness. Subsequently, outside the limits of the Project Area, through assistance to associations of small farmers in the environment. These initiatives will focus not only on training and guidance for farmers in the region but also on raising people's awareness of environmental issues and preserving the forest.

As already mentioned in this document, it is not expected to develop any activity that could lead to the reduction of carbon stocks or the increase of GHG emissions compared to the baseline scenario. If there are significant changes in carbon stock, these activities will be monitored, accounted for and reported. Activities that will cause deforestation within the project area in the baseline case could be displaced outside the project boundary due to the implementation of the AUD project activity. A greater decrease in carbon stocks within the leakage belt during the project scenario than those predicted *ex-ante* would indicate displacement of deforestation activities due to the project.

The *ex ante* activity displacement leakage was calculated based on the anticipated combined effectiveness of the proposed leakage prevention measures and project activities. This was done by multiplying the estimated baseline carbon stock changes for the project area by a “Displacement Leakage Factor” (DLF) representing the percent of deforestation expected to be displaced outside the project boundary, beginning with an index of 5% and decreasing it along the project life-time. The equation is presented below:

$$\Delta CADLK_t = \Delta CBSLP_{At} \times DLF$$

Where,

$\Delta CADLK_t$ Total decrease in carbon stocks due to displaced deforestation at year t; tCO₂e

DLF Displacement leakage factor; %

Thus, a displacement factor of 5% was adopted for the first five years. Then the reduction of the leakage displacement factor is gradual, already considering the influence of the Project in this context. Thus, the leakage displacement factor tends to approach to zero during the 60 years of project implementation.

The actual calculated values for $\Delta CADLK_t$, annually and cumulatively, across the project crediting period are own in *Table 56* below.

Table 56. Ex ante estimated total leakage

Project year	Total ex ante GHG emissions from increased grazing activities		Total ex ante Increase in GHG emissions due to displaced forest fires		Total ex ante decrease in carbon stocks due to displaced deforestation		Carbon stock decrease due to leakage prevention measures		Total net carbon stock change due to leakage within the leakage belt		Total net increase in emissions due to leakage	
	EgLKt annual	EgLK cumulative	EADLKt annual	EADLK cumulative	ΔCADLK _t annual	ΔCADLK cumulative	ΔCLPMLK _t annual	ΔCLPMLK cumulative	ΔCLKt annual	ΔCLK cumulative	ELK _t annual	ELK cumulative
	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e
2024	0,00	0,00	0,00	0,00	1.961	1.961	0,00	0,00	1.961	1.961	0,00	0,00
2025	0,00	0,00	0,00	0,00	2.244	4.205	0,00	0,00	2.244	4.205	0,00	0,00
2026	0,00	0,00	0,00	0,00	2.521	6.726	0,00	0,00	2.521	6.726	0,00	0,00
2027	0,00	0,00	0,00	0,00	2.794	9.520	0,00	0,00	2.794	9.520	0,00	0,00
2028	0,00	0,00	0,00	0,00	3.063	12.583	0,00	0,00	3.063	12.583	0,00	0,00
2029	0,00	0,00	0,00	0,00	2.862	15.445	0,00	0,00	2.862	15.445	0,00	0,00
2030	0,00	0,00	0,00	0,00	3.078	18.523	0,00	0,00	3.078	18.523	0,00	0,00
2031	0,00	0,00	0,00	0,00	3.290	21.812	0,00	0,00	3.290	21.812	0,00	0,00
2032	0,00	0,00	0,00	0,00	3.498	25.310	0,00	0,00	3.498	25.310	0,00	0,00
2033	0,00	0,00	0,00	0,00	2.777	28.087	0,00	0,00	2.777	28.087	0,00	0,00
2034	0,00	0,00	0,00	0,00	2.834	30.922	0,00	0,00	2.834	30.922	0,00	0,00
2035	0,00	0,00	0,00	0,00	2.798	33.719	0,00	0,00	2.798	33.719	0,00	0,00
2036	0,00	0,00	0,00	0,00	2.761	36.481	0,00	0,00	2.761	36.481	0,00	0,00
2037	0,00	0,00	0,00	0,00	1.817	38.298	0,00	0,00	1.817	38.298	0,00	0,00
2038	0,00	0,00	0,00	0,00	1.794	40.092	0,00	0,00	1.794	40.092	0,00	0,00
2039	0,00	0,00	0,00	0,00	1.758	41.850	0,00	0,00	1.758	41.850	0,00	0,00
2040	0,00	0,00	0,00	0,00	1.727	43.578	0,00	0,00	1.727	43.578	0,00	0,00
2041	0,00	0,00	0,00	0,00	848	44.426	0,00	0,00	848	44.426	0,00	0,00
2042	0,00	0,00	0,00	0,00	833	45.259	0,00	0,00	833	45.259	0,00	0,00
2043	0,00	0,00	0,00	0,00	819	46.078	0,00	0,00	819	46.078	0,00	0,00
2044	0,00	0,00	0,00	0,00	781	46.859	0,00	0,00	781	46.859	0,00	0,00
2045	0,00	0,00	0,00	0,00	765	47.624	0,00	0,00	765	47.624	0,00	0,00
2046	0,00	0,00	0,00	0,00	750	48.374	0,00	0,00	750	48.374	0,00	0,00
2047	0,00	0,00	0,00	0,00	734	49.108	0,00	0,00	734	49.108	0,00	0,00
2048	0,00	0,00	0,00	0,00	719	49.827	0,00	0,00	719	49.827	0,00	0,00
2049	0,00	0,00	0,00	0,00	715	50.542	0,00	0,00	715	50.542	0,00	0,00
2050	0,00	0,00	0,00	0,00	703	51.245	0,00	0,00	703	51.245	0,00	0,00
2051	0,00	0,00	0,00	0,00	691	51.936	0,00	0,00	691	51.936	0,00	0,00
2052	0,00	0,00	0,00	0,00	680	52.616	0,00	0,00	680	52.616	0,00	0,00
2053	0,00	0,00	0,00	0,00	668	53.284	0,00	0,00	668	53.284	0,00	0,00
2054	0,00	0,00	0,00	0,00	642	53.926	0,00	0,00	642	53.926	0,00	0,00
2055	0,00	0,00	0,00	0,00	629	54.555	0,00	0,00	629	54.555	0,00	0,00
2056	0,00	0,00	0,00	0,00	616	55.171	0,00	0,00	616	55.171	0,00	0,00
2057	0,00	0,00	0,00	0,00	603	55.774	0,00	0,00	603	55.774	0,00	0,00
2058	0,00	0,00	0,00	0,00	590	56.364	0,00	0,00	590	56.364	0,00	0,00
2059	0,00	0,00	0,00	0,00	524	56.888	0,00	0,00	524	56.888	0,00	0,00
2060	0,00	0,00	0,00	0,00	505	57.392	0,00	0,00	505	57.392	0,00	0,00
2061	0,00	0,00	0,00	0,00	486	57.878	0,00	0,00	486	57.878	0,00	0,00
2062	0,00	0,00	0,00	0,00	467	58.345	0,00	0,00	467	58.345	0,00	0,00
2063	0,00	0,00	0,00	0,00	449	58.795	0,00	0,00	449	58.795	0,00	0,00
2064	0,00	0,00	0,00	0,00	417	59.212	0,00	0,00	417	59.212	0,00	0,00
2065	0,00	0,00	0,00	0,00	401	59.613	0,00	0,00	401	59.613	0,00	0,00
2066	0,00	0,00	0,00	0,00	385	59.998	0,00	0,00	385	59.998	0,00	0,00
2067	0,00	0,00	0,00	0,00	369	60.367	0,00	0,00	369	60.367	0,00	0,00
2068	0,00	0,00	0,00	0,00	354	60.721	0,00	0,00	354	60.721	0,00	0,00
2069	0,00	0,00	0,00	0,00	334	61.055	0,00	0,00	334	61.055	0,00	0,00
2070	0,00	0,00	0,00	0,00	325	61.380	0,00	0,00	325	61.380	0,00	0,00
2071	0,00	0,00	0,00	0,00	315	61.695	0,00	0,00	315	61.695	0,00	0,00
2072	0,00	0,00	0,00	0,00	305	62.000	0,00	0,00	305	62.000	0,00	0,00
2073	0,00	0,00	0,00	0,00	296	62.295	0,00	0,00	296	62.295	0,00	0,00
2074	0,00	0,00	0,00	0,00	288	62.583	0,00	0,00	288	62.583	0,00	0,00
2075	0,00	0,00	0,00	0,00	281	62.864	0,00	0,00	281	62.864	0,00	0,00
2076	0,00	0,00	0,00	0,00	274	63.139	0,00	0,00	274	63.139	0,00	0,00
2077	0,00	0,00	0,00	0,00	268	63.407	0,00	0,00	268	63.407	0,00	0,00
2078	0,00	0,00	0,00	0,00	261	63.668	0,00	0,00	261	63.668	0,00	0,00
2079	0,00	0,00	0,00	0,00	247	63.915	0,00	0,00	247	63.915	0,00	0,00
2080	0,00	0,00	0,00	0,00	241	64.156	0,00	0,00	241	64.156	0,00	0,00
2081	0,00	0,00	0,00	0,00	236	64.391	0,00	0,00	236	64.391	0,00	0,00
2082	0,00	0,00	0,00	0,00	230	64.621	0,00	0,00	230	64.621	0,00	0,00
2083	0,00	0,00	0,00	0,00	225	64.846	0,00	0,00	225	64.846	0,00	0,00

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The net anthropogenic GHG emission reduction of the proposed AUD project activity is calculated as follows:

$$\Delta REDDt = (\Delta CBSLPAt + EBBBSLPAt) - (\Delta CPSPAt + EBBPSPAt) - (\Delta CLKt + ELKt)$$

Where:

$\Delta REDDt$	Ex ante estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t; tCO ₂ e
$\Delta CBSLPAt$	Sum of baseline carbon stock changes in the project area at year t; tCO ₂ e
$EBBBSLPAt$	Sum of baseline emissions from biomass burning in the project area at year t; tCO ₂ e
$\Delta CPSPAt$	Sum of ex ante estimated actual carbon stock changes in the project area at year t; tCO ₂ e Note: If $\Delta CPSPAt$ represents a net increase in carbon stocks, a negative sign before the absolute value of $\Delta CPSPAt$ shall be used. If $\Delta CPSPAt$ represents a net decrease, the positive sign shall be used.
$EBBPSPAt$	Sum of (ex ante estimated) actual emissions from biomass burning in the project area at year t; tCO ₂ e
$\Delta CLKt$	Sum of ex ante estimated leakage net carbon stock changes at year t; tCO ₂ e Note: If the cumulative sum of $\Delta CLKt$ within a fixed baseline period is > 0, $\Delta CLKt$ shall be set to zero.
$ELKt$	Sum of ex ante estimated leakage emissions at year t; tCO ₂ e
t	1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless.

The number of Verified Carbon Units (VCUs) to be generated through the proposed AUD project activity at year t is calculated as follows:

$$VCUt = \Delta REDDt - VBCt$$

$$VBCt = (\Delta CBSLPAt - \Delta CPSPAt) \times Rf_t$$

Where:

$VCUt$	Number of Verified Carbon Units that can be traded at time t; t CO ₂ -e
$\Delta REDDt$	Ex ante estimated net anthropogenic greenhouse gas emission reduction attributable to the AUD project activity at year t; tCO ₂ e

VBC_t	Number of Buffer Credits deposited in the VCS Buffer at time t ; t CO ₂ -e
$\Delta CBSLPAt$	Sum of baseline carbon stock changes in the project area at year t ; tCO ₂ e
$\Delta CPSPAt$	Sum of ex ante estimated actual carbon stock changes in the project area at year t ; tCO ₂ -e ha ⁻¹
RF_t	Risk factor used to calculate VCS buffer credits; %
t	1, 2, 3 ... T, a year of the proposed project crediting period; dimensionless.

The RF_t was estimated using the most recent version of the VCS-approved AFOLU Non-Permanence Risk Tool the resulting value of RF_t was 22%.

The specific summary of GHG reductions and removals in the CULTIVAR PROGRESSO REDD+ PROJECT is included in *Table 57*. The latter table includes estimates of GHG emissions reduction ($REDD_t$), calculations of buffer and leakage, and the resulting calculation of tradeable Verified Carbon Units ($VCUt$).

Table 57. Ex ante estimated net anthropogenic GHG emission reductions (REDDt) and Verified Carbon Units (VCUt)

Project year	Baseline carbon stock changes		Baseline GHG emissions from biomass burning		Ex ante project carbon stock changes		Ex ante project GHG emissions from biomass burning		Ex ante leakage carbon stock changes within the leakage belt		Ex ante leakage GHG emissions		Ex ante net anthropogenic GHG emission reductions		Ex ante VCUs tradable		Ex ante buffer credits	
	ΔCBSLPat annual	ΔCBSLPat cumulative	ΔEBBBSLPat annual	ΔEBBBSLPat cumulative	ΔCPSLPat annual	ΔCPSLPat cumulative	ΔEBBBSPat annual	ΔEBBBSPat cumulative	ΔCLKt annual	ΔCLK cumulative	ELKt annual	ELK cumulative	ΔREDDt annual	ΔREDD cumulative	ΔVCUt annual	ΔVCU cumulative	ΔVBUT annual	ΔVBU cumulative
	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e
2024	39.220	39.220	0	0	1.961	1.961	0	0	5.883	5.883	0	0	31.376	31.376	24.473	24.473	6.903	6.903
2025	43.648	82.868	0	0	2.244	4.205	0	0	6.731	12.614	0	0	34.674	66.050	27.045	51.519	7.628	14.531
2026	48.897	131.765	0	0	2.521	6.726	0	0	7.564	20.178	0	0	38.812	104.862	30.273	81.792	8.539	23.070
2027	53.723	185.488	0	0	2.794	9.520	0	0	8.383	28.560	0	0	42.546	147.408	33.186	114.978	9.360	32.430
2028	58.467	243.955	0	0	3.063	12.583	0	0	9.188	37.748	0	0	46.216	193.624	36.049	151.027	10.168	42.597
2029	68.160	312.115	0	0	2.862	15.445	0	0	10.733	48.482	0	0	54.565	248.189	42.560	193.587	12.004	54.601
2030	72.893	385.008	0	0	3.078	18.523	0	0	11.542	60.023	0	0	58.274	306.462	45.453	239.041	12.820	67.422
2031	77.545	462.554	0	0	3.290	21.812	0	0	12.336	72.360	0	0	61.919	368.381	48.297	287.338	13.622	81.044
2032	82.117	544.671	0	0	3.498	25.310	0	0	13.117	85.477	0	0	65.502	433.884	51.092	338.429	14.410	95.454
2033	86.611	631.282	0	0	2.777	28.087	0	0	13.885	99.361	0	0	69.950	503.833	54.561	392.990	15.389	110.843
2034	88.371	719.652	0	0	2.834	30.922	0	0	14.172	113.534	0	0	71.364	575.197	55.664	448.654	15.700	126.543
2035	87.581	807.234	0	0	2.798	33.719	0	0	13.988	127.522	0	0	70.796	645.993	55.221	503.874	15.575	142.118
2036	86.383	893.617	0	0	2.761	36.481	0	0	13.807	141.329	0	0	69.815	715.808	54.455	558.330	15.359	157.478
2037	85.206	978.823	0	0	1.817	38.298	0	0	13.630	154.959	0	0	69.759	785.567	54.412	612.742	15.347	172.825
2038	84.049	1.062.872	0	0	1.794	40.092	0	0	13.455	168.414	0	0	68.800	854.366	53.664	666.406	15.136	187.961
2039	82.326	1.145.198	0	0	1.758	41.850	0	0	13.187	181.601	0	0	67.380	921.746	52.556	718.962	14.824	202.784
2040	80.841	1.226.039	0	0	1.727	43.578	0	0	12.955	194.556	0	0	66.159	987.906	51.604	770.566	14.555	217.339
2041	79.382	1.305.421	0	0	848	44.426	0	0	12.726	207.281	0	0	65.808	1.053.714	51.330	821.897	14.478	231.817
2042	77.948	1.383.369	0	0	833	45.259	0	0	12.501	219.782	0	0	64.614	1.118.328	50.399	872.296	14.215	246.032
2043	76.539	1.459.908	0	0	819	46.078	0	0	12.280	232.062	0	0	63.440	1.181.768	49.483	921.779	13.957	259.989
2044	72.847	1.532.755	0	0	781	46.859	0	0	11.712	243.774	0	0	60.354	1.242.122	47.076	968.855	13.278	273.267
2045	71.388	1.604.143	0	0	765	47.624	0	0	11.476	255.250	0	0	59.148	1.301.269	46.135	1.014.990	13.012	286.279
2046	69.955	1.674.098	0	0	750	48.374	0	0	11.243	266.493	0	0	57.962	1.359.231	45.210	1.060.200	12.752	299.031
2047	68.546	1.742.644	0	0	734	49.108	0	0	11.015	277.509	0	0	56.796	1.416.027	44.301	1.104.501	12.495	311.526
2048	67.161	1.809.805	0	0	719	49.827	0	0	10.791	288.300	0	0	55.651	1.471.678	43.408	1.147.909	12.243	323.769
2049	66.807	1.876.613	0	0	715	50.542	0	0	10.723	299.022	0	0	55.370	1.527.048	43.189	1.191.098	12.181	335.951
2050	65.697	1.942.309	0	0	703	51.245	0	0	10.544	309.566	0	0	54.450	1.581.498	42.471	1.233.569	11.979	347.930
2051	64.605	2.006.915	0	0	691	51.936	0	0	10.368	319.934	0	0	53.546	1.635.044	41.766	1.275.335	11.780	359.710
2052	63.536	2.070.450	0	0	680	52.616	0	0	10.195	330.129	0	0	52.661	1.687.705	41.075	1.316.410	11.585	371.295
2053	62.481	2.132.932	0	0	668	53.284	0	0	10.025	340.154	0	0	51.788	1.739.493	40.394	1.356.804	11.393	382.688
2054	59.893	2.192.824	0	0	642	53.926	0	0	9.629	349.783	0	0	49.623	1.789.115	38.706	1.395.510	10.917	393.605
2055	58.657	2.251.481	0	0	629	54.555	0	0	9.431	359.213	0	0	48.598	1.837.713	37.906	1.433.416	10.691	404.297
2056	57.442	2.308.923	0	0	616	55.171	0	0	9.236	368.449	0	0	47.590	1.885.303	37.120	1.470.536	10.470	414.767
2057	56.248	2.365.171	0	0	603	55.774	0	0	9.045	377.494	0	0	46.600	1.931.903	36.348	1.506.884	10.252	425.019
2058	55.074	2.420.245	0	0	590	56.364	0	0	8.857	386.351	0	0	45.627	1.977.530	35.589	1.542.474	10.038	435.057
2059	48.501	2.468.746	0	0	524	56.888	0	0	7.856	394.207	0	0	40.121	2.017.651	31.295	1.573.768	8.827	443.883
2060	46.760	2.515.506	0	0	505	57.392	0	0	7.569	401.776	0	0	38.686	2.056.338	30.175	1.603.943	8.511	452.394
2061	45.049	2.560.555	0	0	496	57.878	0	0	7.287	409.063	0	0	37.276	2.093.614	29.075	1.633.019	8.201	460.595
2062	43.366	2.603.921	0	0	467	58.345	0	0	7.009	416.072	0	0	35.890	2.129.504	27.994	1.661.013	7.896	468.491
2063	41.713	2.645.634	0	0	449	58.795	0	0	6.737	422.809	0	0	34.527	2.164.031	26.931	1.687.944	7.596	476.087
2064	38.676	2.684.310	0	0	417	59.212	0	0	6.261	429.070	0	0	31.997	2.196.028	24.958	1.712.902	7.039	483.126
2065	37.201	2.721.511	0	0	401	59.613	0	0	6.016	435.086	0	0	30.784	2.226.812	24.011	1.736.914	6.772	489.899
2066	35.751	2.757.262	0	0	385	59.998	0	0	5.775	440.861	0	0	29.591	2.256.403	23.081	1.759.995	6.510	496.409
2067	34.326	2.791.588	0	0	369	60.367	0	0	5.538	446.399	0	0	28.419	2.284.822	22.167	1.782.161	6.252	502.661
2068	32.925	2.824.513	0	0	354	60.721	0	0	5.305	451.704	0	0	27.266	2.312.088	21.268	1.803.429	5.999	508.659
2069	31.071	2.855.585	0	0	334	61.055	0	0	5.017	456.720	0	0	25.720	2.337.809	20.062	1.823.491	5.658	514.318
2070	30.165	2.885.749	0	0	325	61.380	0	0	4.868	461.588	0	0	24.973	2.362.782	19.479	1.842.970	5.494	519.812
2071	29.274	2.915.024	0	0	315	61.695	0	0	4.721	466.309	0	0	24.238	2.387.020	18.906	1.861.876	5.332	525.144
2072	28.398	2.943.422	0	0	305	62.000	0	0	4.577	470.886	0	0	23.516	2.410.536	18.343	1.880.218	5.174	530.318
2073	27.538	2.970.960	0	0	296	62.295	0	0	4.436	475.322	0	0	22.806	2.433.343	17.789	1.898.007	5.017	535.335
2074	26.815	2.997.775	0	0	288	62.583	0	0	4.319	479.641	0	0	22.208	2.455.550	17.322	1.915.329	4.886	540.221
2075	26.185	3.023.959	0	0	281	62.864	0	0	4.217	483.858	0	0	21.687	2.477.237	16.916	1.932.245	4.771	544.992
2076	25.565	3.049.525	0	0	274	63.139	0	0	4.116	487.974	0	0	21.175	2.498.412	16.517	1.948.761	4.659	549.651
2077	24.957	3.074.481	0	0	268	63.407	0	0	4.017	491.990	0	0	20.672	2.519.084	16.124	1.964.886	4.548	554.199
2078	24.358	3.098.839	0	0	261	63.668	0	0	3.919	495.909	0	0	20.178	2.539.262	15.739	1.980.624	4.439	558.638
2079	22.943	3.121.783	0	0	247	63.915	0	0	3.702	499.612	0	0	18.994	2.558.256	14.815	1.995.440	4.179	562.816
2080	22.421	3.144.203	0	0	241	64.156	0	0	3.617	503.229	0	0	18.563	2.576.819	14.479	2.009.919	4.084	566.900
2081	21.907	3.166.111	0	0	236	64.391	0	0	3.533	506.761	0	0	18.139	2.594.958	14.149	2.024.067	3.991	570.891
2082	21.403	3.187.514	0	0	230	64.621	0	0	3.450	510.212	0	0	17.723	2.612.681	13.824	2.037.891	3.899	574.790
2083	20.907	3.208.421	0	0	225	64.846	0	0	3.369	513.581	0	0	17.313	2.629.994	13.504	2.051.395	3.809	578.599

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	CF _{dc}
Data unit	t C/tonne dry matter
Description	Carbon fraction of density class <i>dc</i>
Source of data	Brazil's FREL for reducing emissions from deforestation in the Amazonia biome for REDD+ results-based payments under the UNFCCC from 2016 to 2020 (2018).
Value applied	0.485
Justification of choice of data or description of measurement methods and procedures applied	The average carbon fraction of dry matter (0.485±0.9%) used by Brazil's FREL is taken from SILVA (2007), as explicit in the FREL document, which is the main reference for carbon emissions inventory in the national level. The CF used by FREL is very similar to those used by Higuchi et al. (1994) after Lima et al. (2007), equal to 0.40 for the average water content in aboveground fresh biomass and 0.47 for the average carbon fraction of dry matter.
Purpose of Data	Indirect calculation of biomass above and belowground, used to estimate the referenced data of carbon stock for the baseline and projections of future deforestation.
Comments	The IPCC default values are 0.5 tonne dry matter/tonne fresh biomass (IPCC 2003); and 0.47 tonne carbon/tonne dry matter (IPCC 2006, Table 4.3), respectively. This demonstrate that the value used by Silva is aligned to conventional references in a universal level.

Data / Parameter	ABSLRR _t
Data unit	ha
Description	Annual area of deforestation in the reference region at year <i>t</i>
Source of data	Calculated
Value applied	Table 23, Section 3.4

Justification of choice of data or description of measurement methods and procedures applied	As stated in Section 3, the method used to project annual deforestation areas in the baseline for the reference region is the Temporal Function approach, using a logarithmic regression equation of the historical data for forest cover loss (i.e., the opposite of deforestation). The use of logarithmic regression is justified by the need to project the deforestation trend in a more realistic way, considering a gradual decrease in the annual deforestation rate as the remaining forest area diminishes. The calculation sequence begins with identifying the remaining forest area at the start of the historical reference period. This is done by subtracting the water-covered areas, non-forest vegetation, and consolidated deforestation area up until 2012 from the total area of the reference region. Once the remaining forest area at the beginning of the reference period is determined, the observed deforestation area is subtracted year by year, creating a historical series of forest cover loss for the reference region.
Purpose of Data	Determination of baseline scenario
Comments	Monitoring – each renewal of baseline validity period

Data / Parameter	ABSLPA _t
Data unit	ha
Description	Annual area of deforestation in the project area at year t
Source of data	Calculated
Value applied	Table 24, Section 3.4
Justification of choice of data or description of measurement methods and procedures applied	Forest cover change due to deforestation will be monitored through assessment of classified satellite imagery covering the three stratum in the project area.
Purpose of Data	This parameter will be used to calculate baseline emissions and project emissions in the project scenario.
Comments	N/A

Data / Parameter	Cabid
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Data unit	tCO2e/ha																																												
Description	Average carbon stock per hectare in aboveground biomass carbon pool of initial forest																																												
Source of data	Estimated from FREL																																												
Value applied	For Reference Region and Leakage Belt: <table><tr><th colspan="5">ΔC_{biol},t</th></tr><tr><th>Year after deforestation</th><th></th><th>Forest formation</th><th>Savanna formation</th><th>Forest-savanna transition</th></tr><tr><th></th><th></th><th>tCO₂e/ha</th><th>tCO₂e/ha</th><th>tCO₂e/ha</th></tr><tr><td>1</td><td>t*</td><td>-69,6</td><td>-88,2</td><td>-64,1</td></tr></table> For Project Area: <table><tr><th colspan="6">ΔC_{biol},t</th></tr><tr><th>Year after deforestation</th><th></th><th>Wooded Savannah</th><th>Savannah Forest</th><th>Steppic-Savanna Forest</th><th>Grassy-Woody Savannah</th></tr><tr><th></th><th></th><th>tCO₂e/ha</th><th>tCO₂e/ha</th><th>tCO₂e/ha</th><th>tCO₂e/ha</th></tr><tr><td>1</td><td>t*</td><td>-95,1</td><td>-26,8</td><td>-35,2</td><td>-48,1</td></tr></table>	ΔC _{biol} ,t					Year after deforestation		Forest formation	Savanna formation	Forest-savanna transition			tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha	1	t*	-69,6	-88,2	-64,1	ΔC _{biol} ,t						Year after deforestation		Wooded Savannah	Savannah Forest	Steppic-Savanna Forest	Grassy-Woody Savannah			tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha	1	t*	-95,1	-26,8	-35,2	-48,1
ΔC _{biol} ,t																																													
Year after deforestation		Forest formation	Savanna formation	Forest-savanna transition																																									
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		tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha	tCO ₂ e/ha																																								
1	t*	-95,1	-26,8	-35,2	-48,1																																								
Justification of choice of data or description of measurement methods and procedures applied	The Carbon stocks per hectare of initial forest existing in the project area were calculated based on official literature regarding specific characteristics of different biomes and vegetation aspects, all referenced on this PD. The calculation has been ran considering an area-weighted average carbon stocks of the post deforestation LU/LC classes existing within each zone are displayed.																																												
Purpose of Data	This parameter was used to calculate the baseline, project and leakage emissions from deforestation occurred in the baseline and project scenarios. Provides an average of the biomass stock per hectare in the above-ground biomass within the project reference region.																																												
Comments	If new and more accurate biomass stock data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.																																												

Data / Parameter	Cbb icl																																																																																	
Data unit	tCO2e/ha																																																																																	
Description	Average carbon stock per hectare in belowground biomass carbon pool of initial forest																																																																																	
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	ΔCbbicl,t																																																																																	
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	8	$t^* + 7$	-4,66	-12,21	-6,50	-1,44
	9	$t^* + 8$	-4,66	-12,21	-6,50	-1,44
	10	$t^* + 9$	-4,66	-12,21	-6,50	-1,44
Justification of choice of data or description of measurement methods and procedures applied	The Carbon stocks per hectare of initial forest existing in the project area were calculated based on official literature regarding specific characteristics of different biomes and vegetation aspects, all referenced on this PD. The calculation has been ran considering an area-weighted average carbon stocks of the post deforestation LU/LC classes existing within each zone are displayed.					
Purpose of Data	This parameter was used to calculate the baseline, project and leakage emissions from deforestation occurred in the baseline and project scenarios. Provides an average of the biomass stock per hectare in the below-ground biomass within the project reference region.					
Comments	If new and more accurate biomass stock data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.					

Data / Parameter	Cbbfcl																								
Data unit	t CO2e/ha																								
Description	Average carbon stock per hectare in aboveground biomass carbon pool of final post-deforestation class fcl																								
Source of data	Brazil's 4th National Forest Reference Emission Level for Results-based Payments for REDD+ under the United Nations Framework Convention on Climate Change.																								
Value applied	<table><tr><th colspan="3">$\Delta C_{bbfcl,t}$</th></tr><tr><th colspan="3">Year after deforestation</th></tr><tr><th></th><th></th><th>tCO₂e/ha</th></tr><tr><td>1</td><td>t*</td><td>0</td></tr><tr><td>2</td><td>t* + 1</td><td>1,83</td></tr><tr><td>3</td><td>t* + 2</td><td>1,83</td></tr><tr><td>4</td><td>t* + 3</td><td>1,83</td></tr><tr><td>5</td><td>t* + 4</td><td>1,83</td></tr></table>	$\Delta C_{bbfcl,t}$			Year after deforestation					tCO ₂ e/ha	1	t*	0	2	t* + 1	1,83	3	t* + 2	1,83	4	t* + 3	1,83	5	t* + 4	1,83
$\Delta C_{bbfcl,t}$																									
Year after deforestation																									
		tCO ₂ e/ha																							
1	t*	0																							
2	t* + 1	1,83																							
3	t* + 2	1,83																							
4	t* + 3	1,83																							
5	t* + 4	1,83																							

6	$t^* + 5$	1,83
7	$t^* + 6$	1,83
8	$t^* + 7$	1,83
9	$t^* + 8$	1,83
10	$t^* + 9$	1,83

Table above: for final class – Agriculture

Year after deforestation		$\Delta C_{bbfcl,t}$
		tCO ₂ e/ha
1	t^*	0
2	$t^* + 1$	2,78
3	$t^* + 2$	2,78
4	$t^* + 3$	2,78
5	$t^* + 4$	2,78
6	$t^* + 5$	2,78
7	$t^* + 6$	2,78
8	$t^* + 7$	2,78
9	$t^* + 8$	2,78
10	$t^* + 9$	2,78

Table above: for final class – Livestock/Pasture

Justification of choice of data or description of measurement methods and procedures applied	The chosen EF for carbon removal (final class) was applied considering a mix of Agriculture and Livestock/Pasture, as stated during Section 3.4. In order to attend the requirements from the methodology, the factor was converted to tCO ₂ /ha, by multiplying the total biomass by the constant (44/12) over a period of 9 years, which is the reasonable timing for carbon sequestration after one year of the deforestation-date.
Purpose of Data	To calculate the carbon emissions in the post-deforestation scenario for the projected deforestation on the main boundaries (PA and LK).
Comments	As it was used as a post-deforestation class, this can be seen as very accurate, since it's connected to the actual LULC proportions, as presented by data of TerraClass (Section 3.4).

Data / Parameter	El
Data unit	%
Description	<i>Ex ante estimated effectiveness index</i>
Source of data	<i>IPCC 2016. Good practice guidance for land use, land-use change and forestry.</i>
Value applied	85
Justification of choice of data or description of measurement methods and procedures applied	<i>The Effectiveness Index (EI) quantifies the anticipated effectiveness of project activities in preventing unplanned deforestation within the project area. This assessment is deliberately conservative, as the actual effectiveness can only be accurately determined through ex post measurements after project activities have been implemented.</i> <i>Given that the EI is intrinsically tied to the scope and execution of project activities, the Project Proponent (PP) determined its value by starting with a baseline of 100 and applying a reduction of 15% to account for internal risks, resulting in an EI of 85%. These internal risks pertain to aspects of the project over which the PP exercises direct influence and control, such as management efficiency, financial stability, opportunity costs, and the long-term sustainability of the initiative. By addressing these risks within the EI calculation, the PP acknowledges their central role in shaping the project's outcomes and integrates a precautionary buffer to enhance reliability.</i>
Purpose of Data	<i>This parameter was used to calculate project emissions in the baseline scenario. Provides an ex ante estimation of the carbon stock changes due to unavoidable unplanned deforestation within the project area, based on the effectiveness of the proposed project activities to reduce the deforestation.</i>
Comments	<i>Ex post monitoring of the project area shall be done to determine deforestation rate and the project emissions. This parameter will be updated at each renewal of fixed baseline period.</i>

Data / Parameter	DLF
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Data unit	%
Description	<i>Displacement Leakage Factor</i>
Source of data	<i>IPCC 2016. Good practice guidance for land use, land-use change and forestry.</i>
Value applied	5
Justification of choice of data or description of measurement methods and procedures applied	<i>The value set for the Displacement Leakage Factor (DLF) was established under the rationale that actions implemented in the leakage management area will effectively enhance income-generating capacity for deforestation agents in the region.</i> <i>The 5% value assigned for the first five years aligns well with historical land-use dynamics, reflecting the similarity in deforestation patterns between the property where the project is located and other areas within the leakage belt. The gradual reduction of the leakage factor to 1% over 17 years further justifies the conservative nature of this approach.</i>
Purpose of Data	<i>This parameter was used to calculate leakage emissions in the baseline scenario due to activity displacement leakage, providing an ex-ante estimation of the decrease in carbon stocks and increase in GHG emissions. This value was calculated based on the percent of deforestation expected to be displaced outside the project boundary due to the implementation of the AUD project activity.</i>
Comments	<i>Ex post monitoring of the leakage belt shall be done to determine deforestation rate outside the project area and consequently, the leakage emissions and carbon stock decrease. This parameter will be updated at each renewal of fixed baseline period.</i>

Data / Parameter	$\Delta C_{abBSLR} t$
Data unit	tCO ₂ e
Description	<i>Changes in the aboveground stock in reference region</i>
Source of data	<i>Calculated</i>
Value applied	Table 47, Section 5.1

Justification of choice of data or description of measurement methods and procedures applied	<i>A carbon stock change factors for initial forest classes in aboveground carbon pool were applied to calculate baseline carbon stock changes.</i>
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	<i>Monitoring – each renewal of baseline validity period</i>

Data / Parameter	$\Delta C_{bbBSLRR} \text{ t}$
Data unit	tCO ₂ e
Description	<i>Changes in the belowground stock in reference region</i>
Source of data	<i>Calculated</i>
Value applied	Table 48, Section 5.1
Justification of choice of data or description of measurement methods and procedures applied	<i>A carbon stock change factors for initial forest classes in belowground carbon pool were applied to calculate baseline carbon stock changes.</i>
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	<i>Monitoring – each renewal of baseline validity period</i>

Data / Parameter	$\Delta C_{abBSLPA} \text{ t}$
Data unit	tCO ₂ e
Description	<i>Changes in the aboveground stock in project area</i>
Source of data	<i>Calculated</i>
Value applied	Table 49, Section 5.1
Justification of choice of data or description of measurement methods and procedures applied	<i>A carbon stock change factors for initial forest classes in aboveground carbon pool were applied to calculate baseline carbon stock changes.</i>
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	<i>Monitoring – each renewal of baseline validity period</i>

Data / Parameter	$\Delta C_{bbBSLPA}$ t
Data unit	tCO ₂ e
Description	Changes in the belowground stock in project area
Source of data	Calculated
Value applied	Table 50, Section 5.1
Justification of choice of data or description of measurement methods and procedures applied	A carbon stock change factors for initial forest classes in belowground carbon pool were applied to calculate baseline carbon stock changes.
Purpose of Data	Calculation of baseline emissions
Comments	Monitoring – each renewal of baseline validity period

Data / Parameter	$\Delta C_{abBSLLK}$ t
Data unit	tCO ₂ e
Description	Changes in the aboveground stock in leakage belt
Source of data	Calculated
Value applied	Table 52, Section 5.1
Justification of choice of data or description of measurement methods and procedures applied	A carbon stock change factors for initial forest classes in aboveground carbon pool were applied to calculate baseline carbon stock changes.
Purpose of Data	Calculation of baseline emissions
Comments	Monitoring – each renewal of baseline validity period

Data / Parameter	$\Delta C_{bbBSLLK}$ t
Data unit	tCO ₂ e
Description	Changes in the belowground stock in leakage belt

Source of data	<i>Calculated</i>
Value applied	<i>Table 53, Section 5.1</i>
Justification of choice of data or description of measurement methods and procedures applied	<i>A carbon stock change factors for initial forest classes in belowground carbon pool were applied to calculate baseline carbon stock changes.</i>
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	<i>Monitoring – each renewal of baseline validity period</i>

Data / Parameter	$\Delta\text{CBSLPA } t$
Data unit	tCO ₂ e
Description	<i>Total baseline carbon stock change within project area at year t</i>
Source of data	<i>Calculated</i>
Value applied	<i>Table 57, Section 5.4</i>
Justification of choice of data or description of measurement methods and procedures applied	<i>The total baseline carbon stock change in the project area at year t is calculated as follows:</i> $\text{CBSLPA}t = \text{CabBSLPA } t + \text{CbbBSLPA } t$ <i>Where,</i> <i>CBSLPA t: Total baseline carbon stock changes in the project area at year t; tCO₂e</i> <i>CabBSLPA t: Total baseline carbon stock change for the above-ground biomass pool in the project area for initial forest class at year t; tCO₂e</i> <i>CbbBSLPA t: Total baseline carbon stock change for the below-ground biomass pool in the project area for initial forest class at year t; tCO₂e</i>
Purpose of Data	<i>Calculation of baseline emissions</i>
Comments	<i>Monitoring – each renewal of baseline validity period</i>

6.2 Data and Parameters Monitored

Data / Parameter	ABSLRR t
Data unit	Ha
Description	Annual area of deforestation in the reference region at year t
Source of data	Remote sensing and GIS
Description of measurement methods and procedures to be applied	Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the reference region
Frequency of monitoring/recording	Annually
Value applied	Table 23, Section 3.4
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing
Purpose of data	This parameter will be used to calculate baseline emissions in the project scenario. Provides the ex post value of the deforested area per forest class within the reference region
Calculation method	Analysis of satellite images and maps
Comments	N/A

Data / Parameter	ABSLPA t
Data unit	Ha
Description	Annual area of deforestation in the project area at year t
Source of data	Remote sensing and GIS
Description of measurement methods and procedures to be applied	Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the project area.

Frequency of monitoring/recording	<i>Annually</i>
Value applied	<i>Table 24, Section 3.4</i>
Monitoring equipment	<i>Remote sensing and GIS</i>
QA/QC procedures to be applied	<i>Best practices in remote sensing</i>
Purpose of data	<i>This parameter will be used to calculate baseline emissions and project emissions in the project scenario. Provides the ex post value of the deforested area per forest class within the project area.</i>
Calculation method	<i>Analysis of satellite images and maps</i>
Comments	<i>N/A</i>

Data / Parameter	ABSLK t
Data unit	<i>Ha</i>
Description	<i>Annual area of deforestation in the leakage belt at year t</i>
Source of data	<i>Remote sensing and GIS</i>
Description of measurement methods and procedures to be applied	<i>Deforestation in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.</i>
Frequency of monitoring/recording	<i>Annually</i>
Value applied	<i>Table 25, Section 3.4</i>
Monitoring equipment	<i>Remote sensing and GIS</i>
QA/QC procedures to be applied	<i>Best practices in remote sensing</i>
Purpose of data	<i>This parameter will be used to calculate leakage emissions in the project scenario. Provides the ex post value of the deforested area within the leakage belt.</i>

Calculation method	<i>Analysis of satellite images and maps</i>
Comments	<i>Where strong evidence can be collected that deforestation in the leakage belt is attributable to deforestation agents that are not linked to the project area, the detected deforestation will not be attributed to the project activity, thus not considered leakage.</i>

Data / Parameter	$\Delta CADLK_t$
Data unit	tCO ₂ e
Description	<i>Total decrease in carbon stocks due to displaced deforestation at year t</i>
Source of data	<i>Remote sensing and GIS.</i>
Description of measurement methods and procedures to be applied	<i>Deforestation in the leakage belt area will be considered activity displacement leakage. Activity data for the leakage belt area will be determined using the same methods applied to monitoring deforestation activity data in the project area.</i>
Frequency of monitoring/recording	<i>Annually</i>
Value applied	<i>Table 56, Section 3.4</i>
Monitoring equipment	<i>Remote sensing and GIS</i>
QA/QC procedures to be applied	<i>Best practices in remote sensing</i>
Purpose of data	<i>This parameter will be used to calculate leakage emissions in the project scenario. Provides the ex post value of the decrease in carbon stocks due to displaced deforestation in the leakage belt.</i>
Calculation method	<i>Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area</i>
Comments	<i>Where evidence can be collected that deforestation in the leakage belt is attributable to deforestation agents that are not linked to the project area, the detected deforestation may not be attributed to the project activity and therefore, not considered leakage.</i>

Data / Parameter	ΔCUDdPA_t
Data unit	tCO ₂ e
Description	Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area
Source of data	Remote sensing and GIS.
Description of measurement methods and procedures to be applied	Forest cover change due to unplanned deforestation will be monitored through assessment of classified satellite imagery covering the project area.
Frequency of monitoring/recording	Annually
Value applied	Table 55, Section 3.4
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	Best practices in remote sensing
Purpose of data	This parameter will be used to calculate project emissions in the project scenario. Provides the ex post value of the change in carbon stocks due to unavoided unplanned deforestation within the project area.
Calculation method	Emissions from deforestation are estimated by multiplying the detected area of forest loss by the average forest carbon stock per unit area
Comments	N/A

Data / Parameter	RF _t
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	- VCS Non-Permanence Risk Report (v3.1) - Remote sensing data and GIS - Supervisor report - Literature data

Description of measurement methods and procedures to be applied	<i>All sources of data from the VCS Non-Permanence Risk Report will be used to measure the various risk factors.</i>
Frequency of monitoring/recording	<i>Annually</i>
Value applied	<i>22</i>
Monitoring equipment	<i>VERRA HUB (AFOLU NPR tool)</i>
QA/QC procedures to be applied	<i>Best practices in remote sensing and GIS</i>
Purpose of data	<i>This parameter represents the non-permanence risk rating of the project, which was used to determine the number of buffer credits that shall be deposited into the AFOLU pooled buffer account.</i>
Calculation method	<i>This parameter was calculated using the AFOLU Non-Permanence Risk Tool. All the risk factors described in the VCS Non-Permanence Risk Report were assessed.</i>
Comments	<i>N/A</i>

6.3 Monitoring Plan

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
TACKLING CLIMATE CHANGE	Monitoring actual carbon stock changes and GHG leakage	Near real-time visualization of the Project Area by using satellite images	PlanetScope: Use of the biggest satellite constellation (about 150 satellites), which collect daily images from any spot on the planet: generating continuous imaging generated from PlanetScope orthorectified images with a 3-meter spatial resolution, accessible with RGB viewing functionality in XYZ format, including levels 15 (4.77 meters spatial resolution) and 16 (2.38 meters spatial resolution) with an 8-bit radiometric resolution.	R: NovaAgri	Visual access for all the area of interest, with daily images. The archive of current images will be archived in the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface. The historical archive of daily PlanetScope images collected from 2017 onwards will also be available for visualization.	An image visualization routine will be implemented daily to check for any significant changes, minor alterations, or early signs of clearing, wildfires, or any other adverse conditions within the forested area.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
TACKLING CLIMATE CHANGE	Monitoring actual carbon stock changes and GHG	Dashboard - Web	Implementation and provision of a Web Dashboard integrated with a robust and structured database, allowing for the retrieval, organization, analysis, quantification, and presentation of information associated with the areas of interest from NovaAgri's database.	R: NovaAgri P.I.: SCCON (supplier)	This is delivered through an interactive panel and the generation of analytical reports directly from the dashboard, reinforcing the governance and transparency for the NovaAgri's projects. The archive of reports will be archived in both dashboard storage system and the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface.	The dashboard will be permanently active (24-7), making the consultation possible for anytime. The users will be configured accordingly to the function on the Project, making certain types of visualisation restrict for some people or parties.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
TACKLING CLIMATE CHANGE	Monitoring actual carbon stock changes and GHG	Deforestation Alerts	Utilization of continuous generation of deforestation alerts throughout the project period. The method involves crossing the fire alert system by INPE with the daily Planet satellite images ´ system in order to confirm the fire spots more precisely. Files provided by IPNPE contain information on forest fire and wildfire incidents at various intervals, ranging from annual to near real-time (10 minutes). These data are available in CSV and KML formats. The alerts will be generated and made available in shapefile format and published on the Web Dashboard for reference. The alerts include before and after images of the occurrence. The alerts cover the following classes: - Clear-cutting; - Access; - Degradation; - Burn scar. A perpetual and unrestricted license for the generated alerts will be provided for use, distribution, and modification.	R: NovaAgri P.I.: SCCON (supplier)	This is delivered through an interactive panel and the generation of analytical reports directly from the dashboard, reinforcing the governance and transparency for the NovaAgri’s projects. As soon as the system alerts a signal, an automatic message is delivered to the team responsible for monitoring. The archive of reports will be archived in both dashboard storage system and the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface.	The alerts will be continually active during the longevity of the project and connected to the permanent web dashboard. The alert system is programmed to detect any signals as immediately as the responsible team runs the images processing, by using geospatial tools and transmitting through the dashboard web.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
TACKLING CLIMATE CHANGE PROTECTING BIODIVERSITY INVOLVING COMMUNITIES	Monitoring actual carbon stock changes and GHG	Training and consciousness activities	Periodic training sessions will be conducted to facilitate effective actions for the protection of forested areas. These sessions will cover practical issues such as firefighting and safety, as well as more general topics like the challenges of climate change and the biodiversity richness. To achieve this, actions such as visual arts, informative brochures, social media posts, and occasional lectures during in-person visits to stakeholders, will be employed.	R: NovaAgri R: Landowner	All activities conducted during the defined crediting period will be documented in the monitoring reports. The documentation will include information such as the number of training sessions, photo records, post views, attendance lists, and other relevant details.	All records will adhere to the company's privacy policy to ensure the protection of third-party data. The mode of communication and any interactions with stakeholders will also have to comply with the Group's Code of Ethics.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
TACKLING CLIMATE CHANGE	Monitoring actual carbon stock changes and GHG	Monitoring LU/LC in the Project Area	In order to monitor and verify any changes in land use within the project area, the responsible team will continuously track land use/land cover (LU/LC) maps of the locality. This will involve utilizing internal information and images, cross-referencing them with external data from organizations that specialize in geoprocessing and are recognized as authorities in Brazil.	R: NovaAgri	Land use change maps and other internal information will be obtained through the web dashboard data, results from the processed satellite images on GIS tools, among others. Land use change maps and other external information will be obtained, for instance, from PRODES and MAPBIOMAS data. Any generated map will be saved and archived for further verification.	It is expected to ensure an annual photography of the LU/LC in order to follow the changes up and updated for future scenarios comparison.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
TACKLING CLIMATE CHANGE	Monitoring actual carbon stock changes and GHG Ex post calculation of net GHG emission reductions	Baseline reassessment	The baseline constructed for understanding the historical rate and also for the future projection of deforestation should be reassessed every six years, as required by the latest VCS Standard.	R: NovaAgri	In this case, all basis elements used for the construction of this baseline, such as primary data, geospatial data, empirical data, peer-review studies, LU/LC maps, and any other considerations, should be reevaluated and published on the agreed-upon dates in Section 3 of this PD, to ensure all relevant evidence for VVB verification.	The baseline will be reassessed compulsorily every 6 years, passing through all control and internal assurance steps as the first PD validation.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
GOVERNANCE AND INTEGRITY	Extra	Land and socioenvironmental compliance	NovaAgri will apply its Sustainability Policy continuously during the course of the project in order to ensure the best integrity for the project area. This includes assessing potential issues such as regarding IBAMA/SEMA Embargoes, debts to IBAMA, Conservation Units, Indigenous and public lands. All the assessment will be done during the project longevity through the current systems used by the proponent.	R: NovaAgri P.I.: SCCON (supplier) P.I.: VEGA (supplier)	Application of updated shapefile layers downloaded from official government and internal data for the generation of overlay maps, in order to alert for any other area that intersects with the project area, especially areas corresponding to public properties or areas designated for preservation/demarcation of any kind. The results will be presented on the systems conventionally used by NovaAgri (LYRA) and also in the Web dashboard. Then, the maps can be managed and edited in GIS software and tools. Any data exported for any reasons will be archived in the internal data storage of NovaAgri - implemented on OneDrive (Microsoft) interface.	The compliance verification is firstly made at the moment of projects feasibility analysis and updated periodically when: 1) any new areas are included into the project area; 2) any changes on documents are communicated by the landowner or observed in the official registry websites (e.g. CAR website); (3) any areas exceptionally excluded from the area belowed by the landowner for any reasons; (4) every time this risk analysis have to be ran by the credit/risks department.

ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
GOVERNANCE AND INTEGRITY	Extra	Legal binding agreement follow-up	The legal binding agreement signed between the project proponent and the project landowner will be periodically reviewed to ensure that all terms and clauses remain valid. It is expected that there will be no changes to the agreements made at the time of signing; however, any changes in legislation or force majeure issues may require the creation of contractual amendment.	R: NovaAgri	Any contractual amendment signed between the parties will be saved alongside the first version of the contract and make it available for the VVB verification.	The changes on contract will only happen in cases of substantial needs' occasions. The proponent's legal department is responsible for any revision, modification and amendment of the contractual draft.

GOVERNANCE AND INTEGRITY	Extra	Avoiding double counting of Credits' sales and/or retirement	The term of the agreement also regards to not engaging in any other remunerated or non-remunerated activities in the same Project Area, generating any credits/assets/values/rights, incurring to double-counting and causing negative impacts on the project and parties' integrity and credibility, under penalty incurring fines. The VERRA registry itself is an essential tool for verification of any credits being registered and retired, thus any unauthorized action will be promptly accused on the website.	R: NovaAgri R: VERRA	The contract is already signed at the time of submitting this PD, so any issues related to the generation and retirement of credits are legally covered through a formal agreement (available for the VVB). The monitoring of records on the VERRA Registry website will also be conducted continuously and reported in the monitoring reports.	Continuous monitoring for checking any needs of contract's updates or actions against double-crediting identified. The proponent's legal team will promptly act in case of any divergence or illegitimate course.
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ASPECT	TASK(s)	PROJECT ACTIVITY	TECHNICAL DESCRIPTION	RESPONSIBLE (R) AND PARTIES INVOLVED (PI)	DATA COLLECTION AND ARCHIVING PROCEDURES	QUALITY CONTROL AND ASSURANCE PROCEDURES
DELIVERING SOCIAL BENEFITS	Extra	Returning benefits for communities by allocating investments	The Grupo Progresso has stated that 100% of the projected financial results from its carbon credits' portion will be entirely allocated to: 1) Social investments in local communities, supporting a series of causes among the population of Piauí, and 2) Reinvesting in the project's activities themselves. All these social investments are genuinely voluntary and run beyond regulatory aspects or government requirements. This is better described in Section 1.12.5. NovaAgri is also committed to allocating its own returns from VCUs to fund and implement impactful social initiatives within the communities surrounding our project area beyond reinvesting in the Projects itself. The social actions may include, for instance, initiatives such as educational support, infrastructure development, access to natural resources etc.	R: NovaAgri R: Landowner	Any evidence regarding the credits pricing, commercialization, total values received, and the distribution of these values among the parties will be documented to enable the verification of the amounts related to NovaAgri's social investments. Any additional funds invested directly in social responsibility will also be considered in the reports, including allocations made by other receiving party (the landowner).	The proponent will rely on its specialized departments, including financial/tax and legal, to ensure a fully compliant process with reference standards. To ensure that the entire investment process is also carried out in a proper manner, due diligence will be conducted on the partners involved - who will benefit from the allocated benefits, as this concern is also essential for the reputation of the proponent and its economic group.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	ABSLRR t
Data unit	Ha
Description	Annual area of deforestation in the reference region at year t
Value applied:	4,457.75
Comments	The RR monitoring was done by using the MapBiomass Alert tool, which has alerts from institutions such as INPE, IMAZON, the University of Maryland, ISA, and others, publishing any loss of native vegetation identified and confirmed with high-resolution satellite imagery. MapBiomass Alert refers to deforestation indicators from several key sources, including DETER/INPE, SAD/IMAZON, GLAD from the University of Maryland, SIRAD-X/ISA, SAD Caatinga by UEFS-Geodatin, SAD Mata Atlântica by SOS Mata Atlântica-Arcplan, and SAD Pantanal by SOS Pantanal-ArcPlan. Of these, the DETER system is the most relevant for monitoring in our areas, as it covers both the Leakage Belt (RR) and Leakage Belt (LK). It is important to highlight DETER/INPE, which is highly responsive to changes in deforestation, degradation, and logging activity, with near-real-time updates provided daily on its platform, accessible at TerraBrasilis. This rapid response and comprehensive coverage make DETER an essential tool for maintaining up-to-date information on deforestation patterns in these areas. This alternative allowed for verification that no deforestation occurred within the project area during the period of: 01/jan/2024 to 22/oct/2023 (last update).
Data / Parameter	ABSLPA t

Data unit	Ha
Description	Annual area of deforestation in the project area at year t
Value applied:	8
Comments	Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the reference region. Through analysis of satellite images and geoprocessed data from deforestation alerts, the applied value was found for the Project Area.

Data / Parameter	ABSLK t
Data unit	Ha
Description	Annual area of deforestation in the leakage belt at year t
Value applied:	512.37
Comments	The RR monitoring was done by using the MapBiomass Alert tool, which has alerts from institutions such as INPE, IMAZON, the University of Maryland, ISA, and others, publishing any loss of native vegetation identified and confirmed with high-resolution satellite imagery. MapBiomass Alert refers to deforestation indicators from several key sources, including DETER/INPE, SAD/IMAZON, GLAD from the University of Maryland, SIRAD-X/ISA, SAD Caatinga by UEFS-Geodatin, SAD Mata Atlântica by SOS Mata Atlântica-Arcplan, and SAD Pantanal by SOS Pantanal-ArcPlan. Of these, the DETER system is the most relevant for monitoring in our areas, as it covers both the Leakage Belt (RR) and Leakage Belt (LK). It is important to highlight DETER/INPE, which is highly responsive to changes in deforestation, degradation, and logging activity, with near-real-time updates provided daily on its platform, accessible at TerraBrasilis. This rapid response and comprehensive coverage make DETER an essential tool for maintaining up-to-date information on deforestation patterns in these areas. This alternative allowed for verification that no deforestation occurred within the project area during the period of: 01/jan/2024 to 22/oct/2023 (last update).

Data / Parameter	$\Delta CADLK_t$
Data unit	tCO ₂ e
Description	Total decrease in carbon stocks due to displaced deforestation at year t
Value applied:	0
Comments	Through combination of satellite images and geoprocessed data from deforestation alerts, the $\Delta CADLK_t$ calculation was ran, considering the same biomass stock from the baseline, and the applied value was found for the Leakage Belt.

Data / Parameter	$\Delta CUDdPA_t$
Data unit	tCO ₂ e
Description	Total actual carbon stock change due to unavoided unplanned deforestation at year t in the project area
Value applied:	961
Comments	Through combination of satellite images and geoprocessed data from deforestation alerts, the $\Delta CADLK_t$ calculation was ran, considering the same biomass stock from the baseline, and the applied value was found for the Project Area.

Data / Parameter	RF _t
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Value applied:	22
Comments	This parameter was calculated using the AFOLU Non-Permanence Risk Tool. All the risk factors described in the VCS Non-Permanence Risk Report were assessed. There was no

change during this monitoring period, which is still part of the first baseline assessment.

7.2 Baseline Emissions

Find below the baseline emissions and/or carbon stock changes for the monitoring period in accordance with the applied methodology. The calculations in the emission reduction and removal were presented in the Sections 3 (Application of Methodology) and 5 (Quantification Of Estimated Ghg Emission Reductions And Removals).

Reference Region

(RR) baseline Projection		
Monitoring Period	Baseline deforestation (ha)	Cumulative deforestation (ha)
01-Jan-2024 to 01-Jan-2025	13.914	13.914

(RR) Monitoring				
Deforestation Forest (ha)	Deforestation Savanna (ha)	Deforestation Forest-Savanna Transition (ha)	Total Deforestation (ha)	Cumulative Deforestation (ha)
0	0	0	0	0

Baseline net carbon stock change of the reference region area		Total ex-post net carbon stock change of the reference region area	
PDD baseline		Monitoring	
tCO2e	tCO2e (cumulative)	tCO2e	tCO2e (cumulative)
950.776	950.776	0	0

Project Area

(PA) baseline Projection		
Monitoring	Baseline deforestation (ha)	Cumulative deforestation (ha)

Period
01-Jan-2024 to 01-Jan-2025
8
7,70

(PA) Monitoring				
Deforestation Wooded Savannah (ha)	Deforestation Savannah Forest (ha)	Deforestation Steppic-Savanna Forest (ha)	Total Deforestation (ha)	Cumulative Deforestation (ha)
0,00	7,70	0,00	7,70	0,00

Total net carbon stock change of the project area		Total ex-post net carbon stock change of the project area	
PDD baseline		Monitoring	
tCO2e	tCO2e (cumulative)	tCO2e	tCO2e
39.220	39.220	961	961

Leakage Belt

(LK) baseline Projection		
Monitoring Period	Baseline deforestation (ha)	Cumulative deforestation (ha)
01-Jan-2024 to 01-Jan-2025	7.499	7.499

(LK) Monitoring				
Deforestation Forest (ha)	Deforestation Savanna (ha)	Deforestation Forest-Savanna Transition (ha)	Total Deforestation (ha)	Cumulative Deforestation (ha)
12,2	500,2	0	512	0

Total net carbon stock change of the leakage belt area	Total ex-post net carbon stock change of the Leakage Belt
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PDD baseline		Monitoring	
tCO ₂ e	tCO ₂ e (cumulative)	tCO ₂ e	tCO ₂ e
482.941	482.941	33.708	33.708

7.3 Project Emissions

Find below the project emission and/or carbon stock changes for the monitoring period in accordance with the applied methodology. The calculations in the emission reduction and removal were presented in Sections 3 (Application of Methodology) and 5 (Quantification Of Estimated Ghg Emission Reductions And Removals).

Project year	Total carbon stock decrease due to planned activities		Total carbon stock increase due to planned activities		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock decrease change in the project case		Total ex ante estimated actual non-CO ₂ emissions from forest fires in the project area	
	ΔCPAdPat annual	ΔCPAdPat cumulative	ΔCPAiPat annual	ΔCPAiPat cumulative	ΔCUDdPat annual	ΔCUDdPat cumulative	ΔCPSPat annual	ΔCPSPat cumulative	EBBPSPat annual	EBBPSPat cumulative
	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e
01-Jan-2024 to 01-Jan-2025	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

7.4 Leakage Emissions

Find below the quantified leakage emissions for the monitoring period in accordance with the applied methodology. The calculations in the emission reduction and removal were presented in the Sections 3 (Application of Methodology) and 5 (Quantification Of Estimated Ghg Emission Reductions And Removals).

Project year	Total ex post GHG emissions from increased grazing activities	Total ex post increase in GHG emissions due to displaced forest fires	Total ex post decrease in carbon stocks due to displaced deforestation	Carbon stock decrease due to leakage prevention measures	Total net carbon stock change due to leakage within the leakage belt	Total net increase in emissions due to leakage
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	EgLKt annual	EgLK cumulative	EADLKt annual	EADLK cumulative	ΔCADLK _i annual	ΔCADLK cumulative	ΔCLPMLK _i annual	ΔCLPMLK cumulative	ΔCLKt annual	ΔCLK cumulative	ELK _i annual	ELK cumulative
	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e	tCO ₂ e
01-Jan-2024 to 01-Jan-2025	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

As a result, the leakage emissions equalled ‘zero’ because the monitoring¹²⁸ demonstrated that the deforestation in the leakage belt showed values below to those projected in the baseline.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Monitoring Period	Summary of GHG Emission Reductions and Removals							
	Baseline carbon stock changes		Ex post project carbon stock changes		Ex post leakage carbon stock changes		Ex post net anthropogenic GHG carbon stock changes	
	Annual tCO ₂ e	Cumulative tCO ₂ e	Annual tCO ₂ e	Cumulative tCO ₂ e	Annual tCO ₂ e	Cumulative tCO ₂ e	Annual tCO ₂ e	Cumulative tCO ₂ e
01-Jan-2024 to 01-Jan-2025	39.220	39.220	-	-	-	-	39.220	39.220

22%	Risk Factor parameter		
Ex post VCUs tradable			
Annual tCO2e	Cumulative tCO2e	Annual tCO2e	Cumulative tCO2e
30.592	30.592	8.628	8.628

Requested information about permanence risk assessment:

State the non-permanence risk rating (%)	22
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e the Long-term Average (LTA).	N/A
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☒ No
If a loss occurred (including a loss event or reversal), state the amount of tCO ₂ e lost:	There was no loss stated for the project area during the first monitoring period.

¹²⁸ Official monitoring report available on attachment.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification
<u>3.5.2</u>	Investments values	in order to protect sensible commercial or strategic information, the values presented above do not represent the actual numbers – all the items were

		shown in a range basis just to demonstrate the magnitude order of those investments
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APPENDIX 2: NON-PERMANENCE RISK REPORT (FROM VERRA HUB AFOLU NPR TOOL)

Last Update data: 24/01/2025 (PD v1.1)

Preview Report

X

Internal Risk

Project Management

Each project management risk factor set out below shall be assessed. Select the checkbox(es) for the risk factors and/or mitigation options that apply to the project and leave those that do not apply empty. Please upload any required documentary evidence using the link [🔗](#) provided to justify your selection. Then click on the 'Save' button at the bottom of the page before moving to the next section.

SCORE

Q1	Does the project have an adaptive management plan in place that includes a monitoring plan? If No, the project fails the risk assessment. If Yes, proceed to Q2.		Yes ☐
Q2	Do any of the following project management risks apply to the project ?		
☐	a	Species planted (where applicable) associated with more than 25 percent of the stocks on which GHG credits have previously been issued are not native nor proven to be adapted to the same or similar agro-ecological zone(s) as the project area.	2
☐	b	Ongoing enforcement to prevent encroachment by outside actors is required to protect more than 50 percent of stocks on which GHG credits have previously been issued.	2
☐	c	Management team does not include individuals with significant experience in all skills necessary to successfully undertake the project activities (i.e., any area of required experience is not covered by an individual with at least five years experience in that area).	2
☐	d	Management team does not maintain a presence in the country or is located more than one day of travel from the project site, considering all parcels or polygons in the project area.	2
☐	e	Management team has previously failed to submit a loss report within two years of detecting a loss event.	2
☐	f	ALM projects: Some or all the farmers participating in the project have not received training on the improved ALM practices implemented as part of the project or the monitoring and reporting procedures implemented during the crediting period.	2
☐	g	ALM projects: Some or all the farmers participating in the project are unaware of the potential for yields to decrease temporarily due to the transition to improved agricultural practices.	2
Q3	Do any of the following mitigations apply to the project?		
☒	h	Mitigation: Management team includes individuals with significant (i.e., more than five years) experience in AFOLU project design and implementation, carbon accounting, and reporting (e.g., individuals who have successfully managed projects through validation, verification, and issuance of GHG credits) under the VCS Program or other approved GHG programs.	-2
☐	i	Mitigation (ALM projects): A comprehensive training plan for all farmers participating in the project is in place, covering implementation of planned ALM practices, monitoring and reporting obligations and their potential costs.	-2
Total Project Management (PM) = [as applicable, (a + b + c + d + e + f + g + h + i)]			0
Total shall not be less than zero.			
Selected Identifiers = [Q1 , h]			

Financial Viability

The financial viability of a project is based on 1) the number of years until breakeven is reached (i.e., the payback period) and 2) the funding secured relative to what is needed to implement and operate the project until reaching breakeven. Select the radio button that corresponds to the correct response for each question. Also, please select the mitigation checkbox if it applies to the project. Please upload any required documentary evidence using the link [🔗](#) provided to justify your selection. Then click on the 'Save' button at the bottom of the page before moving to the next section.

SCORE

Q1	How long is the project's payback period (i.e., how many years will it take to breakeven)?	
☐	a	The payback period is greater than 20 years from the current risk assessment.
☐	b	The payback period is greater than 10 years from the current risk assessment, but less than or equal to 20 years.
☐	c	The payback period is greater than seven and up to and including 10 years from the current risk assessment.

Q3	Does the project have net positive community impacts? A response is only required if the answer to Q1 is Yes.		
☐	g	Net positive community impacts of project activities are not demonstrated.	8
☐	h	Net positive community impacts of project activities are demonstrated.	0
Q4	Do any of the following mitigations apply to the project?		
☒	i	Mitigation: Project is protected by a legally binding agreement (see Section 2.2.4) to continue management practices that protect the credited carbon stocks over the duration of the project crediting period.	-2
☒	j	Mitigation: Project is protected by legally binding agreement (see Section 2.2.4) to continue management practices that protect the credited carbon stocks for at least 100 years.	-4
☒	k	Mitigation: Where there is a potential for revenue loss compared to the most profitable alternative land use activity, project is a non-profit or has additional financial support (e.g., via grants, government funding, ecosystem services payments or SD ViSta assets) to overcome expected revenue loss.	-2
Total Opportunity Cost (OC) = [as applicable, (a, b, c, d, e, f, g or h) + (i or j) + k]			4
Total shall not be less than zero.			
Selected identifiers = [Q1 , a , i , k]			

Project Longevity

Project longevity is the number of years, beginning from the project start date, that project activities will be maintained. Project longevity must be a minimum of 40 years to pass the assessment. Please select the correct response to Q1. If you respond "No" to Q1, complete Q2 and Q3. In Q2, enter the project duration in years. Finally answer Q4. Please upload any required documentary evidence using the link provided to justify your selection. Then click on the 'Save' button at the bottom of the page before moving to the next section.



SCORE

Q1	Did the project request registration on or after 1 January 2024?	Yes v
Q2	Does the project have a legally binding agreement that covers at least a 100-year period from the project's start date? If Yes, the project is given a zero score for this risk category. If No, proceed to Q3.	No v
Q3	What is the project longevity in years? If you answered Yes to Q1 and the project longevity is less than 40 years, the project fails the risk assessment. If you answered No to Q1 and the project longevity is less than 30 years, the project fails the risk assessment. Please note that projects with project longevity of less than 40 years will be ineligible for the Core Carbon Principles label.	60 v
Q4	Does the project have a management, financial and monitoring plan for the entire project longevity? If No, the project fails the risk assessment. If Yes, proceed to Q5.	Yes v
Q5	Is the project an ARR or IFM project with harvesting? If No, proceed to Q7. If Yes, proceed to Q6.	No v
Q6	Can the project demonstrate a commitment to continue the management practice, replant or allow re-growth? If No, the project fails the risk assessment. If Yes, proceed to Q7.	Select v
Q7	Does the project have a legal Agreement or requirement to continue the management practice(s)? If No, proceed to a. If Yes, proceed to b.	Yes v
a	Without legal agreement or requirement to continue the management practice.	0
b	With legal agreement or requirement to continue the management practice.	10
Q8	Is the project a grouped project where contract durations with individual project activity instances are less than the project longevity?	No v

External Risk

Land Tenure and Resource Access/Impacts

Please select the appropriate checkbox(es) for the land/resource access/use rights and the ownership situation applicable to the project. Upload any required documentary evidence using the link [provided](#) to justify your selection. Then click on the 'Save' button at the bottom of the page before moving to the next section.



SCORE

Q1	Has due process been undertaken to discover any disputes over ownership and land, resource access, or use rights? If No, the project fails the risk assessment. If Yes, proceed to Q2.	Yes v	
Q2	Has the project executed a binding legal agreement(s) (such as a contract) securing the legal right to control and operate project activities over the entire project area with all entities that have verified ownership claims or verified land, resource access or use rights (such as customary rights holders)? If No, the project fails the risk assessment. If Yes, proceed to Q3.	Yes v	
Q3	Do the same or different entities hold land ownership and resource access or use rights?		
☒	a	Ownership and resource access or use rights are held by the same entity(ies).	0
☐	b	Ownership and resource access or use rights are held by different entity(ies) (e.g., the government owns the land, and the project proponent holds a lease or concession).	2
Q4	Is the project in a country/jurisdiction with a history of national, sub-national, or local government ("Government") intervention in land or resource use?		
☐	c	Government has previously expropriated significant areas of land (i.e., 10% or more) in the project area in the past 20 years.	10
☐	d	Government has previously changed land rights in the project's jurisdiction (e.g., cancelled, or blocked land titles, expropriated land or issued overlapping land titles) in the past 20 years.	5
☒	e	No instances of Government intervention in the past 20 years or specific instances of expropriation and government intervention in land rights in the project area have been conclusively resolved against the government in a court of competent jurisdiction.	0

Q5	What percent of the project area is affected by disputes over land tenure or ownership?		
☐	f	Disputes exist in more than 5 percent of the project area.	10
☐	g	Disputes exist in up to and including 5 percent of the project area.	5
☒	h	No disputes exist.	0

Q6	What percent of the project area has disputes over access or use rights?		
☐	i	Disputes exist in more than 5 percent of the project area.	10
☐	j	Disputes exist in up to and including 5 percent of the project area.	5
☐	k	No disputes exist.	0

Q7	Have the risks of upstream and sea impacts undermining the carbon stocks of a WRC project been demonstrated as insignificant or effectively mitigated for the ten years that follow the risk assessment?		
☐	l	Potential upstream and sea impacts are not demonstrated to be insignificant or effectively mitigated.	5
☒	m	Potential upstream and sea impacts are demonstrated to be insignificant or effectively mitigated.	0

Q8	Do any of the following mitigations apply to the project?		
☒	n	Mitigation: Project area is protected by a legally binding agreement (e.g., a conservation easement, conservation servitude or protected area) to continue management practices that protect carbon stocks for the duration of the project crediting period.	-2
☐	o	Mitigation: Where disputes over land tenure, ownership or access or use rights exist, documented evidence is provided that demonstrates the project is taking action to try to resolve the disputes or clarify overlapping claims.	-2

Total Land Tenure (LT) = [as applicable, ((a or b) + (c, d, or e) + (f, g or h) + (i, j or k) + (l or m) + n + o)]
Total shall not be less than zero. 0

Stakeholder Engagement

Stakeholder engagement shall be assessed for projects where local populations, including those living within or surrounding the project area (within 20 km of the project boundary), are reliant on the project area, such as for essential food, fuel, fodder, medicines or building materials. Please select the checkbox(es) for the risk factors and/or mitigation options that apply to the project and leave those that do not apply empty. Upload any required documentary evidence using the link [provided](#) to justify your selection. Where local populations are not reliant on the project area, the risk is not relevant to the project, and you may skip this section. Click on the 'Save' button at the bottom of the page before moving to the next section.

SCORE

Q1	Are local populations, including those living in the project area or within 20 km of the boundary of the project area, reliant on the project area? If No, the risk rating for stakeholder engagement is zero and you can proceed to political risk. If Yes, proceed to Q2.		
Yes v			

Q2	Have more or less than 50 percent of stakeholders living within and reliant on the project area been consulted?		
☐	a	Less than 50 percent of stakeholders living within the project area and who are reliant on the project area have been consulted.	10
☒	b	More than 50 percent of stakeholders living within the project area and who are reliant on the project have been consulted.	0

Q3	Have more or less than 20 percent of stakeholders living outside the project area within 20 km of the project area, and who are reliant on the project area been consulted?		
☐	c	Less than 20 percent of stakeholders living outside the project area within 20 km of the project area, and who are reliant on the project area have been consulted.	5
☒	d	More than 20 percent of stakeholders living outside the project area within 20 km of the project area, and who are reliant on the project area have been consulted.	0

Total Stakeholder Engagement (SE) = [as applicable, (a or b) + (c or d)]
Total shall not be less than zero. 0

Selected identifiers = [Q1 , b , d]

Political Risk

The governance score of the country in which this project is located is already populated here. It is calculated from the mean of governance scores across the six indicators of the World Bank Institute's Worldwide Governance Indicators (WGI), averaged over the most recent five years of available data. Please select the mitigation checkbox if it applies to the project. Then, click on the 'Save' button at the bottom of the page before moving to the next section.

SCORE

Q1	What is the governance score for the country?	-0.26
a	Governance score of less than -0.79	6
b	Governance score of -0.79 to less than -0.32	4
c	Governance score of -0.32 to less than 0.19	2
d	Governance score of 0.19 to less than 0.82	1
e	Governance score of 0.82 or more	0
Q2	Does the following mitigation apply to the project?	
☒	Mitigation: The project is in a country that: 1) is party to the Paris Agreement and has submitted an NDC to the UNFCCC Secretariat in the last five years, 2) includes AFOLU commitments (conditional or unconditional) in its NDC, and 3) has a documented and active climate change plan that includes the project activity.	-2

Total Political (PC) = [(a, b, c, d or e) + f]

Total shall not be less than zero.

0

Natural Risk

Natural risk is based on historic natural risk and risks from projected future climate change and sea-level rise impacts. Historic natural risk is based on likelihood (i.e., the historical average number of times the event has occurred in the project area over the last 100 years) and significance (i.e., the average size of each event). Any significant natural risk (i.e., a risk affecting more than 5 percent of the project area) that has occurred over the past 100 years in the project area shall be considered applicable to the project. The frequency and significance of events shall be estimated based on historical records, probabilities, remote sensing data, peer-reviewed scientific literature and/or documented local knowledge, such as survey data in the project areas. Please select the natural risk's historical significance, likelihood, and relevant mitigation factor(s). Then select "yes" or "no" where default responses are not already entered, to indicate if the natural risk category is affected by climate change. If there are additional natural risks than those listed, please add up to three more natural risks by clicking on the "Add New Risk Factor" button below. All AFOLU projects shall assess fire, pest and disease outbreaks, extreme weather events such as droughts and hurricanes, and geological risks such as earthquakes and volcanoes, at a minimum.

Natural Risk Assessment

Fire (F) Historical Significance Major (25 percent to less ... Historical Likelihood More than once every 10 ... Mitigation Factor Both of the above Affected by Climate Change? Yes	Pest and disease outbreaks (PD) Historical Significance Not applicable Historical Likelihood Once every 100 years or l... Mitigation Factor None of the above Affected by Climate Change? Yes	Extreme weather (W) Historical Significance Insignificant (less than 5 ... Historical Likelihood Once every 50 to less tha... Mitigation Factor Demonstrated history of ... Affected by Climate Change? Yes	Geological risk (G) Historical Significance Not applicable Historical Likelihood Once every 100 years or l... Mitigation Factor None of the above Affected by Climate Change? No
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Future Climate Impact

What reference region does your project belong to?

Reference Region	Percentage of project area
N.E.South-America	100

DESCRIPTION	TYPE	CATEGORY INDEX	PROJECTED CHANGE VALUE	CID IMPACT SCORE	CONFIDENCE LEVEL & SIGN OF CHANGE	UNCERTAINTY-WEIGHTED CID IMPACT SCORE	PROJECT IMPACT
		Mean air temperature	1.7	1	High confidence of increase	1.00	Negative
	Heat and cold (hc)	Extreme heat	22.5	1	High confidence of increase	1.00	Negative
		Cold spells	2.2	2	High confidence of decrease	2.00	Does not apply

Future Climate Impact	Wet and dry(wd)	Mean precipitation	4	1	High confidence of decrease	1.00	Negative
		River flood	0.05	4	Low confidence in direction of change	2.00	Negative
		Landslide	N/A	N/A	NaN	0.00	Does not apply
		Soil Moisture	-1	2	Medium confidence of increase	1.50	Does not apply
		Agricultural and ecological drought	6	5	Low confidence in direction of change	2.50	Negative
		Fire weather 1	N/A	2.67	High confidence of increase	2.67	Negative
	Wind(w)	Mean wind speed	2.2	2	Medium confidence of increase	1.50	Negative
		Tropical cyclone	N/A	N/A	NaN	0.00	Does not apply
Projected Future Climate Impact Level Average						2.39	

Adaptive Capacity ¹

Can you demonstrate that the project meets at least five adaptive capacity criteria?

No

ACTION	CRITERIA	ADAPTIVE CAPACITY
☒	1. Variety	(i) Availability of a range of policy options to address climate change that apply directly to the project scope or activity (ii) Participation of different actors, levels, and sectors in the climate change governance process
☐	2. Learning Capacity	(i) Ability to address future climate events and learn from past experiences to improve routines regarding climatic experiences (ii) Evidence of changes in assumptions that support actions regarding climate change experiences.
☐	3. Room for change	(i) Continuous access to climate information at different spatio-temporal scales (e.g. early warning systems, climate change projections)
☒	4. Leadership	(i) Long-term vision on climate change (ii) Stimulus for action, entrepreneurship, and collaboration from different stakeholders
☐	5. Resources	(i) Availability of climate change expertise, knowledge and human labor force in the field of environmental sciences (ii) Availability of financial resources to support policy measures and financial incentives to address climate change
☒	6. Fair governance	(i) Evidence of policies, practices, and behaviors that address climate change effectively. (ii) Availability of accountability procedures to implement climate change orientated-actions
☐	7. Innovation	(i) Availability of novel solutions to reduce the threat or take advantage of climate change.

Future Climate Impact ¹

What reference region does your project belong to?

Reference Region

Percentage of project area ¹

N.E.South-America

100

+

DESCRIPTION	TYPE	CATEGORY INDEX	PROJECTED CHANGE VALUE	CID IMPACT SCORE	CONFIDENCE LEVEL & SIGN OF CHANGE 1	UNCERTAINTY-WEIGHTED CID IMPACT SCORE	PROJECT IMPACT
SLR Impact Level	Coastal(co)	Coastal flood	2.7	5	High confidence of increase	5.00	Does not apply
		Coastal erosion	-150	4	High confidence of increase	4.00	Does not apply
SLR Impact Level Average						4.50	

SLR CID Assessment

SLR risk is assessed only if the project is in a coastal area. Please leave this section blank if the project is not in a coastal area. The Future Climate Impact section above must be completed prior to this assessment. [?](#)

CATEGORY DESCRIPTION	SIGNIFICANCE LEVEL
Ecosystem degradation ?	No degradation
Coastal flooding ?	Without flooding
Coastal Erosion ?	No erosion
Degree of salinization ?	Without saline intrusion

SLR Risk Assessment

Adaptation [?](#)

None of the above

OVERALL SIGNIFICANCE LEVEL ?	Not Applicable	SLR IMPACT LEVEL	5
SLR RISK SCORE ?	0	ADAPTATION SCORE ?	1
SUB-TOTAL RISK SCORE (SLR RISK SCORE * ADAPTATION SCORE)			0.00